Institute of Astronomy

Natural Sciences Tripos

Part III/MASt Astrophysics

Research Project Booklet

2025/26

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Part I: Key Information

Key Dates

Type	Milestone	When?		Format	Length	Where?
		Time	**Date**			
Submission	Interim Progress Report	12:00 noon	Last day of Full Michaelmas Term	PDF - LaTeX/equivalent	1,000 words	Moodle
Submission	Practice Presentation	12:00 noon	Last Monday of Full Lent Term	Microsoft PowerPoint	N/A	Moodle
Oral	Practice Oral Presentation	TBC	Last Tue. – Fri. of Full Lent Term	N/A	30 minutes	Hoyle Committee Room
Submission	Draft Final Report to Supervisor I	N/A	Last day of the Easter Vacation	PDF - LaTeX/equivalent	30 pages	Various
Submission	Final Report	12:00 noon	2nd Monday of Full Easter Term	PDF - LaTeX/equivalent	30 pages	Moodle
	Oral Presentation Slides			Microsoft PowerPoint	N/A	
Oral	Oral Presentation	TBC	2nd Tue. – Fri. of Full Easter Term	N/A	30 minutes	Hoyle Committee Room

The Part III/MASt Calendar

We **strongly advise** adding this calendar to your own, as it will be updated throughout the year with important events, deadlines and more.

- To add to an existing Google Calendar, click here.
- If you use another calendar, right click, copy and paste this link into your own calendar. Please note, simply clicking on the link will download a calendar file or import a copy directly into your calendar (depending on the software you use) but the information will not update as changes are made.

Format

The Written Report

Formal Requirements

- The Interim Report must not exceed 1,000 words in length
- The Final Report must not exceed 30 pages in length, including The Abstract, Figures, Tables, References, and Appendices (where applicable)
- The minimum acceptable font size is 11pt with at least single line spacing
- The text must be in single column format
- Margins must not be less than 2cm
- Figures must be legible when printed on A4 paper

Projects not meeting these requirements will be returned for revision and a penalty may apply for late submission at the discretion of the examiners.

Content

The Interim Progress Report

The Interim Progress Report, which may contain material that can be incorporated in the final project report, must indicate the progress made so far and show preliminary results. It should also give a clear indication of the project aims and a detailed plan of how these aims will be achieved. This is particularly important where the results of the project depend on data that has yet to be analysed.

There is no need for the Interim Progress Report to reiterate the material given in the Project Description.

The Final Report

The Final Report should read as a self-contained document, presented in the style of a scientific research report or paper in a scientific journal. The main sections of the report will describe the work undertaken, the results obtained and an assessment of their significance.

An Abstract, Introduction, Conclusions, References and appendices (where applicable) should also be included within the 30-page limit. Supporting Figures and Tables should be used both as an aid in presenting data and results and also to enhance the clarity of the submission. In some circumstances an appendix containing more extensive tabular material/results may be included.

The reader should be able to understand the context in which the investigation was undertaken, the main features of the project, the results and how they relate to the advancement of the subject. In

addition to the descriptive material, questions a report would be expected to address include, "Why were particular approaches adopted?" – back of the envelope calculations will often be helpful and relevant – "What has been learnt?" and "What information/work would have helped us to learn more?" You should take care to demonstrate that you have tested any analysis packages/codes that you use.

Structure

You will have read many research papers in the course of your project research. Some of these you will have found easy to read and others less so. You therefore already have a good idea of what makes a good write-up.

The Final Report should be logically structured, clear, and complete, while remaining concise. It is usual to set out the write-up in sections that include an introduction, a description of methods, results, and conclusions. The introduction should set out the problem to be solved, including why it is interesting, and previous work done. The methods section should describe what you have done in sufficient detail that the work can be reproduced by a reader. It is important to make clear what new work you have done yourself in this section. In the results section describe what you have found. Try to make it very clear which are the most interesting outcomes of the project. In the conclusion explain whether or not you have solved the problem you set out to solve. If so, explain how and if not then why not. You can also describe future work that might get closer to or verify your solution.

There are some points to particularly bear in mind:

1. Remember that your readers may not be experts in the field of your project. Begin your description from basic physical principles and describe how any observations have been made.
2. Write short sentences. Long and convoluted sentences, with numerous sub-clauses, are hard to read and often grammatically incorrect.
3. Use named references, such as (Eggleton, Fitchett and Tout 1989), in the text. This is the style generally used by astronomers. It is much easier to read than a number reference style that requires continual cross-referencing.
4. Be concise. Well-written reports do not need to fill the page limit.
5. Include a limited number of pertinent figures. A good figure can replace many words, but many similar figures can often be replaced by a few words. Ensure that axes are labelled properly, lines are sufficiently thick, that points and labels are in a large enough font and that the main details of the figure are explained in the caption. Avoid making figures too cluttered and do not include anything that is not relevant to your discussion.
6. Appendices are for additional reading only. The examiners will base their marking on the main report.

The Oral Presentation

Formal Requirements

- The Oral Presentation must consist of a 20-minute description of the project followed by up to 10 minutes of questions from the Part III/MASt Examiners.
- The written presentation must be submitted as a Microsoft PowerPoint (.pptx) file.
- If your Project's Supervisor, Co-Supervisor or University Teaching Officer has been appointed as the Senior Examiner or one of the Examiners, then to mitigate against any conflicts of interest, they will not ask you any questions during your oral presentation.

Referencing and Plagiarism

It is a requirement that the project investigation and the project report are both the work of the candidate alone and no form of collaboration is allowed.

It is a fundamental tenet of scientific research that due acknowledgment is given to the work and ideas of others that form the basis of, or are incorporated in, a research presentation. You must always acknowledge the source of an idea or material you use with a specific reference. Plagiarism, including the use of another individual's ideas, data, or text, is regarded as an extremely serious disciplinary offence by the University. For further information, please see the University's guidelines on Plagiarism and Academic Misconduct.

Guidelines for using Artificial Intelligence-powered tools

"A student using any unacknowledged content generated by artificial intelligence within a summative assessment as though it is their own work constitutes academic misconduct unless explicitly stated otherwise in the assessment brief." Plagiarism and Academic Misconduct

If you decide to use AI-powered tools for the report preparation you must state clearly which portions of the report were created with AI assistance. Note that even advanced versions of these tools can make mistakes and provide biased information. If you choose to use AI agents, it is your responsibility to verify the facts supplied and correct the logic underlying the arguments provided. Ultimately, you are being marked for your effort, so the AI-generated content can only be used to supplement your work not to substitute it. The core of the report must be based on your own ideas, your own analysis, your own conclusions and your own understanding of the topic. Finally, the above guidelines should not be taken as an endorsement or recommendation for use of these tools in your work.

Assessment

The research project element of the NST Part III Astrophysics course accounts for 6 assessment units (equivalent to the marks assigned to two 24-lecture Mathematics Part III lecture courses).

Distribution of Marks

The Interim Progress Report

The Interim Progress Report does not constitute part of the formal assessment but is regarded as an essential part of the monitoring procedure. The Course Coordinator will assess these reports and provide feedback to students and supervisors.

The Mock Oral Presentation

This practice presentation, delivered to an audience including the Part III/MASt Course Coordinator, Project Supervisors and UTOs, is not formally assessed but offers the opportunity to become familiar with the format of the presentation to be assessed by the Part III/MASt Examiners in the Easter Term. Students are strongly encouraged to attend the practice talks of their peers to further develop their understanding of and exposure to various presentation techniques.

The Final Draft Report

The Final Draft Report does not constitute part of the formal assessment but provides students with an opportunity for their work to be read by their Primary Project Supervisor, who will provide feedback during the final supervision.

The Final Report

The research project written report counts for 5 out of the total of 6 units of assessment assigned to the research project i.e., approximately 85% of the marks for the research project will be assigned on the basis of the written final report.

The Final Oral Presentation

The research project oral presentation counts for 1 out of the total of 6 units of assessment assigned to the research project i.e., approximately 15% of the marks for the research project will be assigned on the basis of the assessed oral presentation.

Examiners Criteria for Marking the Project Report and Oral Presentation

If your Project's Supervisor, Co-Supervisor or University Teaching Officer has been appointed as the Senior Examiner or one of the Examiners, then to mitigate against any conflicts of interest, they will not ask you any questions during your oral presentation, and they will also not mark your report.

The Examiners will award marks under three broad headings:

I. Scientific understanding

II. Quality of the research

III. Presentational and communication skills.

The format and timetable for submission form part of the Examination process. In their assessment of the project, the Examiners will take account of any breaches of the guidelines, including exceeding the word limit and late submission of the report.

Oral Presentation

The Examiners assessment will take into account the following:

- Visual Material: including relevance, clarity, attractiveness
- Oral Presentation: including overall structure, clarity, time keeping
- Response to Questions: including grasp of subject material, precision of answers

Written Project Report

- The Examiners will assess the report under the following headings:
- Overall structure and clarity of the report
- Planning, organisation, and prosecution of the research
- Understanding of the physics and the general scientific content
- Technical proficiency
- Analytical and Interpretational skills
- Significance of the results

Extension Requests and Illness

The Institute of Astronomy expects students to meet the advertised deadlines for the submission of all coursework, to ensure fairness to all students taking the course and allow prompt marking by the Department.

Your Part III Project makes up one third of total available Tripos marks and is subject to special arrangements.

The Final Report and Final Presentation Slides are linked with compulsory attendance at the Final Oral Presentation. As such, it is not possible for the Department to grant an extension of more than 48 hours. The following policy builds around point 7 of the University's Policy on managing extensions to Dissertations and coursework policy.

Extension Requests for the Final Report and/or Final Presentation Slides of up to 48 hours

Students may self-certify via this form for an extension of up to 48 hours from the original submission deadline.

Requests must be submitted **no later than 7 days before the original submission deadline** to allow for potential amendments to be made to the Final Oral Presentation schedule.

Extension Requests for the Final Report and/or Final Presentation Slides of over 48 hours

Late submission of over 48 hours is a particularly serious matter, and will require your College to apply to the Examination Access and Mitigation Committee (EAMC) to seek permission for any credit to be allowed. If this is not granted, the late work will not be counted.

The College **must** apply to the Secretary of the EAMC for an extension to a deadline and should **not** approach the Chair of Examiners, Course Coordinator or Senior Examiner directly.

Applications must be submitted **no later than 7 days before the original submission deadline** to enable the Secretary to consult the relevant Chair of Examiners or, where applicable, the Senior Examiner.

Penalties for late submission

You are required to self-certify prior to the original submission deadline (or at the point of submission at the very latest). Retrospective requests will not be accepted.

Where a student does not self-certify and does not submit their Final Report and Final Presentation Slides by the original submission deadline, zero marks will be awarded.

Where a student has self-certified and does not submit their Final Report and Final Presentation Slides by the revised submission deadline, zero marks will be awarded unless a further extension has been granted by the EAMC (see below).

Where a student has applied for a further extension via the EAMC and does not submit their work by the agreed submission date, zero marks will be awarded.

Illness during the Oral Presentations

The department defers to University Policy with regards to illness during an examination, including the Final Oral Presentation.

For full details, see the Policies and Guidance for Students here.

Part II: The Project Descriptions

It is my pleasure to present the Research Project Booklet for the academic year 2025-2026, made possible thanks to generous and enthusiastic contributions of the Cambridge Astronomers.

Each of the projects on offer this year is only described briefly, but combined together these will add up to a lot of information-rich and often rather technical text to digest. I encourage you to take your time and use the weeks before the start of term to read carefully through the project descriptions, taking advantage of the references provided.

If you are interested in a project, please email the supervisors with your questions. It will also give the supervisors a chance to identify the most interested students. Many supervisors will also organise meetings with the students who have expressed interest, be it in person or remotely.

As usual, there are more projects on offer than there are students in the cohort, and the entire spectrum of the Cambridge Astronomy research is represented, so everyone should be able to find a project that interests and suits them the best. Although there are now roughly twice as many projects as students, experience shows that preferences often cluster around a small number of popular topics. As a result, it may not be possible to allocate every student their first choice.

I will ask you to submit a ranked list with your top ten projects in October, just before the beginning of the Michaelmas term. Supervisors will also be invited to express their preference of student if they wish to do so. Student and supervisor preferences will then be fed into a complex computer code which is designed to maximise overall satisfaction (with high weight assigned to student satisfaction). The project allocations will be announced a week after the preferences are collected.

Based on experience, my inkling is that there is more than just one project that would work great for you, but how to find them? Most importantly: be open-minded. This is your chance to learn first-hand what cutting-edge astrophysics is like. Pay less attention to the flavour/topic of research and more to the type of work involved. Projects involving the early Universe, exoplanets, black holes, stars, galaxies, dust and high energy particles will all have something in common: you will be asked to run simulations, analyse data and apply statistics to interpret your results. The absolute majority of our projects require familiarity with at least one programming language - most likely it will be Python, sometimes C/C++.

Singling out one particular project based on a set of fine-grain selection criteria is not a wise strategy, and such considerations will be impossible to satisfy given that the student-project ratio is close to unity. Instead use coarse-grain selection criteria to identify multiple projects you would be excited about, projects that will give you a chance to learn something new.

Please feel free to email me if you have any questions.

Vasily Belokurov, Part III/MASt Astrophysics Course Coordinator, Michaelmas Term 2025
Email: vasily@ast.cam.ac.uk

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Project 1: Testing General Relativity with Gravitational Waves: phenomenology to theory

Supervisor I: Dr Christopher J. Moore (cjm96@cam.ac.uk)

UTO: Dr Christopher J. Moore (cjm96@cam.ac.uk)

Project summary

The dawn of the new era of gravitational wave (GW) of astronomy has created new possibilities for testing fundamental physics, particularly general relativity. But the tests that can be performed have so far been limited by a lack of available predictions in good alternative theories. One possibility that is just starting to become available is to use simulations of merging boson stars; these are compact objects made of a scalar field that can effectively mimic a black hole. This project will explore how effective GW tests of general relativity are by using state-of-the-art simulations of merging boson stars produced by collaborators at DAMTP.

Background

Gravitational waves (GWs), detected for the first time in 2015, provide a new way to observe the strong and dynamical gravitational fields around merging black holes (BHs). They offer many completely new opportunities for probing fundamental physics in regimes inaccessible to all other methods and are an ideal tool for testing the general theory of relativity [1,2].

General relativity (GR) is our best theory of gravity. However, it is known to be incomplete from a theoretical standpoint and a number of outstanding observational puzzles (such as the nature of dark matter and dark energy, and tensions in the standard model of cosmology) motivate searching for observable deviations from the theory with current observations.

Theoretical studies of alternative theories of gravity typically focus on specific theories (or classes of theories) motivated from high-energy physics considerations or with particular mathematical properties, such as well posedness, which make them amenable to numerical simulation. A long-term goal of such work is to understand a theory well enough to be able to model the GW signal produced by the merger of two compact objects in the theory accurately using numerical simulations. And to be able to do this for large numbers of systems in order that they can be compared with GW data. For most theories this is currently not possible. And, to make the situation worse, because we don't have just one good candidate theory to replace GR, we would need to carry out this research programme for a large number of different candidate theories.

Therefore, a lot of effort has focussed on phenomonological tests of GR. To take just one example, the propagation of GWs can be tested by guessing a simple parameterised form for the GW dispersion relation, $k(\omega)$, and fitting this to the data. (In GR, the dispersion relation for all massless particles is simply $k=\omega/c$.) Such crude phenomenological tests are not motivated by any specific alternative theory. But that's the point; in the absence of any good theoretical guide for what sort of deviation from GR to

expect, the hope is that this sort of test will be reasonably sensitive to any theory where GWs don't all travel in the normal way. The LIGO and Virgo collaborations have developed a suite of standard phenomenological tests along these lines that are routinely applied to all suitable binary BH detections [3,4,5]. (This project will focus on tests for binary BHs because there are many more of them than binaries involving one or more neutron stars.) The hope is that, taken together, this suite of standard tests will allow us to tell if GR fails to describe our observations, even without us knowing what the correct theory is.

It is important to understand the limits of this approach. If the real universe deviates from GR in a specific (unknown) way, will these phenomenological tests be able to tell? The answer will depend on the specific deviation from GR considered, the number and type of GW events observed and the quality of the observations. This project aims to study this important question using state-of-the-art simulations of binary boson star mergers as a case study.

Boson stars are stable, self-gravitating objects formed from a scalar field which is posited to exist. Boson stars can be extremely compact (i.e. similar in size to a BH of the same mass) and can effectively mimic BHs for many observations, including the GWs produced when two boson stars merge. This project will use state-of-the-art numerical relativity simulations of the mergers of binary boson stars produced by collaborators in DAMTP [6]. By injecting a number of these simulations into instrumental noise and analysing the signals using the standard suite of tests, this project will answer the question posed above in the special case where all the events that we think are binary BHs are actually binary boson stars. Note that the motivation for using boson stars is not because it is considered likely that they really are producing the signals we see; rather, boson stars provide a concrete example of deviation from the standard picture of BHs in GR for which highly accurate simulations have recently become available [7] and can be used as a case study for general deviations from GR.

During this project, the student will learn GW data analysis using the standard BILBY code library. The student will also learn about testing fundamental physics with GWs, including performing many of the most widely used tests for themselves. The student will also become familiar with using the outputs of numerical relativity simulations and have the opportunity to collaborate with the group in DAMTP producing these simulations.

Project details

1. Literature review. Understand the suite of standard phenomenological tests.

2. Test selection. It will likely not be possible to perform all the standard tests within the time available. Therefore, it will be necessary to prioritise tests. This decision may be based on the different time/resources needed for each test and on the physics interests of the student.

3. GW data analysis. Learn how to inject a signal from a numerical simulation into simulated noise and, using BILBY [8], performing GW parameter estimation with binary BH models.

4. Catalogue construction. In consultation with collaborators in DAMTP, decide on a set of binary boson star parameters to use. Use the numerical simulations of these systems to produce a mock catalogue of GW observations by injecting into simulated noise.

5. Testing GR. Apply the selected tests to all simulations in the mock catalogue.

6. Extra tests. Time permitting, investigate the other standard tests.

7. Report Write up.

Skills required

General Relativity, e.g. at the level of the part II course. Data analysis experience with Python (no experience with Bilby is required). Some familiarity with Bayesian inference preferable, but not essential.

Useful References (List of important papers/review articles relevant to the project)

- [LIGO Scientific Collaboration & Virgo Collaboration (2016), “Tests of general relativity with GW150914” Phys. Rev. Lett. 2016, 116, 221101, arXiv:1602.03841]

General References (List of papers referred to in the project)

---General Background---

[1] N. V. Krishnendu & Frank Ohme (2022), “Testing General Relativity with Gravitational Waves: An Overview”, arXiv:2201.05418

[2] C. Will (2005), “The Confrontation between General Relativity and Experiment” Living Rev. Rel. 9 3 (2014), arXiv:1403.7377

---Collaboration Tests of GR with GWs---

[3] LIGO Scientific Collaboration & Virgo Collaboration (2016), “Tests of general relativity with GW150914” Phys. Rev. Lett. 2016, 116, 221101, arXiv:1602.03841

[4] LIGO Scientific Collaboration & Virgo Collaboration (2019), “Tests of general relativity with the binary black hole signals from the LIGO-Virgo catalog GWTC-1”, Phys. Rev. D 100, 104036, arXiv:1903.04467

[5] LIGO Scientific Collaboration & Virgo Collaboration (2021), “Tests of General Relativity with GWTC-3”, arXiv:2112.06861

---Boson stars and merger simulations---

[6] T. Evstafyeva, R. Rosca-Mead, U. Sperhake & B. Brugmann (2023), “Boson stars in massless and massive scalar-tensor gravity”, Phys. Rev. D 108 10 104064, arXiv:2310.05200

[7] T. Evstafyeva, U. Sperhake, I. Romero-Shaw & M. Agathos (2024), “Gravitational-wave data analysis with high-precision numerical relativity simulations of boson star mergers”, arXiv:2406.02715

---Code libraries---

[8] Link https://lscsoft.docs.ligo.org/bilby/ [see also, G. Ashton et al. (2019), “BILBY: A user-friendly Bayesian inference library for gravitational-wave astronomy”, Astrophys. J. Suppl. 241 2 27, arXiv:1811.02042

Project 2: The spatial distribution of GW sources

Supervisor I: Dr Christopher J. Moore (cjm96@cam.ac.uk)

UTO: Dr Christopher J. Moore (cjm96@cam.ac.uk)

Project summary

Are gravitational wave sources isotropically distributed on the sky? Given that LIGO and Virgo now routinely detect GW signals out to distances $z \lesssim 1$ the cosmological principle would certainly suggest that they should be. The upcoming GW catalog, expected to include several hundred sources, presents an opportunity to test this with unprecedented precision. However, a unique challenge of GW astronomy is that source locations are typically not well constrained, and it will be necessary to develop techniques to handle the wide array of complex sky maps found throughout the catalog.

Future GW detectors promise to detect GW sources in ever greater numbers and it may eventually become possible to use their locations as tracers of the underlying distribution of galaxies and of large-scale structure. Furthermore, the subtle effects of gravitational lensing can leave anisotropic imprints on the distribution of sources. This project can be extended to consider the prospects for detecting such effects in future catalogs.

Background

GW sources - mostly binary black hole (BH) mergers but also including several mergers involving neutron stars (NSs) - are expected to be produced in galaxies, either by binary stars or in dense stellar environments. Therefore, the distribution of GW sources is expected to follow the underlying distribution of galaxies, which in turn should be homogeneous and isotropic on sufficiently large scales.

This is difficult to test directly because the 3-dimensional localisation of GW sources is typically poorly constrained. A typical GW sky map is shown in Figure 1. A notable exception is the binary NS merger GW170817 which was accompanied by a gamma ray burst and a kilonova allowing it to be confidently associated with the elliptical galaxy NGC 4993. The third Gravitational Wave Transient Catalog (GWTC-3 [1]) contains around 100 observations from the LIGO and Virgo detectors. The next GWTC-4 catalog, which is expected to be published around the time work on this project starts, is expected to approximately double the number of sources. The next generation of GW detectors, such as the Einstein Telescope and Cosmic Explorer, will detect a GW event once every few minutes, quickly producing catalogs containing 105-106 events [4].

Given a sufficiently large catalog of GW signals, it should be possible to use the localisations of the events as a tracer of the underlying distribution of galaxies. Similarly, small effects such as gravitational lensing of source by the Sun might be expected to leave an anisotropic imprint on the source distribution. These questions have not been studied in detail and there are many open questions. For example, it is not known what the prospects are for being able to detect such effects, how large a

catalog might be required, or what data analysis techniques may be needed to handle the uncertain sky locations during the analysis.

Project details

In the very early days of GW astronomy, back when we only had 10 GW observations, a first attempt to quantify the isotropy of the GW source population was made [2]. This first attempt was limited by the small number of sources. The first part of this project will be to reproduce the study in Ref. [2] using the greatly expanded number of sources now available.

The second part of the project will consider what will become possible with future catalogs and investigate how the results scale with the number of events in the catalog. This part of the project might involve investigating the use of GW sources as tracers of large-scale structure, or the effects of gravitational lensing; this can be guided by the interests of the student.

Prerequisite Skills required

Some theoretical background in general relativity is required; no specific prior experience in GW astronomy is necessary. Strong computing and data processing skills.

Other skills used in the project

Working with data products in GW astronomy. Bayesian inference and working with stochastic samples from posterior probability distributions. These are not prerequisites; these skills can be learned during the project.

Acceptable Programming Languages

Python, Julia or a similar programming language suitable for data science applications. The overwhelming majority of GW data analysis is done using Python.

Useful References (List of important papers/review articles relevant to the project)

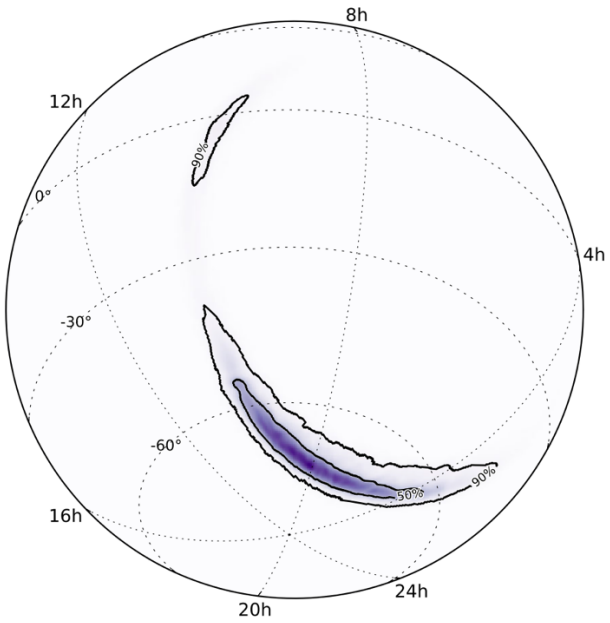
[1] LIGO & Virgo collaborations (2023) “GWTC-3: Compact Binary Coalescences Observed by LIGO and Virgo during the Second Part of the Third Observing Run”, Phys. Rev. X 13 041039, arXiv:2111.03606

[2] R. Stiskalek, J. Veitch & C. Messenger (2020) “Are stellar-mass binary black hole mergers isotropically distributed?”, Monthly Notices of the Royal Astronomical Society 501 1, arXiv:2003.02919

[3] LIGO & Virgo collaborations (2016) “Properties of the Binary Black Hole Merger GW150914”, Phys. Rev. Lett. 116 241102, arXiv:1602.03840

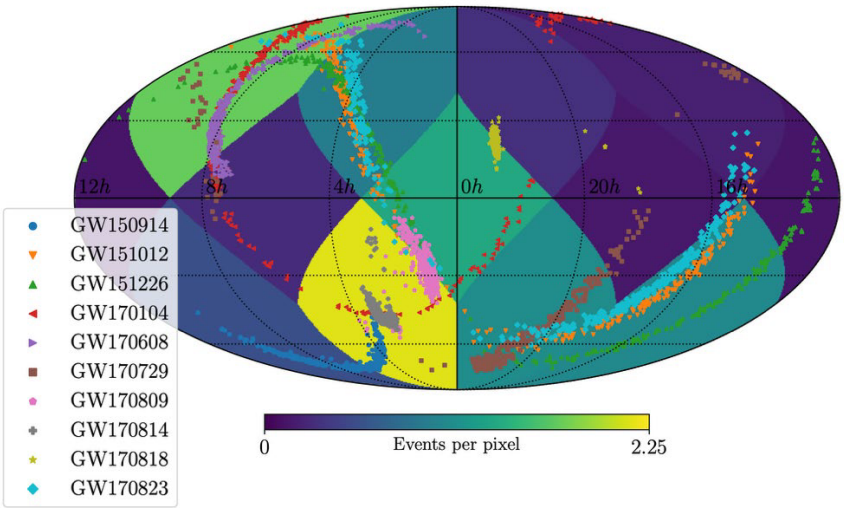
[4] Einstein Telescope collaboration (2025) “The Science of the Einstein Telescope”, Report ET-0036C-25, arXiv:2503.12263

Figure 1:



Caption: Example GW sky localisation. Figure reproduced from Ref [3]. The sky localisation for the first event GW150914. The coordinates ra and dec are plotted in orthographic projection and the contours show the 50% and 90% credible regions.]

Figure 2:



Caption: Figure reproduced from Ref. [2]. A map of the sky divided into just 12 pixels of equal area and showing the measured locations of the first 10 GW events. This can be used as the basis of a simple test of the isotropy of the GW source distribution. With GWTC-4, we will have more than a one order of magnitude increase in the number of GW sources with which to perform such a test and to look for anisotropies in the distribution.

Project 3: Disentangling the host galaxies of massive black hole mergers with LISA

Supervisor I: Dr Alex Jenkins (acj46@cam.ac.uk)

Supervisor II: Professor Hiranya Peiris (hiranya.peiris@ast.cam.ac.uk)

UTO: Professor Hiranya Peiris (hiranya.peiris@ast.cam.ac.uk)

Project summary

This project will investigate the population of massive black hole binaries (MBHBs) that will be detected by the future gravitational-wave observatory LISA. By combining existing gravitational-wave data from pulsar timing arrays and electromagnetic observations of active galactic nuclei, the student will generate simulated data-informed catalogues of LISA MBHBs, and will explore the exciting range of science questions that will be probed with this population.

Background

Giant black holes weighing millions or even billions of solar masses are believed to reside in the centres of almost all large galaxies. As these galaxies merge with each other through cosmic time, their black holes become bound to each other in binaries that are thought to be some of the loudest sources of gravitational waves (GWs) in the Universe. These massive black hole binaries (MBHBs) are the most likely source of the GW spectrum recently detected by various pulsar timing array (PTA) collaborations, and are one of the key observational targets of the future space-based GW observatory LISA, which will capture their final death throes as they merge to form individual remnant black holes.

As well as their loud GW signals, these black holes generate powerful electromagnetic (EM) emission via accretion of surrounding matter, and are observed as active galactic nuclei (AGN). While huge numbers of AGN have been identified, including in large spectroscopic surveys such as SDSS, the majority of these are orders of magnitude more massive than the MBHBs that will be detected by LISA. This means that relatively little is known observationally about the galaxies that host this LISA population. Understanding this population and its evolution throughout cosmic history is essential to unlocking key science questions such as the origin and formation mechanisms of massive black holes and the astrophysical processes that drive them to merge.

Project details

The starting point will be to use the publicly-available software **holodeck** to generate simulated populations of MBHBs, ensuring that these are consistent with the GW spectrum that has been detected by PTAs. The student will then investigate ways of comparing this population against known demographics of AGN, including modelling the accretion and growth of black hole merger remnants from high redshift to the present day using numerical tools such as the **TRINITY** model. They will then

extract the sub-population of merging MBHBs that will be detectable with LISA, and will use the links they have developed between GW and EM datasets to investigate the host galaxy demographics of these sources. This is an open-ended project that can evolve in multiple directions depending on the interests and abilities of the student. Questions that could be explored include the possibility of multi-messenger identification of the hosts of individual MBHBs due to radio/X-ray emission from the merger, or prospects for discriminating between different black hole formation channels (i.e. whether MBHs grow from “heavy seeds” or “light seeds”) and binary hardening processes (i.e. what physics brings MBHBs to small enough separations to merge, thereby solving the “last parsec problem”).

Prerequisite Skills required
High levels of independence and motivation

Strong verbal and written communication skills

Good numerical skills, particularly Python

Basic familiarity with concepts from gravitational-wave astronomy, as well as galaxy formation and evolution

Previous knowledge of relativity, cosmology, and galaxies at the level of a Part II course or equivalent will be extremely useful. Relevant Part III courses include Formation of Galaxies and Astrostatistics.

Other skills used in the project

Finding, critically reading and reviewing scientific literature

Acceptable Programming Languages
Python (essential), Fortran and/or C++ (desirable)

Useful References **(List of important papers/review articles relevant to the project)**
LISA Definition Study Report, arXiv:2402.07571 (particularly section 3.2)

The NANOGrav 15 yr Data Set: Constraints on Supermassive Black Hole Binaries from the Gravitational-wave Background, Astrophys.J.Lett. 952 (2023) 2, L37, arXiv:2306.16220

The quest for dual and binary supermassive black holes: A multi-messenger view, New Astron.Rev. 86 (2019) 101525, arXiv:2001.06293

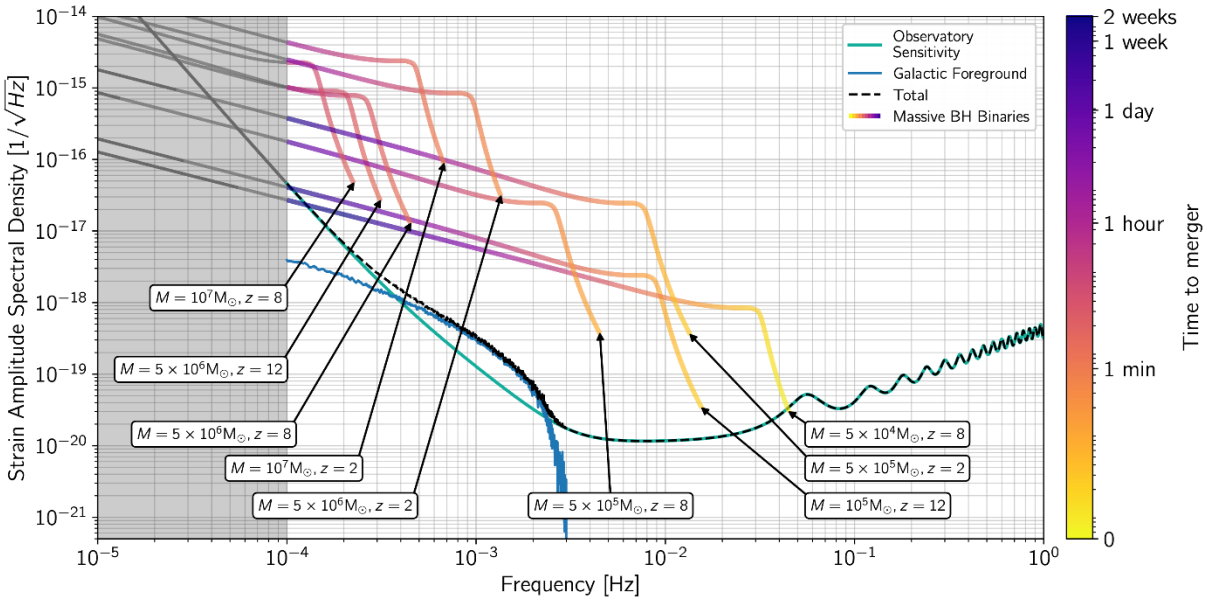


Figure 1: Strain amplitude spectral density of the GW signal of merging MBHBs versus the observer-frame frequency, for systems with mass ratio 0.5. Source-frame masses and redshifts are indicated in round boxes (coloured traces). Lines are colour coded according to the time to merger at each frequency. Solid teal, solid blue and dashed black lines denote LISA's instrumental strain amplitude, the Galactic foreground, and their sum, respectively. (Source: LISA Definition Study Report, arXiv: 2402.07571)

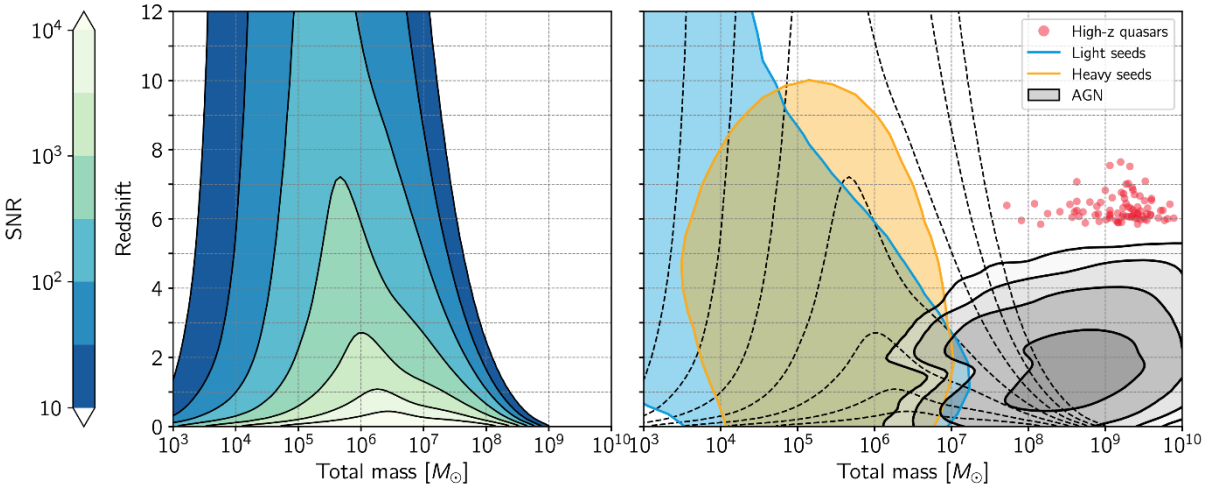


Figure 2: Left panel: contour lines of constant signal-to-noise ratio (SNR) from MBHBs. Binaries have mass ratio 0.5 and aligned spins of amplitude 0.2. Right panel: Overlaid on the dashed contour lines of constant SNR is the distribution of two simulated populations of MBHB mergers resulting from light seeds (light blue) and heavy seeds (yellow). The grey shaded areas show the current distribution of the AGN population, and red dots indicate quasars observed at 6.3 < z < 7.5. For AGN and quasars, the mass M on the x-axis is that of a single MBH. (Source: LISA Definition Study Report, arXiv: 2402.07571)

Project 4: The chemical fingerprint of supermassive black hole seed formation

Supervisor I: Dr Antti Rantala (ajr249@cam.ac.uk)

Supervisor II: Professor Debora Sijacki (deboras@ast.cam.ac.uk)

Supervisor III: Professor Roberto Maiolino (rm665@cam.ac.uk)

UTO: Professor Debora Sijacki (deboras@ast.cam.ac.uk)

Project summary

This project will connect state-of-the-art theoretical models of supermassive black hole (SMBH) seed formation to the latest observations of metal enriched high redshift galaxies by the James Webb Space Telescope (JWST). The student will lead a project which looks to model the chemical fingerprint of SMBH seed formation through runaway stellar collisions in early, dense and clumpy stellar environments. This work will be crucial for paving the way towards unravelling the most likely physical scenario for the SMBH origins using the JWST.

Background

The origin of SMBHs at early cosmic epochs remains a long-standing unresolved question in modern astrophysics (Rees 1984), and the recent JWST observations of accreting SMBHs in active galactic nuclei (AGN) at redshifts z>6-10 have only deepened the puzzle (Maiolino+2024, 2025). Recently, JWST observations of high-redshift galaxies and AGN with anomalous nitrogen enrichment at low metallicities and high star formation densities (e.g. Bunker+2023, Schaerer+2024, Isobe+2025, Ji+2025) as well as ancient globular star clusters (Bastian+2018) have renewed the interest in the formation of SMBH seeds through runaway stellar collisions in dense early star clusters. In the scenario, stars may substantially grow by repeated collisions and at low metallicities collapse forming SMBH seeds up to 10000 solar masses (Portegies Zwart+2004, Vergara+2025) at the end of their lives. The scenario is especially attractive as it may simultaneously explain both the SMBH seed formation and the anomalous enrichment pattens at high redshifts as stellar collisions and winds of the collisionally growing stars enrich their surroundings with metals (e.g. Wang+2019, Charbonnel+2023). Recently, a novel simulation project (Rantala+2024, Rantala+2025ab) has incorporated the observed clumpiness of early, dense star formation (Adamo+2024, Fujimoto+2024) into a GPU accelerated simulation framework (Rantala+2023) including post-Newtonian gravitational dynamics, single and binary stellar evolution processes as well as prescriptions for stellar collisions and gravitational wave driven black hole mergers. Most interestingly, the more realistic clustered setups boost the SMBH seed formation compared to older simplified models.

Project details

The student will be provided a large sample (1500 models) of latest state-of-the-art numerical simulations (Rantala+ in prep.) including stellar evolution and relativistic post-Newtonian gravitational dynamics. The simulations focus on the formation and early growth of SMBH seeds in realistic clustered environments through stellar collisions, tidal disruption events and gravitational wave driven black hole mergers in a wide range of stellar densities and metallicities. The key aim of the proposed project is to characterise the expected amount of metal enrichment in the runaway collisional scenario for SMBH formation. Using a detailed model for chemical abundances of stars, the student will estimate the chemical composition of the massive star winds and collisional ejecta. Together with the simulation sample data, the student will subsequently measure the evolving enrichment rates of metals including nitrogen, oxygen and carbon recycled back into the surroundings of the SMBH progenitors. Finally, the modelling results can be compared with the latest JWST observations (Maiolino+2025), revealing if the anomalous high redshift enrichment patters can be the chemical fingerprint of the SMBH seed formation.

Prerequisite Skills required

Knowledge of Python or a similar programming language is a strong advantage for the project and enthusiasm to acquire new coding skills for this programming-intensive project is crucial. The supervisors will be happy to provide support with finding relevant resources. Prior detailed knowledge about astrophysical simulations, galaxy formation, stellar evolution or black hole physics is not strictly required but some familiarity with the topics is a clear advantage.

Required Part II courses: Astrophysical Fluid Dynamics, Stellar Dynamics and Structure of Galaxies, Structure and Evolution of Stars

Other skills used in the project

Relevant skills for the project include the ability to synthesize complex and evolving literature, physical interpretation of simulation results, and the ability to link the latest theoretical models with state-of-the-art observations.

Acceptable Programming Languages

Python or similar language for data analysis and visualisation.

Useful References (List of important papers/review articles relevant to the project)

- Charbonnel+2023
- Isobe+2025
- Ji+2025
- Maiolino+2024
- Maiolino+2025

- Rantala+2023
- Rantala+2024
- Rantala+2025a
- Rantala+2025b
- Wang+2019

General References (List of papers referred to in the project)

- Adamo+2024
- Bastian+2018
- Bunker+2023
- Fujimoto+2024
- Portegies Zwart+2004
- Rees 1984
- Schaerer+2024
- Vergara+2025
-

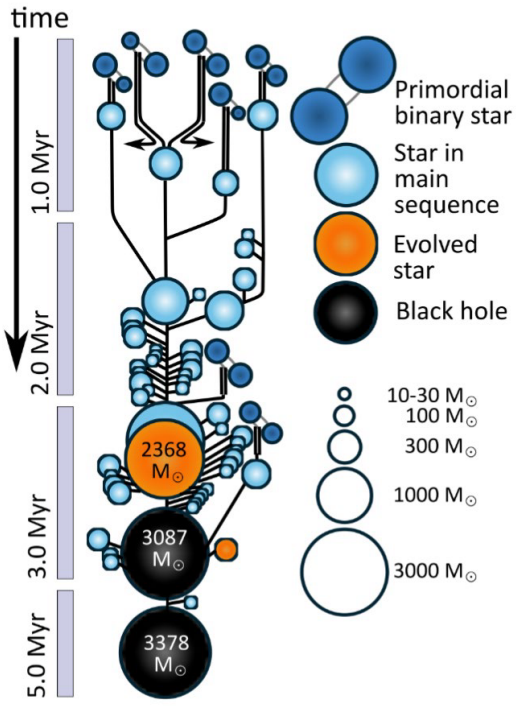


Figure 1: The merger tree leading to a formation of a single SMBH seed of >3000 solar masses in a single, isolated star cluster. Similar merger trees in realistic hierarchically assembling star clusters may contain >600 stellar collisions per formed SMBH seed. Mass loss in the substantial number of stellar collisions will enrich the surroundings of the forming SMBH seed.

Project 5: Are chemical inhomogeneities in wide binary stars a consequence of hidden triple systems?

Supervisor I: Professor Cathie Clarke (cclarke@ast.cam.ac.uk)

Supervisor II: Dr GyuChul Myeong (gm564@cam.ac.uk)

Supervisor III: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

UTO: Professor Cathie Clarke (cclarke@ast.cam.ac.uk)

Project Summary

The majority of wide binary systems are chemically homogeneous, suggesting that their components arise from the same well mixed reservoir of star forming gas. However, in a minority of systems, the metallicity of the two components differs by more than 0.1 dex (Hawkins et al 2020). It has been suggested that this puzzling result is a sign that one of the stars has ingested a planetary companion and that this has changed the stellar composition from its original value (Liu et al 2021). In this project an alternative explanation is explored, i.e. that one of the components of the wide binary system is actually a close (unresolved) binary, making the system a triple star system. In this case, the spectrum of the unresolved binary is a blend of the spectra from the two members of the close pair which, if mistakenly interpreted as coming from a single star, would affect the stellar metallicity inferred. To test this hypothesis, the student will synthesise spectra from the close pair and will then use a chemical retrieval code to discover how much the inferred metallicity differs from its true value. The results of this exercise will be compared with the observed statistics on wide binary pairs with discrepant metallicities in order to determine if the `hidden triple' explanation fits the data.

Background

Recent studies have revealed that a minority of wide binary stars exhibit discrepant abundances of heavy elements between their components (Hawkins et al 2020). This is surprising because the abundances measured in star forming clouds are rather homogeneous and one expects that both components of a wide binary system should form from the same cloud. A possible explanation is that one of the two components of the wide binary has at some stage ingested a planet (Liu et al 2021 and references therein). Just as in the solar system we know that the chemical composition of planets is not the same as the Sun, the addition of material from a planet to the stellar photosphere would be expected to change the chemical composition of the star. While this is an interesting hypothesis, it does not always give the correct prediction for the detailed differences in the abundances of different chemical species within some wide binary systems. An alternative, and simpler, explanation is that some wide binary stars are in fact triple systems, comprising one component which is itself a close binary system, which is not resolved by the observations used to measure abundances. When the spectra of the two stars in the close binary system are blended together by the finite resolution of the telescope, an observer who tries to interpret this spectrum as a single star will derive an incorrect metallicity for the pair (El Badry et al 2018), so that it could appear chemically distinct from its wide binary companion,

even though all components in the triple system share the same metallicity. Does this effect explain chemical inhomogeneities in wide binary stars?

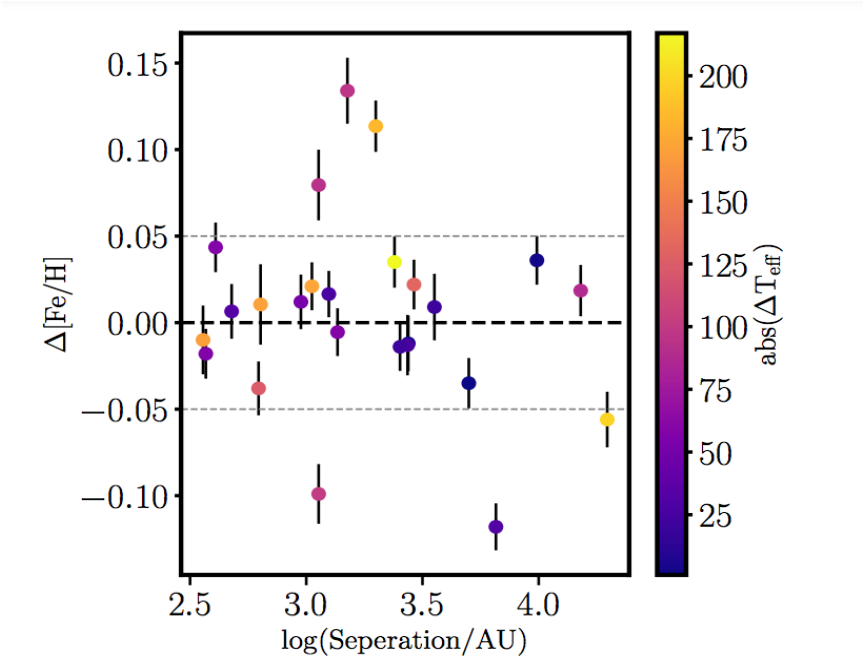


Figure 1.:Chemical differences between components of wide binaries as a function of separation (Hawkins et al 2020.) colour coded according to the difference in photospheric temperature. The dashed lines represent typical errors in observational determinations.

Project Details

The project will involve first generating a synthetic catalogue of `apparent wide binary star systems' i.e. systems that are a mixture of genuine wide binaries and those that are in reality triple systems containing a close binary pair that is not resolved by the Gaia satellite. In creating such a synthetic catalogue, it will be assumed that the properties and abundances of triple systems are those derived by Clarke 2020, which successfully reproduced the anomalous kinematics of wide binaries in the Gaia wide binary sample and which also matched triple star statistics resulting from dedicated surveys in small survey volumes (Riddle et al 2015). In order to test the null hypothesis (i.e. that all the stars in the system are chemically homogeneous), we will assume chemical homogeneity in all the systems in the 'apparent wide binary catalogue' and focus on any apparent chemical inhomogeneities in the triple systems. To do this, the student will use a stellar spectral synthesis code, MOOG (2019 version: https://www.as.utexas.edu/~chris/moog.html ; Sneden 1973), and synthesise the individual spectra of each of the members of the close binary pair. These individually synthesised spectra will be combined together to produce a composite spectrum. The next step is to use a retrieval code, e.g., q2 (Ramirez et al. 2014), on this composite spectrum to derive the metallicity that an observer would infer, if observing this close pair without knowledge of the individual systems. The retrieved metallicity will then be compared with the metallicity of the triple companion. When repeated for all the triple stars in the synthetic catalogue it will be possible to assess whether this effect explains the 10% of wide binary

pairs that appear chemically inhomogeneous. Depending on progress in the project, there are several possible extensions. For example, the student could examine the patterns of abundances of individual elements (rather than a bulk metallicity signature) and compare this with the observational data (Liu et al 2021). Alternatively, the student could synthesise the so-called RUWE diagnostic (Lindegren et al 2018) which contains information on the errors in the single-source astrometric fit and is available for all Gaia data: this would indicate whether the hypothesis of hidden triples should predict a correlation between RUWE value and chemical inhomogeneity, which could be investigated in the observational data.

Prerequisite Skills Required
The student will use spectral synthesis and stellar parameter and abundance retrieval codes and will need to create scripts to run these codes as well writing any necessary data analysis codes in Python or a language of their choice.

Other Skills Used in the Project
The Stars course in Part II Astrophysics would provide suitable background knowledge, though is not essential and attendance of the Stars course at Part III is recommended.

Acceptable Programming Languages
Python or language of choice.

Useful References (List of important papers/review articles relevant to the project)

- The distribution of relative proper motions of wide binaries in Gaia DR2: MOND or multiplicity? Clarke, C., 2020. MNRAS 491,L72-L75
- Signatures of unresolved binaries in spectral spectra: implications for spectral fitting. El Badry, K. et al., 2018. MNRAS 476,528
- Identical or fraternal twins? The chemical homogeneity of wide binaries from Gaia DR2. Hawkins, K. et al., 2020. MNRAS 492,1164
- Detailed elemental abundances of binary stars: searching for signatures of planet formation and atomic diffusion. Liu, F. et al, 2021. MNRAS 508,1227
- The Solar Twin Planet Search. I. Fundamental parameters of the stellar sample. Ramirez, I. et al., 2014. A&A 572A, 48R
- Carbon and Nitrogen Abundances in Metal-Poor Stars. Sneden, C.A., 1973. Ph.D. Thesis, The University of Texas at Austin

General References (List of papers referred to in the project)

- Gaia Data Release 2: the astrometric solution. Lindegren, L. et al., 2018 A&A 616,A2

- A survey of the high order multiplicity of solar type binary stars with Robo-AO. Riddle, R. et al 2015. ApJ 799,4

Project 6: Photochemical Modelling of Hycean Worlds

Supervisor I: Dr Gregory Cooke (gjc53@cam.ac.uk)

Supervisor II: Professor Nikku Madhusudhan (nmadhu@ast.cam.ac.uk)

UTO: Professor Nikku Madhusudhan (nmadhu@ast.cam.ac.uk)

Project summary

This project will utilise an existing photochemical model to investigate the atmospheric composition of Hycean worlds. Hycean worlds are a recently proposed class of theoretical exoplanets (planets outside our solar system) which have a liquid water ocean beneath a hydrogen-dominated atmosphere (Madhusudhan et al. 2021, Madhusudhan 2024). Whilst they have no rocky surface like the solar system terrestrial planets (Mercury, Venus, Earth, and Mars), their liquid water ocean means they have the potential to host extraterrestrial life.

There are several hycean candidates, each with different host stars, densities, and surface gravities. The project will focus on how various planetary and stellar parameters affect the predicted chemical composition and any potential biosignatures for these Hycean atmospheres. The project will also aim to estimate the detectability of various molecules in these simulated scenarios.

Background

Hycean worlds are sub-Neptune exoplanets, being more massive and larger than Earth but smaller in size when compared to Neptune (see Figure 1). Hycean oceans may be up to 1000 km deep (Rigby & Madhusudhan, 2024), with habitable surface conditions and a sufficient supply of nutrients for extraterrestrial life (Madhusudhan et al., 2023a)

Because there are no sub-Neptunes in the solar system, we cannot take in-situ measurements of super-Earths, Hycean worlds, or mini-Neptunes. Therefore, when observing such exoplanets, we currently have no comprehensive understanding of the planetary processes due to no existing solar system analogue.

No Hycean exoplanet has yet been confirmed, there exist only candidates (Madhusudhan et al., 2021). However, recent JWST observations of K2-18 b, a Hycean candidate, showed a chemical composition that is consistent with prior photochemical predictions for Hycean atmospheres (Madhusudhan et al., 2023b). The JWST transmission spectrum of K2-18 b is shown in Figure 2. Similar inferences were made for TOI-270 d, another Hycean candidate (Holmberg & Madhusudhan, 2024).

Confirming the existence of Hycean Worlds is important because they are easier to characterise when compared to classically habitable, terrestrial exoplanets, which are smaller. If Hyceans exist, these planets may be the first place we find extraterrestrial life, should it exist. Understanding the photochemical and chemical processes occurring in Hycean exoplanet atmospheres is crucial in aiding to confirm an exoplanet as a Hycean world, and then further, to characterise their atmospheric and surface conditions.

Project details
The project will begin with a literature review, and this will involve reading the few published papers on Hycean worlds. A general understanding of planetary processes and atmospheric chemistry should be acquired through reading textbooks and/or relevant journal articles.

The project will involve setting up, running, and analysing photochemical simulations under varying conditions. The simulations will be done with VULCAN (Tsai et al., 2017), Photochem (Wogan et al., 2024), Atmos (Arney et al., 2017), or any other suitable photochemical model.

Taking inspiration from Madhusudhan et al. (2023a) – see Figure 3 – the simulations will be set up on a 'grid', such that a suitable parameter space is covered for each Hycean exoplanet candidate in order for a comparative study to be made. Ultimately, the study is flexible and can be adapted to the interests and skills of the candidate.

Several questions of interest are:

- How do different host stars and the uncertainty in the measured host star flux affect photochemical model predictions?
- What is the role in the pressure-temperature profile for different Hycean candidates?
- What role does the surface pressure play for various molecules?
- How does convective inhibition in the temperature profile alter the expected photochemical predictions?
- What fluxes of biological molecules are required to be observable by JWST?

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Skills required
Basic python (or other equivalent programming language) and experience with Linux (or desire to learn).

Useful References (List of important papers/review articles relevant to the project)

- Madhusudhan, N., Moses, J.I., Rigby, F. and Barrier, E., 2023. Chemical conditions on Hycean worlds. Faraday Discussions, 245, pp.80-111., https://pubs.rsc.org/en/content/articlehtml/2023/fd/d3fd00075c
- Madhusudhan, N., Piette, A.A. and Constantinou, S., 2021. Habitability and biosignatures of Hycean worlds. The Astrophysical Journal, 918(1), p.1., https://iopscience.iop.org/article/10.3847/1538-4357/abfd9c/meta
- Rigby, F.E. and Madhusudhan, N., 2024. On the ocean conditions of Hycean worlds. Monthly Notices of the Royal Astronomical Society, 529(1), pp.409-424. https://academic.oup.com/mnras/article/529/1/409/7617367
- Madhusudhan, N., 2024. The Hycean Paradigm in the Search for Life Elsewhere. arXiv preprint arXiv:2406.12794. https://arxiv.org/abs/2406.12794

- Madhusudhan, N., Sarkar, S., Constantinou, S., Holmberg, M., Piette, A.A. and Moses, J.I., 2023. Carbon-bearing molecules in a possible Hycean atmosphere. The Astrophysical Journal Letters, 956(1), p.L13. https://iopscience.iop.org/article/10.3847/2041-8213/acf577

General References (List of papers referred to in the project)

- Holmberg, M. and Madhusudhan, N., 2024. Possible Hycean conditions in the sub-Neptune TOI-270 d. Astronomy & Astrophysics, 683, p.L2. https://doi.org/10.1051/0004-6361/202348238.
- Wogan, N.F., Batalha, N.E., Zahnle, K.J., Krissansen-Totton, J., Tsai, S.M. and Hu, R., 2024. JWST observations of K2-18b can be explained by a gas-rich mini-Neptune with no habitable surface. The Astrophysical Journal Letters, 963(1), p.L7., https://iopscience.iop.org/article/10.3847/2041-8213/ad2616/meta
- Arney, G.N., Meadows, V.S., Domagal-Goldman, S.D., Deming, D., Robinson, T.D., Tovar, G., Wolf, E.T. and Schwieterman, E., 2017. Pale orange dots: the impact of organic haze on the habitability and detectability of Earthlike exoplanets. The Astrophysical Journal, 836(1), p.49. https://iopscience.iop.org/article/10.3847/1538-4357/836/1/49/meta
- Tsai, S.M., Lyons, J.R., Grosheintz, L., Rimmer, P.B., Kitzmann, D. and Heng, K., 2017. VULCAN: an open-source, validated chemical kinetics Python code for exoplanetary atmospheres. The Astrophysical Journal Supplement Series, 228(2), p.20. https://iopscience.iop.org/article/10.3847/1538-4365/228/2/20/meta

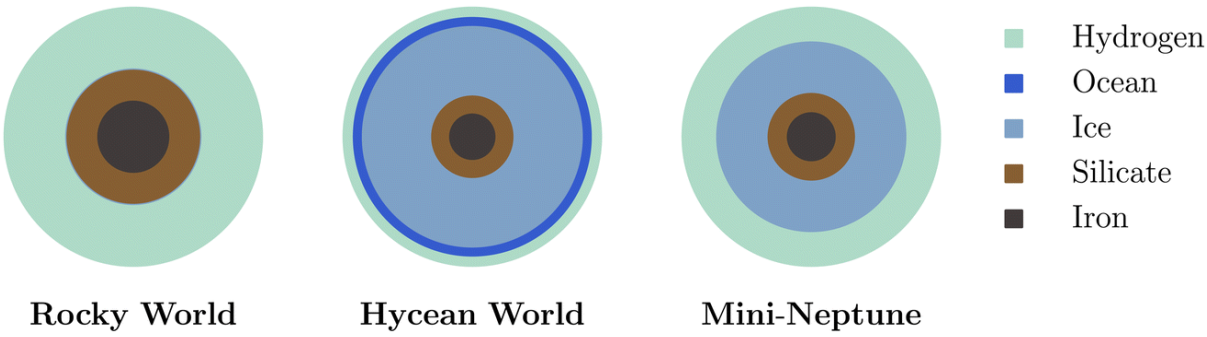


Figure 1: This is figure 1 from Madhusudhan et al. (2023a): Schematic of degeneracy between internal structures of Hycean worlds with those of other possible planet types with H2-rich atmospheres.

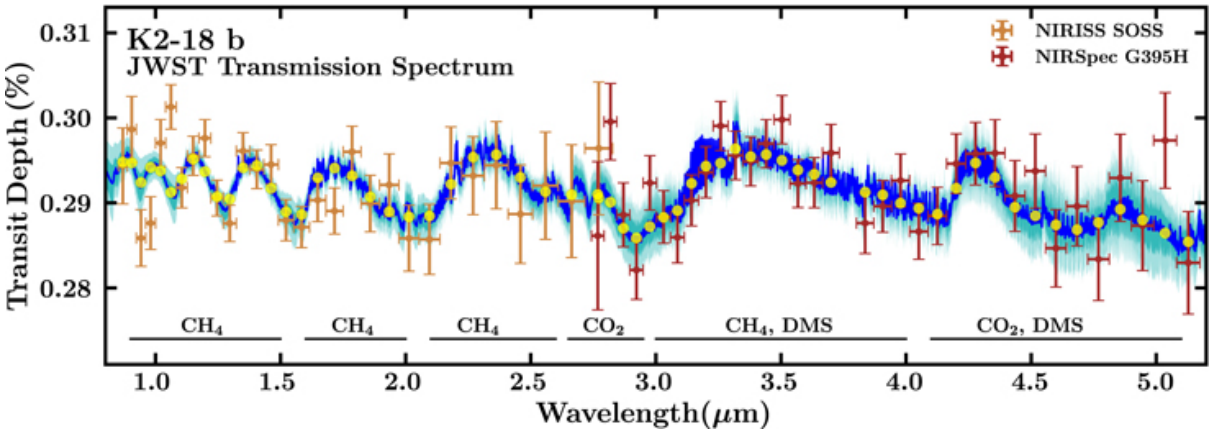


Figure 2: This is figure 3 from Madhusudhan et al. (2023b): The transmission spectrum of K2-18 b. The observed JWST spectrum and retrieved model fits are shown for the one-offset retrieval case discussed in Section 3. The data in orange show our NIRISS spectrum between 0.9 and 2.8 μm, and those in dark red show our NIRSpec G395H spectrum between 2.8 and 5.2 μm. The spectra are binned to R 25 and R 55 for NIRISS and NIRSpec, respectively, for visual clarity. The retrievals are conducted on the native resolution spectra, resulting in the best-fit reduced The NIRSpec spectrum is vertically offset by −41 ppm, corresponding to the median retrieved offset in the one-offset retrieval case. The blue curve shows the median retrieved model spectrum, while the medium- and lighter-blue contours denote the 1σ and 2σ intervals, respectively. Yellow points correspond to the median spectrum binned to match the observations. The prominent molecules responsible for the features in different spectral regions are labelled.

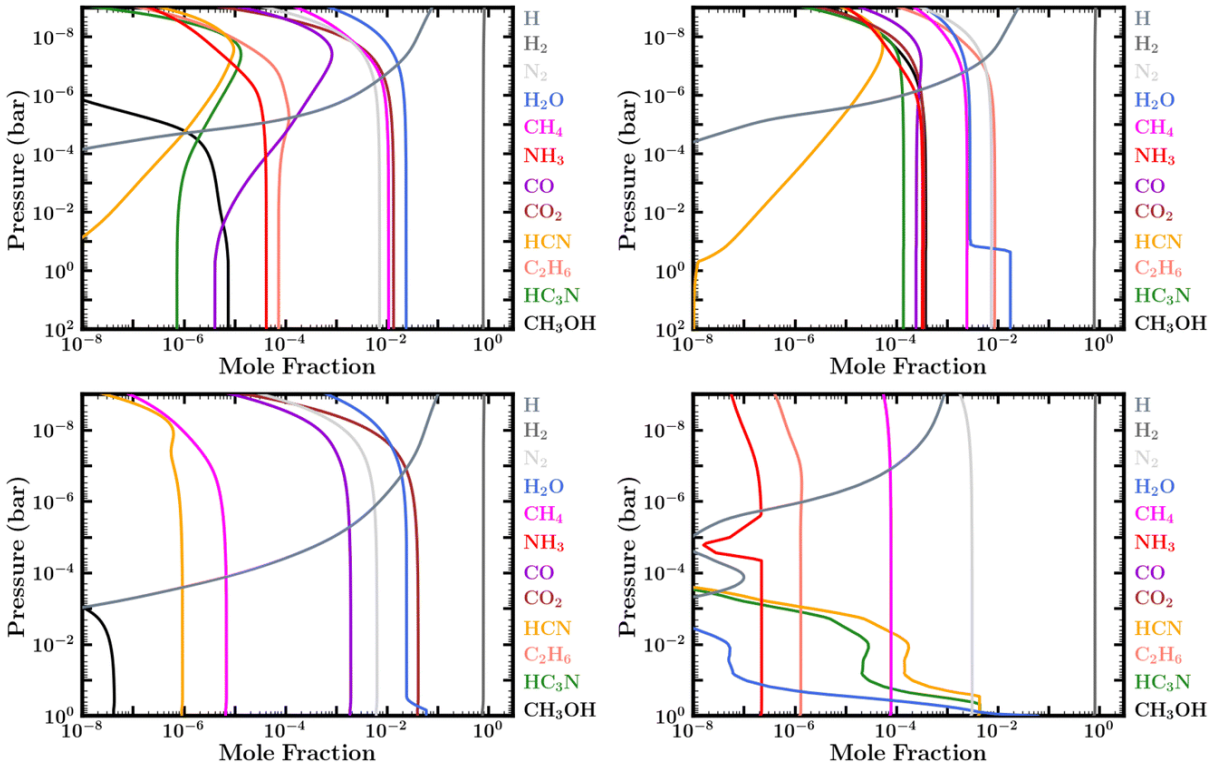


Figure 3: This is figure 3 from Madhusudhan et al. (2023a): Fully evolved (steady state) volume mixing-ratio profiles of prominent volatile species with zero-flux boundary conditions (inactive surface) for a 100× solar composition atmosphere and surface pressure of (top) 100 bar and (bottom) 1 bar: (top left) for the PT1 temperature profile in Fig. 2, corresponding to case 1 in Table 1; (top right) for the PT3 temperature profile, corresponding to case 3; (bottom left) for the PT1 temperature profile, corresponding to case 4; (bottom right) for the PT4 temperature profile, corresponding to case 6. Note that the sharp decline in H2O, HCN, HC3N, and NH3 with altitude in case 6 is due to condensation.

Project 7: Disentangling Eccentric White Dwarf Binaries in LISA: Population Inference Challenges

Supervisor I: Dr Nikhil Sarin (ns2043@cam.ac.uk)

Supervisor II: Dr Alex Jenkins (acj46@cam.ac.uk)

Supervisor III: Dr James Alvey (jbga2@cam.ac.uk)

UTO: Dr Will Handley (wh260@cam.ac.uk)

Project summary

This project investigates a potential source of degeneracy in gravitational-wave astronomy: whether the Laser Interferometer Space Antenna (LISA) mission can correctly distinguish between eccentric double white dwarf binaries (DWDs) emitting gravitational waves at multiple frequencies amongst a population of circular DWDs emitting at a single frequency. The project will combine aspects of gravitational-wave signal modeling, Bayesian inference, and population inference to address a critical question for one of LISA's primary science goals.

Background

White dwarf binaries represent the most numerous gravitational wave sources for LISA, with tens of thousands expected to be detected during its mission lifetime. These systems serve as tracers of stellar evolution and the structure of the Milky Way. While most DWDs are expected to have nearly circular orbits due to the effects of mass transfer and tidal interactions during their evolution, a significant fraction may retain non-zero eccentricity depending on their formation channels.

Gravitational wave emission from circular binaries occurs primarily at twice the orbital frequency. However, eccentric binaries emit at multiple harmonics of the orbital frequency, with the power distribution across harmonics depending on the eccentricity. This creates a potential degeneracy: the signal from a single eccentric binary might be misinterpreted as coming from multiple circular binaries at different frequencies, or vice versa.

LISA's measurement precision for individual binaries will depend on their signal-to-noise ratio, and for many sources, the constraints may not be tight enough to definitively distinguish between these scenarios on a source-by-source basis. This uncertainty at the individual source level could propagate to significant biases in population-level inferences, affecting our understanding of white dwarf binary formation and evolution in the galaxy.

Project details

The student will systematically investigate how prior assumptions about the eccentricity distribution of white dwarf binaries affect population inference with LISA data. The project will progress through the following stages:

Signal modeling: Implement simplified models for gravitational wave signals from both circular and eccentric DWDs.

Review literature and population synthesis publications to understand the range of DWD populations possible in the Milky Way.

Injection studies: Create simulated LISA data containing populations of DWDs with various eccentricity distributions. Initially focus on simplified scenarios with a small number of sources to understand basic degeneracies.

Parameter recovery: Apply Bayesian inference techniques to recover source parameters from the simulated data under different prior assumptions about eccentricity distributions.

Population inference: Develop a hierarchical Bayesian framework to infer population properties from the collection of individual detections, quantifying how eccentricity distribution misspecification affects population-level conclusions.

Degeneracy analysis: Quantitatively characterize the degree to which eccentric binaries can be confused with multiple circular binaries across the LISA frequency band and for different signal-to-noise ratios.

The student will have the opportunity to apply these methods to realistic simulated LISA data catalogs and develop recommendations for handling eccentricity uncertainty in future LISA data analysis.

The student will be supervised by Nikhil Sarin, Alex Jenkins, and James Alvey. They will additionally have the opportunity to interact with Will Handley's research group, as well as access to computational resources to support the simulation and inference work.

Prerequisite Skills required
Basic understanding of general relativity and gravitational wave physics

Basic understanding of gravitational dynamics

Experience with statistical inference, preferably Bayesian methods

Other skills used in the project
Literature review and synthesis

Gravitational wave data analysis techniques

Hierarchical Bayesian inference and population statistics

Acceptable Programming Languages
Python

Useful References (List of important papers/review articles relevant to the project)

- Nelemans et al. (2001), "Population synthesis for double white dwarfs", A&A, 365, 491

- Peters & Mathews (1963), "Gravitational Radiation from Point Masses in a Keplerian Orbit", Physical Review, 131, 435

General References (List of papers referred to in the project)

- Amaro-Seoane et al. (2017), "Laser Interferometer Space Antenna", arXiv:1702.00786
- Korol et al. (2017), "Prospects for detection of detached double white dwarf binaries with Gaia, LSST and LISA", MNRAS, 470, 1894

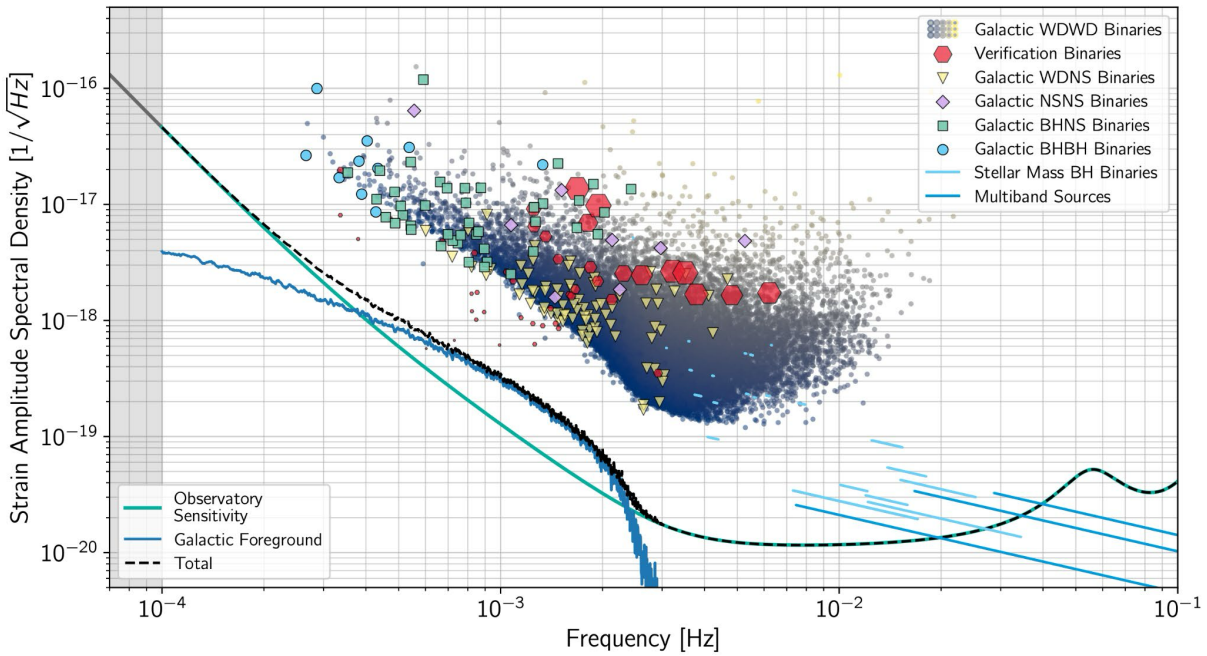


Figure 1: Simulations of different classes of galactic binaries whose gravitational-wave emission will be resolved by LISA. The cloud of dark blue/yellow points denotes galactic DWDs. Solid teal, solid blue and dashed black lines denote the LISA's instrumental strain amplitude, the stochastic galactic foreground, and their sum, respectively. (Source: LISA Definition Study Report, arXiv:2402.07571)

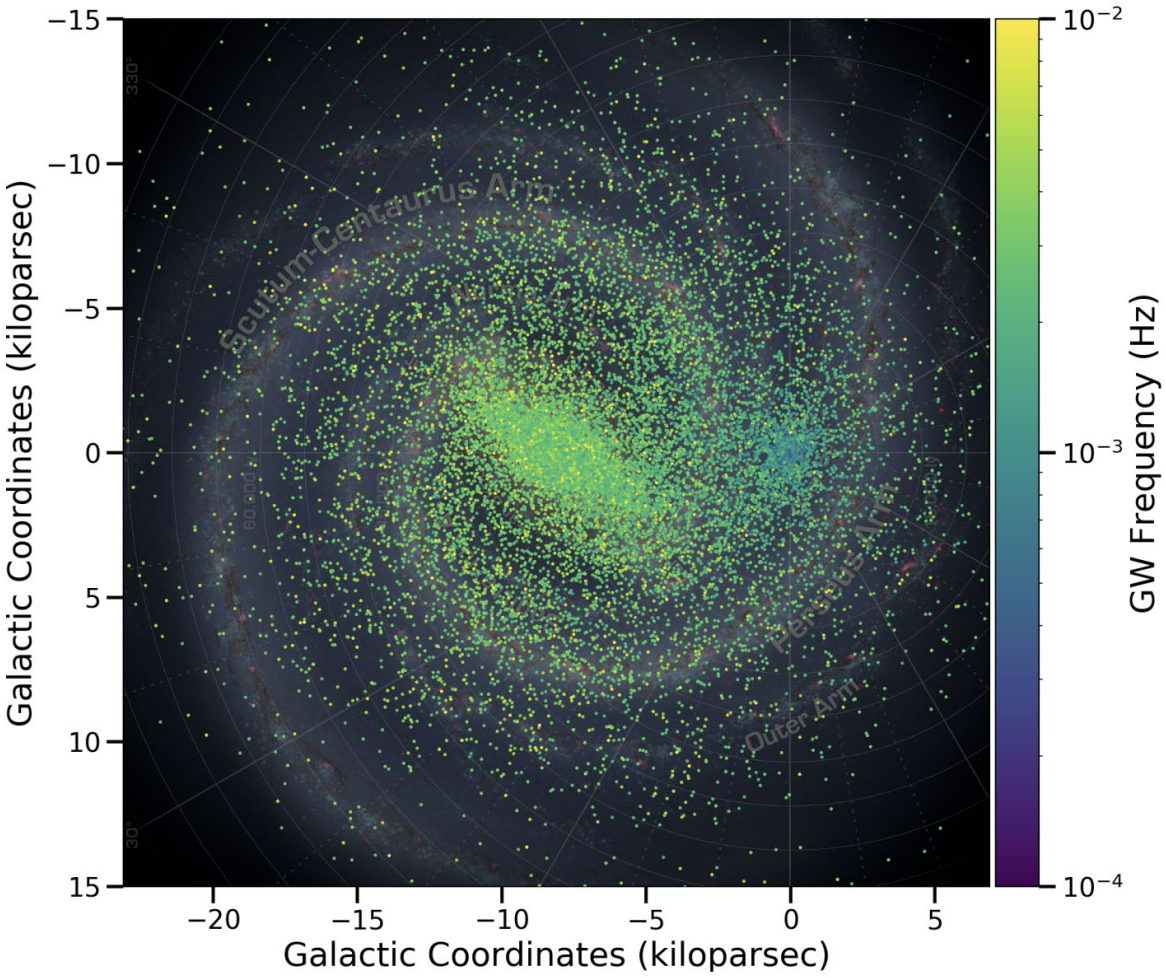


Figure 2: Simulated spatial distribution of numerous individual DWDs (dots, coloured according to GW frequency) projected onto the galactic plane. LISA's position is at (0,0). In the background is an artist's impression of the Milky Way. (Source: LISA Definition Study Report, arXiv:2402.07571)

Project 8: Probing the impact of thermal electrons in Jetted TDE and GRB afterglows

Supervisor I: Dr Nikhil Sarin (ns2043@cam.ac.uk)

UTO: Dr Will Handley (wh260@cam.ac.uk)

Project summary

This project investigates how thermal electrons influence our understanding of relativistic jets from black holes in tidal disruption events (TDEs) and gamma-ray bursts (GRBs). These phenomena represent some of the most energetic explosions in the universe, where relativistic jets produce intense multiwavelength emission through synchrotron radiation. While standard models assume all electrons are accelerated into a power-law distribution, growing evidence suggests a significant thermal electron population may also exist, potentially altering our interpretation of these events. Using thermal electron models implemented in the Redback framework, the student will analyze observations of jetted TDEs and GRB afterglows to determine how the inclusion of thermal electrons affects inferred jet parameters and our understanding of the central engine properties.

Background

When a star is disrupted by a supermassive black hole (TDE) or when a massive star collapses to form a black hole (GRB), powerful relativistic jets can be launched, producing some of the brightest transient phenomena in the universe. These jets generate radiation primarily through synchrotron emission as a result of shocks forming when these ultra-relativistic jets interact with the surrounding interstellar medium.

The standard approach to modeling this emission assumes all electrons at the shock are accelerated into a non-thermal power-law distribution. However, observations of events like the jetted TDE AT2022cmc (Andreoni et al. 2022) have shown emission features difficult to explain with conventional models, suggesting the need to reconsider our modelling assumptions.

Recent work by Margalit & Quataert (2024) has shown that including thermal electrons can significantly alter the predicted lightcurves and spectrum from these relativistic jets, potentially leading to very different interpretations of the underlying physics of these systems. This project aims to quantify these effects and reassess our understanding of the mechanisms powering these extreme cosmic explosions.

Project details

The student will investigate how thermal electrons in synchrotron emission models affect our interpretation of black hole-powered jets. Using the thermal electron models already implemented in the Redback framework, the project will proceed through these stages:

Literature review: Develop a solid understanding of synchrotron emission physics in relativistic jets and recent developments regarding thermal electron populations, with particular focus on the framework developed by Margalit & Quataert (2024).

Model exploration: Gain proficiency with the thermal electron synchrotron models in the Redback framework, understanding the key parameters and how they differ from standard models.

Parameter exploration: Conduct a systematic study of how the thermal electron fraction affects the observed emission across multiple wavelengths, identifying key observational signatures that distinguish thermal from purely non-thermal models.

Case studies: Apply both standard and thermal electron models to well-observed jetted TDEs (such as AT2022cmc) and GRB afterglows with multi-wavelength data. Compare the inferred physical parameters and assess how including thermal electrons changes our understanding of the jet properties and central engine.

Physical implications: Evaluate how the revised parameter estimates affect our understanding of black hole jet physics, particle acceleration in relativistic shocks, and the interpretation of these events.

The student will use Bayesian inference to quantify parameter uncertainties and employ model selection techniques to determine whether the data favor models that include thermal electron populations.

The student will be supervised by Nikhil Sarin. They will additionally have the opportunity to interact with Will Handley's research group, as well as access to computational resources to support the simulation and inference work.

Prerequisite Skills required
Basic understanding of radiation processes in astrophysics, particularly synchrotron emission

Experience with basic programming and data analysis

Interest in high-energy transient astrophysics

Other skills used in the project

Literature review

Familiarity with relativistic jets and shock physics

Bayesian inference and parameter estimation

Spectral energy distribution modelling

Acceptable Programming Languages
Python

Useful References (List of important papers/review articles relevant to the project)

- Sarin et al. 2024 - https://arxiv.org/abs/2308.12806
- Andreoni et al 2022 - https://arxiv.org/abs/2211.16530

- Margalit and Quataert 2024 - https://arxiv.org/abs/2403.07048

General References (List of papers referred to in the project)

- Sari et al. 1998 - https://arxiv.org/abs/astro-ph/9712005

Project 9: Probing reionization with Lyman-alpha emitters

Supervisor I: Professor Martin Haehnelt (haehnelt@ast.cam.ac.uk)

UTO: Professor Martin Haehnelt (haehnelt@ast.cam.ac.uk)

Project summary

Lyman-alpha emission from high-redshift galaxies is a sensitive probe of foreground neutral hydrogen and can be used as a probe of how the hydrogen in the Universe became reionized.

The project will predict how different models for the timing of reionization affect the properties of high redshift Lyman-alpha emitters.

Background

The James Webb Space Telescope (JWST) is looking back to the earliest epochs of the Universe and has observed galaxies at a time when the Universe was only 400 million years old. Before the first galaxies formed, the intergalactic gas had cooled sufficiently for hydrogen and helium to become neutral. Stars in the first galaxies emitted intense ultraviolet radiation, gradually (re-)ionizing the hydrogen in the surrounding intergalactic gas. The resulting bubbles of ionized hydrogen grew in time and eventually overlapped to encompass the entire Universe when the Universe was about 1 billion years old (see Fig.1) . In its neutral state, hydrogen, the most abundant element in the intergalactic gas, efficiently scatters Lyman-alpha light, preventing it from reaching us. Many of the earliest galaxies detected by JWST nevertheless shine brightly in the Lyman-alpha line, indicative of newly formed stars. The strength of the Lyman-alpha emission from the first galaxies is thus a probe of how the Universe became reionized.

Project details

The aim of the project is to use timelines of neutral hydrogen obtained from existing super-computer simulations for different models of the timing of reionization (see Figure 1) to calculate the evolution and distribution of the transmissivity of Lyman-alpha emission. The student will write their own code to calculate the line-of-sight transmissivity for a range of assumptions for the Lyman-alpha emission of high-redshift galaxies. This will then be used to predict the evolution of the luminosity function and equivalent with distribution of Lyman-alpha emitters for the different models during the epoch of reionization.

Skills required

The project will require programming in Python. Knowledge of C would be beneficial.

(Some of) the content of the Part II course “Introduction to Cosmology” will be relevant.

Useful References (List of important papers/review articles relevant to the project)

- Asthana et al., arXiv:2404.06548
- Keating et all. 2024,MNRAS, 532, 1646
- Miralda-Escudé J., 1998, ApJ, 501, 15
- Weinberger et al. 2018, MNRAS, 479, 2564
- Weinberger et al. 2019, MNRAS, 485, 1350

General References (List of papers referred to in the project)

- Dayal P., Ferrara A., 2018, Physics Reports, 780-782, 1
- Ryden B., Introduction to Cosmology, 2016 (2nd edition), CUP
- Peebles P.J.E., Principles of Physical Cosmology, 1993, Princeton

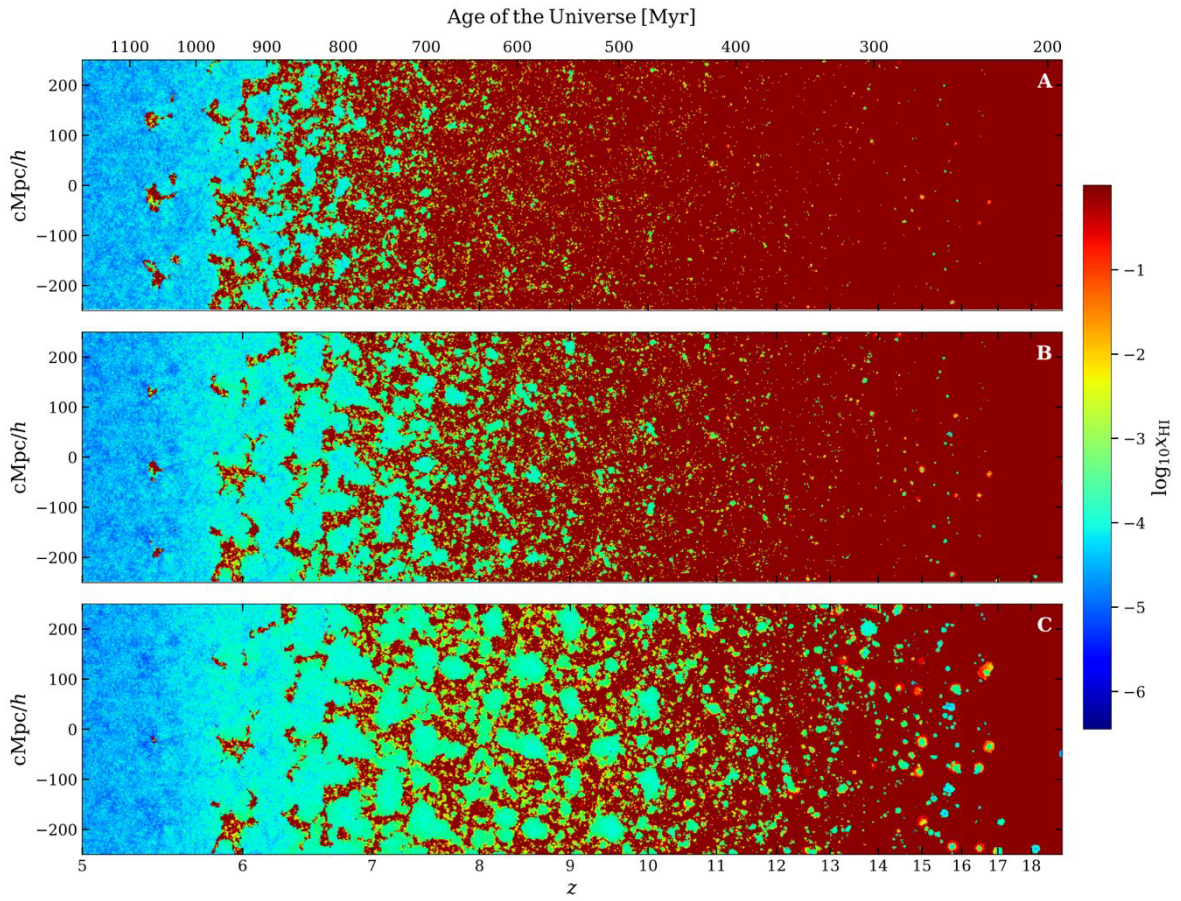


Figure 1: Timeline of the neutral hydrogen fraction for three different models of reionization

Project 10: GPU-Accelerated Nested Sampling for Binary Neutron Star Inference

Supervisor I: Dr James Alvey (jbga2@cam.ac.uk)

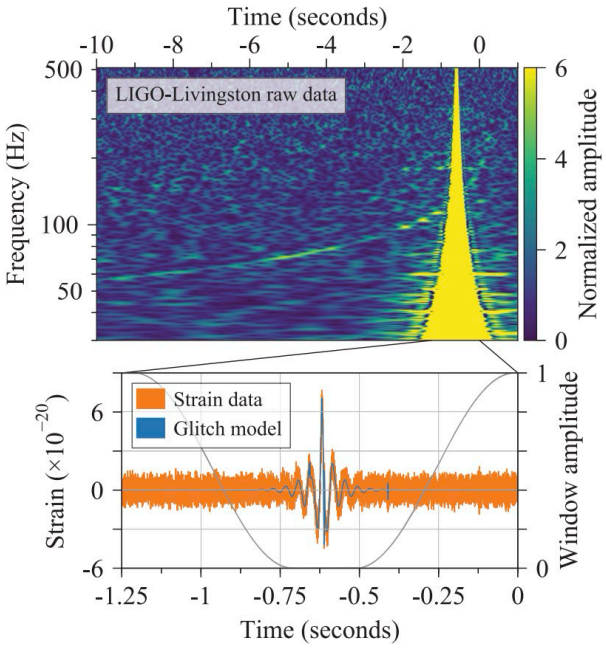
Supervisor II: Metha Prathaban (myp23@cam.ac.uk)

Supervisor III: Dr David Yallup (dy297@cam.ac.uk)

UTO: Dr Will Handley (wh260@cam.ac.uk)

Project Summary

This project aims to extend the benchmark work on GW150914 to analyze the first binary neutron star (BNS) event, GW170817, using the GPU-accelerated BlackJAX nested sampling algorithm. We will derive detailed performance metrics including runtime, effective sample size, convergence behaviour, and accuracy, by comparing BlackJAX against the standard sampling pipelines currently accepted within the LIGO collaboration. By tackling the challenges inherent to the longer and more complex BNS signal, the project will validate BlackJAX in a new astrophysical context and set the stage for future real-time follow-up applications.



Background

Gravitational wave astronomy has seen rapid advancement with the detection of binary black hole, binary neutron star, and neutron star-black hole mergers. While early benchmark studies on events such as GW150914 demonstrated that GPU-accelerated nested sampling via BlackJAX can outperform classical CPU-based samplers, the characteristics of binary neutron star signals (e.g. long signal duration, complex tidal effects) require careful reassessment of these methods. The flagship GW170817 event, with its unique rich multi-messenger context provides an excellent testbed for exploring performance trade-offs and algorithmic robustness when transitioning from BBH to BNS analysis. Establishing reliable performance metrics for BlackJAX not only enhances our understanding of its capabilities but also informs strategies for deploying fast inference during live follow-up campaigns.

Project Details

The project will proceed in the following stages:

1. Reproducing the Benchmark

   Adapt and validate the existing BlackJAX nested sampling pipeline for the GW170817 data, taking into account the extended signal duration and the more complex waveform features present in BNS events.

   Set up the analysis using state-of-the-art BNS waveform models (e.g. IMRPhenomD_NRTidal, SEOBNRv4T) and standard BNS likelihood formulations employed by the LIGO collaboration.

   Establish baseline performance metrics such as overall runtime, effective sample size per second, and convergence diagnostics.

2. Comparative Analysis with Standard Techniques

   Run parallel inference analyses using classical CPU-based sampling techniques (such as those implemented within bilby alongside dynesty) that are currently in accepted use by the LIGO collaboration.

   Compare quantitative performance metrics between BlackJAX and these standard methods, focusing on improvements in computational speed, sampling efficiency, and robustness in exploring the complex BNS parameter space.

Extension Studies and Further Directions

1. Future Detector Prototypes

   Apply the BlackJAX algorithm to prototype events simulated for next-generation observatories (e.g. the Einstein Telescope), allowing for insights into scalability and performance under conditions expected in the near future.

2. Likelihood and Waveform Benchmarking

   Benchmark various likelihood formulations and waveform models in the BNS case. This will help identify model-dependent influences on computational performance and highlight potential areas for further method refinement.

3. Fast Follow-Up and Live Inference

   Investigate the role of rapid inference methods in real-time follow-up scenarios by considering reduced datasets (e.g. tracking the early inspiral phase).

   Assess the feasibility of employing the fast GPU-accelerated inference approach to provide live sky localisation estimates during ongoing events, facilitating prompt electromagnetic follow-ups.

4. Additional Directions

   Explore integration of machine learning surrogates at the likelihood evaluation level.

Consider adaptive tuning of the slice sampling proposals in BlackJAX to better handle the geometry of BNS parameter spaces.

Potential Impact

This project will yield valuable insights into the performance and efficiency of a GPU-accelerated nested sampling algorithm in the challenging context of binary neutron star signals. By providing a rigorous performance benchmark against standard techniques, the study will contribute to the evolution of real-time data analysis pipelines necessary for prompt multi-messenger follow-ups. The extension studies envisaged also position BlackJAX as a versatile tool for next-generation gravitational wave astronomy, potentially influencing how rapid inference is conducted as both hardware capabilities and signal complexities increase.

Prerequisite Skills Required

Proficiency in Python, with an emphasis on scientific computing libraries.

Familiarity with Bayesian inference and nested sampling algorithms.

An interest in gravitational wave astronomy and data analysis.

Willingness to learn and work with GPU programming environments and libraries (e.g. using JAX).

Useful References

Yallup, D., et al., *Nested Slice Sampling (2025) [*https://openreview.net/pdf?id=ekbkMSuPo4*]*

LIGO Collaboration, *GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral* [https://arxiv.org/abs/1710.05832]

Project 11: Clustering in cosmological galaxy cluster count analyses with neural simulation-based inference

Supervisor I: Dr Iñigo Zubeldia (iz221@cam.ac.uk)

Supervisor II: Professor Anthony Challinor (a.d.challinor@ast.cam.ac.uk)

Supervisor III: Dr Boris Bolliet (bb667@cam.ac.uk)

UTO: Professor Anthony Challinor (a.d.challinor@ast.cam.ac.uk)

Project summary

The abundance of galaxy clusters as a function of mass and redshift is a powerful cosmological probe from which fundamental information about the Universe can be inferred. Galaxy clusters can be detected from observations at millimetre wavelengths through the imprint they leave on the cosmic microwave background (CMB) via the thermal Sunyaev-Zeldovich (tSZ) effect. These tSZ-detected clusters can, in turn, be used for cosmological inference in what is known as "cluster number count" analyses, for which the statistics of the observed cluster abundance must be accurately modelled. Clusters are tracers of the cosmic density field and, as a consequence, are not randomly distributed across the Universe: they cluster, with a given cluster having on average, more clusters near it than a randomly chosen point in space.

Cluster clustering has so far been ignored in cosmological number count analyses of tSZ-detected clusters, as it has been a negligible effect due to the size of past and current cluster catalogues, with hundreds of objects. This, however, may no longer the case with the advent of the Simons Observatory (SO), an upcoming millimetre observatory forecasted to detect about 20,000 clusters within its first few years of operation, starting in 2025 (see Figure 1): if accurate cosmological inference is to be made with the clusters that SO is going to detect, cluster clustering may need to be properly modelled. This project aims to answer this question (must cluster clustering be taken into account for accurate cosmological inference with the SO clusters?), to implement a simulation-based model to account for clustering if found to be significant, and to quantify the significance at which the cluster clustering signal itself will be detectable.

Background

The abundance of galaxy clusters, the largest gravitationally bound objects in the Universe, as a function of mass and redshift is a powerful cosmological probe, sensitive to a number of cosmological parameters and models (see Allen et al. 2011 and Kravtsov & Borgani 2012 for reviews on the topic). This cosmological information can be extracted by comparing the observed cluster abundance, as obtained through X-ray, optical, or mm observations, to its theoretical prediction, in what are known as "cluster number count" analyses (see, e.g., Zubeldia & Challinor 2019 for an example).

In mm observations, which are the focus of this project, clusters can be detected via the imprint they leave on the CMB through the thermal Sunyaev-Zeldovich (tSZ) effect. The tSZ effect is a spectral

distortion of the CMB spectrum that occurs when (very cold) CMB photons go through the very hot, ionised gas that fills the intra-cluster medium (ICM). Some of these photons interact with the ICM electrons via inverse Compton scattering and are, on average, upscattered towards higher energies, leading to a small distortion of the CMB blackbody around the locations of clusters in the sky. By searching for this distortion, clusters can be detected in a blind way from the data to relatively high redshifts (see, e.g., Zubeldia et al. 2024). The number of detected clusters as a function of redshift and tSZ signal (a quantity that scales with the cluster mass) can then, in turn, be used for cosmological inference.

In a cluster number count analysis, the statistics of the counts must be accurately modelled. As tracers of the cosmic density field, galaxy clusters themselves cluster, which has an impact on the cluster number count statistics. However, clustering has been ignored so far in cosmological number count analyses of tSZ-detected clusters. This approximation has been valid to date, as the available cluster samples, with a few hundred objects, have been small enough for the statistics to be approximately Poissonian, i.e., for Poisson shot noise to dominate over the sample variance due to cluster clustering (see Hu & Kravtsov 2002). However, upcoming millimetre experiments such as the Simons Observatory (SO) are set to dramatically increase the number of detected clusters, with SO forecasted to detect 20,000 clusters within its first few years of operation, starting in 2025 (see Figure 1 and Section 7 of Simons Observatory 2019). With more objects, Poisson shot noise is reduced, and sample variance due to clustering becomes increasingly important. Thus, sample variance due to clustering has the potential to become a significant effect to be accounted for in the analysis of the SO clusters. In addition, the cluster clustering signal may become a useful cosmological probe in its own right through the measurement of the cluster two-point correlation function.

This project aims first to assess whether cluster clustering will be a significant effect to consider in cosmological number count analyses of the SO clusters. If so, the goal will be to develop and implement a simulation-based method to account for it. Time permitting (or if cluster clustering turns out not to be significant), the project will quantify the significance at which the cluster clustering signal itself will be detectable with SO.

Project details

After absorbing the relevant background literature, you will use an existing suite of cosmological simulations to produce a synthetic set of SO-like galaxy cluster catalogs. With this simulated dataset in hand, you will quantify the magnitude of the covariance in the data due to cluster clustering in data space and compare it to the Poisson shot noise also present in the data. If cluster clustering is found to be significant, you will then use the cluster number count theory code cosmocnc (github.com/inigozubeldia/cosmocnc, see Zubeldia & Bolliet 2024) to produce a theoretical prediction for the statistics (more precisely, the likelihood) of the cluster number counts in your simulated catalogs, modifying the code to account for cluster clustering. You will also estimate the likelihood using a deep neural simulation-based inference (SBI) method and will compare the analytical likelihood prediction

with the SBI-based one (see Zubeldia et al. 2025, where a similar analysis was done ignoring the effect of clustering). Next, you will use your two likelihood estimates and the Markov chain Monte Carlo (MCMC) code Cobaya to infer the values of several cosmological parameters with your simulated cluster catalogs, investigating whether neglecting sample variance in the analysis leads to any significant biases in the inferred parameter constraints. Finally, you will directly probe the cluster clustering signal by measuring the cluster two-point correlation function in your simulated catalogs, producing forecasts for the significance of such measurements with SO. If time permits (e.g., if sample variance due to clustering is found to be negligible), you will translate these forecasts into forecasts of the constraints on cosmological parameters that could be achieved with the SO cluster two-point function, using a combination of analytical and neural SBI-based methods.

Skills required

- Good programming skills, with good knowledge of Python, basic notions of statistics and interest in machine learning methods. Basic notions of Bayesian inference would be helpful.
- Familiarity with the material covered in the Part II Introduction to Cosmology course and the Part III Cosmology course is advisable.

Useful References (List of important papers/review articles relevant to the project)

- Allen, Evrard & Mantz, 2011. Cosmological parameters from observations of galaxy clusters. arXiv:
- Kravtsov & Borgani, 2012. Formation of galaxy clusters. arXiv:1205.5556.
- Hu & Kravtsov, 2002. Sample variance considerations of cluster surveys. arXiv: 0203169.
- Simons Observatory Collaboration, 2019. The Simons Observatory: Science goals and forecasts (see, in particular Section 7). arXiv: 1808.07445.
- Zubeldia & Bolliet, 2024. cosmocnc: A fast, flexible and accurate galaxy cluster number count likelihood code for cosmology. arXiv: 2403.09589.
- Zubeldia et al. 2025. Extracting cosmological information from the abundance of galaxy clusters with simulation-based inference. arXiv: 2504.10230

General References (List of papers referred to in the project)

- Zubeldia, Melin, Chluba & Battye, 2024. The Planck SZiFi catalogues: a new set of Planck catalogues of Sunyaev-Zeldovich-detected galaxy clusters.
- Zubeldia & Challinor, 2019. Cosmological constraints from Planck galaxy clusters with CMB lensing. arXiv: 1904.07887.

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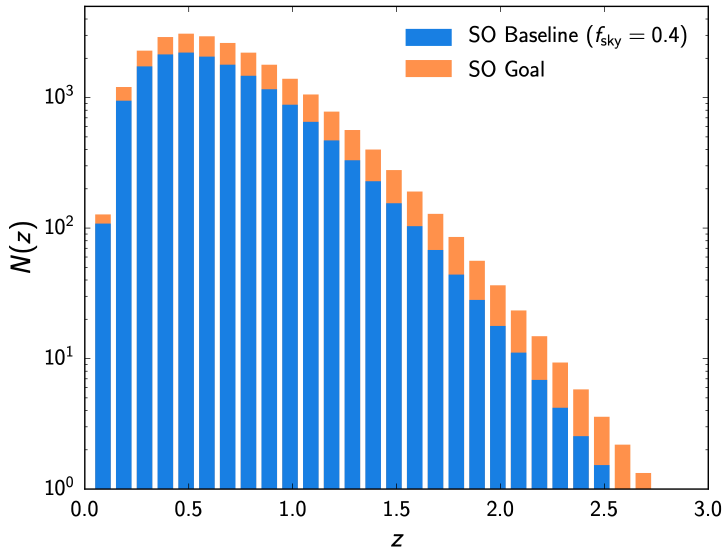

Figure 1: Cluster number counts as a function of redshift forecasted to be detected by the upcoming Simons Observatory, for two possible noise levels configurations (“baseline” and “goal”).

Project 12: Stellar Disc Instabilities and Bar Growth in a Simulated Milky Way Halo

Supervisor I: Floor van Donkelaar (current: floor.vandonkelaar@uzh.ch)

UTO: Debora Sijacki (deboras@ast.cam.ac.uk)

Project Summary

Stellar bars are common in disc galaxies and play a crucial role in redistributing angular momentum, funnelling gas toward the galactic centre, and influencing both star formation and black hole growth. This project will investigate the formation and evolution of bar structures in the high-resolution cosmological zoom-in simulation *GigaEris* (Tamfal et al. 2022), which follows the build-up of a Milky Way–mass galaxy from the early Universe to z = 4.4. *GigaEris* combines very high spatial resolution with a rapid early build-up of the galaxy, enabling the formation of a bar at much earlier times compared to other simulation. By identifying and characterizing the onset of bar formation, this project will be able shed light on the physical conditions and dynamical triggers that drive bar formation in realistic galaxy environments, but also the impact of such a bar on the general evolution of the galaxy.

Background

Stellar bars are dynamically important features in disc galaxies, capable of funnelling gas from the outskirts inward, where it can fuel central starbursts and feed supermassive black holes, as shown in both observations and theory (e.g. Levy et al. 2022). Bars play a central role in redistributing angular momentum, triggering morphological changes, and are often linked to quenching in red, gas-poor systems. They also interact with the dark matter halo and can scramble chemical abundance patterns across the disc, making them important markers of dynamical and chemical evolution. Although bars are commonly viewed as long-lived structures, their origin, particularly at early cosmic times, is not yet fully understood.

The *GigaEris* simulation offers a unique opportunity to study bar formation under realistic conditions. It builds on the original *Eris* suite using the ChaNGa code, capturing the early formation of a stable disc galaxy with exceptional resolution to z=4.4. Gas inflow to the centre is clearly visible (van Donkelaar et al. 2024, see the figure), producing a spiky star formation rate in the core that is consistent with bar-driven dynamics. Compared to other suites like *Auriga*, where bars typically form later and are harder to resolve in the early universe, *GigaEris* allows a detailed investigation of early bar formation and its consequences.

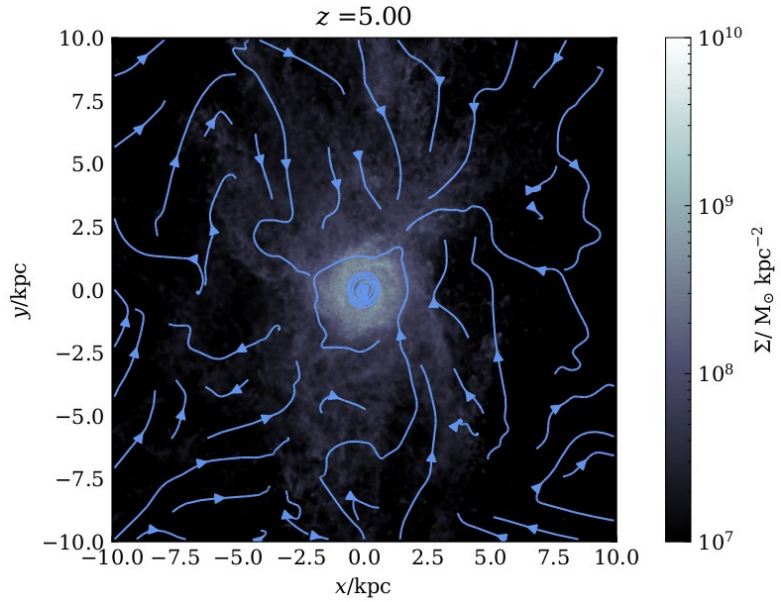

Figure 1: Gas surface density map of the main galaxy withing the GigaEris simulation at z = 5.0, with the velocity vectors of the gas overlaid. The galaxy is centred face-on on the disc stars from the main galaxy halo.

This project will explore the timing and mechanisms of bar formation. By doing so, it will help clarify how bars shape galaxies over time and demonstrate that these features can emerge already by z=4, something only made possible by the extreme resolution and early disc growth captured in *GigaEris*.

Project details

1. **Literature Review**
   Study theoretical and observational literature on bar formation, disc instabilities, and angular momentum transport in galaxies.

2. **Simulation Analysis Setup**
   Learn to work with GigaEris simulation data. Access and extract relevant properties such as stellar mass, disc kinematics, circular velocity profiles, and potential fields.

3. **Bar Identification**
   Identify the formation time and strength of the stellar bar along different redshifts and track the evolution of the bar, and link the to the evolving properties of the host galaxy, such as disc mass, velocity dispersion, and merger history.

4. **Interpret Physical Triggers**
   Investigate which physical factors, e.g. disc mass growth, gas inflows, minor mergers, or central mass concentrations, correlate with bar onset or bar growth.

5. **Optional Extensions (if time allows 😉)**

   Analyse how the presence of the bar affects gas inflow to the galactic centre, central star formation rates, or angular momentum profiles. Compare with observational bar statistics in the local Universe.

Prerequisite Skills required

- Python programming and data analysis (NumPy, Matplotlib, h5py, etc.)
- Part II courses Astrophysical Fluid Dynamics, Stellar Dynamics and Structure of Galaxies
- Some exposure to astrophysical simulations is helpful, but not required

Useful References (List of important papers/review articles relevant to the project)

- van Donkelaar F., Mayer L., Capelo P. R., Madau P., 2025, MNRAS, 539, 1259. doi:10.1093/mnras/staf597 https://ui.adsabs.harvard.edu/abs/2025MNRAS.539.1259V/abstract
- Kormendy J., Kennicutt R. C., 2004, ARA&A, 42, 603. doi:10.1146/annurev.astro.42.053102.134024 https://ui.adsabs.harvard.edu/abs/2004ARA%26A..42..603K/abstract
- Rosas-Guevara Y., Bonoli S., Puchwein E., Dotti M., Contreras S., 2025, A&A, 698, A20. doi:10.1051/0004-6361/202453160 https://ui.adsabs.harvard.edu/abs/2025A%26A...698A..20R/abstract
- Geron T., Smethurst R. J., Dickinson H., Fortson L. F., Garland I. L., Kruk S., Lintott C., et al., 2025, arXiv, arXiv:2505.01421. doi:10.48550/arXiv.2505.01421 https://ui.adsabs.harvard.edu/abs/2025arXiv250501421G/abstract
- Pritchett K., Jogee S., Guo Y., 2025, ASPC, 540, 27 https://ui.adsabs.harvard.edu/abs/2025ASPC..540...27P/abstract

General References (List of papers referred to in the project)

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- van Donkelaar F., Mayer L., Capelo P. R., Tamfal T., Quinn T. R., Madau P., 2024, MNRAS, 529, 4104. doi:10.1093/mnras/stae804 https://ui.adsabs.harvard.edu/abs/2024MNRAS.529.4104V/abstract
- Levy R. C., Bolatto A. D., Leroy A. K., Sormani M. C., Emig K. L., Gorski M., Lenkic L., et al., 2022, ApJ, 935, 19. doi:10.3847/1538-4357/ac7b7a https://ui.adsabs.harvard.edu/abs/2022ApJ...935...19L/abstract

Project 13: Cleaning out dust from the Cosmic Microwave Background maps with galaxies: Simulation-based Study

Supervisor I: Dr Ola Kusiak (ak2628@cam.ac.uk)

Supervisor II: Dr Niall MacCrann (nm746@cam.ac.uk)

UTOs: Professor Anthony Challinor (a.d.challinor@ast.cam.ac.uk), Professor Blake Sherwin (bds30@cam.ac.uk)

Project summary

The cosmic microwave background (CMB) is a snapshot of the early Universe and a powerful probe of cosmology. However, the observed CMB signal is not pure — it is "contaminated" by secondary anisotropies arising from interactions of the CMB photons with cosmic structures along the line of sight, such as gravitational lensing by dark matter, scattering off electrons in galaxy clusters, and emission from dusty star-forming galaxies. These secondary effects complicate the extraction of the primary CMB signal, which encodes key early-Universe physics.

In this project we aim to use galaxy surveys, which trace the same structures that generate these secondaries, to help disentangle the different components that form the observed CMB. We will investigate a range of statistical ways of using this galaxy data, perhaps involving machine learning methods, to generate maps that correlate highly with CMB secondary components and thus can be used to remove them from the primary CMB.

This work is particularly motivated by the challenge of separating the Cosmic Infrared Background (CIB) radiation, which originates from dusty galaxies across a wide redshift range and is a large contaminant to the CMB, but is very challenging to model accurately. We would like to use large-scale structure (LSS) data, e.g., galaxies, that correlate with the CIB, to improve foreground cleaning, in combination with traditional multifrequency methods.

Background

Modern CMB experiments such as Planck, ACT, or the Simons Observatory measure the microwave sky across multiple frequencies to extract cosmological signals. While primary anisotropies in the CMB are well-characterized and originate from redshift ~1100, secondary anisotropies are generated more recently and are correlated with large-scale structure.

Foreground signals such as the CIB are particularly troublesome. Traditional component separation methods (e.g., internal linear combination or ILC techniques, Remazeilles et al. 2011) rely on distinct frequency spectra to disentangle foregrounds from the CMB. But the CIB lacks a well-constrained spectrum, motivating alternative approaches.

Recent studies (e.g. Kusiak et al. 2023) have proposed using galaxy number density maps as templates for CIB foregrounds, under the assumption that galaxies trace the same structures that generate the CIB, while remaining uncorrelated with the primary CMB. Kusiak et al. proposed three new methods of including galaxy maps in the standard component separation (e.g., ILC) pipelines to help remove the CIB contamination from the primary CMB temperature maps. These methods vary in effectiveness and induced noise-penalty but, importantly, do not require exact assumptions about CIB modeling. This project will build on such ideas using simulations to test the effectiveness of various galaxy-based tracers and weighting schemes.

The project will focus on working with mock data from cosmological simulations (e.g., the AGORA simulation suite (Omori 2022) or the Backlight Simulations (Coulton et al. in prep).), enabling controlled testing of methods without the complexities of real-world systematics. The student will experiment with statistical techniques to optimize the construction of galaxy-based tracer maps for use in component separation. A code pipeline to include galaxy maps in the ILC methods is publicly available.

Project details
A possible outline of the project would be:

1. Introduction to CMB observations, CMB secondaries (the SZ effect), CIB, power spectra, and basic component separation techniques such as ILC. Overview of cosmological simulations, e.g. the AGORA simulation suite (Omori 2022) or the Backlight Simulations (Coulton et al.).
2. Learn to work with simulated CMB and galaxy maps. Use tools such as healpy, astropy to compute the cross- and auto-correlations between different fields.
3. Construct optimal galaxy/halo templates from simulations that correlate with the CIB.
4. Integrate the galaxy templates into the ILC component separation pipelines from Kusiak et al. (https://github.com/olakusiak/deCIBing) using simulated multifrequency CMB data. Compare cleaned maps with input primary CMB to evaluate performance.
5. Time permitting and depending on student's interest:
   - Extend the projects to other LSS tracers that correlate with CIB, e.g., CMB lensing
   - Try the cleaning methods on real data with ACT DR6 maps and the unWISE galaxy catalog.
   - Use machine learning in constructing an optimal LSS tracer

Prerequisite Skills required
Basic familiarity with cosmology (e.g. from the Part III Cosmology course or equivalent) would be helpful.

Programming experience in Python and Jupyter Notebooks, as well as shell scripting is advantageous, but all required skills can be developed during the project.

Useful References (List of important papers/review articles relevant to the project)

- Kusiak et al. 2023 (https://arxiv.org/abs/2303.08121)

- https://github.com/olakusiak/deCIBing

General References (List of papers referred to in the project)

- Remazeilles et al. 2011 (https://arxiv.org/abs/1103.1166)

- Omori 2022 (https://arxiv.org/abs/2212.07420, AGORA simulations)

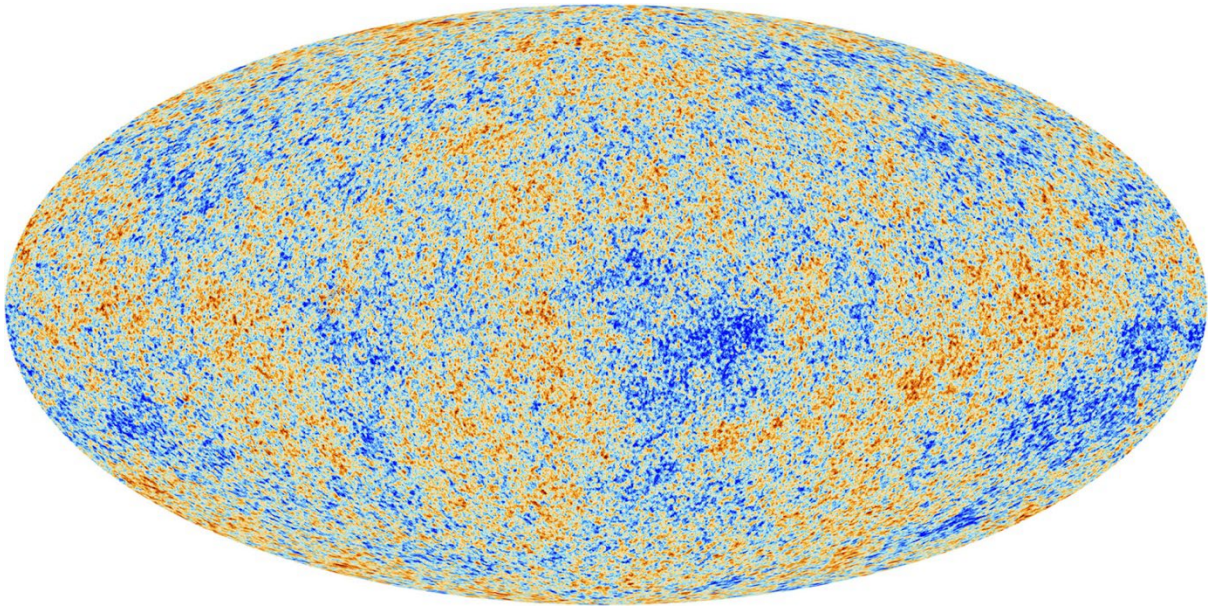


Figure 1: The Planck map of cosmic microwave background temperature anisotropies. Its power spectrum (Figure 2.) enables extremely precise constraints on our cosmological model. Image credit: ESA (https://www.esa.int/ESA_Multimedia/Images/2013/03/Planck_CMB)]

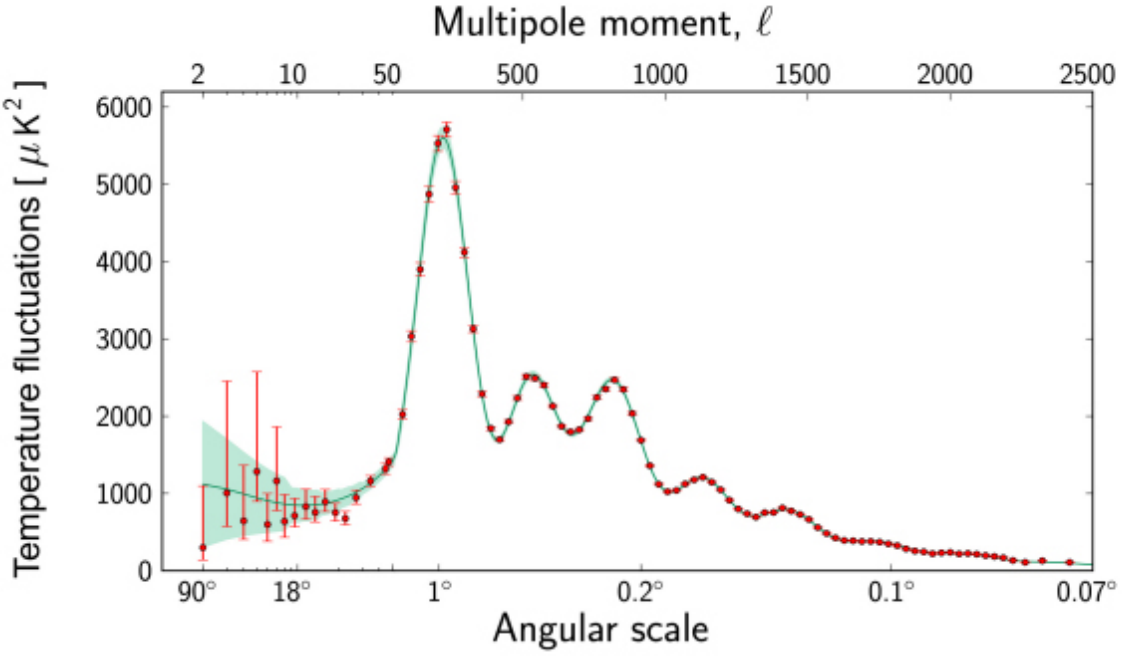


Figure 2: The CMB temperature power spectrum as measured by the Planck satellite. It is an extremely rich source of information for constraining the cosmological model. Image credit: ESA (https://sci.esa.int)

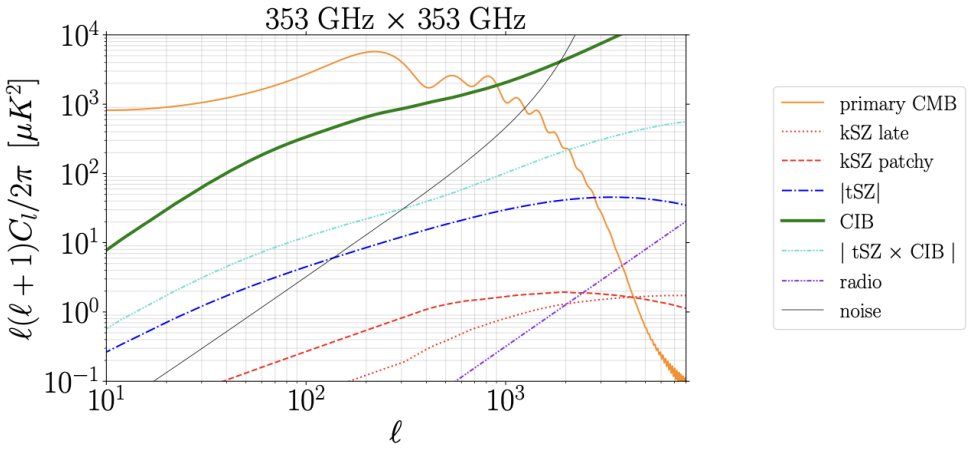


Figure 3: Microwave sky as observed at 353 GHz. It consists of the primary CMB anisotropies, secondary anisotropies (kSZ, tSZ), and foregrounds, e.g., the CIB or radio emission, as well as instrumental noise. At higher frequencies, the CIB becomes the dominant foreground, overtaking primary CMB at multipole moment ~1000. Figure from Kusiak et al. 2023.

Project 14: Magneto-Rotational Instability: convergence and connection to large-scale magnetic field generation

Supervisor I: Prasun Dhang (prasundhang@gmail.com)

Supervisor II: Roman Rafikov (rrr34@cam.ac.uk)

UTO: Roman Rafikov (rrr34@cam.ac.uk)

Project summary

The magneto-rotational instability (MRI) is thought to be responsible for the accretion of matter onto a central object in sources like X-ray binaries (XRBs) and Active Galactic Nuclei (AGNs). In this project, we will explore the convergence of MRI simulations in a zero net flux stratified isothermal local shearing box, a numerical approach often used to study this instability. Additionally, we will also investigate the generation of large-scale magnetic fields, which are interconnected to convergence and play a crucial role in producing jets and outflows in accreting sources.

Background

It is thought that most of the powerful sources (e.g. XRBs, AGNs, GRBs) in the Universe are energized by the extraction of gravitational energy of the material falling into the deep potential well of the compact objects like black holes (BHs), neutron stars (NSs). In astronomy, this process is called accretion. In rotationally supported accretion discs, accretion is inevitably accompanied by the angular momentum transport. Microscopic diffusivity alone is inadequate to provide the angular momentum transport required to match the observed luminosity of the accreting sources. Shakura and Sunyaev (1973) explained the angular momentum transport in terms of an emergent turbulent viscosity. However, a convincing source of turbulence was unidentified until the rediscovery of a local linear magnetohydrodynamic (MHD) instability in Keplerian flows, the magneto-rotational instability (MRI), by Balbus and Hawley (1991).

Although the linear MRI ensures outward angular momentum transport, understanding the instability in the nonlinear regime is essential. To date, the nonlinear regime has been studied extensively in the local shearing box and global simulations. A widely used configuration in local MRI studies is the zero-net-flux (ZNF) shearing box model, where the disc is threaded by a vertical magnetic field with zero net flux.

It is important to note that the reliability of numerical simulations is often assessed through convergence studies. In the case of ideal MHD—i.e., in the absence of explicit physical dissipation such as viscosity and resistivity—convergence in simulations of the magnetorotational instability (MRI) is typically

evaluated by examining whether volume-integrated quantities, such as the accretion stress driven by MRI-induced turbulence, remain independent of numerical resolution. However, with recent advances in computational power, very high-resolution ZNF shearing box simulations of MRI in the ideal MHD regime have revealed a lack of convergence (Ryan et al., 2017), raising important questions about the reliability of results obtained from earlier ZNF MRI simulations.

We propose to carry out stratified, isothermal, local shearing box simulations with zero net magnetic flux (ZNF) at varying numerical resolutions to revisit the longstanding question of convergence in MRI-driven turbulence. The primary objective of this study is to determine whether the non-convergence reported in previous works reflects a genuine physical effect or is merely a consequence of inadequate numerical resolution. A key focus will be the generation of large-scale magnetic fields—believed to play a crucial role in launching jets and outflows from accreting systems—and their potential maintenance by an MRI-driven dynamo. We aim to elucidate the intricate relationship between this dynamo activity and convergence behaviour in MRI simulations. To this end, we will compute turbulent dynamo coefficients from the simulations across a range of effective Reynolds numbers, which, in the framework of ideal MHD, are inherently controlled by numerical resolution.

Project details

A crucial component of this project involves conducting shearing box simulations with varying resolutions using PLUTO, a Godunov-type code. While simulations with the highest (~1024^3) and lowest (~128^3) resolutions have already been completed, the student's primary responsibility will be to run intermediate-resolution simulations with the help of supervisors and analyze all the data. The following tasks outline the student's responsibilities in detail:

Literature Review: Conduct a comprehensive literature survey on accretion physics, with a specific focus on magneto-rotational instability and dynamo.

Simulation Runs: Utilize the PLUTO code to run simulations with intermediate resolutions.

Data Analysis and Diagnostics: Analyze the simulated data, performing required diagnostics focussing on accretion physics to test convergence. The student is also encouraged to develop novel diagnostics to enhance the analysis.

Turbulent Dynamo Coefficient Measurement: Extract the turbulent dynamo coefficient from the simulated data.

Project Write-up and Manuscript Preparation: Prepare a detailed write-up of the project, and if time permits, draft a manuscript for submission to a journal.

Prerequisite Skills required

Basic knowledge of numerical techniques using Python and C.

Acceptable Programming Languages

Python and C.

Useful References (List of important papers/review articles relevant to the project)

- Balbus & Hawley, 1998, Reviews of Modern Physics, Volume 70, Issue 1, January 1998, pp.1-53
- Brandenburg & Subramanian, 2005, Physics Reports, Volume 417, Issue 1-4, p. 1-209.

General References (List of papers referred to in the project)

- Shakura & Sunayev, 1973, Astronomy and Astrophysics, Vol. 24, p. 337 – 355
- Balbus & Hawley, 1991, Astrophysical Journal v.376, p.214
- Ryan et al., The Astrophysical Journal, Volume 840, Issue 1, article id. 6, 14 pp. (2017).

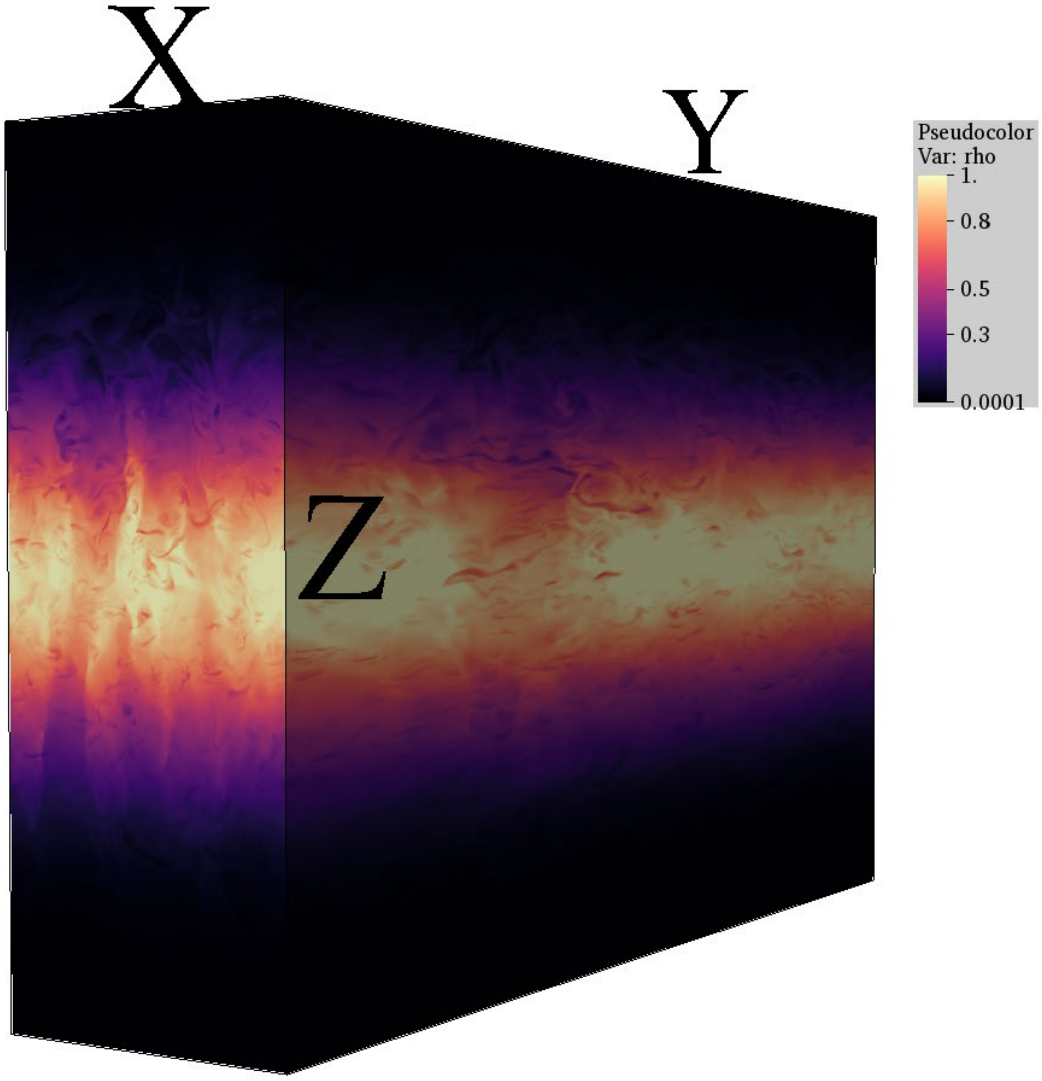


Figure 1: Volume rendering of the density in a stratified shearing box, depicting a dense disc midplane and a tenuous atmosphere.

Project 15: Origins scenarios for early Earth: the potential role of late delivery

Supervisor I: Dr Amy Bonsor (abonsor@ast.cam.ac.uk)

Supervisor II: Dr Paul Rimmer (pbr27@cam.ac.uk)

Supervisor III: Richard Anslow (rja92@cam.ac.uk)

UTO: Dr Amy Bonsor (abonsor@ast.cam.ac.uk)

Project summary

Impacts play a key role during the late stages of planet formation. There is great debate as to whether impacts play an important role in the origin of life on Earth. This project focuses on whether material present in impacts survives the dramatic events that brought these bodies to Earth. Building on recent observations that found organic molecules, amino acids and nucleobases on the asteroids Ryugu and Bennu this project will assess whether organic molecules could arrive onto early Earth during impacts.

Background

If we are to determine how life originated on Earth, it is key to assess the viability of various pathways to life. The history of early Earth was dominated by a period of intense bombardment. Many of these impactors are expected to contain many key feedstock molecules required for prebiotic chemistry. This project focuses on the ability of these impactors to deliver key precursor molecules for origins of life scenarios.

One potential pathway to prebiotic chemistry is known as the cyanosulfidic pathway. This relies on chemical feedstock molecules, including notably hydrogen cyanide, as well as chemistry activated by UV light, to build the amino acids and nucleobases required for life (Patel et al, 2015). One place where this chemistry could take place is in warm ponds on the surface of early Earth.

The chemical feedstock molecules are commonly present in comets and potentially asteroids in the outer Solar System. Whilst hydrogen cyanide could be synthesised on Earth, this project focuses on its potential delivery during impacts. This follows on from several key papers in our group, including McDonald et al., 2025, in press, who present simulations of cometary impactors onto a solid, rocky surface, mapping comets' internal pressure and temperature distributions during hypervelocity impact and Anslow et al., 2025 in prep who investigate the propensity of HCN to survive impact across a representative population of comets, considering Earth's bombardment history and the atmospheric evolution of the comets.

The destruction of aliphatic organics has previously been investigated for asteroidal and cometary impacts onto the dwarf planet Ceres (Bowling et al., 2020). Bowling et al., consider a range of impactor sizes, two impact angles (30° and 45°), and two representative impact velocities (4.8km/s and 10km/s) appropriate for Ceres.

Project details

The student will process an existing set of impact simulations to determine the chemistry on-going during the impact. The simulations represent rocky asteroidal bodies hitting early Earth, Mars or other Solar System body. The chemical network STAND will be used to analyse the survival of organics or other precursor molecules, relevant to origins scenarios. The project will provide key conclusions regarding the ability of impacts to provide chemical feedstock molecules for origins scenarios.

Prerequisite Skills required

Basic computational skills, programming

Other skills used in the project

Chemical reaction networks, Shock physics

Acceptable Programming Languages

Python would be ideal.

Useful References (List of important papers/review articles relevant to the project)

- Anslow, R.~J., Bonsor, A., Rimmer, P.~B.\ 2023.\ Can comets deliver prebiotic molecules to rocky exoplanets?.\ Proceedings of the Royal Society of London Series A 479. doi:10.1098/rspa.2023.0434
- Anslow, R.~J. et al, 2025.\ The atmospheric entry of cometary impactors.\ Monthly Notices of the Royal Astronomical Society 539, 376–392. doi:10.1093/mnras/staf507
- Changela, H.~G. and 20 colleagues 2024.\ The evolution of organic material on Asteroid 162173 Ryugu and its delivery to Earth.\ Nature Communications 15. doi:10.1038/s41467-024-50004-w
- Glavin, D.~P., et al., 2024.\ Abundant ammonia and nitrogen-rich soluble organic matter in samples from asteroid (101955) Bennu.\ Nature Astronomy 9, 199-210.\ doi:10.1038/s41550-024-02472-9
- Bowling T.~J. et al., 2020.\ An endogenic origin of cerean organics.\ Earth and Planetary Science Letters 534 116069.\ doi:10.1016/j.epsl.2020.116069
- Sasselov, D.~D., Grotzinger, J.~P., Sutherland, J.~D.\ 2020.\ The origin of life as a planetary phenomenon.\ Science Advances 6, eaax3419. doi:10.1126/sciadv.aax3419]

General References (List of papers referred to in the project)

- Patel, B.~H., Percivalle, C., Ritson, D.~J., Duffy, C.~D., Sutherland, J.~D.\ 2015.\ Common origins of RNA, protein and lipid precursors in a cyanosulfidic protometabolism.\ Nature Chemistry 7, 301–307. doi:10.1038/nchem.2202

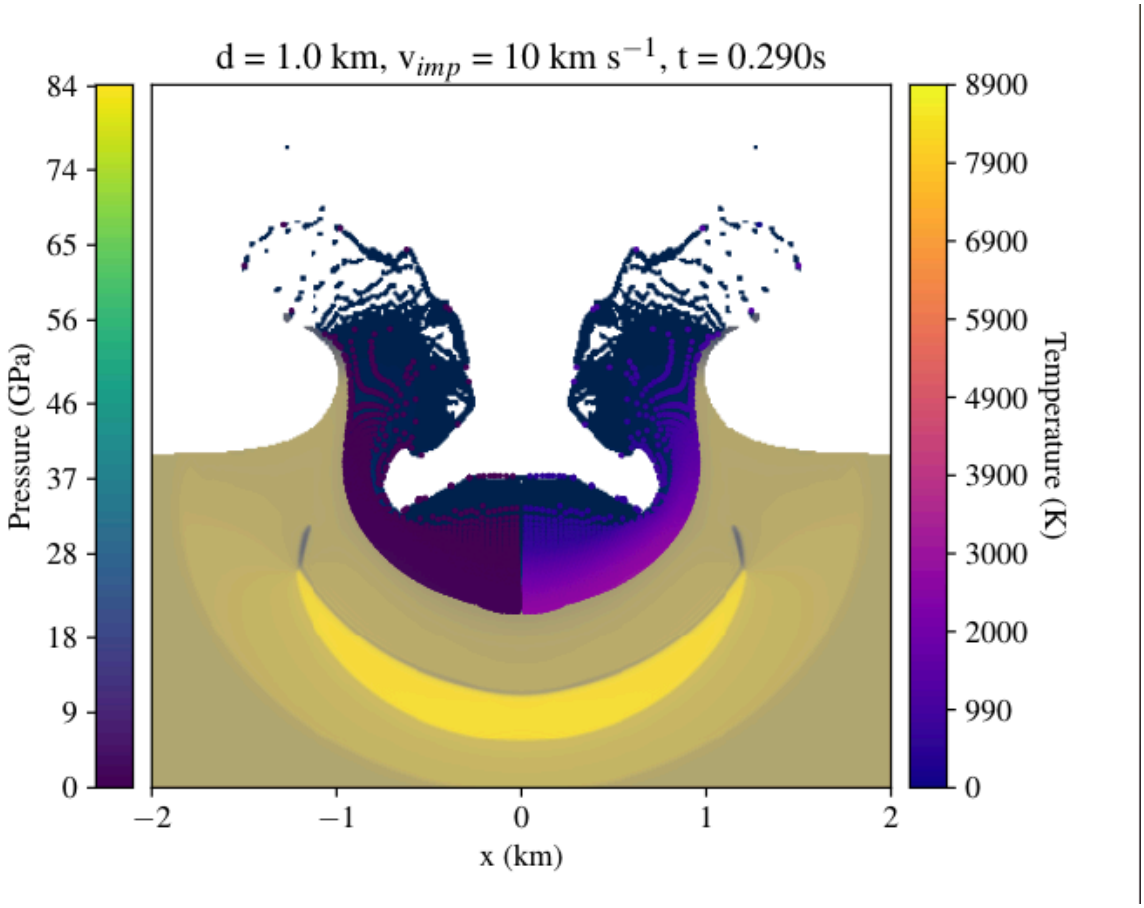


Figure 1: A cometary impactor of diameter 1km hitting early Earth (pure rock) at 10km/s. The impactor (and some target material) is shock heated to high temperatures and pressures (shown by the scale bars).

Project 16: The Future of Exoplanetary Composition Detection

Supervisor I: Dr Amy Bonsor (abonsor@ast.cam.ac.uk)

Supervisor II: Dr Mariona Badenas-Agusti (mb2716@cam.ac.uk)

UTO: Dr Amy Bonsor (abonsor@ast.cam.ac.uk)

Project summary

Exoplanet interiors are hard to probe observationally. Spectroscopic observations of white dwarfs provide unique insights regarding the bulk composition of exoplanetary material. Their high surface gravity renders white dwarfs the perfect laboratory, as their atmospheres should be pure hydrogen or helium. Material from surviving planetary systems enters the white dwarf atmosphere and shows up in their spectra. The availability of high-resolution spectra in the era of the Extremely Large Telescope (ELT) will improve white dwarf atmospheric characterisation. This project investigates the improvement to white dwarf atmospheric characterisation and will determine how much better new observations will be necessary to characterise the diversity of exoplanetary compositions.

Background

Almost all observed exoplanets orbit stars that will one day end their lives as white dwarfs. Whilst the inner regions of these planetary systems can be lost during the giant branch evolution, the outer planetary system should survive and orbit the white dwarfs. Such outer planetary systems can continue to evolve dynamically, with occasionally asteroids or comets being scattered into the inner system. Any body that approaches within a solar radius of the white dwarf is torn to pieces by the strong tidal forces and accreted onto the white dwarf. Photospheric metals from these planetary materials are seen in the white dwarf atmospheres, whilst infrared observations probe the dusty material accreting on to the white dwarf. Metals such as C, O, Mg, Ca, Cr, Ni seen in the white dwarf photosphere provide some of the best constraints available on the bulk composition of exoplanetary material. Planetary material in the atmospheres of white dwarfs has been used to probe the geology of exoplanets – with evidence for the formation of iron cores and the prevalence of radiogenic heating.

Project details

Spectroscopic observations of white dwarfs are key to providing the bulk composition of exoplanets: the more elements that can be detected, the better the insights regarding the exoplanetary material. This project will use current spectra of white dwarfs, alongside predictions from white dwarf atmospheric models, to predict the ability of future observations to probe the composition of exoplanetary material.

Spectroscopic observations of polluted white dwarfs provide a direct window into the bulk composition of extrasolar material: the more elements we detect, the deeper our insights into the geology and

chemistry of their accreted material. This project will use current spectra of polluted white dwarfs, alongside predictions from white dwarf atmospheric models, to predict the ability of future instruments to probe the composition of extrasolar bodies. A possible outline of the project is the following: first, the student will compile a library of observed and model spectra and use the ANDES ETC tool to compute the signal-to-noise (S/N) as a function of wavelength for a set of representative systems and exposure times. From these S/N metrics, the student will determine detection limits for important metal lines at various significance thresholds, enabling predictions of how many white dwarfs can show multiple heavy elements in future ELT-class observations. Second, the student will quantify joint-detection probabilities for key elemental abundance pairs under realistic observing scenarios (e.g. Mn-Na, Fe-Ni-Cr), providing a valuable roadmap to perform detailed abundance studies with next-generation ELT programs.

Prerequisite Skills required

The project will require basic computational and numerical skills, such as programming in Python. Attendance at the Exoplanets Part III Option would be helpful.

Other skills used in the project

Knowledge of stellar spectroscopy and Bayesian statistics would be beneficial but is not essential to the project.

Acceptable Programming Languages

Python

Useful References (List of important papers/review articles relevant to the project)

1. **Bonsor, A.** (2024). *White Dwarf Systems: The Composition of Exoplanets* [PDF]. In I. Mandel (Ed.-in-Chief) & D. Veras (Sec. Ed.), *Encyclopedia of Astrophysics*.
2. **Blouin, S.** (2024). *White dwarf fundamentals* [arXiv preprint]. arXiv:2409.03941. https://doi.org/10.48550/arXiv.2409.03941
3. **Marconi, A., et al.** (2024). ANDES, the high resolution spectrograph for the ELT: Science goals, project overview and future developments. *Proceedings of SPIE, 13096*, 1309613. https://arxiv.org/pdf/2407.14601
4. **Xu, S., Rogers, L. K., & Blouin, S.** (2024). The chemistry of extra-solar materials from white dwarf planetary systems [arXiv preprint]. arXiv:2404.15425. https://doi.org/10.48550/arXiv.2404.15425

General References (List of papers referred to in the project)

- **Bédard, A.** (2024). *The spectral evolution of white dwarfs: Where do we stand?* Astrophysics and Space Science, 369, Article 43. https://doi.org/10.1007/s10509-024-04307-5 link.springer.com

- **Harrison, J. H. D.** (2020). *Polluted white dwarfs: Insights into ancient extrasolar planetary systems* (Doctoral dissertation, University of Cambridge). https://doi.org/10.17863/CAM.58441 repository.cam.ac.uk
- **Saumon, D., Blouin, S., & Tremblay, P.-E.** (2022). *Current challenges in the physics of white dwarf stars.* Physics Reports, 988, 1–63. ui.adsabs.harvard.edu
- **Veras, D.** (2016). *Post-main-sequence planetary system evolution.* Royal Society Open Science, 3, 150571. https://doi.org/10.1098/rsos.150571
- **Xu, S., Bonsor, A**. (2021) *Exogeology from Polluted White Dwarfs,* Elements 17, 241. doi:10.2138/gselements.17.4.241

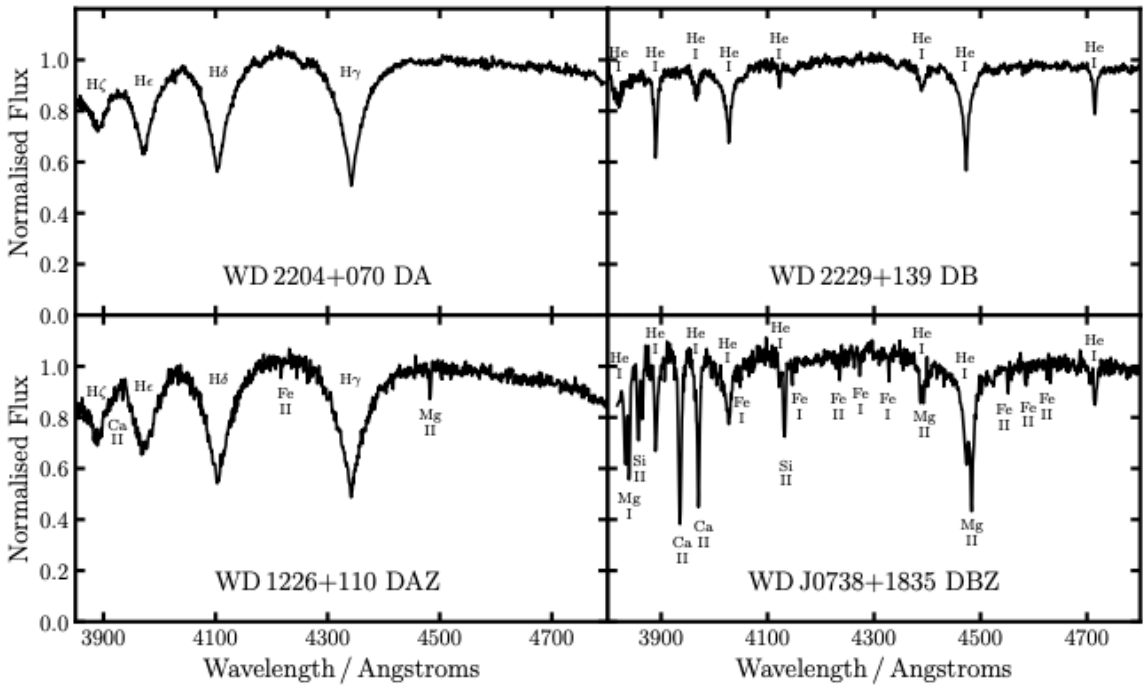


Figure 1: The optical spectra of a DA, DAZ, DB, and DBZ white dwarf. Polluted white dwarfs are easily identified by the strong Ca (and other metal) absorption features in their optical spectra (Harrison et al, 2020)

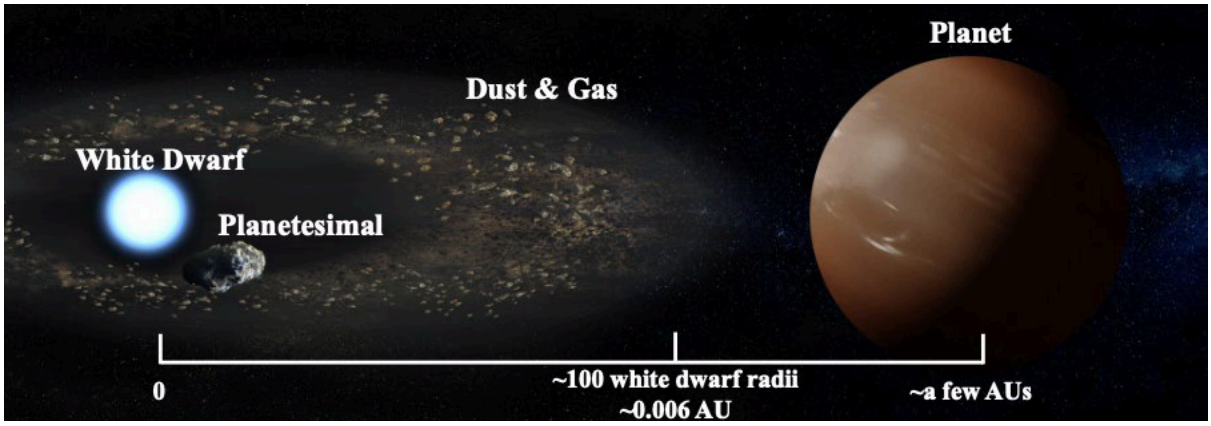


Figure 2: A schematic view of a white dwarf planetary system. Planets scatter asteroids or comets from a suriving outer planetary system in towards the star, where they are torn apart by the strong tidal forces and accreted. (Xu et al, 2021)

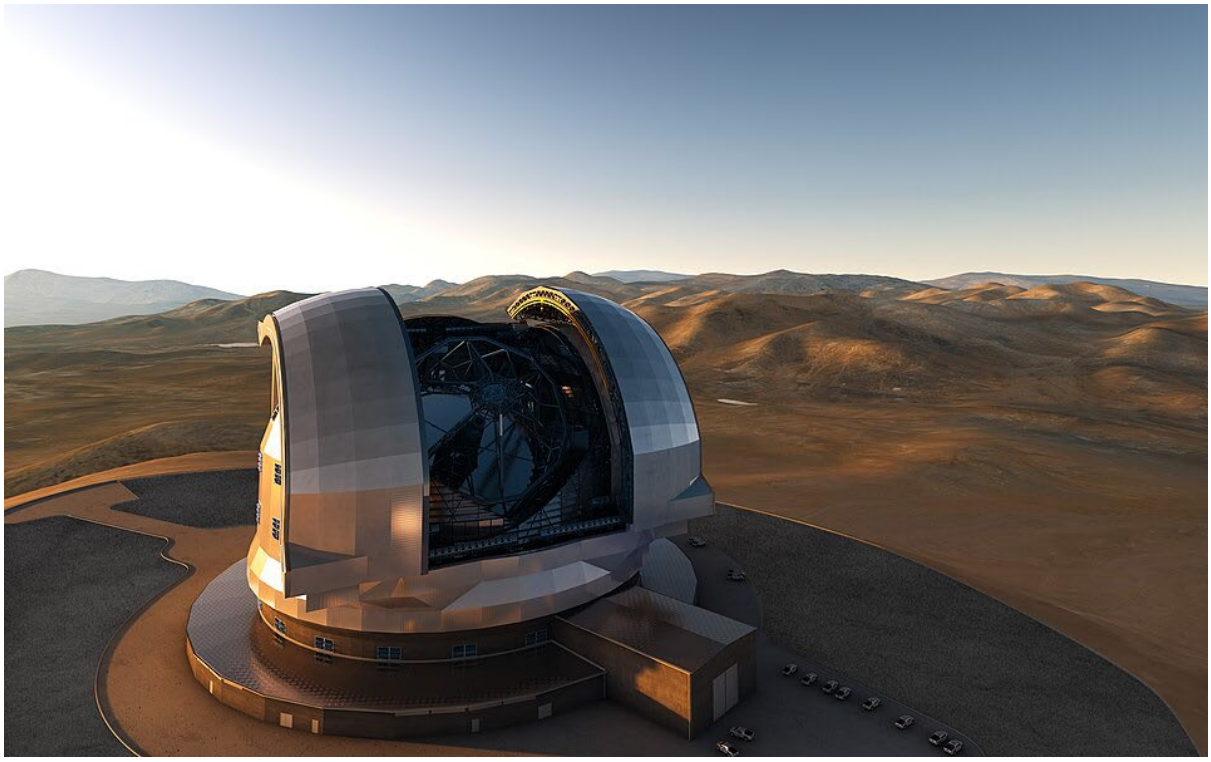


Figure 3: Artist's impression of the ELT.

Project 17: The calm before the burst: investigating accretion rate variability in protoplanetary discs

Supervisor I: Dr Cristiano Longarini (cl2000@cam.ac.uk)

Supervisor II: Professor Cathie Clarke (cclarke@ast.cam.ac.uk)

UTO: Professor Cathie Clarke (cclarke@ast.cam.ac.uk)

Project summary

Protostellar discs are gas and dust structures around young stars, playing a crucial role in the early stages of stellar evolution and in the process of planet formation. These astrophysical objects are accretion discs, meaning that angular momentum is transported throughout them and their material is delivered to the central protostar. The physical mechanism responsible for angular momentum transport in protoplanetary discs remains debated, with gravitational instabilities, magnetohydrodynamic processes, and turbulence all proposed as contributors. It is possible to work around this problem by assuming that the angular momentum is transported through a viscous-like mechanism, whose efficiency is described by a parameter called α. α-viscosity is not a viscous theory, but just a number which describes the efficiency of angular momentum transport (and hence the ratio of accretion rate to disc mass) regardless of its origin.

The aim of the project is to test whether the wide range of inferred alpha-viscosities in observed discs can be explained by temporal variability in the process of accretion onto the star, rather than by intrinsically different global transport mechanisms in each system. This is the *accretion bottleneck* hypothesis: discs may share a similar intrinsic efficiency of angular momentum transport, but this is manifest in short-lived bursts of efficient accretion onto the star

Background

The primary physical mechanism driving mass accretion and angular momentum transport in discs is still not well understood. Considering only the viscosity from microscopic collisions results in a viscous timescale that is too long to account for the formation of planetary systems. Therefore, it is generally accepted that angular momentum transport is driven by turbulent motions generated by hydrodynamical instabilities (Balbus & Papaloizou 1999). To quantify the efficiency of angular momentum transport, the alpha-viscosity prescription (Shakura & Sunyaev 1973) is usually employed, where α is a dimensionless parameter that determines the strength of viscosity. In general, the value of α is difficult to constrain.

A practical way to estimate α is to use observed accretion rates onto the central protostar, along with assumptions about disc properties such as mass, size, and temperature. This method has been applied in several works (e.g. Andrews et al. 2009, Rafikov 2017, Ansdell et al. 2019, van der Marel et al. 2021), all reporting a widespread in α values from 1E-4 up to 1E-1. More recently, Longarini et al. 2025a found the same result in a smaller sample (15 sources), for which the discs' properties are constrained very well, making possible a precise estimate of the α-viscosity.

One possible interpretation, as proposed in Longarini & Clarke (in prep.), is that this broad distribution of α does not reflect true differences in angular momentum transport efficiency, but rather traces variability in the rate of accretion onto the star. This hypothesis is supported by the finding that inferred α values shows no correlation with structural disc properties (mass, size, aspect ratio.

The aim of the project is to investigate the hypothesis of accretion rate variability in a large sample of protoplanetary discs (Manara et al. 2023). Through determining the distribution of α values in this large homogeneously analysed sample, new constraints can be placed on the typical pattern of temporal variability of circumstalle discs during the epoch of planet formation

Project details

The student involved in this project will investigate the hypothesis of accretion rate variability of protoplanetary discs within the Manara et al 2023 sample.

The main steps of the project consist of:

Extracting the relevant quantities from the Manara23 sample, such as mm-fluxes, disc sizes and accretion rates;

Converting mm-fluxes into disc masses, taking into account the different regimes of optical depth;

Computing a distribution of α values for the sample;

Using the distribution of derived α values to place constraints characteristic variability patterns in protoplanetary discs.

Prerequisite Skills required

A background of fluid dynamics, for example, the Part II course “Astrophysical fluid dynamics” is desirable.

Other skills used in the project

The project will require programming in a language of the student’s choice.

Acceptable Programming Languages

Python (preferable), C, fortran.

Useful References (List of important papers/review articles relevant to the project)

- Rosotti (2023), New Astronomy Reviews, Volume 96, ID 101674
- Lodato (2008), New Astronomy Reviews, Volume 52, ID 101016
- Manara et al. (2023), Protostar and planets VII, 539]

General References (List of papers referred to in the project)

- Balbus & Papaloizou (1999), The Astrophysical Journal, Volume 521, Issue 2, pp. 650-658.

- Shakura & Sunyaev (1973), Astronomy and Astrophysics, Vol. 24, p. 337 – 355
- Andrews et al. (2009), The Astrophysical Journal, Volume 700, Issue 2, pp. 1502-1523
- Ansdell et al. (2018), The Astrophysical Journal, Volume 859, Issue 1, article id. 21
- Van der Marel et al. (2021), The Astrophysical Journal, Volume 914, 161, 33
- Longarini et al. (2025), The Astrophysical Journal Letters, Volume 984, Issue 1, id. L17

Figure 1: Discs within the Manara23 sample in the accretion rate vs disc mass plane.. This data will be used by the student to compute the distribution of α values across the sample.

Figure 2: Results of Longarini & Clarke in prep., showing that the value of alpha does not correlate with structural properties of the discs, such as their mass, size or temperature, but it is strongly determined by the accretion rate.

Project 18: Galactic Neutron Star–White Dwarf Binaries as LISA Sources

Supervisor I: Dr Avishai Gilkis (agilkis@ast.cam.ac.uk)

UTO: Dr Christopher Moore (cjm96@cam.ac.uk)

Project summary

Binary systems composed of neutron stars (NSs) and white dwarfs (WDs) are promising gravitational wave (GW) sources in the low-frequency band targeted by the upcoming *Laser Interferometer Space Antenna* (LISA) mission. They also provide a clean probe of neutron star formation physics, such as natal kicks and common envelope evolution. Recent work has predicted the Galactic population of such systems using rapid population synthesis models. This project will revisit these predictions by constructing a synthetic population based on detailed stellar evolution models, enabling improved modelling of mass transfer and common envelope evolution phases. The main goal of the project is to assess the contribution of NS–WD binaries to the LISA signal and explore how predictions depend on assumptions about stellar and binary physics.

Background

Neutron star–white dwarf (NS–WD) binaries are remnants of stellar evolution and are expected to be present in abundance in the Milky Way. These systems emit gravitational waves (GWs) at millihertz frequencies, placing them in the sensitivity band of the upcoming *Laser Interferometer Space Antenna* (LISA) mission. While double white dwarfs (DWDs) are expected to be the dominant population of LISA sources – forming a strong GW foreground and the majority of individually resolvable systems – the relatively underexplored NS–WD binaries offer unique scientific opportunities. Unlike DWDs, which involve no supernovae, NS–WD binaries typically form through a single supernova, making them valuable laboratories for probing supernova physics and neutron star natal kicks. They also provide a cleaner evolutionary context than double neutron star systems, which involve two supernovae and are generally more complex.

Using rapid binary evolution models, it has recently been shown that dozens to hundreds of NS–WD binaries may be individually resolved by LISA, and that their frequency, chirp mass, and eccentricity distributions are sensitive to both natal kicks and common envelope prescriptions [1]. However, rapid population synthesis methods can suffer from oversimplifications. This project will explore the impact of incorporating the state-of-the-art stellar physics in modelling binary interaction phases leading to the formation of NS–WD binaries, based on detailed evolution models [2].

Project details
The student will construct a synthetic Galactic population of NS–WD binaries using detailed binary stellar evolution simulations carried out with the open-source stellar evolution code MESA (Modules for Experiments in Stellar Astrophysics [3]). The main aims of the project are:

(1) Investigate the key formation channels of NS–WD binaries and generate synthetic populations under varying assumptions about supernova physics.

(2) Compute the gravitational wave signal from these populations and assess their detectability by LISA.

(3) Quantify the sensitivity of predicted NS–WD populations and their GW characteristics to variations in stellar and binary interaction physics.

(4) Compare the resulting parameter distributions (chirp mass, eccentricity, spatial distribution) to those predicted by rapid population synthesis approaches.

A limitation of detailed binary evolution is its relatively high computational cost compared to rapid synthesis methods. To make the project manageable, a library of precomputed detailed binary evolution tracks covering a wide range of initial parameters is already available and can serve as a starting point [2]. The student will build upon this dataset and extend it where needed to explore specific trends or refine particular formation channels.

Prerequisite Skills required
Some knowledge of stellar evolution is essential. The part-II course on Structure and Evolution of Stars suffices but the part-III course is desirable.

Other skills used in the project
Use of scientific software (MESA), construction and analysis of synthetic Galactic populations, interpretation of GW observability criteria and source population modelling for LISA, data analysis and visualisation.

Acceptable Programming Languages
Basic coding skills with Python will be useful.

Useful References (List of important papers/review articles relevant to the project)
- “The demographics of neutron star - white dwarf mergers. Rates, delay-time distributions, and progenitors”, Toonen et al., 2018, A&A, 619, 53

General References (List of papers referred to in the project)
[1] “Neutron star - white dwarf binaries: probing formation pathways and natal kicks with LISA”, Korol et al., 2024, MNRAS, 530, 844

[2] “The landscape of binary core-collapse supernova progenitors and the late emergence of Wolf-Rayet winds”, Gilkis et al., 2025, arXiv e-prints, p. arXiv:2503.01993

[3] “Modules for Experiments in Stellar Astrophysics (MESA)”, Paxton et al., 2011, ApJS, 192, 3]

t/Myr	$M_1/M_\odot$	Type		$M_2/M_\odot$	Type	$a/R_\odot$
0	8.5	MS		6.5	MS	50
33.1	8.4	HG		6.5	MS	50.3
33.2	1.72	He MS		13.2	MS	290
40.2	1.62	He G		13.2	HG	292
40.2	1.1	WD		13.65	HG	569
40.2	1.1	WD		13.6	CHeB	570
40.3	1.1	WD		3.4	He MS	8.5
42.5	1.1	WD		3.0	He G	9.3
42.5	1.1	WD		1.3	NS	3.4

Figure 1: Example of the evolution of a NS–WD binary (from Toonen et al. 2018). Orbital and stellar parameters are given in the columns beside the illustration of the binary. The extent of the Roche lobes and stars in the illustration are approximate and are not drawn to scale. The abbreviations of the stellar types are the following: **MS** – main sequence star; **HG** – Hertzsprung-gap star; **He MS** – star on the equivalent of the main sequence for hydrogen-poor helium-burning stars; **He G** – hydrogen-poor helium-burning giant; **CHeB** – helium burning giant with a hydrogen envelope; **WD** – white dwarf; **NS** – neutron star.

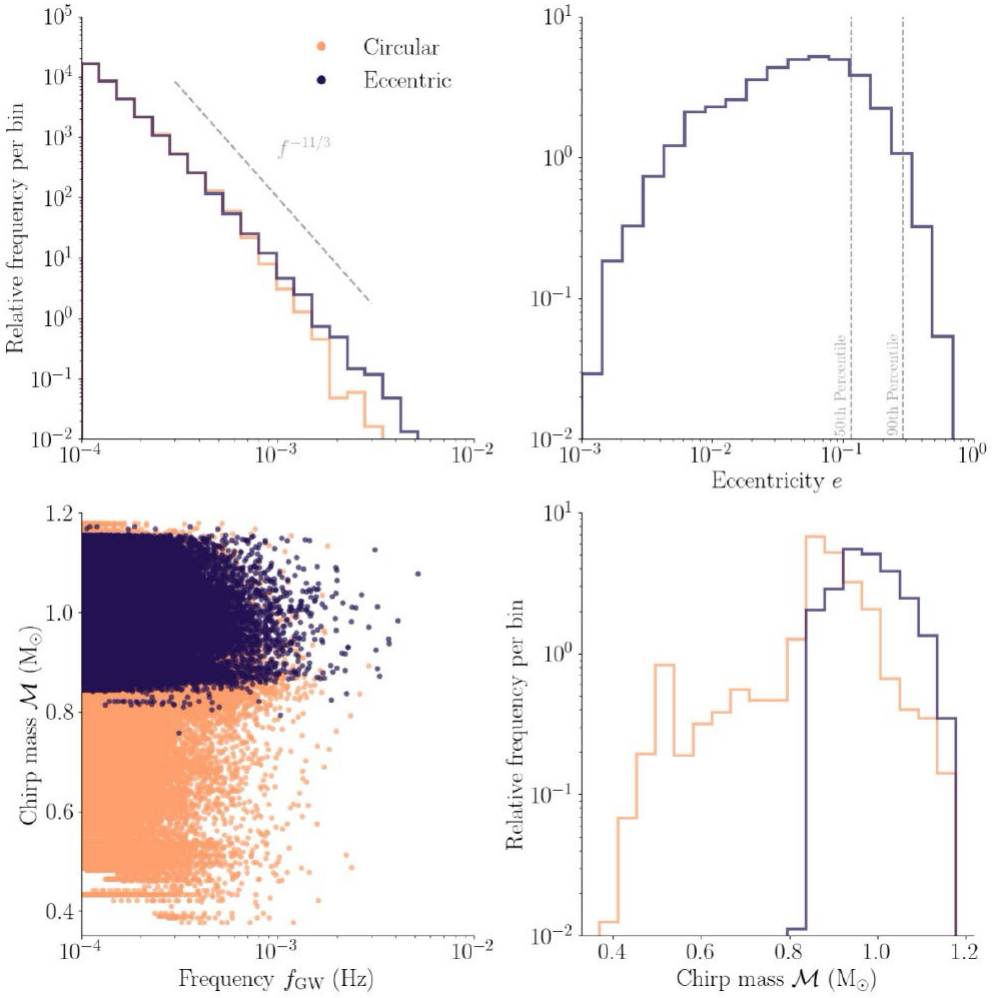


Figure 2: Model of the present-day Galactic NS–WD population (from [1]), represented in the frequency–chirp mass parameter space (bottom left), with projected frequency and chirp mass distributions per bin presented in the top left and bottom right panels, respectively. Binaries are categorised by colour: circular ($e = 0$, pale orange) and eccentric ($e \neq 0$, purple). The top right panel showcases the distribution of eccentricities. The dashed grey line in the top left panel shows the expected $\propto f^{-11/3}$ distribution in frequency, with fewer sources at higher frequencies because of the accelerating, or chirping, inspiral.

Project 19: Investigating Dust Particle Swarms within Magnetospheres

Supervisor I: Dr Max Sommer (ms3078@cam.ac.uk)

Supervisor II: Professor Mark Wyatt (wyatt@ast.cam.ac.uk)

UTO: Professor Mark Wyatt (wyatt@ast.cam.ac.uk)

Project summary

This planetary science project will analyse time-resolved data from the Geostationary Orbit Impact Detector (GORID) to investigate episodic swarms of dust impacts detected in Earth's magnetosphere. The student will examine correlations between impact timing, particle charge, geomagnetic activity, and spacecraft location to test the hypothesis that these swarms result from the electrostatic disruption of meteoroids. The findings will contribute to our understanding of dust–plasma interactions and support the interpretation of future in-situ dust measurements, such as those from *Europa Clipper*.

Background

The study of cosmic dust is an increasingly relevant field in astronomy and planetary science, offering insights into processes ranging from planetary formation to the dynamics of space environments. In addition to observing cosmic dust remotely through the light it emits or scatters, in-situ dust sensors aboard spacecraft enable detailed studies of individual particles. For instance, the Europa Clipper mission will use its dust analyzer to investigate material ejected from the subsurface oceans of Jupiter's moon Europa, shedding light on its potential habitability (Kempf et al. 2025).

This project revisits data from the Geostationary Orbit Impact Detector (GORID), which detected clusters of dust particle impacts in geostationary orbit from 1997 to 2002. While initial studies attributed these swarms to human-generated debris, a more recent perspective points toward a natural origin linked to geomagnetic storm activity (Sommer 2024). Preliminary reports of dust swarms within the Saturn magnetosphere recorded by the Cassini Dust Detector (CDA) have now added a new dimension to this phenomenon (Altobelli et al. 2025). This project aims to explore the correlation between the swarm occurrence and magnetospheric activity in greater detail. A hypothesized mechanism to generate the swarms of micron-sized particles involves the rapid charging and subsequent electrostatic disruption of meteoroids as they pass through excited parts of the magnetosphere—an intriguing but poorly understood process (Mendis 1984). This project may provide valuable evidence to support or refine theoretical models. Understanding the GORID data will help decipher the interaction between cosmic dust / meteoroids and magnetospheric plasmas, which could prove vital for the interpretation of dust data returned from Jupiter's magnetosphere once Europa Clipper arrives in 2030.

Project details

This project will involve the analysis of time-resolved impact data from the Geostationary Orbit Impact Detector (GORID), with the aim of identifying and characterising dust swarm events. The student will first familiarise themselves with the data set, which includes timing, dust particle charge, and detector orientation for each event. These will be cross-referenced with time-resolved (~hourly) geomagnetic activity indices (e.g. Kp, AE, Dst) available from public space weather databases, as well as with the spacecraft's position relative to key magnetospheric structures.

The analysis will focus on identifying statistically significant correlations between swarm occurrence and geomagnetic activity, and on examining how particle charge distributions vary within and outside of these events. The student will develop tools to visualise the data in time and space, and may also consider detector pointing geometry to constrain source directions of the incoming particles, which could in turn help to link observed events to specific meteoroid populations.

The analysis framework may also be extended to data from the Cassini Dust Analyzer (CDA), which also detected dust swarms in Saturn's magnetosphere (Altobelli et al. 2025; data publicly available). This offers an opportunity to compare phenomena across different planetary environments using consistent methodology.

If time allows, the project may also explore simple physical models for the electrostatic disruption of meteoroids in varying plasma conditions, to infer what local environments would be required for such a mechanism to operate.

Prerequisite Skills required

Familiarity with scripting languages.

Attendance at the Part III Planetary System Dynamics course would be beneficial.

An interest in space probes and in-situ instrumentation would be an asset.

Acceptable Programming Languages

Python, Matlab

Useful References (List of important papers/review articles relevant to the project)

- Grün, E. et al. (2019), "The Dawn of Dust Astronomy", Space Sci. Rev., 215:46, https://doi.org/10.1007/s11214-019-0610-1
- Hsu, H. et al. (2024), "In Situ Cosmic Dust Detection for Heliophysics", Whitepaper #176 in Decadal Survey for Solar and Space Physics 2024-2033, https://doi.org/10.3847/25c2cfeb.b165eaae
- Matzka, J. et al. (2021), "The Geomagnetic Kp Index and Derived Indices of Geomagnetic Activity", Space Weather 19(5) https://doi.org/10.1029/2020SW002641]

General References (List of papers referred to in the project)

- Kempf, S. et al. (2025), Space Sci. Rev., 221-1, 10, https://doi.org/10.1007/s11214-025-01134-0
- Sommer, M. (2024), Phil. Trans. A, 382-2273. https://ui.adsabs.harvard.edu/abs/2024RSPTA.38230370S/abstract
- Mendis, D.A. (1984), Adv. in Space Res, 4-9. https://doi.org/10.1016/0273-1177(84)90015-2
- Altobelli, N. (2025), MNRAS, 539-4, 3551-3567. https://doi.org/10.1093/mnras/staf656

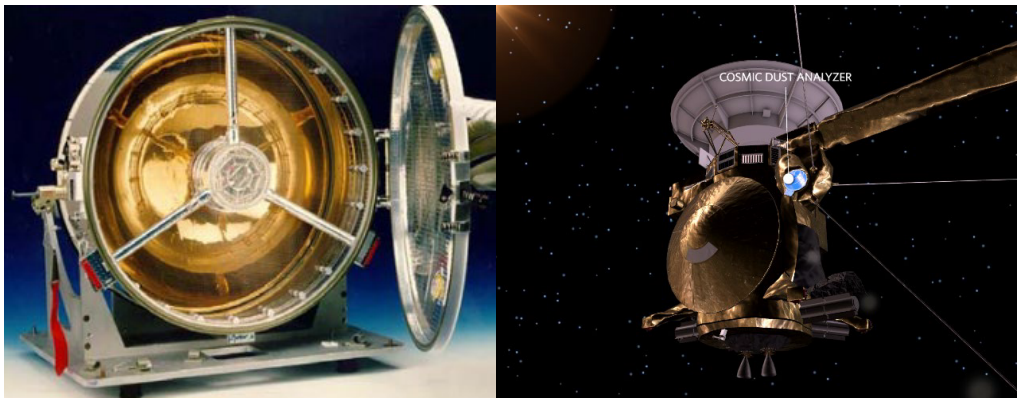


Figure 1: The GORID and CDA instruments.

Figure 2: The Cosmic Dust Analyzer onboard the Cassini spacecraft.

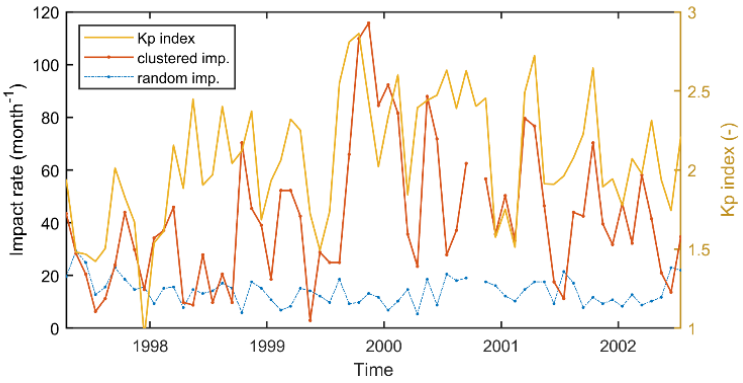


Figure 3: Monthly mean cluster occurrence with respect to monthly mean geomagnetic activity (Kp index). Preliminary analysis of the GORID dataset.

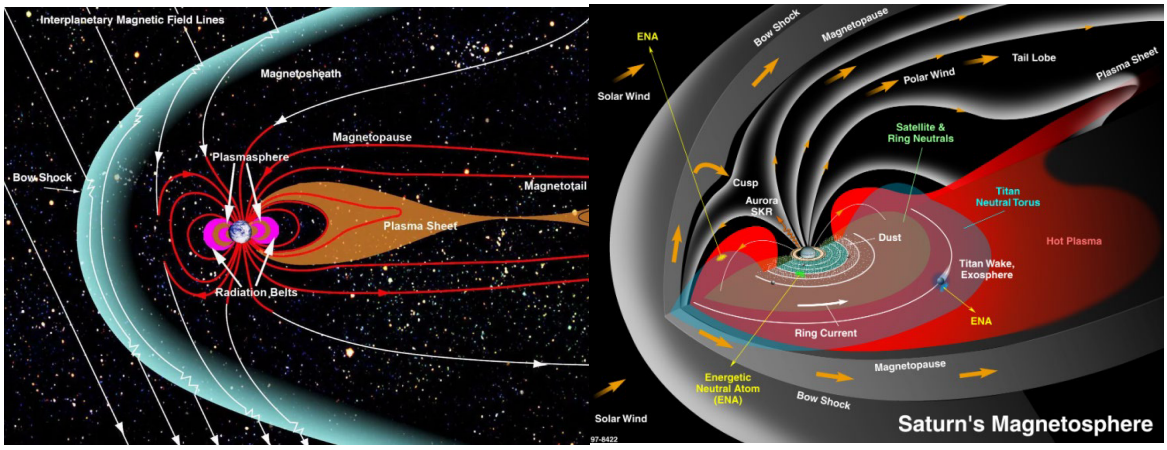


Figure 4: The Magnetospheres of Earth and Saturn. Swarm occurrence shall be analyzed with respect to magnetospheric region.

Project Title: Searching for Systematics in the Cosmic Microwave Background with Simulation-Based Inference

Supervisor I: Dr William Coulton (wrc27@cam.ac.uk)

Supervisor II: Dr James Alvey (jbga2@cam.ac.uk)

Supervisor III: Dr David Yallup (dy297@cam.ac.uk)

UTO: Dr Will Handley (wh260@cam.ac.uk)

Project summary

When a new paper makes exciting claims about a cosmological or astrophysical process, why should we trust them? A necessary, but not sufficient, piece of evidence is the use of “null and consistency tests”; these are simple tests that are designed to demonstrate that systematic effects are not unduly influencing the results. There is a wealth of Bayesian and Frequentist tools that underpin these tests. Simulation based-inference (SBI) is an exciting new paradigm where the explicit statistical model of classical approaches is replaced by an implicit model comprised of simulations. The goal of this project is to understand how to translate the suite of null and consistency tests to this new framework and to apply these tools to a cosmic microwave background dataset (Fig 1).

Background

Modern cosmological and astrophysical experiments are highly complex and, typically, astrophysicists are looking for subtle signals in these datasets. A key challenge is to understand how to differentiate the signal of interest from an unaccounted for instrumental, observational or computational (i.e. a bug) effect. The current gold standard is to construct an extensive suite of “null and consistency tests”; these tests typically involve dividing the data in parts and asking are these statistically consistent. Each of these tests is designed to isolate a potential source of systematic bias and passing these tests provides confidence that the result is robust to the tested sources of bias. Despite their apparent simplicity, these are proven tools for routing out unexpected biases.

The complexity of modern datasets often means that it is not possible to write down an analytical form for the probability distribution of our observables (consider the distribution of matter in the Universe). The leading approach to address this challenge is simulation-based inference (SBI). In this class of methods, we replace the analytical probability distribution with an implicit distribution formed by a large ensemble of simulations. A further benefit of this approach is that it also that is also directs researcher’s time to be focused on understanding the physics of the processes (and implementing that in the simulations) rather than developing complex (and approximate) analytical probability distributions. Machine learning methods are increasingly used to enable fast and precise SBI analyses; however, the black box nature of these methods is a source of concern for many. How can we trust these analyses if we do not “understand” all the intervening steps?

Project details
The goal of this project is to translate the ideas of “null and consistency” tests to the SBI framework. To demonstrate the efficacy of the developed ideas, we will use a set of mock cosmic microwave background (CMB) observations. Null and consistency tests are extensively used in that field — leading analyses include hundreds of them! — and it is simple to generate simulated datasets that contain systematics.

Specifically, the project will have three stages: first we will implement a simple SBI analysis for the cosmic microwave background power spectrum. Second, we will introduce divisions of the data, called splits, with plausible systematics for this analysis and demonstrate how a naïve analysis would miss the systematics and produce a biased inference. Finally, we will implement our SBI-null tests to demonstrate that we can identify the systematic effect. Our starting point for SBI-null tests will be to compute the probability for one data split given a second split, in the SBI framework.

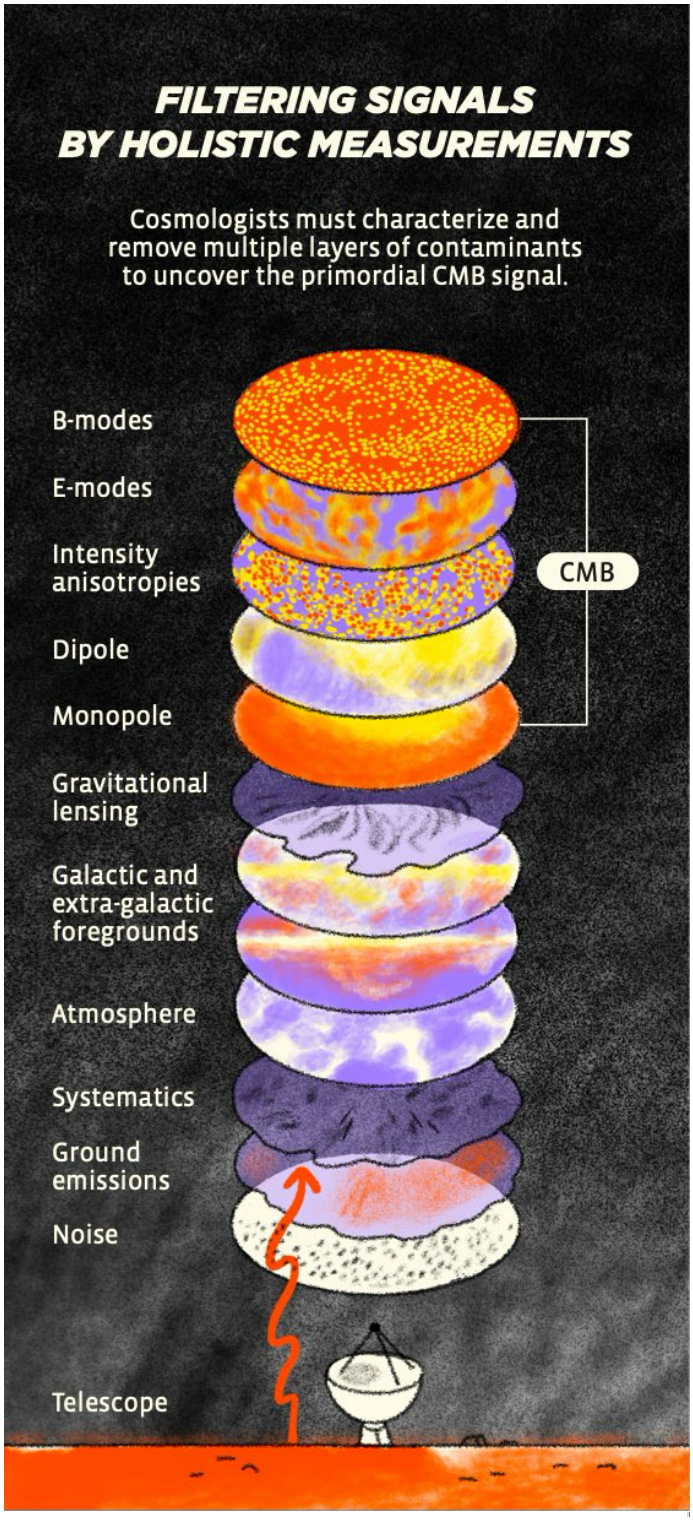
Prerequisite Skills required
The key skills are reasonable programming skills, rudimentary knowledge of cosmology, statistics and machine learning.

Other skills used in the project
An understanding of Bayesian statistical inference or experience with any ML framework (Tensorflow, pytorch, JAX…) is desirable, but not necessary.

Acceptable Programming Languages
Python

Useful References (List of important papers/review articles relevant to the project)

- The Atacama Cosmology Telescope: DR6 Power Spectra, Likelihoods and ΛCDM Parameters – The ACT collaboration (arXiv: 2503.14452)
- The frontier of simulation-based inference Kyle Cranmer, Johann Brehmer, Giles Louppe (arXiv: 1911.01429)
- Evidence Networks: simple losses for fast, amortized, neural Bayesian model comparison Niall Jeffrey, Ben Wandelt (arXiv: 2305.11241)
- Tests for model misspecification in simulation-based inference: from local distortions to global model checks, Noemi Anau Montel, James Alvey, Christoph Weniger (arXiv: 2412.15100)



Schematic of CMB systematics (credit Josquin Errard)

Figure 1: Schematic of CMB systematics (credit Josquin Errard)

Project 21: Ultra-stripped Supernova Progenitors – Formation, Evolution, and Gravitational Wave Signatures

Supervisor I: Dr Avishai Gilkis (agilkis@ast.cam.ac.uk)

UTO: Professor Christopher Tout (cat@ast.cam.ac.uk)

Project summary

Ultra-stripped supernovae (USSNe) are the endpoints of binary systems in which an initially massive star loses all its hydrogen and nearly all of its helium envelope before exploding. These systems might be critical for forming close double neutron stars – key sources of gravitational waves for ground-based observatories. This project will use detailed stellar evolution modelling to explore the full evolutionary pathway from zero-age main sequence binaries to ultra-stripped progenitors. Emphasis will be placed on mapping the parameter space leading to USSNe, quantifying how results depend on assumptions about common envelope evolution and supernova physics, and investigating the role of ultra-stripped systems as gravitational wave progenitors.

Background

Ultra-stripped supernovae (USSNe) occur in close binaries where a compact object, typically a neutron star, strips its companion through a combination of common envelope evolution and Roche lobe overflow. The resulting core is left with an extremely thin envelope and explodes with a low ejecta mass. These events produce neutron stars with low natal kicks and are thought to be a promising channel for forming merging double neutron stars. Previous studies (e.g. [1], [2]) used detailed stellar evolution simulations of helium stars starting from the helium zero-age main sequence, without fully simulating the entire evolution since the birth of the binaries. Recently, Gilkis et al. (2025) presented ultra-stripped progenitors evolved from zero-age hydrogen-burning main sequence binaries, but only for a small set of representative systems [3]. The proposed project will build upon previous results and systematically explore the range of binary parameters leading to USSN progenitors, as well as the physical assumptions, including common envelope efficiency and the time scale of envelope ejection. Key aims of the project will include modelling the detailed structure of USSN progenitors, computing their formation rate and assessing their contribution to the rate of merging neutron stars which are sources of gravitational waves as well as potential sites of heavy element formation.

Project details

The student will use detailed stellar evolution calculations to model the evolution of binary systems leading to ultra-stripped supernovae. The project goals include:

(1) Extending the set of initial binary conditions (masses, separations) leading to USSNe using detailed simulations.

(2) Investigating the sensitivity of USSN progenitor formation to common envelope physics (e.g. ejection efficiency, time scales) and neutron star natal kicks.

(3) Evaluating which systems produce merging double neutron stars and estimating gravitational wave signals for detectors like LIGO/Virgo.

(4) Assessing whether some ultra-compact progenitors may emit low-frequency gravitational waves before explosion, similar to NS–WD binaries in the LISA band, albeit briefly and rarely.

Prerequisite Skills required

Some knowledge of stellar evolution is essential. The part-II course on Structure and Evolution of Stars suffices but the part-III course is desirable.

Other skills used in the project

Use of scientific software, data analysis and visualisation

Acceptable Programming Languages

Basic coding skills with Python will be useful.

Useful References (List of important papers/review articles relevant to the project)

"Formation of Double Neutron Star Systems", Tauris et al., 2017, ApJ, 846, 170

General References (List of papers referred to in the project)

[1] "Ultra-stripped Type Ic Supernovae from Close Binary Evolution", Tauris et al., 2013, ApJ, 778, L23

[2] "Ultra-stripped supernovae: progenitors and fate", Tauris et al., 2015, MNRAS, 451, 2123

[3] "The landscape of binary core-collapse supernova progenitors and the late emergence of Wolf-Rayet winds", Gilkis et al., 2025, arXiv e-prints, p. arXiv:2503.01993]

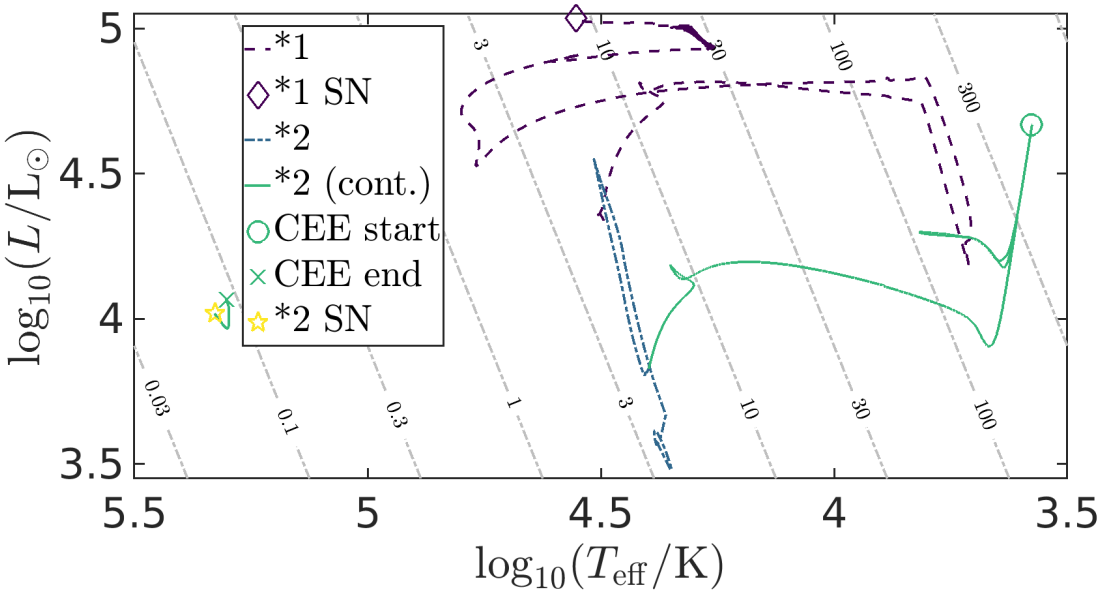


Figure 1: Example of the detailed evolution from a zero-age main sequence binary system through the first supernova and to the formation of an ultra-stripped progenitor after a common envelope evolution (CEE) phase involving the secondary star and the neutron star formed from the explosion of the primary star (from [3]). The diagram presents the effective temperature and luminosity evolution of the two stellar components in a system with a Solar-like metallicity (Z=0.014), initial masses M_1=16$M_\odot$ and M_2=8.8$M_\odot$, and initial orbital period P_i=201d. Lines of constant radius according to the temperature-luminosity relation are plotted, with the numbers indicating the radius in units of $R_\odot$.**]**

Project 22: Tracing the chemical and dynamical structure of the Milky Way disk with open clusters

Supervisor I: Dr GyuChul Myeong (gm564@cam.ac.uk)

Supervisor II: Professor N.W. Evans (nwe@ast.cam.ac.uk)

Supervisor III: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

UTO: Professor N.W. Evans (nwe@ast.cam.ac.uk)

Project summary

Open clusters have been providing a useful arena for studying the stellar formation and evolution. An open cluster typically consists of a single stellar population, with its stars born at the similar time and location. It is also sparse in density, making it easier to study the individual member stars. As there are an order of thousand open clusters known in our Galaxy so far, open clusters can help us to trace the chemistry and the stellar age distribution of the Galactic disk in detail. In addition, open clusters' orbit can be traced to seek their original birthplace.

Large-scale spectroscopic and astrometric surveys such as APOGEE, GALAH, and Gaia are providing rich data for studying the chemistry and dynamics of our Galaxy, on a scale that only has become available recently. While this provides us a valuable opportunity to study our Galaxy to a new extent, a suitable multi-dimensional data analysis is essential.

Based on this multi-dimensional information, we can study potential associations between the clusters and study the chrono*-chemo-dynamical structure of the Galactic disk.

*age estimates for open clusters from Cantat-Gaudin et al. (2020)

Background

A typical open cluster consists of a single stellar population as its stars are formed at the similar time and location, from the same materials. Because of its relative sparseness in spatial distribution, its member stars can be resolved more easily and studied individually. This leads to a better constraint on its properties such as age, chemical abundances, and distance than for the case of some isolated stars.

With such advantages, open clusters have been useful tracers for studying the chemical and dynamical structure of the Galactic disk. For example, the Galaxy's chemical abundance gradients (e.g., negative [Fe/H] gradient along Galactic radius) and the effects of radial migration in the disk are important for understanding the chemical evolution history of our Galaxy (see e.g., Doner et al. 2018, Netopil et al. 2021, Myers et al. 2022), but until recently there have been limitations on such study mostly due to the limit on the sample size and inhomogeneity in data.

The recent progress in large-scale spectroscopic and astrometric surveys, such as APOGEE (Blanton et al. 2017), GALAH (Martell et al. 2017), and Gaia (Gaia Collaboration et al. 2016), has revolutionised

the field with large and comprehensive new datasets. In addition, the Open Cluster Chemical Abundances and Mapping (OCCAM; Myers et al. 2022) survey provides a list of stars in 150 open clusters, with 16 chemical elemental abundances based on APOGEE DR17. Cantat-Gaudin et al. 2020 also provides useful parameters for open clusters including the cluster radius and age estimates. With various tools for Galactic Dynamics such as AGAMA (Vasiliev 2018), and suitable multi-dimensional clustering algorithms, the chemistry and orbital dynamics of individual stars and clusters will be studied in great detail which will expand our understanding on the chemo-dynamical evolution of the Galactic disk.

Project details
We will use a combination of spectroscopic and astrometric data from APOGEE, GALAH, and Gaia. A combination of APOGEE's OCCAM and Gaia provides a list of stars in ~100 "high quality" open clusters with chemical and astrometric information which will serve as a useful starting point for the project. The age of open clusters (Cantat-Gaudin et al. 2020) will also be considered in our analysis. The project will start from acquiring necessary information from the available survey datasets, and from deriving useful parameters (e.g., stellar orbits) from the observables (e.g., astrometric information).

With this rich dataset in hand, we will perform multi-dimensional analysis based on clustering algorithms to study the chrono-chemo-dynamical trends among the open clusters which will help us to understand the chemical and dynamical structure of the Galactic disk, such as chemical abundance gradients and radial migration.

Prerequisite Skills required
Familiarity with Python is desired.

Acceptable Programming Languages
Python, SQL

Useful References (List of important papers/review articles relevant to the project)

- The Open Cluster Chemical Abundances and Mapping Survey: VI. Galactic Chemical Gradient Analysis from APOGEE DR 17. Myers et al. (2022), AJ, 164, 85M
- The Galactic metallicity gradient shown by open clusters in the light of radial migration. Netopil et al. (2021), MNRAS, 509, 421N
- Painting a portrait of the Galactic disc with its stellar clusters. Cantat-Gaudin et al. (2020), A&A, 640A, 1C
- The Open Cluster Chemical Abundances and Mapping Survey: VII. APOGEE DR17 [C/N]-Age Calibration. Spoo et al. (2022), AJ, 163, 229S
- Mapping radial abundance gradients with Gaia-ESO open clusters: Evidence of recent gas accretion in the Milky Way disk. Palla et al. (2024), A&A, 690, A334

- OCCASO. V. Chemical-abundance trends with Galactocentric distance and age. Carbajo-Hijarrubia et al., A&A, 687, A239

General References (List of papers referred to in the project)

- Cantat-Gaudin et al. (2020), A&A, 640A, 1C
- Netopil et al. (2021), MNRAS, 509, 421N
- Donor et al. (2018), AJ, 156, 142
- OCCAM: Myers et al. (2022), arXiv:2206.13650
- APOGEE: Blanton et al. (2017), AJ, 154, 28
- GALAH: Martell et al. (2017), MNRAS, 465, 3203
- Gaia: Gaia Collaboration et al. (2016), A&A, 595, A1
- Vasiliev (2019), MNRAS, 482, 1525V

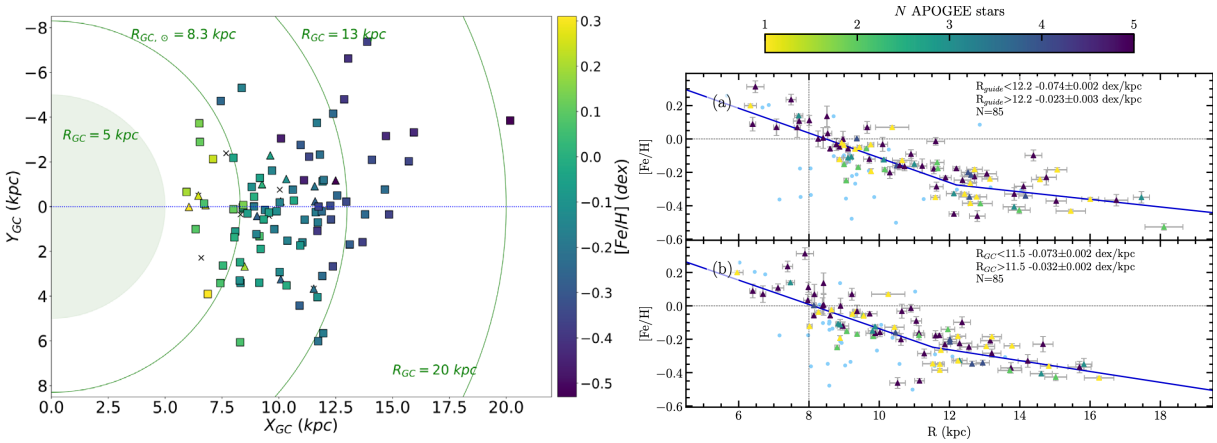


Figure 1: From Myers et al. (2022). *Left*: The Galactic X–Y distribution of a sub-sample of open clusters in OCCAM catalogue, colour-coded with their mean metallicity. The “high quality” clusters are marked with squares, lower quality clusters with triangles and crosses. *Right*: Metallicity gradients mapped as a function of guiding radius (upper panel), and current radius (bottom panel), along with a bilinear fit. Clusters flagged as potentially unreliable are shown as light blue circles. Colour-coding indicates the number of OCCAM member stars per cluster.

Project 23: Supernovae in Dustpedia Galaxies

Supervisor I: Dr Lisa Kelsey (lgk29@cam.ac.uk)

UTO: Professor Kaisey Mandel (km723@cam.ac.uk)

Project summary

This project will involve compiling a sample of type Ia supernova (SNe Ia) host galaxies from the Dustpedia galaxy archive, to investigate the unresolved correlation between SN brightnesses and the properties of their host galaxies, which is the largest uncertainty in modern SN cosmology. Dustpedia provides a wealth of galaxy data that is as-yet unexplored by the SN community and may provide insights on the underlying cause of this brightness difference. Depending on interest, the work can extend to comparisons with other supernova classes in Dustpedia hosts, re-deriving host properties with PROSPECTOR, or probing sub-galactic ("local") environments around each SN.

Background

SNe Ia, the powerful explosions of carbon-oxygen white dwarfs in binary systems, have characteristically low dispersion (~0.35mag) in their peak absolute magnitudes, making them ideal for use as cosmological distance indicators. To obtain the most precise cosmology, empirical corrections are made to further reduce their brightness dispersion, based on light-curve shape and colour. After these, ~0.08-0.10mag dispersion remains that cannot be explained by observational uncertainties.

It is likely that this dispersion is due to differences in SN Ia progenitors and explosion physics, which motivates searches for correlations with the closest proxy we have for their progenitor stellar populations, their host galaxies. After the empirical corrections, SN Ia brightnesses have a strong dependence on host galaxy properties: SNe Ia in high-mass, passive hosts are brighter than those in lower-mass, star-forming hosts (e.g., Sullivan et al. 2010). This is commonly referred to as the "mass step", but is also observed in other environmental properties, such as star formation rate and galaxy colour. This effect is the largest unknown in modern SN cosmology (Vincenzi et al. 2024). Furthermore, this effect is shown to be stronger when considering sub-galactic regions around the SNe, rather than the galaxy as a whole (e.g. Kelsey et al. 2021,2023).

Figure 1: An example of the host galaxy mass step; Figure 2 from Sullivan et al. (2010) The x axis shows the host galaxy stellar mass, split into high and low mass. The y axis is the difference in magnitude between observation and a model cosmology (the Hubble residual). The red markers represent the bin mean Hubble residuals and show a step in Hubble residual of ~0.06mag between low and high mass galaxies.

There are ongoing debates as to the cause of this difference: some consider it to be due to intrinsic differences between SNe Ia in different stellar populations, whilst others consider it to be due to differences in dust extinction along the line of sight to these SNe absorbing the SN brightness. Of course, the most likely scenario is that it is a combination of these effects. When considering SN host galaxies however, we are dealing with extended objects and so dust along the line of sight may not best evaluate the dust (Duarte et al. 2023). It is valuable to consider whether the brightness offset can be explained by host galaxy dust properties.

The Dustpedia archive (Davies et al. 2017) provides access to multiwavelength imagery and photometry in up to 41 wavelengths from the UV to the radio for 875 nearby galaxies observed by the Herschel Space Observatory. Also provided are model-derived parameters for each galaxy fitted with CIGALE using the THEMIS physical model for galaxy dust, including the star formation rate, stellar mass, dust mass, FUV attenuation and dust temperature. This valuable dataset provides opportunity to study stellar population properties that are not yet considered by the SN Ia community.

Project details

The aim of this project is to investigate the properties of SN Ia host galaxies in the Dustpedia galaxy sample. The student will:

- Undertake a literature review to familiarise themselves with supernova cosmology, and the ongoing debate on the cause of the unresolved dependence of SN Ia luminosities on environmental properties.
- Cross-check between the Transient Name Server, the Pantheon+ supernova sample (Scolnic et al. 2022) and Dustpedia galaxy sample to compile a sample of Dustpedia SN Ia host galaxies.
- Perform population analysis to compare the SN hosts to the full Dustpedia sample.
- Use the Pantheon+ subsample of cosmology SN Ia to measure the differences in brightness, and SN empirical parameters such as stretch and colour, with respect to host galaxy environmental properties from Dustpedia. For example, stellar mass, star formation rate, dust masses and dust properties. This sample can be supplemented by fitting more additional SNe with a light-curve fitter such as SALT (Guy et al. 2010, Kenworthy et al. 2021) or BayeSN (Mandel et al. 2022).

Depending on the student's interest, this project can continue in several directions. A non-exhaustive list of options are as follows:

- Compare the environments of SN Ia to other classes of SN in Dustpedia hosts – do different classes of SN explode in different environments?
- Compare the published Dustpedia properties to those obtained using the PROSPECTOR SED-fitting code. Does this change anything?
- Investigate sub-galactic differences in environments of SN Ia, using HostPhot and a SED-fitting code such as CIGALE or PROSPECTOR to obtain "local" properties. How do these compare to the galaxies as a whole?

Prerequisite Skills required
Python programming experience is required.

Interest in supernova astrophysics and data analysis

Relevant Part II and III courses: Introduction to Astrophysics, Structure and Evolution of Stars, Introduction to Cosmology

Other skills used in the project
Working with large tables and databases. The student will use libraries like numpy, pandas and astropy, and query and cross-match public catalogues.

Light-curve fitting.

Statistical and population analysis.

Data visualisation.

Literature review and scientific writing skills

Acceptable Programming Languages
Python

Useful References (List of important papers/review articles relevant to the project)

- Jha, Maguire and Sullivan. (2019) “Observational properties of thermonuclear supernovae” https://www.nature.com/articles/s41550-019-0858-0

General References (List of papers referred to in the project)

- Davies et al. (2017) “DustPedia: A Definitive Study of Cosmic Dust in the Local Universe” https://iopscience.iop.org/article/10.1088/1538-3873/129/974/044102
- Duarte et al. (2023) “A sample of dust attenuation laws for Dark Energy Survey supernova host galaxies” https://www.aanda.org/articles/aa/full_html/2023/12/aa46534-23/aa46534-23.html
- Guy et al. (2007) “SALT2: using distant supernovae to improve the use of type Ia supernovae as distance indicators” https://www.aanda.org/articles/aa/abs/2007/16/aa6930-06/aa6930-06.html
- Kelsey et al. (2021) “The effect of environment on Type Ia supernovae in the Dark Energy Survey three-year cosmological sample” https://academic.oup.com/mnras/article/501/4/4861/6043221
- Kelsey et al. (2023) “Concerning Colour: The Effect of Environment on Type Ia Supernova Colour in the Dark Energy Survey” https://academic.oup.com/mnras/article/519/2/3046/6939845
- Kenworthy et al. (2021) “SALT3: An Improved Type Ia Supernova Model for Measuring Cosmic Distances” https://iopscience.iop.org/article/10.3847/1538-4357/ac30d8
- Mandel et al. (2022), “A hierarchical Bayesian SED model for Type Ia supernovae in the optical to near-infrared” https://academic.oup.com/mnras/article/510/3/3939/6448478
- Scolnic et al. (2022) “The Pantheon+ Analysis: The Full Data Set and Light-curve Release” https://iopscience.iop.org/article/10.3847/1538-4357/ac8b7a
- Sullivan et al. (2010) “The dependence of Type Ia Supernovae luminosities on their host galaxies” https://academic.oup.com/mnras/article/406/2/782/998229
- Vincenzi et al. (2024) “The Dark Energy Survey Supernova Program: Cosmological Analysis and Systematic Uncertainties” https://iopscience.iop.org/article/10.3847/1538-4357/ad5e6c

Project 24: Exploring transients discovered by citizen scientists via Kilonova Seekers

Supervisor I: Dr Lisa Kelsey (lgk29@cam.ac.uk)

UTO: Professor Kaisey Mandel (km723@cam.ac.uk)

Project summary

In this project, the student will analyse approximately two years of volunteer transient discoveries from the Kilonova Seekers citizen science project from the GOTO collaboration, by compiling and fitting photometric and spectroscopic data to characterize each event and analyse the overall population. Additionally, the student will investigate volunteer behaviour, such as discovery biases, classification effort per detection, recovery rates, and response times, to inform improvements for the next generation of Kilonova Seekers. Throughout, they will engage with Kilonova Seekers operations, collaborating with the project team and gaining valuable experience in public engagement alongside scientific analysis.

Background

In the current era of time-domain astronomy, we are close to the limit of by-eye validation of potential transients due to the vast numbers of observations made per night. The TB-per-night data volumes of current and upcoming all-sky surveys highlight the need for automated, machine-learned source classification to vet and follow-up candidates quickly.

Transients are mainly discovered via difference imaging, where ‘reference’ images are subtracted from new ‘science’ images to remove non-varying sources from the image, resulting in a ‘difference’ image. Anything that remains in the difference image is something that has changed between the reference and science images, such as the appearance of a potential transient. However, these residuals need validating to ensure they are not artefacts, known as ‘bogus’ sources. Machine learning techniques are used to tackle the challenge of separating the real from the bogus, based on pre-labelled samples of training objects vetted by the community. However, as surveys grow larger, more efficient classifiers are required, meaning larger labelled samples are required to train these algorithms and fully sample the diversity of detections seen as new surveys come online. These data volumes quickly outpace small teams of scientists to effectively label. A complementary approach, which can be used to create a human-labelled data set for training machine-learning based classifiers, is to use citizen science.

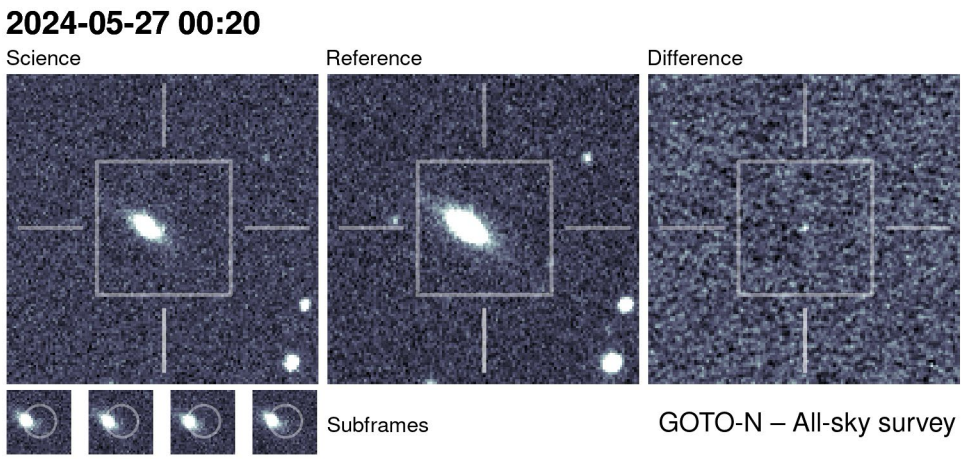


Figure 1: Example subject from Kilonova Seekers. The science, reference and difference images from the GOTO telescopes are plotted, along with subframes (individual exposures that are stacked to make the science image) and event information indicating which telescope took the science image, and the survey mode it was operating in. This example shows SN2024jih, a Type Ia Supernova discovered by the citizen science volunteers.

Citizen science enables collaboration between researchers and members of the public, by engaging the public to participate in research tasks and make scientific discoveries. For tasks such as vetting of candidate transients, the person-power increase of opening this task up to the public is highly significant. Transient astronomy projects on the Zooniverse citizen science platform (the predominant platform for citizen science) such as Galaxy Zoo Supernovae (Smith et al. 2011) and Supernova Hunters (Wright et al. 2017) have had great success involving the public in this way. In both cases, volunteers were provided with a set of target, reference, and difference images for a candidate transient that had been flagged as interesting by a computer algorithm and were asked to determine if the observation was real or bogus. Alongside the direct benefits for scientific analysis, citizen science provides an opportunity for public engagement by enabling members of the public to help in scientific discovery.

We launched the Kilonova Seekers citizen science project (Killestein and Kelsey et al. 2024) in June 2023, providing an opportunity for members of the public to help the GOTO collaboration (Steeghs et al. 2022, Dyer et al. 2024) in the discovery of transient events that may have been otherwise missed or overlooked, and enabling them to participate in cutting-edge science in near real-time. To date, the volunteers have reported >100 transients to the Transient Name Server (TNS), many of which would have been entirely missed by our current machine learning classifier. A prime example of this is GOTO0650 (Killestein et al. 2025), a bright dwarf nova discovered by volunteers only 3.5 hours after it was imaged by the GOTO telescopes, for which they are now the co-authors on the publication. We will soon be redeveloping the project to coincide with the end of the LIGO O4 observing run (the primary aim of GOTO is to follow-up alerts from these detectors), so now is the prime time to take stock of what the volunteers have found to date and gain further insights into how they interact with the platform to inform the future of Kilonova Seekers.

Project details

The project will begin with a literature review of the history of finding transients, and the requirement for new methods to improve efficiency of detection as we enter the era of large surveys, including the role of citizen science to complement modern machine learning methods. Knowledge and understanding of different transient phenomena, from detection to classification and analysis, will be gained through this review.

At time of writing, Kilonova Seekers has been operating for 2 years, with >100 transients discovered by volunteers having been reported to the TNS. This number is constantly growing. The main aim of this project is to analyse this sample of transients to document the dataset and understand what the volunteers have discovered. This will involve:

1. Querying public transient brokers, such as Lasair (Williams et al. 2024), to obtain their photometric data.
2. Obtain their spectral data (if available) from the TNS and fit using SNID (Blondin and Tonry, 2007) or Superfit (Howell et al. 2005) to confirm reported classifications.
3. Fit photometric data to models using REDBACK (Sarin et al. 2024) to estimate transient parameters.
4. Perform population analysis of the transient sample, and more in-depth analysis of anything particularly unique or interesting to the community.

In addition to understanding the transients that have been found by the citizen scientist volunteers to date, we are keen to understand the context of these discoveries in the Zooniverse framework. Key questions for the Kilonova Seekers team are:

- Do volunteers preferentially discover transients of a particular type?
- How does the GOTO survey operation mode impact discoveries?
- How many classifications do volunteers make on average before they make a discovery?
- What percentage of potential discoveries in this period did the volunteers recover?
- How soon after initial GOTO observation was the candidate agreed to be a real transient? Has this changed with time?

This will help the Kilonova Seekers team to prepare for the next generation of the project as we pivot to the new multiclass classifier after the ongoing LIGO O4 observing run.

Alongside the scientific analysis, the student will have the opportunity to gain experience at engaging the public with astronomy by becoming involved with the regular operations of the Kilonova Seekers project on Zooniverse and the Kilonova Seekers - LCO: STAR Global Sky Partners program.

Prerequisite Skills required
Python programming experience is desirable.

Some experience with, or willingness to learn, SQL is useful but not essential.

Interest in transient astrophysics, citizen science and public engagement.

Relevant Part II and III courses: Introduction to Astrophysics, Structure and Evolution of Stars

Other skills used in the project
- Data handling with Python libraries such as numpy, pandas, astropy.

- API and catalog querying.

- Use of spectral classification tools, and photometric fitting (eg. Using REDBACK or similar packages)

- Statistical and population analysis.

- Data visualisation.

- Literature review and scientific writing.

- Communication and coordination with the wider Kilonova Seekers working group in the GOTO collaboration.

- Public engagement and communication with the public on Zooniverse (optional but encouraged).

Acceptable Programming Languages
Python

Useful References (List of important papers/review articles relevant to the project)
- Kilonova Seekers project page – https://kilonova-seekers.org/
- Kilonova Seekers – LCO: STAR Global Sky Partners page - https://lco.global/education/partners/kilonova-seekers/
- Lintott et al. (2008), “Galaxy Zoo: morphologies derived from visual inspection of galaxies from the Sloan Digital Sky Survey”, https://academic.oup.com/mnras/article/389/3/1179/1017183
- Spiers et al. (2019) “Everyone counts? Design considerations in online citizen science” https://jcom.sissa.it/article/pubid/JCOM_1801_2019_A04/

General References (List of papers referred to in the project)
- Blondin and Tonry (2007), “Determining the Type, Redshift, and Age of a Supernova Spectrum” https://iopscience.iop.org/article/10.1086/520494

- Dyer et al. (2024), “The Gravitational-wave Optical Transient Observer (GOTO)” https://doi.org/10.1117/12.3018305
- Howell et al. (2005), “Gemini Spectroscopy of Supernovae from the Supernova Legacy Survey: Improving High-Redshift Supernova Selection and Classification”
- https://iopscience.iop.org/article/10.1086/497119
- Killestein and Kelsey et al. (2024), “Kilonova Seekers: the GOTO project for real-time citizen science in time-domain astrophysics” https://academic.oup.com/mnras/article/533/2/2113/7735340
- Killestein et al. (2025), “GOTO065054+593624: a 8.5 mag amplitude dwarf nova identified in real time via Kilonova Seekers”, Preprint available at: https://arxiv.org/abs/2501.11524
- Sarin et al. (2024) “Redback: A Bayesian inference software package for electromagnetic transients) https://academic.oup.com/mnras/article/531/1/1203/7671143
- Smith et al. (2011) “Galaxy Zoo Supernovae” https://academic.oup.com/mnras/article/412/2/1309/1747130
- Steeghs et al. (2022) “The Gravitational-wave Optical Transient Observer (GOTO): prototype performance and prospects for transient science” https://academic.oup.com/mnras/article/511/2/2405/6505141
- Williams et al. (2024) “Enabling science from the Rubin alert stream with Lasair” https://academic.oup.com/rasti/article/3/1/362/7712474

Project 25: Constraining physical properties of the population of type Ia supernovae

Supervisor I: Dr Nikhil Sarin (ns2043@cam.ac.uk)

Supervisor II: Dr Matt Grayling (mg2102@cam.ac.uk

UTO: Professor Kaisey Mandel (km723@cam.ac.uk)

Project summary

Type Ia supernovae (SNe Ia) are powerful cosmological probes, allowing for precise estimates of distance to faraway galaxies using various empirically derived relations to 'standardise' their observed properties. This approach first unveiled the accelerating expansion of the universe caused by 'dark energy', the nature of which remains one of the greatest mysteries in science. However, there is still much that is not known about the physical nature of these events, for example the exact details of their progenitor systems and physical processes which cause the explosion.

To improve our understanding of these events, recent work has focused on the application of advanced statistical techniques such as hierarchical Bayesian modelling to large samples of SNe Ia. However, this work is grounded in empirical, data-driven models which are agnostic to the physical processes driving these explosions. Previous studies applying physical models to SNe Ia have tended to focus on individual objects or comparatively small samples, and do not attempt to constrain distributions of physical properties across the population.

In this project, the student will work to combine these two approaches. They will apply physically-motivated models for the power sources driving SNe Ia to large new samples of SNe Ia such as the second data release of the Zwicky Transient Facility (ZTF), using hierarchical Bayesian inference to constrain the distribution of physical properties of SNe Ia across the population. This will be the first analysis of this kind, with a dataset containing thousands of SNe Ia, and will provide insight into how SNe in different environments vary and the nature of their progenitor systems.

Background

Type Ia supernovae are the explosions of white dwarf stars located in a binary system and are of wide interest as they can be used in cosmology as distance indicators – after applying a number of empirical relations to 'standardise' them, these objects display remarkable consistency across the population. However, these empirical relations are derived purely from observations and there remains much that we do not know about the physics of these events. For example, the exact nature of their progenitor stars - whether these binary systems contain a white dwarf with a massive companion, or two interacting white dwarfs - remains unknown. Additionally, we observe that type Ia supernovae located in different environments have different properties but the physical cause of this environmental dependence remains uncertain.

Beyond these empirical models, there are physically-motivated models for the explosions of type Ia supernovae which include treatments of effects such as radioactive decay of nickel-56. Though these

models are imperfect and incomplete, they can still be illustrative of some of the properties of these explosive events such as the mass of nickel formed in the explosion and the mass of the ejected material. Studying these physical characteristics can inform us about the nature of the progenitor systems leading to these explosions. Given that we observe an environmental-dependence across the population, constraining the distributions of physical properties of SNe Ia in different types of environment may give some more insight into what is driving those differences..

Project details
The purpose of this project is to constrain the distribution of physical properties of SNe Ia across the population using hierarchical Bayesian modelling. Specifically, the student will:

- Perform a literature review to become familiar with type Ia supernova astrophysics and cosmology.
- Investigate different physical models for SNe Ia e.g. the Arnett model, based on the radioactive decay of Nickel-56.
- Use hierarchical Bayesian modelling to uncover the physical population properties using a homogenous literature sample of Ia SNe.

Prerequisite Skills required
[Required: An interest and motivation to learn Python programming and analysis of real supernova data

Desired (though not essential): experience coding with Python, some familiarity with Bayesian statistics]

Other skills used in the project
Literature review. Radiation modelling in astrophysical outflows

Acceptable Programming Languages
Python

Useful References (List of important papers/review articles relevant to the project)
- Arnett 1982 - https://ui.adsabs.harvard.edu/abs/1982ApJ...253..785A/abstract
- Scalzo et al 2019 - https://ui.adsabs.harvard.edu/abs/2019MNRAS.483..628S/abstract
- Ruiter and Seitenzahl - https://ui.adsabs.harvard.edu/abs/2025A%26ARv..33....1R/abstract
- Maoz et al 2014 - https://ui.adsabs.harvard.edu/abs/2014ARA%26A..52..107M/abstract

General References (List of papers referred to in the project)
- Sarin et al. 2024 - https://arxiv.org/abs/2308.12806f

Project 26: Calibrating Euclid galaxy cluster masses with CMB lensing

Supervisor I: Dr Louis Legrand (ll783@cam.ac.uk)
Supervisor II: Dr Laura Salvati
Supervisor III: Dr Inigo Zubeldia (iz221@cam.ac.uk)
UTO: Professor Blake Sherwin (bds30@cam.ac.uk)

Project summary
In March 2025, the Euclid collaboration released its first dataset (Q1), covering approximately 50 deg², just a fraction of the 14,000 deg² that the full mission will eventually survey. Despite this limited area, over 400 galaxy clusters have already been detected. In this project, the student will carry out the first mass calibration of Euclid-detected galaxy clusters using gravitational lensing of the Cosmic Microwave Background (CMB), using publicly available CMB lensing maps.

Euclid is mapping the sky with unprecedented precision, with the goal of observing billions of galaxies and identifying hundreds of thousands of clusters, including many at redshifts z>1. In this high-redshift regime, conventional mass calibration methods using galaxy weak lensing become less effective due to decreasing background galaxy density. CMB lensing offers a robust alternative, providing an unbiased probe of total cluster mass, including dark matter, at higher redshifts.

Background
Galaxy clusters are the most massive gravitationally bound systems in the Universe. Their abundance as a function of mass and redshift provides a powerful probe of cosmology, particularly of the growth of structure and the matter content of the Universe. However, measuring cluster masses accurately is a challenging task.

While clusters contain hundreds to thousands of galaxies, most of their baryonic mass resides in hot intracluster gas. This baryonic component accounts for only ~20% of the total mass—the rest being dark matter.

Cluster mass estimates can be derived from observations of the hot intracluster medium via X-ray and millimeter measurements. However, these methods rely on assumptions such as hydrostatic equilibrium and require external calibration to account for non-thermal pressure support and to accurately recover the total mass, including the full dark matter component.

Gravitational lensing, in contrast, is sensitive to the total mass along the line of sight. Using the CMB as a backlight allows one to measure the subtle distortions caused by galaxy clusters in the CMB image, yielding an unbiased estimate of their mass. This technique is particularly well-suited for high-redshift clusters and complements Euclid's observational strategy.

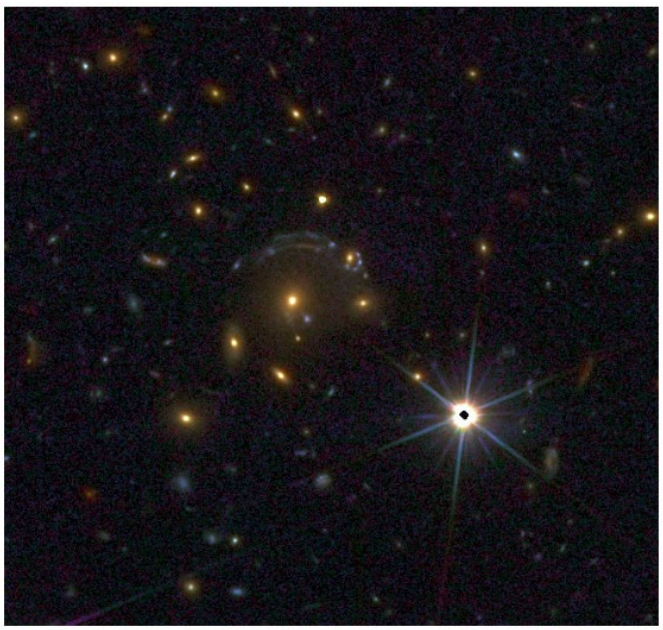


Figure 1: Zoom in on a galaxy cluster from the Euclid Q1 sample, at a redshift of z=0.58, where we can also see a strong lensing arc (Bhargava et al 2025).

Project details

The student will calibrate the masses of Euclid Q1 galaxy clusters using CMB lensing maps from the Atacama Cosmology Telescope (ACT). The project can be structured as follows:

Background Familiarization:

- Understand cluster detection and calibration techniques.
- Study the Euclid Q1 cluster catalogue (e.g., Bhargava et al. 2025) and the ACT CMB lensing products.
- Learn the basics of CMB lensing reconstruction, with the quadratic estimator method.

Data Analysis:

- Use standard Python-based tools to stack the CMB lensing convergence maps at the locations of Euclid clusters.
- Assume a standard cluster mass profile (e.g., NFW) to fit the stacked signal and estimate the average mass.

Methodological Refinements (optional):

- Explore improved CMB lensing reconstruction techniques to reduce known biases (e.g., Hu et al. 2007).
- Investigate matched-filter approaches to enhance signal-to-noise (e.g., Zubeldia & Challinor 2019).

Cosmological Implications (optional):

- Use Bayesian inference to constrain cosmological parameters from the calibrated cluster catalogue.
- Combine results with complementary mass estimates from SZ or X-ray observations.

Skills required

- It would be helpful if the student took the Part III cosmology module
- Some experience coding in python and jupyter notebooks would be helpful (but not essential - this can be learned as they go along).

Useful References (List of important papers/review articles relevant to the project)

- Euclid Collaboration, S. Bhargava, C. Benoist, A. H. Gonzalez, M. Maturi, J. -B. Melin, S. A. Stanford, et al. “Euclid Quick Data Release (Q1). First Detections from the Galaxy Cluster Workflow.” arXiv, 2025. https://doi.org/10.48550/ARXIV.2503.19196
- Zubeldia, Íñigo, and Anthony Challinor. “Cosmological Constraints from Planck Galaxy Clusters with CMB Lensing Mass Bias Calibration.” Monthly Notices of the Royal Astronomical Society 489, no. 1 (October 11, 2019): 401–19. https://doi.org/10.1093/mnras/stz2153.
- Salvati, L., A. Saro, S. Bocquet, M. Costanzi, B. Ansarinejad, B. A. Benson, L. E. Bleem, et al. “Combining Planck and SPT Cluster Catalogs: Cosmological Analysis and Impact on Planck Scaling Relation Calibration.” arXiv, September 12, 2022. https://doi.org/10.48550/arXiv.2112.03606.

General References (List of papers referred to in the project)

- Allen, Steven W., August E. Evrard, and Adam B. Mantz. “Cosmological Parameters from Observations of Galaxy Clusters.” arXiv, June 10, 2011. https://doi.org/10.48550/arXiv.1103.4829.
- Medezinski, Elinor, Nicholas Battaglia, Keiichi Umetsu, Masamune Oguri, Hironao Miyatake, Atsushi J. Nishizawa, Cristóbal Sifón, et al. “Planck Sunyaev-Zel’dovich Cluster Mass Calibration Using Hyper Suprime-Cam Weak Lensing.” arXiv, January 10, 2018. https://doi.org/10.48550/arXiv.1706.00434.
- Saha, Sayan, Louis Legrand, and Julien Carron. “Cluster Profiles from Beyond-the-QE CMB Lensing Mass Maps.” arXiv, October 3, 2023. https://doi.org/10.48550/arXiv.2307.11711.
- Hu, Wayne, Simon DeDeo, and Chris Vale. “Cluster Mass Estimators from CMB Temperature and Polarization Lensing.” arXiv, January 10, 2007. https://doi.org/10.48550/arXiv.astro-ph/0701276.
- Raghunathan, S., S. Patil, E. Baxter, B. A. Benson, L. E. Bleem, T. L. Chou, T. M. Crawford, et al. “Mass Calibration of Optically Selected DES Clusters Using a Measurement of CMB-Cluster

Lensing with SPTpol Data." arXiv.org, October 25, 2018. https://doi.org/10.3847/1538-4357/ab01ca.

- Geach, James E., and John A. Peacock. "Cluster Richness–Mass Calibration with Cosmic Microwave Background Lensing." Nature Astronomy 1, no. 11 (October 9, 2017): 795–99. https://doi.org/10.1038/s41550-017-0259-1.

Project 27: Tracing Cosmic Gas with Precision: Integrating Cutting-Edge Data and Simulations

Supervisor I: Dr Boryana Hadzhiyska (bth26@cam.ac.uk)

Supervisor II: Lurdes Ondaro (lurdes.ondaro@dipc.org)

UTO: Dr Boryana Hadzhiyska (bth26@cam.ac.uk)

Background and Project Summary

One of the key open problems in cosmology and astrophysics is understanding the distribution of baryonic (gas) matter in the universe. In astrophysics, this is known as the missing baryon problem, which arises because a significant fraction of gas is in a low-density, low-temperature state, making it difficult to detect. In cosmology, this challenge is critical because key observational probes, such as weak gravitational lensing, depend on accurately modeling the relationship between baryons and dark matter. Recently, a promising new approach (Schaan et al. 2021) has emerged using the kinematic Sunyaev-Zel'dovich (kSZ) effect, which occurs when cosmic microwave background (CMB) photons scatter off moving gas surrounding galaxy groups and clusters, leaving a measurable imprint on CMB maps. By combining CMB data with galaxy surveys, we have made a high signal-to-noise measurement of the gas distribution (Hadzhiyska et al. 2024, Liu et al. 2025), revealing hints that gas extends much farther than dark matter. If confirmed, this would have major implications for cosmology, particularly for weak lensing analyses and constraints on the clumpiness of matter (σ_8) (Amodeo et al. 2021, Bigwood et al. 2024). However, to reach definitive conclusions, further work is needed using state-of-the-art simulations and improved theoretical models.

In this project, the student will work with high-resolution cosmological simulations to interpret the latest gas and dark matter measurements and determine whether the missing baryon problem can be resolved using these new observations. Additionally, the student will explore how baryons impact weak lensing observables and whether these effects could bias cosmological constraints. Depending on progress and interest, the project could also include preliminary data analysis to improve constraints on key cosmological parameters, such as σ_8. This project offers an exciting opportunity to work at the intersection of theory, simulations, and observational cosmology, contributing to one of the most pressing challenges in modern astrophysics.

Project Details

New exciting results from kinematic Sunyaev-Zel'dovich (a probe measuring gas density), thermal Sunyaev-Zel'dovich (a probe measuring gas pressure), X-rays (a probe measuring gas density and temperature) are all pointing towards the presence of strong baryonic feedback (i.e., large outflow and heating of the surrounding gas) around group-sized halos, which could have important consequences for weak gravitational lensing with the upcoming billion-dollar experiment, Vera Rubin Observatory (LSST). Here we will construct a model, using hydrodynamical and N-body simulations, that is able to

explain all of these observations self-consistently. We will then delve into the implications of such a model both on our understanding of astrophysics as well as our ability to undercover the fundamental cosmological model of the Universe. No familiarity with simulations or cosmological data is required prior to the start of the project. The student will not need to run new simulations and instead be using already existing ones: namely, the N-body simulation suites AbacusSummit and the hydro simulations IllustrisTNG, SIMBA, Flamingo and Illustris, to which Supervisor I has access (all but Flamingo are as of now in the public domain). The project will focus on their postprocessing and tuning to match the observations in the data in a self-consistent and physically motivated manner. As such our computational requirements are medium, at ~1000 node hours

Prerequisite Skills
Basic knowledge of thermodynamics, statistics

Basic skills with Python

Additional Skills
Basic scientific writing skills

Presenting in public

Programming Languages
Python

Useful and General References

1. Schaan, E., “Atacama Cosmology Telescope: Combined kinematic and thermal Sunyaev-Zel'dovich measurements from BOSS CMASS and LOWZ halos”, Physical Review D, vol. 103, no. 6, Art. no. 063513, APS, 2021. doi:10.1103/PhysRevD.103.063513.
2. Hadzhiyska, B., “Evidence for large baryonic feedback at low and intermediate redshifts from kinematic Sunyaev-Zel'dovich observations with ACT and DESI photometric galaxies”, arXiv e-prints, Art. no. arXiv:2407.07152, 2024. doi:10.48550/arXiv.2407.07152.
3. Liu, R. H., “Measurements of the Thermal Sunyaev-Zel'dovich Effect with ACT and DESI Luminous Red Galaxies”, arXiv e-prints, Art. no. arXiv:2502.08850, 2025. doi:10.48550/arXiv.2502.08850.
4. Amodeo, S., “Atacama Cosmology Telescope: Modeling the gas thermodynamics in BOSS CMASS galaxies from kinematic and thermal Sunyaev-Zel'dovich measurements”, Physical Review D, vol. 103, no. 6, Art. no. 063514, APS, 2021. doi:10.1103/PhysRevD.103.063514.
5. Bigwood, L., “Weak lensing combined with the kinetic Sunyaev-Zel'dovich effect: a study of baryonic feedback”, Monthly Notices of the Royal Astronomical Society, vol. 534, no. 1, OUP, pp. 655–682, 2024. doi:10.1093/mnras/stae2100.

Project 28: Shadows of Motion: Projected Dynamics of Orbits

Supervisor I: Professor N Wyn Evans (nwe@ast.cam.ac.uk)

Supervisor II: Dr GyuChul Myeong(gm564@cam.ac.uk)

UTO: Professor N Wyn Evans (nwe@ast.cam.ac.uk)

Project summary

The aim of the project is to test a new method and identify the degeneracies in inferring the mass distribution and flattening of a galaxy from a stream or orbit. The outputs will be corner plots that reveal correlations between parameters for hot and cold streams viewed around external galaxies, which will verify the validity and effectiveness of a new novel method of measuring galaxies' dark matter distribution.

Background

Stellar streams are elongated trails of stars that stretch across the halos of galaxies like the Milky Way and Andromeda (M31), formed when smaller satellite galaxies or globular clusters are torn apart by tidal forces. These streams preserve the orbital paths of their progenitors, offering valuable insights into the gravitational potential and dark matter distribution of their host galaxies. In the Milky Way, prominent examples include the Sagittarius and GD-1 streams, while M31 hosts a complex web of streams, such as the Giant Stellar Stream, revealing a rich history of galactic mergers. Studying these structures helps astronomers reconstruct the formation and evolution of large galaxies. Given that the available observable information is limited in most cases, we propose a new method which can overcome such problems.

Project details

Streams around external galaxies are observed in projection. We will work with (i) exact orbits and (2) streams generated from N body disruptions, in both cases assuming a simple triaxial potential (an anisotropic harmonic oscillator). The likelihood function controls the inference on the posterior distribution of parameters given the prior. In this problem, the likelihood function we want to propose and test is analytic, as are the projected orbits. This enables us to understand the degeneracies in the problem in a systematic way.

Specifically, we assume we have a single stream or orbit observed in projection, so that the stars have two-dimensional positions and velocities. We would like to infer (i) the mass enclosed M, (ii) the orientations of the principal axes of the potential θ and ϕ and (ii) the flattenings or the axis ratios p and q.

This will depend on the eccentricity of the stream or orbit e, its extent L and its pericentric position in the Galaxy D. So the first task is to visualise the joint posterior distributions (e.g., using corner plots) to

reveal correlations between parameters and highlight regions of high probability, indicating possible degeneracies.

The second task is repeat the analysis for streams (rather than orbits). In the limit of low velocity dispersion of the progenitor (σ -> 0), streams become orbits. To systematically investigate, we want to study a suite of simulations with different velocity dispersions, tracking how the degeneracies broaden as the hotness of the progenitor changes.

Prerequisite Skills required
Some knowledge of Hamiltonian Dynamics and/or Galactic Astronomy

Other skills used in the project
Computing skills

Acceptable Programming Languages
Python

Useful References (List of important papers/review articles relevant to the project)

- Deason A.J., Belokurov V., 2024, NewAR, 99, 101706. doi:10.1016. New Astronomy Reviews, 2024.101706
- Dey A., Najita J.R., Koposov S.E., Josephy-Zack J., Maxemin G., Bell E.F., Poppett C., et al., 2023, ApJ, 944, 1
- Nibauer J., Cranmer M., Belokurov V., Goodman J., Ho S., 2022, DDA, 54, 203.05

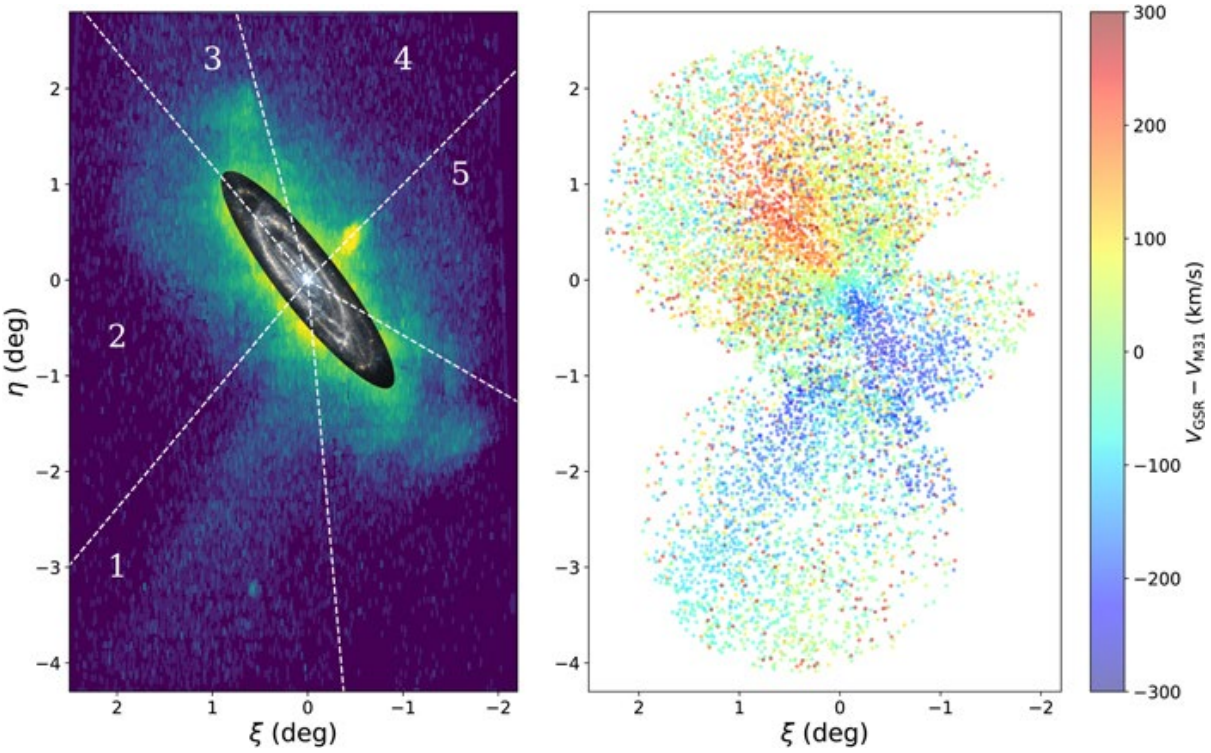


Figure 1: Left: Spatial density distribution of stars in the inner halo of M31 from the PAndAS catalog superposed for the central galaxy, M31. Right: spatial distribution of the subset of DESI targets. The points are colour-coded by radial velocity relative to

M31's recession velocity. Highly redshifted and blueshifted stars extend far from the disk. The GSS appears as a stream of blueshifted stars approaching the disk from the south. (Dey et al. 2023)]

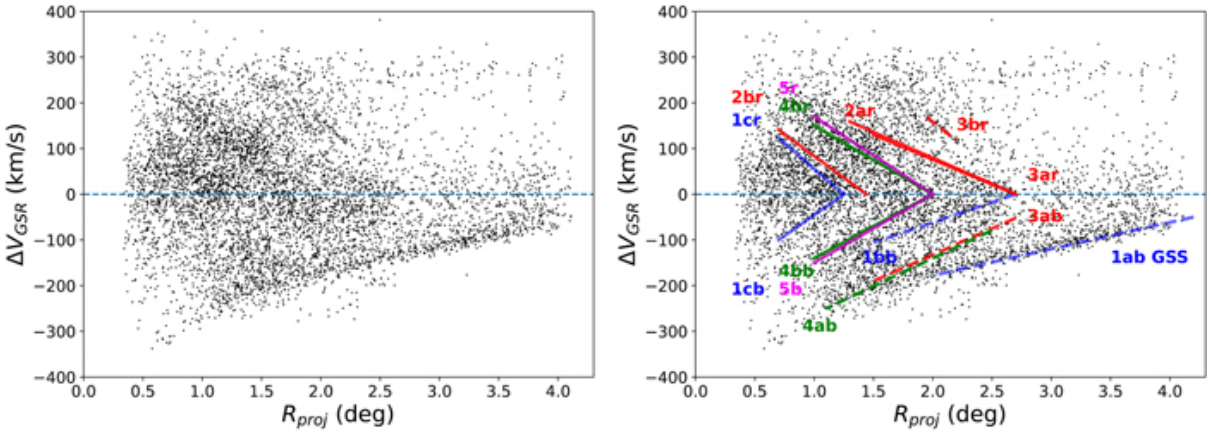


Figure 2: Combined line-of-sight positions and velocities for M31 stars. The right panel also shows the main stellar stream features appearing in phase space. (Day et al. 2023)]

Project 29: Accelerated Strong lens modelling with GPU based sampling

Supervisor I: Dr David Yallup (dy297@cam.ac.uk)

Supervisor II: Dr Natalie Hogg (nbh25@cam.ac.uk)

UTO: Dr Will Handley (wh260@cam.ac.uk)

Project summary

Gravitational lensing is a powerful tool for probing astrophysics and cosmology, but next-generation surveys like Euclid will produce vast datasets containing hundreds of thousands of lensed systems. Traditional analysis methods are too slow to process this data deluge, creating a significant bottleneck. This project aims to solve this problem by developing a new, ultra-fast analysis pipeline. We will combine a recently developed class of GPU-accelerated lens models with a state-of-the-art GPU-native sampler. The goal is to build and validate a robust pipeline capable of rapid, automated analysis of strong lensing systems, thereby unlocking the immense scientific potential of large-scale astronomical surveys.

Background

Light from distant astrophysical sources is deflected by the presence of mass along the line of sight, in the form of dark matter haloes which contain galaxies and galaxy clusters. If a galaxy of sufficient mass lies in near-perfect alignment with a source, multiple, distorted images of that source can be produced. This is known as strong gravitational lensing. By modelling the "lens" and source galaxies, we can infer a wealth of cosmological information from these images, such as the distribution of dark matter within the lens galaxy, and, if the source varies over time (for example, the source galaxy may contain a quasar or supernova), the expansion rate of the Universe.

One way to attack this problem is to fit parametric lens models to the observed lensed images. The accuracy and speed of these models are critical. The challenge is dramatically exacerbated by the imminent arrival of data from next-generation surveys like Euclid [Fig 1], which are expected to discover hundreds of thousands of new strong lenses. Current CPU-based inference methods are far too slow to handle this volume of data, threatening our ability to exploit these incredible new datasets.

Recent works [1] have demonstrated a class of lens models that can effectively leverage Graphics Processing Units (GPUs) to accelerate the calculation to unprecedented speed relative to classical CPU based inference models. This project will build directly on these advances to address the data analysis challenge. The full potential of these models is limited by the inference algorithms, however we have recently developed a new GPU native sampling tool [2] that could extract even more performance from these models.

Project details

This project will focus on creating an, end-to-end inference pipeline for strong lens analysis that is fully accelerated on GPUs. This project will investigate combining these models, with a new state of the art GPU native accelerated nested sampler [2], to probe the speed and robustness of this inference algorithm in this context, and build a pipeline for rapid analysis of lensed systems that can apply to incoming large data sets. This will require working with JAX (a machine learning framework in Python), to develop and run GPU accelerated analysis code. We will explore numerical algorithms to sample jointly in large parameter spaces, and investigate the statistical arguments and visualisation tools we need to make accurate inference here.

There are a variety of emerging GPU accelerated models, that incorporate more features [3]. Investigating application of the skills and technology developed to this wider range of models will allow us to potentially extract richer information about the underlying physics, and push the rapid inference algorithm to it's limits. There is some opportunity to pivot to working on innovations in the sampling algorithms themselves if we can establish a good baseline of performance on realistic physics problems, leveraging machine learning algorithms to accelerate the inference for example.

Prerequisite Skills required

Reasonable coding skills primarily in python, some Unix experience, usage of ML frameworks (Torch, JAX etc.) and GPUs

Other skills used in the project

Coding, probabilistic inference and statistical knowledge

Acceptable Programming Languages

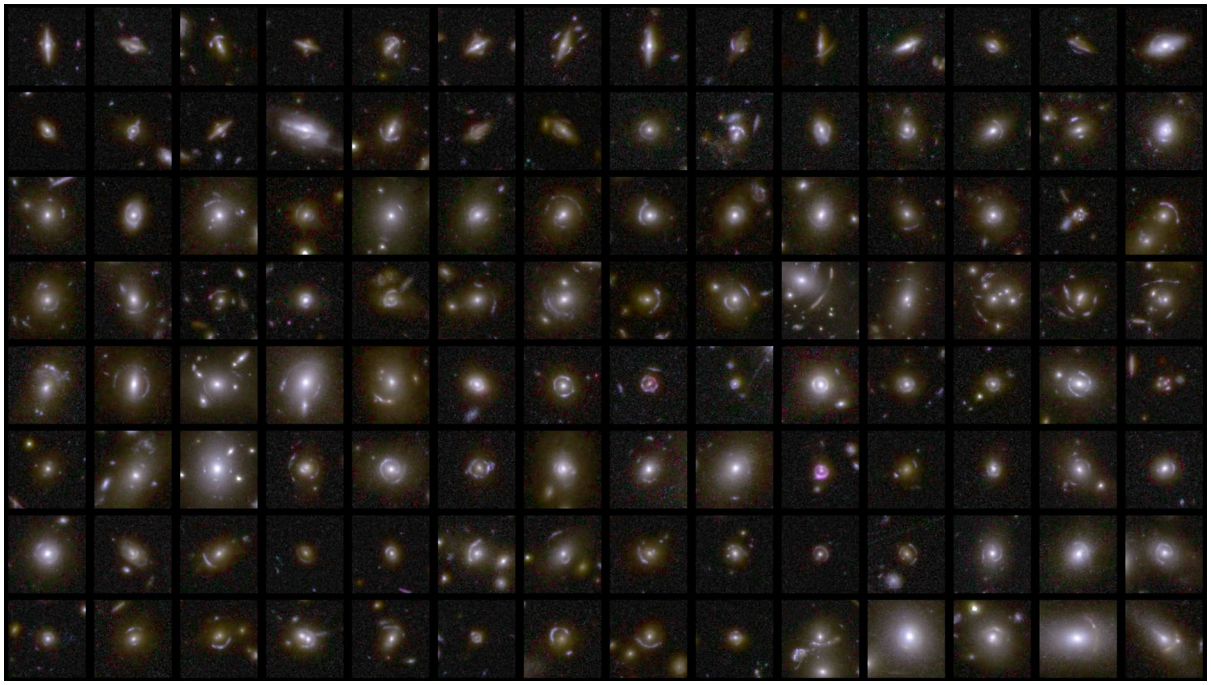
Python (JAX)

Useful References (List of important papers/review articles relevant to the project)

- [1] CSST Strong Lensing Preparation: Fast Modeling of Galaxy-Galaxy Strong Lenses in the Big Data Era: arxiv 2503.08586
- [2] Nested Slice Sampling: openreview.net/forum?id=ekbkMSuPo4
- [3] JAXtronomy: https://github.com/lenstronomy/JAXtronomy

General References (List of papers referred to in the project)

- [4] Attached figure: https://www.esa.int/ESA_Multimedia/Images/2025/03/Strong_gravitational_lenses_captured_by_Euclid



Project 30: Euclidifying Gaia

Supervisor I: Diana Harrison (dlh@ast.cam.ac.uk)

Supervisor II: Dr Simon Hodgkin (sth@ast.cam.ac.uk)

Supervisor III: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

UTO: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

Project summary

The goal of this project is to use machine learning models to classify objects in the Gaia DR3 catalogue, focusing on non-point-like objects. Firstly, we would like to identify them as such, and secondly we want to be able to apply labels such as “galaxy nucleus”, “binary star”, “star-forming region”, etc. As the full-frame images from Gaia are not downloaded to ground, the data we have is a collection of different statistics on each Gaia source provided in the DR3 catalogue. However, for a subset of the Gaia data we have Euclid catalogues (and images) from which we can establish which of the Gaia catalogue statistics are useful in identifying and labelling non-point-like objects.

Background

Gaia and Euclid are two ESA flagship space observatories, equipped with cameras of similar resolution. Euclid sends its enormous (~100 times larger than HST’s) images to the ground, but Gaia does not. This is because Gaia’s field of view is even larger than Euclid’s but the image depth is shallower, so quite often, most of the image is dark, and hence not worth the ground station time to download. Instead, objects on the Gaia focal plane, such as stars, galaxies, quasars, asteroids, are detected on board and only small, compressed cut-outs are sent to Earth. As Gaia’s main goal is to measure properties of stars, the on-board detection algorithm is optimised for point-sources, however no algorithm is perfect, and many non-stellar objects are detected. These include for example, the nuclei of galaxies, bits of galaxies ’spiral arms, chunks of planetary nebulae etc.

Gaia continuously scans the sky, coming back to the same spot many times as it builds up its full-sky catalogue. The catalogue statistics we are interested in are derived from these multiple observations of each source, as these will likely be different for point-like and extended sources. These differences arise from Gaia’s increased resolution in the along-scan direction over that of the across-scan direction. This means that for extended objects their orientation with respect to Gaia’s scan direction will affect their measurements and even whether they are detected on that pass at all.

Although most diffuse structures on the sky are not recorded by Gaia, many high surface brightness knots and blobs are. The goal of this project is to train a machine learning model to reconstruct the origin and the context of the non-stellar objects measured by Gaia using their properties in the published catalogue with the help of high-resolution Euclid images.

Project details

You will need to cross-match Euclid (ERO, Q1) and Gaia DR3 catalogues to build the training and the validation data-sets.

The simplest way to start to understand which Gaia catalogue statistics are helpful in identifying non-point like sources may be by using the XGBoost package, which provides a simple and efficient means of carrying out supervised learning. To this end, it will be beneficial to familiarise yourself with the contents of the Gaia DR3 catalogue and understand how the catalogue statistics are generated. You will need to choose which catalogue statistics from the Gaia catalogue to use as inputs.

Once it is understood how to find non-point like sources in Gaia, the Euclid images can be used to see if these sources can be further classified. What information about non-stellar objects can we actually extract? Will it be just the crude classes such as “galaxy nucleus”, “binary star”, “star forming region” or can we provide more useful details on the origin and properties of the source, as well as its surroundings?

A large variety of ML methods are currently available, and your task will be to identify the most suitable ones to achieve the goals of the project. To deal with the missing data and/or data with large error-bars in the Gaia statistics gracefully, and to provide uncertainties for the classification, we might investigate building a model based on Gaussian processes. It may also be useful to consider unsupervised techniques, such as AstroCLIP (Parker et al 2023).

Skills required

The project will be computationally based; knowledge of Python is highly desirable together with knowledge of /or willingness to learn how to query a PostgreSQL database. Support will be provided.

Useful References (List of important papers/review articles relevant to the project)

- Cuillandre et.al. https://arxiv.org/abs/2405.13496
- Euclid Q1 data release https://www.cosmos.esa.int/en/web/euclid/euclid-q1-data-release
- The Gaia Archive https://gea.esac.esa.int/archive/
- Gaia DR3 documentation https://gea.esac.esa.int/archive/documentation/GDR3/index.html
- AstroCLIP https://arxiv.org/abs/2310.03024
- XGBoost https://xgboost.readthedocs.io/en/stable/index.html
- UMAP https://umap-learn.readthedocs.io/en/latest/

General References (List of papers referred to in the project)

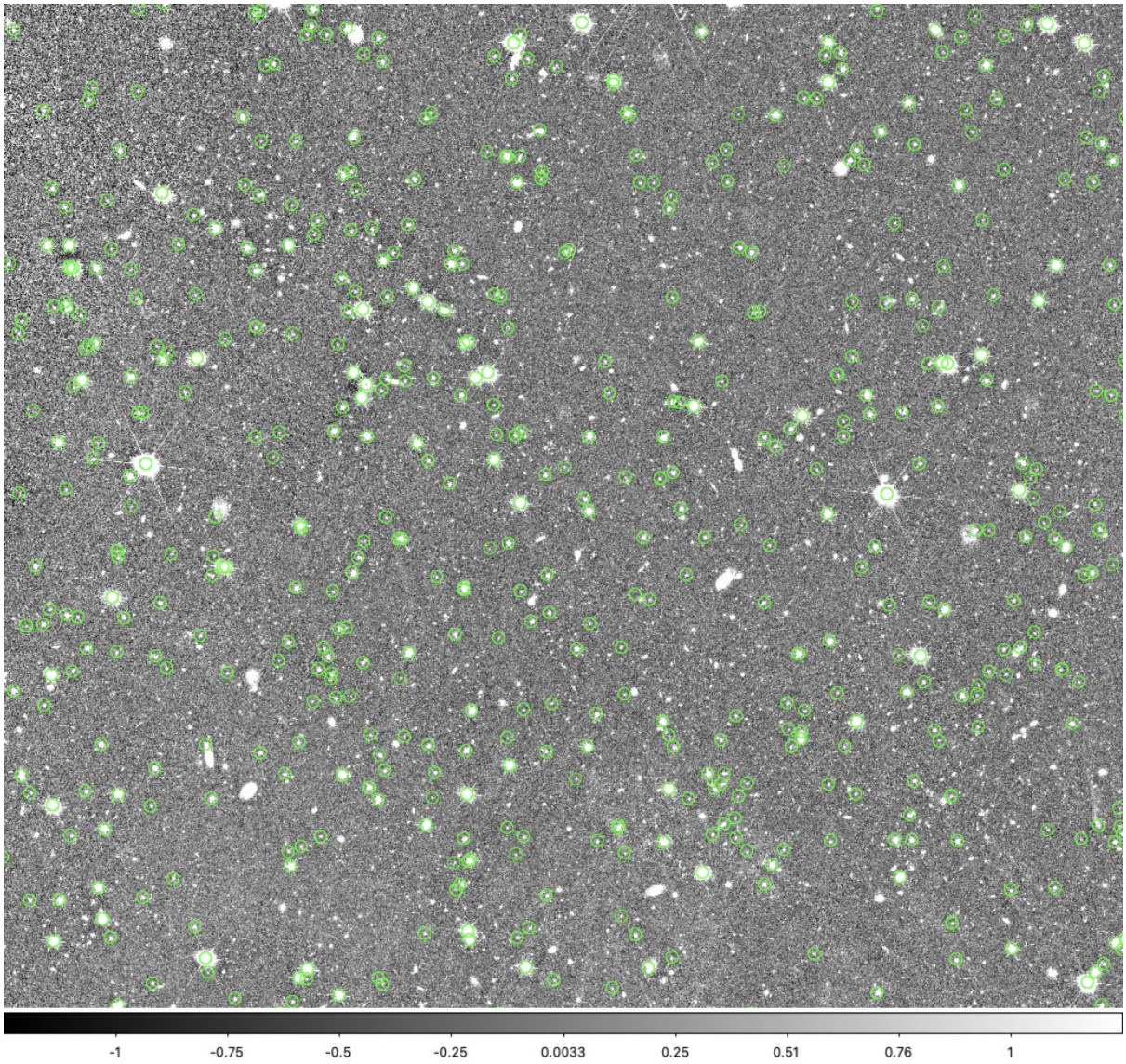


Figure 1: Gaia DR3 source positions overlaid on a small part of the Euclid Image of Abell 2390. The greater depth of the Euclid image is clearly apparent, and if you look carefully you can find Gaia detections connected with extended emission.

Project 31: Convective instabilities and turbulence in the intra-cluster medium

Supervisor I: Dr Henrik Latter (hl278@cam.ac.uk)

Supervisor II: Dr David Hosking (dnh26@cam.ac.uk)

UTO: Dr Henrik Latter (hl278@cam.ac.uk)

Project summary

This project will explore, using local numerical simulations, plasma turbulence generated by a certain convective instability prevalent in the cores of galaxy clusters (the Heat-Flux-Buoyancy instability – HBI). Previous work has suggested the instability saturates by thermally isolating the cluster core. We will test this idea with better resolved simulations and a more comprehensive sweep of the parameter space.

Background

Galaxy clusters are gravitationally bound astrophysical structures comprising hundreds, even thousands, of galaxies. Well known examples 'near' us include the Virgo, Coma, and Hercules clusters. A characteristic feature of these structures is the extremely hot (~ 10^7 K) ionised gas that permeates the space between the galaxies. This gas, referred to as the intracluster medium, is weakly collisional and, as a consequence, the conduction of heat and momentum is anisotropic, aligning itself with the local magnetic field. The anisotropy of the heat transport, in particular, is associated with two unusual 'convective' instabilities: the magnetothermal instability and the HBI (Balbus 2000,2001; Quataert 2008), the former operating in the outskirts of clusters, and the latter in their cores. Preliminary numerical work on the HBI indicates that, by swaddling the core in magnetic fields, little to no heat can be transported between it and the rest of the ICM (Parrish & Quataert 2008, Bogdanovic et al. 2009). If true, this likely exacerbates the, so called, cooling flow problem (Peterson & Fabian 2006), and, on smaller scales, may have implications for the sharpness of cold fronts. Previous simulations were confined to a small, and potentially unrepresentative, region of parameter space, and were not able to probe the fine-scales needed to adequately describe the various turbulent cascades. It is time for the problem to be revisited.

Project details

You will construct, run, and analyse a suite of numerical simulations using the spectral Boussinesq code, Snoopy, which has been adapted to include the heat flux along magnetic fields, super time-stepping, and most necessary diagnostics (Perrone & Latter 2022). Initially, we will explore the evolution of the HBI in two-dimensions at high resolutions, with different background field configurations and themodynamic equilibria. The first goal is to reproduce field wrapping and to assess its prevalence and dependence on the parameters. Next, is to compare these saturated states with, and to extend, extant

theories of metastable MHD equilibrium (Hosking et al. 2025). Any fully turbulent states we encounter we will then explore, testing recent phenomenological theories of Braginskii turbulence (Perrone & Latter 2022) while calculating the turbulent heat flux these states can support. Finally, if time permits, you will conduct a small number of three-dimensional simulations to assess the possibility of large-scale magnetic field generation and dynamo.

Prerequisite Skills required
Prior exposure to coding is essential for this project, and experience with the coding language C would be very helpful. Familiarity with numerical hydrodynamics, or magnetohydrodynamics, would be useful. It is highly recommended that you attend the Part III course Astrophysical Fluids Dynamics.

Other skills used in the project
N/A

Acceptable Programming Languages
C, Matlab, Python

General References (List of papers referred to in the project)

- Balbus, 2000, ApJ, 534, 420.
- Balbus 2001, ApJ, 562, 909.
- Bogdanovic et al., 2009, ApJ, 704, 211.
- Hosking et al., 2025, JPP, 91, 35.
- Quataert, 2008, ApJ, 673, 758.
- Parrish et al., 2009, ApJ, 703, 96.
- Perrone & Latter, 2022, MNRAS, 513, 4605.
- Peterson & Fabian, 2006, Physics Reports, 427, 1.

Project 32: Hunting Supermassive Black Hole binaries with Gaia

Supervisor I: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

Supervisor II: Dr Christopher Moore (cjm96@cam.ac.uk)

Supervisor III: Júlia Laguna Miralles (jl2268@cam.ac.uk

UTO: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

Project summary

Gaia DR4 will open a new observational window on the most elusive stage of galaxy evolution: the moment when two super-massive black holes are still circling one another but have not yet merged. For the first time, the catalogue will contain epoch-by-epoch positions and brightnesses for hundreds of thousands of quasars. If even a fraction of those quasars hide a binary nucleus, the subtle, variability-induced wobble of their photocentres – varstrometry - will betray their presence.

This project positions the student at the forefront of that discovery. By combining a forensic trawl through existing GaiaDR3 data with a state-of-the-art Monte-Carlo simulator that anticipates the DR4 data stream, the work will deliver both a vetted pre-release candidate list and the software toolkit required to mine DR4 the instant it appears. Success means not just finding the first statistically significant sample of sub-parsec SMBH binaries, but also laying the empirical groundwork for future multi-messenger studies that will link Gaia's astrometric wobbles to the gravitational-wave signals LISA will detect a decade later.

This project aims to use Gaia to fill in a gap in our observations of supermassive BH evolution between dual-core active radio galaxies that probe these objects at wider separations, before they form a binary, and pulsar timing which gives details about the population at smaller separations when they are deep in the GW-dominated phase of their evolution.

Background

Most galaxies host a supermassive Black Hole (SMBH, e.g. Kormendy & Ho 2013). How and when such SMBHs were seeded is a cutting-edge question (Volonteri 2010). The subsequent growth of SMBHs is also an open question (Yu & Tremaine 2002). When their host galaxies merge, the SMBHs should, in principle, also merge, forming first a binary SMBH system and eventually fusing together into a single BH (see e.g. Begelman et al 1980, Merritt & Milosavljević 2005). In the future, with e.g. LISA, these mergers will be observable through the accompanying gravitational wave emission (see Jaffe & Backer 2003, Wyithe & Loeb 2003), however, currently these close SMBH binaries are too difficult to "see", yielding only a handful of discoveries (see e.g. Komossa et al 2003). The details of the merger process of these SMBHs is unknown. This project will test and improve upon a novel technique of identifying close pairs of SMBH whilst they are still in their active quasar phase. The method relies on

detecting small shifts of the observed photo-centre of two QSOs using astrometric data from the Gaia satellite, using a technique known as varstrometry (see Hwang et al 2020).

If the binary QSO is unresolved by Gaia, i.e. it appears as a single object, the centroid of the light from two individual sources will move about if the flux contribution from one or both sources is changing with time. Most QSOs are indeed variable sources (see e.g. Hook et al 1994), making the hunt for the unresolved binary QSO pairs promising. This project will build on the ideas developed in the published varstrometry papers (e.g. Hwang et al 2020) and will improve the selection of promising binary QSO candidates by i) using the new Gaia DR3 data (Hwang et al 2020 used Gaia DR2), ii) using new, larger catalogues of QSOs (e.g. that by Storey-Fisher et al 2023), iii) working with actual Gaia DR3 quasar lightcurves, iv) analysing QSO hosts with Euclid QR1 images and v) creating mock varstrometry observations for Gaia DR4.

Project details

The work naturally divides into two complementary avenues. The first is an exercise in data-mining and discovery with the information that already exists. The student will start from the full Gaia DR3 quasar catalogue and identify those objects whose astrometric parameters betray an unusually large excess noise or reduced-chi-square. Each suspect source will then be scrutinised through its multi-epoch Gaia light curve, public optical photometry, radio counterparts and high-resolution images collected by HST, JWST, Euclid, Pan-STARRS and DESI. By confronting this ensemble of diagnostics with the catalogue of known contaminants - stellar blends, white-dwarf binaries, jet-dominated AGN, gravitational lenses and asymmetric host galaxies - the student will iteratively refine the sample until only the most credible, unresolved SMBH binaries remain. The outcome of this first strand is a ranked list of pre-DR4 binary candidates, each accompanied by a quantitative probability of being genuine and, where appropriate, a recommendation for follow-up with adaptive optics or space-based imaging.

The second avenue looks to the future and answers a different question: what will genuine binaries look like when GaiaDR4 delivers epoch-resolved astrometry? Here the student will build a Monte-Carlo simulator that draws pairs of accreting black holes from plausible distributions in mass, separation, redshift and variability amplitude. Independent light curves will be generated for each component using damped-random-walk models, after which the instantaneous photocentre and integrated flux of the unresolved system will be computed for every planned Gaia scan. By folding these theoretical positions through the spacecraft’s point-spread function, scanning law and end-of-mission noise model, the code will deliver mock DR4 time series that are indistinguishable—format-wise—from the data the consortium will soon release. Running the discovery algorithm developed in the first strand on this synthetic sky will reveal the completeness, purity and parameter-recovery accuracy that can realistically be achieved, and will indicate which regions of binary-parameter space remain beyond reach without auxiliary data such as Euclid imaging or next-generation thirty-metre telescopes.

Taken together, the two avenues transform the project into a closed discovery loop. The first creates the best-possible candidate list that can be assembled before DR4, while the second predicts the bounty that DR4 should bring and calibrates the tools needed to harvest it the moment the catalogue appears.

Prerequisite Skills required

Coding, data analysis

Acceptable Programming Languages

Python, or any other programming language e.g. C.

Useful References (List of important papers/review articles relevant to the project)

- Hwang et al 2020 https://ui.adsabs.harvard.edu/abs/2020ApJ...888...73H/abstract
- Chen et al 2022 https://ui.adsabs.harvard.edu/abs/2022ApJ...925..162C/abstract
- Storey-Fisher et al 2023 https://ui.adsabs.harvard.edu/abs/2023arXiv230617749S/abstract

General References (List of papers referred to in the project)

- Kormendy & Ho 2013 https://ui.adsabs.harvard.edu/abs/2013ARA%26A..51..511K/abstract
- Volonteri 2010 https://ui.adsabs.harvard.edu/abs/2010A%26ARv..18..279V/abstract
- Yu & Tremaine 2002 https://ui.adsabs.harvard.edu/abs/2002MNRAS.335..965Y/abstract
- Begelman et al 1980 https://ui.adsabs.harvard.edu/abs/1980Natur.287..307B/abstract
- Jaffe & Backer 2003 https://ui.adsabs.harvard.edu/abs/2003ApJ...583..616J/abstract
- Komossa et al 2003 https://ui.adsabs.harvard.edu/abs/2003ApJ...582L..15K/abstract
- Merritt & Milosavljević 2005 https://ui.adsabs.harvard.edu/abs/2005LRR.....8....8M/abstract
- Wyithe & Loeb 2003 https://ui.adsabs.harvard.edu/abs/2003ApJ...590..691W/abstract
- Hook et al 1994 https://ui.adsabs.harvard.edu/abs/1994MNRAS.268..305H/abstract

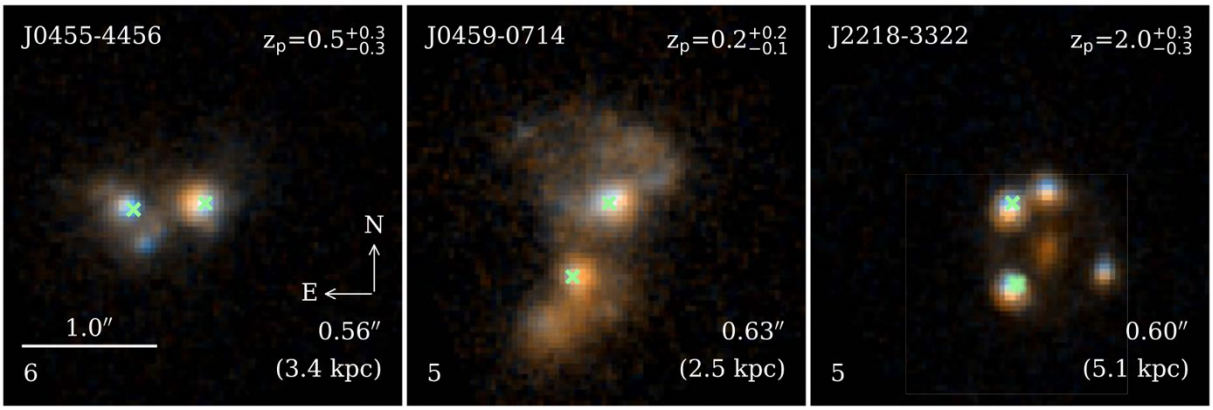


Figure 1. HST/WFC3 colour composite images of three highlighted targets. Left: a triple quasar candidate with an irregular tidal feature. Middle: a dual quasar candidate with an irregular tidal feature. Right: a quadruply lensed quasar. This is a reproduction of Figure 8 from Chen et al 2022

Project 33: Revealing Phase-Space Chevrons with RR Lyrae: The Galactic Bar and Its Interaction with the Dark Matter Halo

Supervisor I: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

Supervisor II: Hanyuan Zhang (hz420@cam.ac.uk)

Supervisor III: Adam Dillamore (amd206@cam.ac.uk)

UTO: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

Project summary

Phase-space chevrons — V-shaped overdensities traced in the plane of Galactocentric radius versus radial velocity — have emerged as clear signatures of the Milky Way's rotating bar (see Figure 1). Their geometry encodes both the present pattern speed of the bar and Galactic potential, and they may also be sensitive to the bar's deceleration due to dynamical interactions with dark matter. Current chevron maps are confined to the Solar vicinity because they rely on stars that possess full 6D phase-space data (including distances, proper motions and line-of-sight velocities). This project aims to overcome that limitation by using the more distant Gaia RR Lyrae variable stars with only 5D data. Starting from idealised or cosmological simulations, mock observations of chevrons in 5D can be constructed and compared to the data. This approach should allow the chevrons to be traced to larger radii and other Galactic locations. This will a) test predictions made by the theory that they are created by the bar; and b) possibly place constraints on properties of the bar, such as its deceleration rate.

Background

The physical mechanism behind chevrons was first formalised in six-dimensional action -angle language, which revealed that successive wraps of tidal debris fold into nested caustics; when projected into (r, v_r) these folds appear as a series of nested V-shapes. Gaia DR3 has revealed exactly such features in the local halo, but with a pronounced excess of stars on the inward (negative-v_r) branches. This asymmetry is striking: if the chevrons were the result of an ancient merger event, as initially proposed, one would expect the tidal debris to produce chevrons that are broadly symmetric about v_r = 0. The sharp imbalance observed in the data is therefore difficult to reconcile with a purely accretion-driven origin, unless the accretion happened recently. This prompted a reconsideration of the chevrons' origin and pointed instead toward a dynamical mechanism involving the Galactic bar. An alternative interpretation showed that halo orbits trapped by the bar's resonances reproduce every aspect of the pattern, including its one-sidedness near the Sun, thus recasting chevrons as live tracers of bar dynamics rather than relics of an ancient merger.

That reinterpretation carries a clear prediction: at radii beyond roughly fifteen kiloparsecs the imbalance should reverse, with the outward (positive radial velocity) branches becoming dominant. This is caused

by the geometry of resonant orbits with respect to the bar. Detecting this flip in asymmetry is therefore a key test of the theory that the chevrons are generated by the bar. Similarly, the asymmetry should flip when crossing the principal axes of the bar. If the bar is decelerating due to dynamical friction with the dark matter halo, the resonant orbits are expected to twist with respect to the bar's axes. Any discrepancy between the orientations of the flips and the bar's axes may therefore be an indication of deceleration.

RR-Lyrae are known to be good standard candles in the Milky Way that trace the kinematics of old stars to great distances (see figure 2). To date, there are more than 10,000 halo RRL that reside 15 kpc away from the centre of the Milky Way. Therefore, RRL serves as a perfect tracer to study the phase-space chevrons at far distances.

Project details

The first task is to assemble a mock RR Lyrae data set from a set of simulations. These will include those run with and without a rotating bar, with possible extensions to include cosmological simulations. The resulting 5D mock data can be compared across different simulations, and signatures of asymmetric chevrons at different Galactic locations can be identified from proper motions.

These insights will be applied to the quarter of a million Gaia RR Lyrae that currently lack spectroscopy. Because these stars are standard candles and thus have measurable distances, they should trace the chevrons morphology beyond r = 15 kpc and to different Galactic azimuths.

If signatures of chevrons are found in the data, the extended chevron map can be used to place constraints on the bar's current and past pattern speed, and the Galactic potential to large radii. This project will therefore expand chevron studies from a Solar-neighbourhood experiment to a halo-wide test of Galactic dynamics.

Possible simplifications and extensions include the following. DESI RR Lyrae with known l.o.s. velocities can be analysed together with the sample of RR Lyrae that Gaia RVS l.o.s. velocities (see Figure 2). Such a sample will have approximately 10,000 stars with full 6D phase-space information. These two sources of RR Lyrae l.o.s. information are preferred as they take into account the change in the RR Lyrae l.o.s. velocity with phase of pulsation.

Prerequisite Skills required

Experience with Python, some knowledge of stellar dynamics and statistics

Other skills used in the project

Analysing observed and simulated kinematic data

Acceptable Programming Languages

Python

Useful References (List of important papers/review articles relevant to the project)

- A 6D view of stellar shells https://ui.adsabs.harvard.edu/abs/2022MNRAS.510..230D/abstract
- Energy wrinkles and phase-space folds of the last major merger https://ui.adsabs.harvard.edu/abs/2023MNRAS.518.6200B/abstract
- Radial halo substructure in harmony with the Galactic bar https://ui.adsabs.harvard.edu/abs/2024MNRAS.532.4389D/abstract
- Dynamical streams in the local stellar halo https://ui.adsabs.harvard.edu/abs/2025arXiv250302926D/abstract
- Resonance sweeping by a decelerating Galactic bar https://ui.adsabs.harvard.edu/abs/2021MNRAS.500.4710C/abstract
- Filling in the blanks: A method to infer the substructure membership and dynamics of 5D stars https://ui.adsabs.harvard.edu/abs/2024A%26A...690A..98C/abstract]

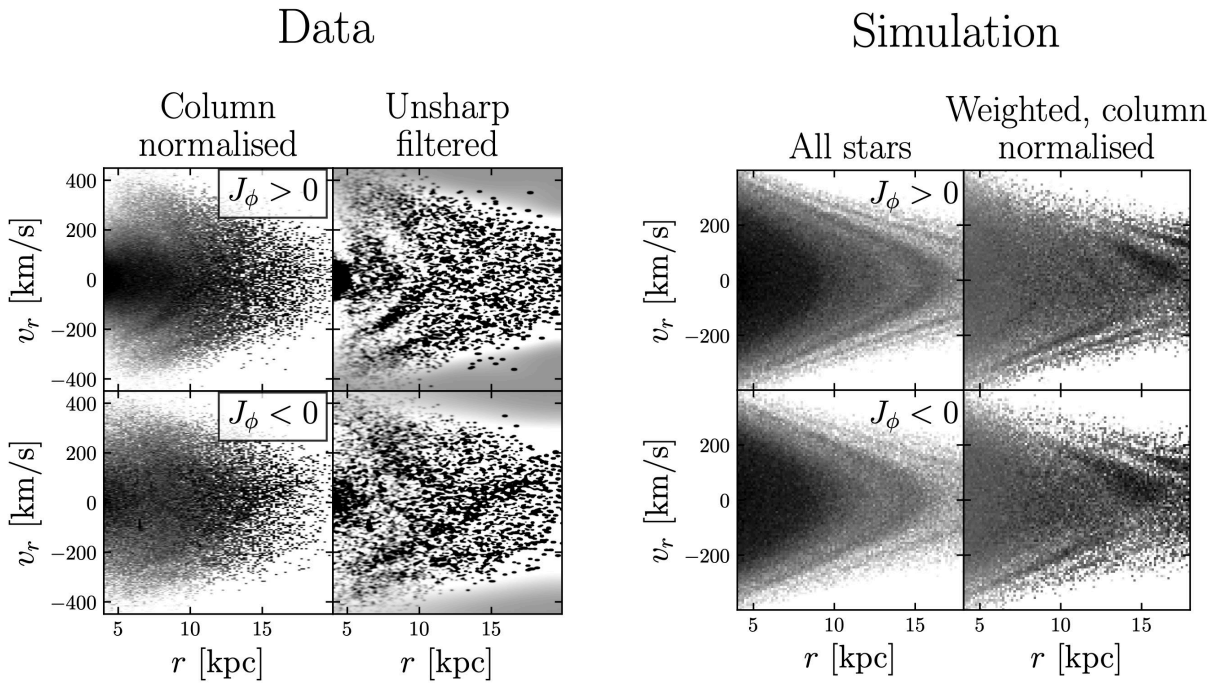


Figure 1: Radial phase space chevrons in Gaia data (left) and in a simulation with a bar (right).

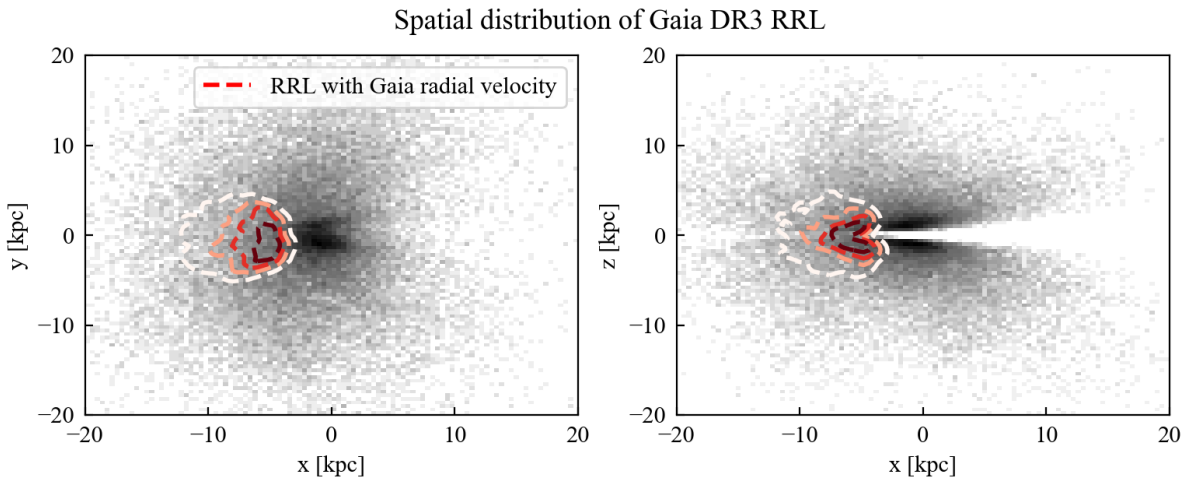


Figure 2: Spatial distribution of RRL in the Milky Way, where the contour shows the distribution of those RRL with Gaia DR3 radial velocity.

Project 34: Understanding the environmental dependence of type Ia supernovae

Supervisor I: Dr Matthew Grayling (mg2102@cam.ac.uk)

UTO: Professor Kaisey Mandel (km723@cam.ac.uk)

Project summary

Type Ia supernovae (SNe Ia) are powerful cosmological probes, allowing for precise estimates of distance to faraway galaxies using various empirically derived relations to 'standardise' their observed properties. This approach first unveiled the accelerating expansion of the universe caused by 'dark energy', the nature of which remains one of the greatest mysteries in science. However, there is still much that is not known about the physical nature of these events, for example the exact details of their progenitor systems and physical processes which cause the explosion.

The biggest mystery relating to SNe Ia is how their properties are impacted by the environment in which they explode. Post-standardisation, we find that SNe Ia in high-mass galaxies are brighter than those in low-mass galaxies; the mechanism responsible for this is unknown, and this result has major implications for both astrophysics and cosmological analyses using SNe Ia. Recent work has demonstrated that this result is maximised by studying SNe Ia located in the inner regions of their host galaxies, while the offset almost disappears when looking at SNe Ia solely located in the outermost regions of their galaxies.

In this project the student will apply BayeSN, a hierarchical Bayesian model for SNe Ia developed in Cambridge, to investigate what is driving the difference in environmental trend between inner and outer regions of galaxies. This work will extend previous BayeSN analyses which have aimed to disentangle intrinsic differences between SNe Ia in different environments from those caused by extrinsic effects i.e. dust extinction.

Background

Type Ia supernovae are the explosions of white dwarf stars located in a binary system and are of wide interest as they can be used in cosmology as distance indicators – after applying a number of empirical relations to 'standardise' them, these objects display remarkable consistency across the population. However, these empirical relations are derived purely from observations and there remains much that we do not know about the physics of these events. Most notably, we find that post-standardisation SNe Ia in high-mass galaxies (heavier than 10 billion solar masses) are slightly brighter than those in low-mass galaxies. This effect is known as the 'mass step'. Going forward, obtaining more precise constraints on cosmology and understanding the nature of dark energy is reliant on improved physical understanding of these events.

The exact cause of this is unknown; some studies indicate that there are intrinsic differences between SNe Ia in different environments which would have exciting astrophysical implications, while others

suggest that material along the line of sight to these SNe – dust – absorbs the SN light and causes this apparent difference. This has led to a lot of work in the literature analysing the properties of dust in the galaxies that host SNe Ia. There is not yet a common consensus about the cause of the mass step although a picture is emerging that there are some intrinsic differences involved in explaining the mass step; in reality, both effects may play a role.

An exciting recent development is the finding that the mass step is enhanced when focusing solely on SNe Ia in the inner regions of galaxies and reduced when considering only those in the outer regions of galaxies. This finding has not yet been explained and working to understand it will undoubtedly provide insight into the overall cause of the environmental dependence of SNe Ia.

The last few years have seen the development of BayeSN (Mandel et al. 2022), a hierarchical Bayesian model which separately captures the intrinsic variation and effect of dust extinction across the population of SNe Ia – this is a major development over the previous state-of-the-art, which combined these two effects into one. BayeSN analyses have demonstrated that are intrinsic differences between SNe which explode in different environments (Graying et al. 2024, Grayling & Popovic 2024), but have not yet considered differences between those that explode in the inner regions of galaxies and those that explode in the outer regions of galaxies. There is exciting scope to extend this work to investigate these findings.

Project details

The purpose of this project is to investigate the investigate the environmental-dependence of SNe Ia using BayeSN, the probabilistic SED model for SNe Ia developed in Cambridge, expanding on previous work in Grayling et al. (2024). BayeSN is written in Python and leverages GPU-accelerated hierarchical Bayesian inference to constrain the effect of dust reddening jointly with the intrinsic properties of SNe Ia. The student will apply BayeSN to investigate differences between SNe Ia that explode in the inner regions of galaxies compared with those in the outer regions. Specifically, the student will

- Perform a literature review to become familiar with supernova cosmology and ongoing debates around the impact of dust.
- Reproduce some existing supernova light curve fits using BayeSN to build familiarity with the code and data
- Extend BayeSN model to constrain populations of SNe Ia in inner and outer regions of galaxies separately
- Gather samples for analysis of SNe Ia in inner/outer regions of galaxies (e.g. from the Dark Energy Survey and Zwicky Transient Facility), and apply the model to infer differences between inner and outer regions.
- If time allows, explore training new BayeSN models on larger samples to improve our model in NIR/possibly NUV.

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Prerequisite Skills required
Required: An interest and motivation to learn Python programming and analysis of real supernova data

Desired (though not essential): experience coding with Python, some familiarity with Bayesian statistics
Support will be provided in the case of limited programming and/or statistics experience

Other skills used in the project
Numpyro/jax

Acceptable Programming Languages
Python

Useful References (List of important papers/review articles relevant to the project)

- Toy et al. (2024), "Reduction of the type Ia supernova host galaxy step in the outer regions of galaxies", https://arxiv.org/abs/2408.03749
- Ward et al. (2023), "BIRD-SNACK: Bayesian inference of dust law RV distributions using SN Ia apparent colours at peak", https://arxiv.org/abs/2310.07753
- Vincenzi et al. (2024), "The Dark Energy Survey Supernova Program: Cosmological Analysis and Systematic Uncertainties", https://arxiv.org/abs/2401.02945

General References (List of papers referred to in the project)

- Grayling et al. (2024), "Scalable hierarchical BayeSN inference: investigating dependence of SN Ia host galaxy dust properties on stellar mass and redshift", https://arxiv.org/abs/2401.08755
- Grayling & Popovic (2024) "BayeSN and SALT: A Comparison of Dust Inference Across SN Ia Light-curve Models with DES5YR", https://arxiv.org/abs/2410.13747
- Mandel et al. (2022), "A hierarchical Bayesian SED model for Type Ia supernovae in the optical to near-infrared", https://arxiv.org/abs/2008.07538

Project 35: Detecting outlier stellar spectra in DESI spectroscopic survey data

Supervisor I: Professor Sergey Koposov (skoposov@ed.ac.uk)

Supervisor II: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

UTO: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

Project summary

We propose to study the population of millions of newly released stellar spectra from the DESI survey to find rare objects, such as extremely metal-poor stars, binaries, and objects with unusual abundances. As part of this project, we aim to map the multi-dimensional structure of these stellar spectra to understand their sensitivity to the abundances of multiple chemical elements, interstellar extinction, and binarity. This is a project in Data Intensive Science and Machine Learning.

Background

Stellar spectra contain a wealth of information about stars and the intervening interstellar medium. While forward-modeling of spectra using theoretical stellar models is a promising way to extract this information, we are often limited by the inadequacy of these models. This can be because they lack sufficient flexibility, have a limited number of free parameters, or do not incorporate the correct physics. Furthermore, some objects may be true outliers and completely unexpected.

Unsupervised learning techniques and data-driven methods offer a way to address these challenges. These approaches can learn the data distribution in high-dimensional spaces, determine the effective dimensionality of the data (i.e., the most important parameters), and identify stellar spectra that deviate significantly from the general population (i.e., outliers).

The DESI survey is a spectroscopic survey primarily aimed at observing large-scale structure tracers, such as luminous red galaxies. However, during bright time and in poor observing conditions, it also observes a large number of stars. To date, DESI has observed more than 12 million stars and has publicly released spectra for over 4 million of them as part of its First Data Release (DR1).

Each DESI spectrum has approximately 12,000 pixels, making it a point in a 12,000-dimensional space. While existing analysis of these spectra primarily focuses on extracting a few basic stellar properties—such as radial velocity, surface gravity, effective temperature, and metallicity—the spectra must contain significantly more information. This includes the abundance of elements like carbon, details about interstellar extinction, indications of binarity, and much more.

Project details

In this project, we will work with the spectra from the public DESI DR1 release.

This project has several possible pathways. One avenue focuses on the dimensionality reduction of DESI spectra, transforming them into a latent space and interpreting the physical meaning of the key features within it. This could begin with simpler methods, such as PCA or UMAP, and progress to more complex approaches like auto-encoders and normalizing flows. These data-driven techniques can be complemented by synthesized theoretical models (for instance, using the Korg spectral synthesis code).

Alternatively, the project could take a supervised learning approach by building a data-driven set of models using external training labels (i.e., a method analogous to DD-Payne).

Another possible direction is to focus purely on outlier identification through dimensionality reduction and outlier detection techniques (e.g., OPTICS, isolation forests, and similar methods).

Prerequisite Skills required

Data wrangling in python, some experience/understanding/interest in neural networks/ machine learning.

Acceptable Programming Languages

Python

Useful References (List of important papers/review articles relevant to the project)

- Cooper, A.~P., Koposov, S.~E., Allende Prieto, C., et al.\ 2023, \apj, 947, 1, 37. doi:10.3847/1538-4357/acb3c0
- Liang, Y., Melchior, P., Lu, S., et al.\ 2023, \aj, 166, 2, 75. doi:10.3847/1538-3881/ace100
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- Zhong, F., Napolitano, N.~R., Heneka, C., et al.\ 2024, , arXiv:2412.21130. doi:10.48550/arXiv.2412.21130
- Nicolaou, C., Nathan, R.~P., Lahav, O., et al.\ 2025, , arXiv:2506.17376. doi:10.48550/arXiv.2506.17376
- Ciuca, I. \& Ting, Y.-S.\ 2022, Machine Learning for Astrophysics, 17. doi:10.48550/arXiv.2207.02785
- Horta, D., Price-Whelan, A.~M., Hogg, D.~W., et al.\ 2025, \aj, 169, 6, 314. doi:10.3847/1538-3881/adcc20
- Zhang, X., Green, G.~M., \& Rix, H.-W.\ 2023, \mnras, 524, 2, 1855. doi:10.1093/mnras/stad1941
- Neitzel, A.~W., Campante, T.~L., Bossini, D., et al.\ 2025, \aap, 695, A243. doi:10.1051/0004-6361/202451718
- Xiang, M., Ting, Y.-S., Rix, H.-W., et al.\ 2019, \apjs, 245, 2, 34. doi:10.3847/1538-4365/ab5364

General References (List of papers referred to in the project)

- Ting, Y.-S., Conroy, C., Rix, H.-W., et al.\ 2019, \apj, 879, 2, 69. doi:10.3847/1538-4357/ab2331
- Yanny, B., Rockosi, C., Newberg, H.~J., et al.\ 2009, \aj, 137, 5, 4377. doi:10.1088/0004-6256/137/5/4377
- Baron, D. \& Poznanski, D.\ 2017, \mnras, 465, 4, 4530. doi:10.1093/mnras/stw3021
- Reis, I., Rotman, M., Poznanski, D., et al.\ 2021, Astronomy and Computing, 34, 100437. doi:10.1016/j.ascom.2020.100437
- Fabbro, S., Venn, K.~A., O'Briain, T., et al.\ 2018, \mnras, 475, 3, 2978. doi:10.1093/mnras/stx3298

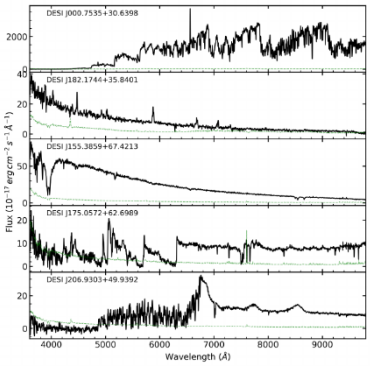

Figure 1: Smoothed spectra of a few rare objects from the backup program of DESI survey. From top to bottom, the panels show a Carbon Star, a AM CVn type cataclysmic variable, an metal enriched white dwarf with strong Ca lines, a broad absorption line QSO at z = 3.077 and a high-redshift QSO at z = 4.526. (Plot from Dey, Koposov+2025, arxiv: 2505.17230) **The green lines show the noise level in these spectra.**

Project 36: Probing binary star population in the Milky Way stellar halo with DESI spectroscopy

Supervisor I: Professor Sergey Koposov (skoposov@ed.ac.uk)

Supervisor II: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

UTO: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

Project summary

This project has two principal science goals. The first is to design and test a new method for detecting and unscrambling binary stars using stacked spectra from the Dark Energy Spectroscopic Instrument (DESI). This technique, based on decomposing a single spectrum into two constituent stellar spectra, is targeted at the poorly explored regime of binaries with low differential radial velocities.

The second goal is to apply this method to probe the binary fraction of the Milky Way's stellar halo. By creating a catalogue of probable binary systems, complete with mass ratio estimates, we will investigate how binarity depends on metallicity and location within the Galaxy. This focuses on a more distant population of stars where traditional astrometric methods for detecting binaries are less effective, providing a new window into the properties of these ancient stellar systems.

Background

It is well-established that many stars are born in clusters and frequently in binary systems. The binarity of stars plays a crucial role in their evolution, affecting the chemical enrichment of galaxies, the dynamical evolution of globular clusters, and the occurrence of various rare events, such as supernova explosions. Stellar binarity is becoming better understood for stars in the solar neighborhood, primarily due to data from the Gaia satellite, which detects the astrometric effects of binary motion. However, these studies are biased towards disk stellar populations with solar-like chemical abundances. In contrast, the properties of binaries within the Milky Way's stellar halo—a population formed from multiple accreted galaxies of much lower metallicity, or binaries formed in the very early phases of the Milky Way formation —are significantly less understood.

Project details

In this project, we will work with spectra from the DESI spectroscopic survey. While primarily designed to probe the large-scale structure of the universe through galaxy observations, DESI has also observed a large number of stars. The survey has been operating for over three years, and its first public data release was made available this year. This latest release includes approximately 4 million stellar spectra. These spectra have already been processed, and parameters such as radial velocity and stellar properties have been measured. However, these measurements were made under the assumption that each spectrum belongs to a single star. We will search for spectra that are better described as a linear

combination of spectra from stars with different luminosities, temperatures, and a relative radial velocity offset.

We will then analyse the fraction of identified binaries in relation to their metallicity and position within the Galaxy. For stars with multiple radial velocity measurements, we will aim to validate the spectroscopic decomposition and estimate the binary parameters. The key advantage of our proposed spectroscopic approach over existing studies based on Gaia data is that those studies are limited. They either rely on astrometric wobbles, which restricts them to distances up to 1 kpc from the Sun, or on Gaia radial velocities, which are only available for very bright stars. In this project, we will be able to probe much fainter and more distant stars and thus will have access to a very wide set of different Galactic stellar populations.

An additional avenue for this project is to focus on a subset of stars with multi-epoch observations by DESI. While the approach of using co-added spectra is most sensitive to binaries where the mass ratio is close to one and both components are luminous, radial velocity variations between multiple epochs can reveal binaries even when the secondary component is dark (e.g., a white dwarf, neutron star, or black hole). With nearly a million stars in the DESI survey having more than one observation, we expect to detect a large number of binaries through this method. For this approach, we will rely on both summary statistics for each binary star, such as the standard deviation of radial velocities, and on specialized codes (provided for the project). Given a handful of epochs and radial velocities, these codes can provide possible solutions for the binary parameters, such as period, semi-major axis, and masses (see e.g., Price-Whelan+2017). With these measurements, we willl be able to characterize a large number of binaries observed by DESI completely independently from the first approach that relied on stacked DESI spectra.

The key outcome of this work (from either or both approaches) is to learn the properties of binary stars down to much lower metallicities and across a much broader range of stellar populations than previously studied.

Prerequisite Skills required

Data wrangling in python, some statistics knowledge, understanding of stellar spectra

Acceptable Programming Languages

Python

Useful References (List of important papers/review articles relevant to the project)

- Liu, J., Zhang, B., Wu, J., et al.\ 2024, \apjs, 275, 2, 40. doi:10.3847/1538-4365/ad856a (arxiv: 2410.07637)
- Bashi, D., Belokurov, V., \& Hodgkin, S.\ 2024, \mnras, 535, 1, 949. doi:10.1093/mnras/stae2385
- Cooper, A.~P., Koposov, S.~E., Allende Prieto, C., et al.\ 2023, \apj, 947, 1, 37. doi:10.3847/1538-4357/acb3c0

- Price-Whelan, A.~M., Hogg, D.~W., Rix, H.-W., et al.\ 2020, \apj, 895, 1, 2. doi:10.3847/1538-4357/ab8acc
- Kounkel, M., Covey, K.~R., Stassun, K.~G., et al.\ 2021, \aj, 162, 5, 184. doi:10.3847/1538-3881/ac1798
- Koposov et al. 2025 (arxiv: 2505.14787)
- Price-Whelan, A.~M., Hogg, D.~W., Foreman-Mackey, D., et al.\ 2017, \apj, 837, 1, 20. doi:10.3847/1538-4357/aa5e50
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- Raghavan, D., McAlister, H.~A., Henry, T.~J., et al.\ 2010, \apjs, 190, 1, 1. doi:10.1088/0067-0049/190/1/1
- Duch{\^e}ne, G. \& Kraus, A.\ 2013, \araa, 51, 1, 269. doi:10.1146/annurev-astro-081710-102602

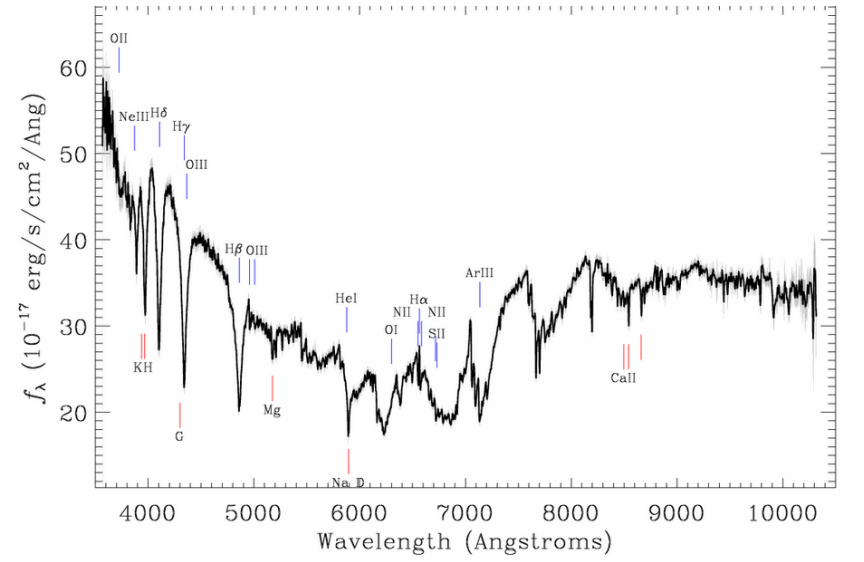

Figure 1: Example spectrum from the Sloan Digital Sky Survey. This spectrum appears to be a blend of two distinct spectra with very different effective temperatures, likely a white dwarf (hot, SED peaking in the UV) and an M-dwarf (cold, SED peaking in the near-IR)

Project 37: Invariant similarity metrics for Machine Learning on galaxy images

Supervisor I: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

Supervisor II: Dr Elisabeth Sola (es2074@cam.ac.uk)

Supervisor III: Professor Sergey Koposov (skoposov@ed.ac.uk)

UTO: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

Project summary

The imminent flood of optical and near-infrared imagery from the Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST) and ESA's Euclid spacecraft will push the census of resolved galaxies from the present tens of millions to many billions. Each postage-stamp cut-out in these archives records a viewpoint-dependent facet of a galaxy's true morphology; rotation on the sky, parity flips, cosmological dimming, pixel resampling, and the vagaries of atmospheric and instrumental point-spread functions all conspire to alter the arrangement of light in ways unrelated to the underlying structure. The central aim of this project is to discover, implement, and rigorously benchmark image-similarity measures that remain faithful to "what a galaxy is" while being largely indifferent to "how it happens to be seen." By combining analytic descriptors, optimal-transport distances, and group-equivariant deep networks, the study will establish which approaches can deliver survey-scale speed without sacrificing morphological sensitivity, culminating in an open-source toolkit and a set of best-practice recommendations for the exabyte era of galaxy astronomy.

Background

LSST will revisit the southern sky every few nights for a decade, yielding on the order of 50 petabytes of calibrated images and uncovering structures from giant stellar disks to the faintest dwarf companions. Euclid, meanwhile, is mapping a comparable cosmic volume from space, free of atmospheric seeing, with the specific goal of measuring the shapes of more than a billion galaxies to unprecedented precision. In such datasets, a single spiral may appear in thousands of rotated, shifted, and resampled "look-alikes," and a naïve pixel-by-pixel comparison treats each version as distinct. Mean-squared error, peak signal-to-noise ratio, or even the more perceptually motivated Structural Similarity Index punish the slightest mis-registration; they are therefore ill-suited to morphological matching when the imaging geometry is uncontrolled.

Classical optimal-transport metrics such as the Earth Mover's (Wasserstein-1) distance respect the spatial organization of flux and are robust to partial occlusion, yet they, too, respond sensitively to rigid motions unless one performs an explicit alignment or integrates the cost over all possible transforms. Moment-based invariants, most notably the seven Hu moments, overcome translation, uniform scaling, and in-plane rotation at negligible computational cost; nevertheless, their power is limited to clean,

silhouette-like shapes and they cannot capture the rich internal texture that distinguishes, say, a flocculent spiral from a lenticular disk. Fourier phase-correlation techniques offer alignment-free matching by exploiting the shift theorem, but their efficacy wanes in the presence of noise and field crowding—common traits of deep survey cut-outs.

The central challenge, then, is to identify image similarity metrics that balance three critical demands: robustness to the low signal-to-noise ratios typical of deep extragalactic imaging, invariance to geometric distortions and observational artefacts, and computational efficiency suitable for petascale data processing. This project will undertake a systematic evaluation of both classical and machine learning–based approaches to determine which metrics offer the best trade-off between sensitivity to intrinsic morphology and insensitivity to viewing conditions. By benchmarking these methods across controlled perturbations of real galaxy images, the study will isolate those descriptors that are not only theoretically elegant but also practically viable for deployment in upcoming large-scale surveys.

Project details

The project will investigate a range of analytic descriptors designed to compare galaxy images while being invariant to transformations such as rotation, translation, and uniform scaling. Classical moment-based techniques, such as Hu moments, are low-dimensional shape descriptors that provide invariance to translation, rotation, and scale through combinations of central geometric moments. While computationally efficient and interpretable, Hu moments are primarily sensitive to the global shape outline and are known to degrade under significant noise or when rich internal texture is present. Zernike moments, defined over the unit disk and forming an orthogonal basis, offer improved discrimination by encoding more localized shape information while retaining rotational invariance; however, they similarly assume clean, centered, and properly scaled input, making them susceptible to noise and pixelation artefacts unless pre-processed carefully.

The Fourier-Mellin transform achieves full similarity invariance by converting rotation and scaling into translations via a log-polar remapping, followed by phase correlation in the Fourier domain. It is theoretically elegant and has been widely used in registration problems, but its reliance on global transformations makes it sensitive to image borders, occlusions, and non-rigid variations. In contrast, wavelet scattering transforms, such as those implemented in Kymatio, cascade complex wavelet filters with modulus and low-pass operations to construct hierarchical image embeddings that are stable to small deformations and partially invariant to translation. With appropriate angular pooling or spherical harmonics, scattering transforms can be extended to achieve rotational and limited scale invariance. Unlike learned methods, they require no training and preserve interpretability, but they typically offer only approximate invariance and their robustness at low signal-to-noise ratios depends on the choice of wavelet scales and the averaging scheme.

To establish a gold standard for comparison, the project will implement a brute-force similarity search over a 3D grid of relative transformations: rotation angle, translation offsets, and scaling factor. For each pair of galaxy images, the misalignment that yields the minimum pixel-wise mismatch (e.g., mean squared error or Earth Mover’s Distance) will be identified, defining the best possible alignment-

independent similarity score by exhaustive search. This baseline, while computationally expensive, serves as a reference against which the efficiency and accuracy of all invariant descriptors—Hu, Zernike, Fourier-Mellin, and wavelet scattering—can be benchmarked. Metrics will be evaluated on their invariance fidelity, sensitivity to morphological detail at low S/N, and computational viability for survey-scale deployment.

If time permits, the project will be extended to include neural network–based approaches that can learn invariances from data. These may include Siamese or triplet networks trained on augmented galaxy pairs, Spatial Transformer Networks that learn to align inputs before comparison, and group-equivariant convolutional networks (e.g. using the *e2cnn* or *escnn* libraries) that build in rotational and reflectional symmetries by design.

Prerequisite Skills required

Coding, data wrangling, no prior ML knowledge required

Acceptable Programming Languages

Python

Useful References (List of important papers/review articles relevant to the project)

- Hu Moments: Hu, M. K. (1962). *Visual Pattern Recognition by Moment Invariants*. IRE Transactions on Information Theory, **8**(2), 179–187. link
- Zernike Moments:Teague, M. R. (1980). *Image Analysis via the General Theory of Moments*. Journal of the Optical Society of America, **70**(8), 920–930. link
- Fourier-Mellin Transform and Phase Correlation: Reddy, B. S., & Chatterji, B. N. (1996). *An FFT-based technique for translation, rotation and scale-invariant image registration*. Pattern Recognition, **26**(8), 1141–1160. link
- Wavelet Scattering Transform: Andén, J., & Mallat, S. (2014). *Deep Scattering Spectrum*. IEEE Transactions on Signal Processing, **62**(16), 4114–4128. link
- Kymatio Library (Python implementation of the scattering transform) Andreux, M. et al. (2020). *Kymatio: Scattering Transforms in Python*. Journal of Machine Learning Research, **21**(60), 1–6. URL: https://www.kymat.io/
- *Equivariant Wavelets: Fast Rotation and Translation Invariant Wavelet Scattering Transforms* link

General References (List of papers referred to in the project)

- LSST Overview: Ivezić, Ž., et al. (2019). *LSST: From Science Drivers to Reference Design and Anticipated Data Products*. The Astrophysical Journal, **873**(2), 111. link
- Euclid Mission Overview: Laureijs, R., et al. (2011). *Euclid Definition Study Report*. ESA/SRE(2011)12. link

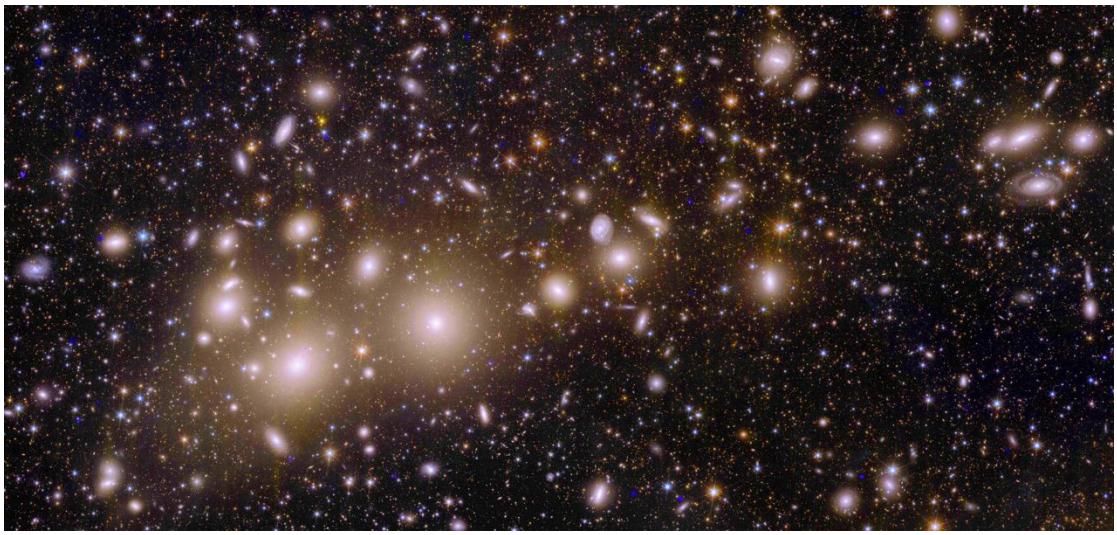

Figure 1: Euclid view of the Perseus galaxy cluster. This appears to be a rich and dense environment with thousands of galaxies of different masses and sizes. On further examination, it is clear that only two main classes of self-similar galaxies are present: 1) yellow-ish in appearance,quasi-spherical, smooth galaxies with little gas and little star-formation and 2) blue-ish galaxies with gas and on-going star formation, showing small-scale structure.

Project 38: A New Era of Quasar Light-Curve Science with Gaia

Supervisor I: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

Supervisor II: Professor Sergey Koposov (skoposov@ed.ac.uk)

Supervisor III: Júlia Laguna Miralles (jl2268@cam.ac.uk)

UTO: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

Project summary

Space-based time-domain astronomy has just vaulted into a new regime. Gaia's third data release supplies calibrated epoch photometry for more than eleven million sources, each measured through a stable point-spread function that sweeps the sky with sub-percent repeatability. When those light-curves are combined with the recently published all-sky Quaia quasar catalogue, one obtains roughly half a million bona-fide QSOs and AGNs with multi-band, milli-magnitude-precision light-curves recorded from space—an order of magnitude larger than any previous homogeneous sample of active nuclei and entirely free of the seeing fluctuations that plague ground-based surveys (see examples of Gaia G-band QSO lightcurves in the left panel of Figure 1). The sheer number of objects, the regular cadence determined by Gaia's scanning law and the uniformly calibrated G, BP and RP fluxes together create an unprecedented laboratory for studying the physics of accretion, for mapping empirical variability classes and for hunting strongly lensed quasars. The project described here invites a Part III student to capitalise on this uniquely rich data set, weaving together complementary analyses that promise to set new benchmarks for the field.

Background

Quasar variability has fascinated astronomers since the discovery of 3C 273 in 1963, when Maarten Schmidt's realisation that rapid optical fluctuations originated in a cosmologically distant source supplied the first concrete evidence for super-massive black holes accreting at prodigious rates. Early photographic monitoring and photo-electric work in the 1960s and 1970s established that quasars flicker on timescales from days to decades, with amplitudes that seemed to diminish toward higher luminosity, but these pioneering studies were limited to a few dozen bright objects and were hampered by heterogeneous calibration.

The modern era began with CCD-based synoptic surveys. Long, high-quality sequences from projects such as SDSS Stripe 82, Pan-STARRS and the Catalina Real-time Transient Survey finally allowed ensemble statistics on thousands of quasars. Analysis of those data established the damped-random-walk (DRW) model - mathematically an Ornstein-Uhlenbeck process - as a remarkably successful statistical description of optical quasar light-curves (see right panel of Figure 1). Its two parameters, a characteristic damping time-scale and an asymptotic amplitude, were shown to correlate (albeit with large scatter) with black-hole mass, luminosity and rest-frame wavelength. Subsequent work

introduced more flexible continuous-time autoregressive moving-average (CARMA) processes, demonstrating that some light-curves demand higher-order stochastic structure and hinting at quasi-periodic signatures that could reflect disc instabilities or orbital features. Yet even the largest ground-based samples - on the order of ten thousand objects - still suffer from atmospheric systematics, irregular cadence and seasonal gaps.

Gaia DR3 changes the landscape. Its space-borne vantage point delivers uniform, high-cadence coverage of the entire sky, while its line-spread function achieves milli-mag photometric stability for point sources. At the same time, the Quaia catalogue leverages Gaia's low-resolution BP/RP spectra and unWISE infrared colours to compile more than a million high-purity quasars with well-characterised selection effects, of which about half overlapped with the epoch-photometry subset at release. Together these products furnish the first data set large enough and clean enough to revisit decades-old questions How does the damping time-scale truly scale with luminosity? Do distinct variability "species" exist? with decisive statistical power, while also opening novel avenues such as the detection of galaxy-scale strong lenses purely from unresolved light-curves via autocorrelation-based time-delay searches. The work of Shu, Belokurov & Evans has already proved the principle on much smaller samples; scaling it to half a million objects will place strong-lens discovery on an industrial footing.

Seen against this historical arc -from the first single-object studies to Gaia's all-sky cadence the project offers a rare opportunity: for the first time, a student can explore quasar variability with a data set that is simultaneously vast, homogeneous and space-quality, enabling analyses that push well beyond the current state of the art.

Project details

The half-million quasar light-curves obtained by cross-matching Gaia DR3 epoch photometry with the all-sky Quaia catalogue provide a uniquely versatile playground for a Part III Project. To give the project both structure and flexibility, four complementary avenues are proposed; the student can pursue any combination of them, adjusting the emphasis as their interests and skills evolve.

The first strand tackles photometric de-contamination. Genuine nuclear variability must be isolated from scan-angle–dependent artefacts caused by host-galaxy light leaking into Gaia's one-dimensional point-spread function. The geometry of every observation can now be reconstructed from the publicly available scanning-law predictions, and deep imaging from the DESI Legacy Surveys and the newest Euclid releases resolves host structure with sub-arcsecond detail. Forward-modelling that structure through the Gaia line-spread function and subtracting/de-correlating it epoch-by-epoch will yield the first large, object-by-object catalogue of clean quasar light-curves. Such a catalogue would immediately improve every study that uses Gaia variability statistics and would set the benchmark for future space-based time-domain surveys.

A second line of enquiry explores whether quasars fall into distinct variability "species". Optical variability is thought to trace disc turbulence and reprocessing, yet recent machine-learning work on Gaia has shown that unsupervised auto-encoders trained on multi-modal data naturally recover meaningful latent spaces in which different kinds of variables separate cleanly. Embedding the cleaned

Gaia light-curves alongside unWISE mid-infrared colours will allow the student to identify clusters that may correspond to different accretion states, orientation angles or dust geometries. Demonstrating such a data-driven taxonomy at the half-million-object scale would move the field beyond traditional one-parameter descriptions of quasar flickering and provide a new empirical map of AGN phenomenology.

The third avenue brings physical modelling to the fore. The damped-random-walk representation is now the standard baseline for quasar variability, while higher-order continuous-time autoregressive moving-average (CARMA) models capture a richer range of stochastic behaviour. Fitting DRW and low-order CARMA processes to every Gaia light-curve -and then correlating the inferred damping timescales and driving amplitudes with luminosity, redshift and black-hole-mass proxies -will test whether the canonical scaling relations survive in a vastly larger, space-quality data set. A positive or negative result will either solidify or overturn current theoretical expectations about the coupling between disc size, thermal time-scale and radiative output.

Finally, the project can convert the Gaia sample into a lens-finding engine. Recent work showed that two unresolved quasar images sharing the same stochastic signal but offset by a time delay leave a distinctive imprint in the autocorrelation function of the joint light-curve. Applying that technique at scale will generate a ranked list of lens candidates, many of which will be below the resolution limit of standard imaging searches. Confirming even a modest fraction would enlarge the census of small-separation lenses and furnish new systems for time-delay cosmography, directly advancing efforts to measure the Hubble constant and to probe dark-matter substructure.

By weaving these strands together -or focusing deeply on a subset -the student will not only refine the Gaia quasar data set itself but also deliver fresh scientific insight into accretion physics, variability demographics and strong-lens discovery, pushing each of these fields beyond their current state of the art.

Prerequisite Skills required
Data wrangling, statistics

Acceptable Programming Languages
Python, or any other programming language

Useful References (List of important papers/review articles relevant to the project)

- Quaia, the Gaia-unWISE Quasar Catalog: An All-sky Spectroscopic Quasar Sample https://ui.adsabs.harvard.edu/abs/2024ApJ...964...69S/abstract
- Quantifying Quasar Variability as Part of a General Approach to Classifying Continuously Varying Sources https://ui.adsabs.harvard.edu/abs/2010ApJ...708..927K/abstract
- Learning novel representations of variable sources from multi-modal Gaia data via autoencoders https://ui.adsabs.harvard.edu/abs/2025arXiv250516320H/abstract

- Discovering strongly lensed QSOs from unresolved light curves
  https://ui.adsabs.harvard.edu/abs/2021MNRAS.502.2912S/abstract

General References (List of papers referred to in the project)

- Gaia Data Release 3. Summary of the variability processing and analysis.
  https://ui.adsabs.harvard.edu/abs/2023A%26A...674A..13E/abstract
- DESI Legacy Imaging Survey Data Release Description
  https://www.legacysurvey.org/dr9/description/
- Euclid Quick Data Release (Q1) -- Data release overview
  https://ui.adsabs.harvard.edu/abs/2025arXiv250315302E/abstract
- Time Variability of Quasars: the Structure Function Variance
  https://ui.adsabs.harvard.edu/abs/2008AIPC.1082..282M/abstract

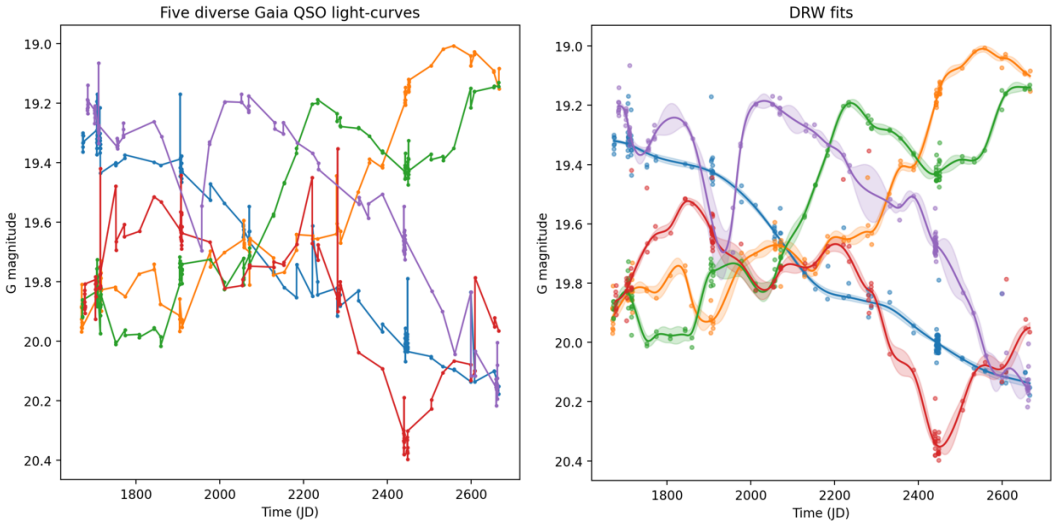

Figure 1:

Five diverse QSO/AGN lightcurves in the Gaia DR3 dataset. Left: Gaia G band magnitude as a function of time. Right: Damped Random Walk model fits to the Gaia data.

Project 39: How hot will exoplanets be?

Supervisor I: Professor Oliver Shorttle (os258@cam.ac.uk)

Supervisor II: Dr Amy Bonsor (ahcb2@cam.ac.uk)

Supervisor III: Yuqi Li (yl817@cam.ac.uk)

UTO: Professor Oliver Shorttle (os258@cam.ac.uk) & Dr Amy Bonsor (ahcb2@cam.ac.uk)

Project summary
A key parameter governing Earth's evolution and habitability has been the temperature of its rocky interior, its mantle. However, how hot will the interiors of rocky exoplanets be? We can potentially constrain the temperature of exoplanet interiors by recognising the role that composition plays in setting their dynamics. In this project, we will use available stellar composition data and thermodynamic and dynamic modelling of convection in rocky planet interiors to place constraints on the likely range of temperatures of exoplanet interiors.

Background
The temperature of rocky planet interiors, their 'mantles', is a key driver of their evolution and habitability. High mantle temperatures causes mantle melting, which leads to volcanism and provides the ultimate supply of nutrients to planetary biospheres and gasses to the atmospheres. However, little is known about how hot planetary mantles are going to be, and so for the rates of these key planetary processes.

Project details
In this project the student will implement a parameterised convection model to predict the thermal evolution of planets. Stellar compositional data will be used to define how diverse planetary mantles will melt, and be used as input to the parameterised convection modelling. Translation of composition to melting properties of planets will be by thermodynamic modelling. The result will be a distribution of exoplanet mantle temperatures and rates of volcanism simply from their inherited compositional variation from their stars.

Prerequisite Skills required
The primary prerequisite is the ability to code, as simple numerical models will need to be built and data assimilated.

Other skills used in the project
The project explores both fluid dynamical ("geophysical") processes in the context of exoplanetary science. An enthusiasm for multi-disciplinary science is key.

Acceptable Programming Languages
Python and Julia preferred, but any language the student is comfortable with.

Useful References (List of important papers/review articles relevant to the project)

Fraeman & Korenaga (2010). The influence of mantle melting on the evolution of Mars. Icarus

Hauck & Phillips (2002). Thermal and crustal evolution of Mars. JGR Planets

Guimond et al. (2024). From Stars to Diverse Mantles, Melts, Crusts, and Atmospheres of Rocky Exoplanets. Reviews in Mineralogy and Petrology.

Project 40: Exoplanetary Ring Emission

Supervisor I: Professor Mark Wyatt (wyatt@ast.cam.ac.uk)

Supervisor II: Dr Max Sommer (ms3078@cam.ac.uk)

UTO: Professor Mark Wyatt (wyatt@ast.cam.ac.uk)

Project summary

This project aims to develop a model for how the presence of a circumplanetary ring affects an extrasolar planet's thermal emission spectrum. This will be applied to JWST observations of imaged planets to determine the ring properties required to explain any excess infrared emission.

Background

With over 5000 extrasolar planets now known, a major focus is now on the characterisation of those planets. This can be achieved by taking a spectrum of the planet to infer its bulk properties and atmospheric composition, which is possible either when the planet is seen in direct imaging, or when the planet passes in front of the star. However, such measurements not capture only the planet, but also the contribution from any circumplanetary material (Teachey 2024). Solar system planets also host populations of regular and irregular satellites, as well as ring systems. Some of these, like the individual small moons, present a small perturbation to the planet's spectrum. However, the fact that Saturn's rings are readily discernable in a small telescope to have a scale similar to the planet itself is a clear indication that the planet's spectrum could be significantly affected by such a component.

There is as yet no definitive detection of an extrasolar circumplanetary ring. However, their existence could explain some features of exoplanets inferred from transit observations, such as the low density of planets known as super-puffs (Piro & Vissapragada 2020; Lu et al. 2025), an interpretation which is supported in some cases by the spectrum (Ohno et al. 2022). Transits of ring material in front of the star have also been inferred (Mamajek et al. 2012; Kenworthy et al. 2015), but without repeat transits, presumably due to the long period of the parent body, further investigation of this interpretation is challenging. Observations of directly imaged exoplanets often show emission levels above that expected from the planet itself, particularly in the mid-IR (e.g., Matthews et al. 2024; Beichman et al. 2025). This could arise from a circumplanetary ring, but could also be down to deficiencies in the planet spectrum models. In the case of young ~5Myr protoplanets, the emission has been inferred to originate in a circumplanetary disk (Isella et al. 2019), out of which it might be expected that satellite and ring systems may form. Overall it is expected that exoplanetary rings will be detected in the near future, if not already, and this project aims to be prepared for the interpretation of such detections.

Project details

This project will develop a model for the emission from a circumplanetary ring that can be used for the interpretation of the spectrum of directly imaged exoplanets. In the case that the ring is optically thin such a model would be straight-forward: a physically motivated size distribution could be assumed and Mie theory used in combination with dust optical properties to determine its temperature and emission properties (e.g., Kennedy & Wyatt 2011). In the case of an optically thick ring there are additional complications to take into consideration. The radiative transfer of an optically thick ring can be treated in a simplified way (Wyatt et al. 2018), but would require the equations to be developed to account for the appropriate geometry. This can build on the development of such equations for an exoplanetary ring's scattered light properties (Sucerquia et al. 2020; Ohno & Fortney 2022; Zuluaga et al. 2022), which shows that in addition to the obliquity and inclination to our line-of-sight, shadowing effects from the planet may also be important.

The goal of this first step is to create a model such that the emission from the ring can be predicted given various input parameters, with the most important degeneracies between those parameters explored. Consideration would also be needed for the physical plausibility for where a ring might form and reside (e.g., Schlichting & Chang 2011). This would then be used to determine the evolution of the emission spectrum around the planet's orbit, given that the geometry both to instellation and to our line-of-sight changes along the orbit.

The second step is to apply this model to JWST observations. This would require an understanding of the observational process and how to work with the limited information with which to constrain both the orbit of the planet and its spectrum. This usually means that the problem is unconstrained, but it is expected that new data will be obtained during the project with which to improve the constraints, and/or the predictions can be used to identify the observational set-up that would be most constraining.

Prerequisite Skills required

Attendance at the Part III Planetary System Dynamics course would be beneficial to understand the processes being modelled.

Acceptable Programming Languages

Code could be written in any language. Python would allow to connect to existing code to model dust optical properties, but is not a requirement.

Useful References (List of important papers/review articles relevant to the project)

- Beichman et al. 2025, ApJ, submitted
- Isella et al. 2019, ApJL, 879, L25
- Kennedy & Wyatt 2011, MNRAS, 412, 2137
- Kenworthy et al. 2015, MNRAS, 446, 411

- Lu et al. 2025, ApJ, 980, 39
- Mamajek et al. 2012, AJ, 143, 72
- Matthews et al. 2024, Nature, 633, 789
- Ohno & Fortney 2022, ApJ, 930, 50
- Ohno et al. 2022, ApJL, 940, L30
- Piro & Vissapragada 2020, AJ, 159, 131
- Schlichting & Chang 2011, ApJ, 734, 117
- Sucerquia et al. 2020, MNRASL 496, L85
- Teachey 2024, Handbook of Exoplanets, arXiv:2401.13293
- Wyatt et al. 2018, MNRAS, 473, 5286
- Zuluaga at al. 2022, AC, 40, 100623

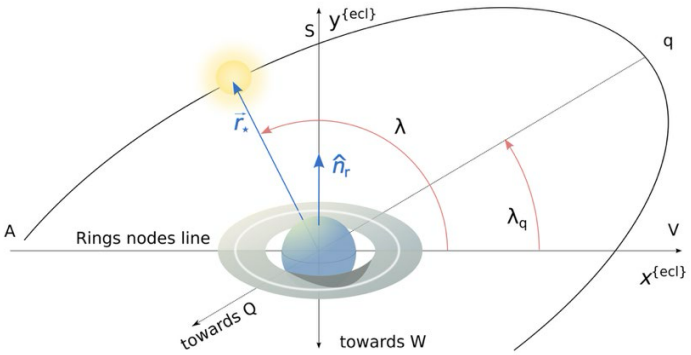

Figure 1: Geometry of a circumplanetary ring (Zuluaga et al. 2022). The inclination of the ring plane to the star along with shadowing effects set the thermal balance, while the ring's inclination to our line-of-sight sets the emission received at the Earth. This geometry is continually changing as the planet orbits the star.

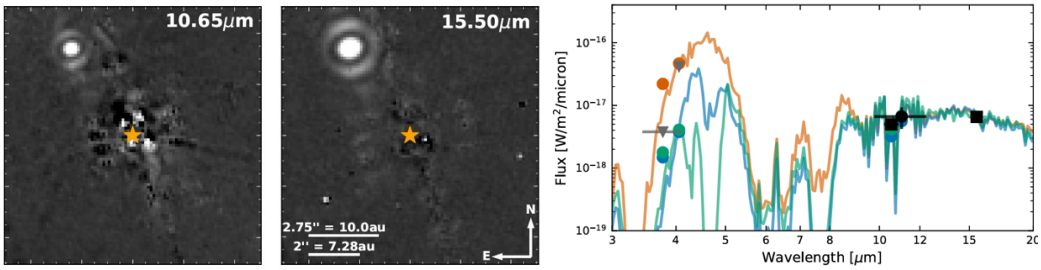

Figure 2: JWST imaging of the planet orbiting eps Ind A (left), and its spectrum (right) (Matthews et al. 2024). The planet's spectrum can be fitted, but is overluminous, potentially indicating an additional emission component from a circumplanetary ring, as also inferred for other directly imaged exoplanets (Beichman et al. 2025).

Project 41: CMB-lensing mass calibration of galaxy clusters: Sunyaev–Zel'dovich selection biases

Supervisor I: Professor Anthony Challinor (a.d.challinor@ast.cam.ac.uk)

Supervisor II: Dr Inigo Zubeldia (iz221@cam.ac.uk)

UTO: Professor Anthony Challinor (a.d.challinor@ast.cam.ac.uk)

Project summary

The abundance of galaxy clusters as a function of mass and redshift is a powerful probe of late-time cosmology. Cosmic microwave background (CMB) surveys are now detecting thousands of clusters via the distinctive scattering imprints they leave across multi-frequency mm-wave maps. Calibrating the mass of these clusters is essential to extract cosmological information, with weak gravitational lensing being the method of choice. For high-redshift clusters, where there is a dearth in background galaxies, lensing of the CMB has an important role to play in mass calibration. An unexplored systematic effect may be important for this approach: cluster searches in multi-frequency maps may preferentially detect clusters in regions where the primary CMB fluctuations are negative, as this reinforces the characteristic temperature decrement due to scattering in the cluster. This atypical nature of the CMB around detected clusters may interfere with their lensing mass calibration. In this project, you will explore this effect by analysing simulated maps to detect clusters and perform the mass calibration. If significant, you will investigate mitigation strategies such as searching for clusters in multi-frequency maps in a way that is independent of the primary CMB fluctuations.

Background

Galaxy clusters are the largest gravitationally bound objects in the Universe. They form in the tail of the distribution of fluctuations in the matter density and so their number density depends sensitively on the amplitude of matter clustering and other cosmological parameters (Battye & Weller 2003). By constructing samples of galaxy clusters with well-understood selection functions, and with measured redshifts and masses, we can compare the abundances to theoretical predictions to place powerful constraints on cosmology. Clusters can be detected in various ways, such as via their X-ray emission or by identifying associations of galaxies in optical surveys. Alternatively, they can be detected in mm-wave maps from cosmic microwave background (CMB) surveys by making use of the distinct spectral signature imprinted in these maps by the CMB being Compton up-scattered off the electrons in the hot, ionized gas in the intracluster medium. This process, known as the thermal Sunyaev–Zel'dovich (tSZ) effect, allows the detection of clusters to high redshift with a very well-understood selection function (e.g., Carlstrom et al. 2002). With data from the Planck satellite, hundreds of clusters were detected via this route at high signal-to-noise (Planck Collaboration 2016). Catalogues from current ground-based

surveys, with their superior angular resolution, extend to 1000s of clusters (e.g., Hilton et al. 2021), and forthcoming surveys such as the Simons Observatory will detect 20,000 or so (Ade et al. 2019).

Connecting the SZ observable to the cluster mass empirically is essential to extract cosmology, since theory only allows us to predict accurately the cluster abundance as a function of mass and redshift. This mass calibration is best done with weak gravitational lensing, either by measuring the coherent distortion in the shapes of background galaxies or searching for cluster-centric dipole-like signatures imprinted on the primary fluctuations in the CMB; see Fig. 1 and, e.g., Lewis & King (2006). Examples of cosmological analyses using these approaches are Bocquet et al. (2024) for galaxy lensing and Zubeldia & Challinor (2019) for CMB lensing. The CMB lensing route has less signal-to-noise per cluster at low redshift but is particularly powerful for high-redshift clusters, where there are few background galaxies and measuring their lensing and redshifts is difficult. Moreover, the CMB is behind every galaxy cluster in the Universe and so can be useful if there is only partial overlap between the CMB survey used for cluster detection and the optical imaging survey used for galaxy lensing (as in the Bocquet et al. 2024 analysis).

The tSZ spectral signature is a decrement below 217 GHz and an increment above. For ground-based surveys, where the maximum frequency is limited by the atmosphere to around 280 GHz, clusters therefore appear as negative “shadows” over most of the frequency range. Clusters just below the detection threshold in the absence of background CMB fluctuations may be boosted above the threshold if they happen to lie on top of a negative fluctuation, meaning that clusters may be preferentially selected in regions of negative fluctuations in the primary CMB. This is one of the first things you will explore in the project, using cluster-detection algorithms such as SZiFi (Zubeldia et al. 2025 and references therein).

If selected clusters do reside preferentially in regions of negative CMB fluctuations, this may be expected to interfere with mass calibration by CMB lensing. The background CMB on which a lensing dipole is imprinted by the cluster is atypical, due to the selection, and this may bias the cluster lensing reconstruction and mass estimation. Investigating this bias, assessing its significance, and exploring modifications to the cluster detection to mitigate it are key aims of the project.

Project details

You will start by getting to grips with the ideas of cluster cosmology, the multi-frequency matched-filter algorithm – as implemented in, e.g., SZiFi (Zubeldia et al. 2025) – used to detect clusters in mm-wave maps via their tSZ signature, and lensing reconstruction from the CMB (Hu 2001, Lewis & Challinor 2006). We will then verify that clusters are indeed preferentially selected in regions of negative primary CMB fluctuations by running the detection algorithm on simulated temperature maps containing the tSZ signal and primary CMB fluctuations (and instrument noise). Stacking the primary CMB on the locations

of the detected clusters should reveal the selection effect. The dependence on frequencies used in the cluster detection should also be explored at this stage. We will then move onto explore the impact on CMB lensing reconstruction. We will reconstruct the lensing signal from simulated lensed maps, including cluster lensing, and stack on the locations of detected clusters. This stacked signal can be compared to expectations in the absence of a selection effect. We will also aim to develop analytic models for the bias, based on what is seen in the stacks of the primary CMB. Finally, if time allows, we can explore mitigation strategies such as deprojecting the CMB fluctuations during the cluster detection stage or modifications to the lensing reconstruction algorithm.

Prerequisite Skills required

The project will use several software packages in Python, some of which will require modification. Strong mathematical skills are also required, and an affinity for statistical analyses. Attendance at the Part-III courses Cosmology and Field Theory in Cosmology will be very helpful.

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Acceptable Programming Languages

You will mostly be using software packages written in Python.

Useful References (List of important papers/review articles relevant to the project)

Hu, W. Mapping the Dark Matter through the cosmic microwave background damping tail. ApJL 557, L79, 2001

Lewis, A., King, L. Cluster masses from CMB and galaxy weak lensing. Phys. Rev. D73, id. 063006, 2006

Lewis, A., Challinor, A. Weak gravitational lensing of the CMB. Phys. Rep. 429, 1, 2006

Zubeldia, I., Challinor, A. Cosmological constraints from Planck galaxy clusters with CMB lensing mass bias calibration. MNRAS 489, 401, 2019

Zubeldia, I., Melin, J.-B., Chluba, J., Battye, R. The Planck SZiFi catalogues: a new set of Planck catalogues of Sunyaev–Zel'dovich-detected galaxy clusters. MNRAS 539, 185, 2025

General References (List of papers referred to in the project)

Ade, P. et al. The Simons Observatory: science goals and forecasts. JCAP 02, 056, 2019

Battye, R., Weller, J. Constraining cosmological parameters using Sunyaev–Zel'dovich cluster surveys. Phys. Rev. D68, id. 083506, 2003

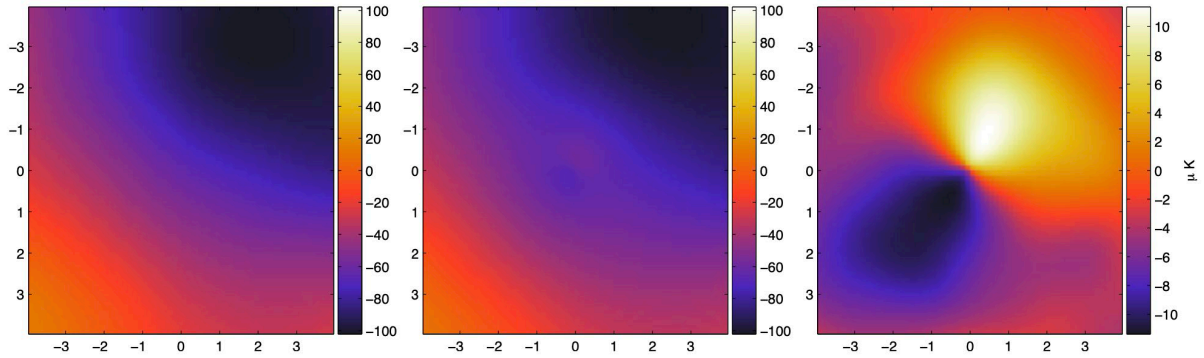
Bocquet, S. et al. SPT clusters with DES and HST weak lensing. II. Cosmological constraints from the abundance of massive halos. Phys. Rev. D110, id.083510, 2024

Carlstrom, J., Holder, G., Reese, E. Cosmology with the Sunyaev–Zel'dovich effect. ARAA 40, 643, 2002

Hilton, M. et al. The Atacama Cosmology Telescope: A catalog of >4000 Sunyaev–Zel'dovich galaxy clusters. ApJS 253, 3, 2021

Planck Collaboration. Planck 2015 results. XXIV. Cosmology from Sunyaev–Zel'dovich cluster counts. A&A 594, A24, 2016

Figure 1: [



Simulation of weak gravitational lensing of the CMB by a galaxy cluster. The panels show the unlensed CMB fluctuations (left), lensed CMB fluctuations (middle) and the dipole-like signature induced by lensing (right), which is aligned with the gradient of the CMB..]

Project 42: Ionising Radiation from the First Stars and Its Impact on Cosmic Reionisation

Supervisor I: Dr Avishai Gilkis (agilkis@ast.cam.ac.uk)

UTO: Professor Anastasia Fialkov (afialkov@ast.cam.ac.uk)

Project summary

The radiation from the 21-centimetre transition of atomic hydrogen in the high-redshift early Universe is expected to be affected by stellar sources emitting high-energy photons. These stellar sources – the first generation of stars – cannot be observed directly, and the 21-centimetre signal can be used to constrain their nature. Several different stellar processes can contribute to this high-energy radiation during the Epoch of Reionisation; in this project, the focus will be on modelling and quantifying contributions from various stellar sources, including the extreme UV and X-ray radiation produced by low-mass stellar coronae, and comparing them with other ionising sources such as massive stars and X-ray binaries.

Background

The first stars in the Universe, known as Population III stars, formed shortly after the Big Bang from pristine gas composed almost entirely of hydrogen and helium. Because these stars have not been observed directly, our understanding of their properties and evolution relies on theoretical modelling and indirect constraints. One promising probe is the 21-centimetre signal from neutral hydrogen, which is sensitive to ionising radiation in the early Universe. High-energy photons from various stellar sources influence this signal. Known contributors include UV radiation from hot massive stars [1], and X-rays from binaries formed by early stellar remnants [2]. Another candidate for additional ionising input is the coronae of low-mass Population III stars [3], which may emit X-rays due to magnetic heating processes. The relative importance of these sources, and their cumulative impact on the 21-centimetre signal, remains an open question.

Project details

The student will develop a numerical framework to model Population III stars and quantify their high-energy photon output under a range of assumptions. The project will involve: 1) Developing and coding prescriptions for estimating ionising fluxes from various stellar sources, including coronae. 2) Running zero-metallicity stellar evolution models for a range of masses using the MESA (Modules for Experiments in Stellar Astrophysics) code [4]. 3) Comparing high-energy flux predictions from different stellar channels, such as low-mass stellar coronae and X-ray binaries, over time since star formation. 4) Applying these models to assess the influence of Population III radiation on the cosmic 21-centimetre signal.

Prerequisite Skills required

Some knowledge of stellar evolution is essential. The part-II course on Structure and Evolution of Stars suffices but the part-III course is desirable.

Other skills used in the project

Use of scientific software (MESA, prior experience not required), data analysis and visualisation.

Acceptable Programming Languages

Basic coding skills with Python will be useful.

Useful References (List of important papers/review articles relevant to the project)

- “Cosmology at low frequencies: The 21 cm transition and the high-redshift Universe”, Furlanetto et al., 2006, PhR, 433, 181
- “The observable signature of late heating of the Universe during cosmic reionization”, Fialkov et al., 2014, Nature, 506, 197

General References (List of papers referred to in the project)

- [1] “Impact of the primordial stellar initial mass function on the 21-cm signal”, Gessey-Jones et al., 2022, MNRAS, 516, 841
- [2] “Population III X-ray binaries and their impact on the early universe”, Sartorio et al., 2023, MNRAS, 521, 4039
- [3] “Coronal properties of low-mass Population III stars and the radiative feedback in the early universe”, Washinoue & Suzuki, 2021, MNRAS, 506, 1284
- [4] “Modules for Experiments in Stellar Astrophysics (MESA)”, Paxton et al., 2011, ApJS, 192, 3

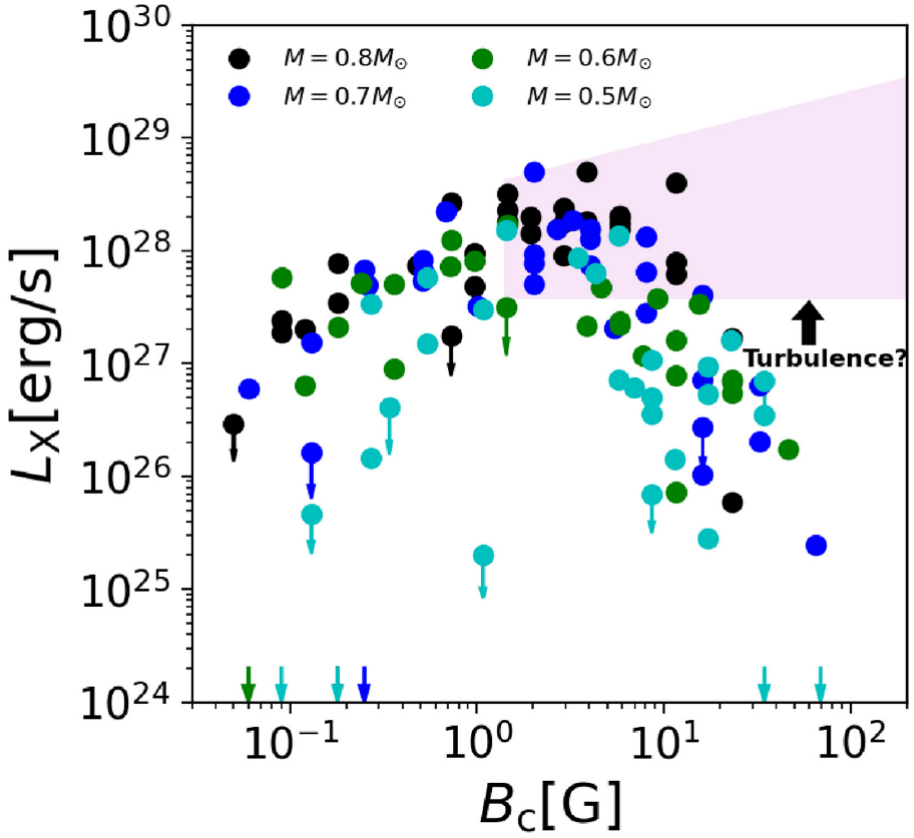


Figure 1:

X-ray luminosity in stellar models as function of magnetic field strength (from [3]). Colours of the points show the different stellar masses. The points with arrow are the upper limits. The pink shaded region is a speculated tendency if heating through Alfvén wave turbulence is active.

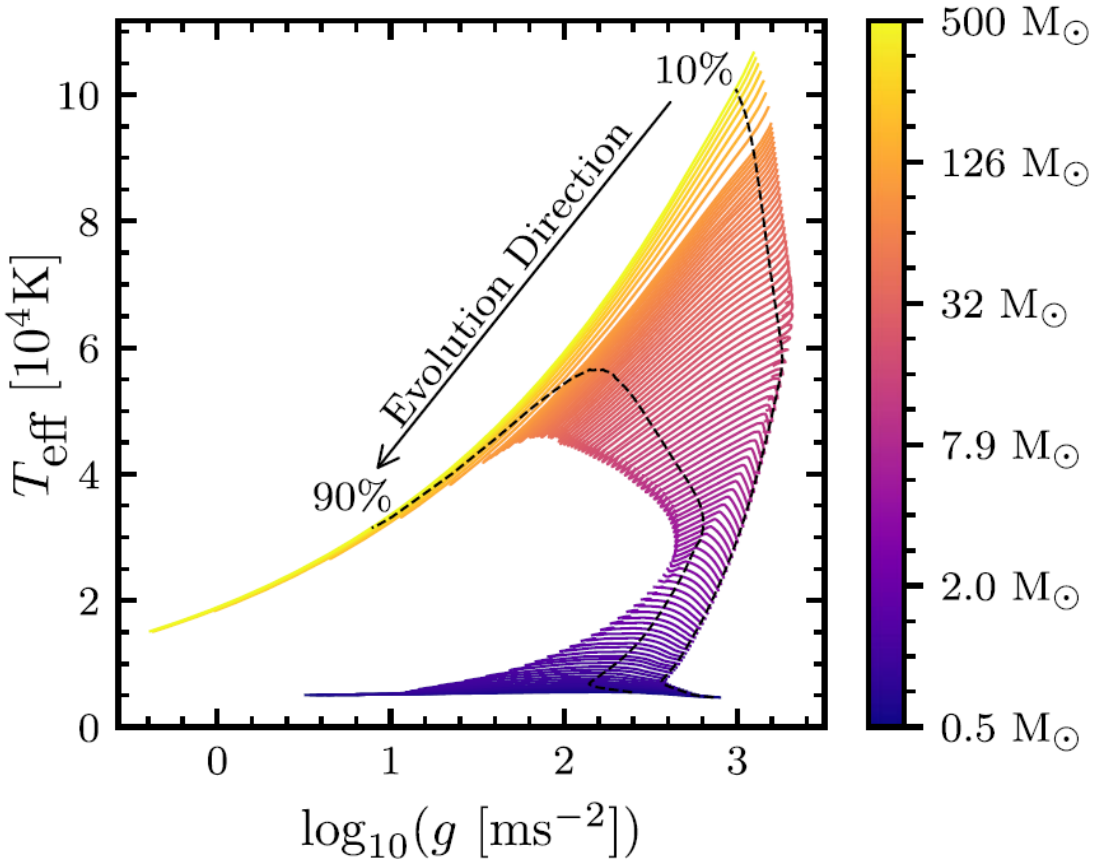


Figure 2: Evolutionary tracks of individual initially metal-free stars projected into effective temperature T_{eff} versus logarithm of surface gravity g space (from [1]). Each line is colour-coded with the stellar mass (colour bar on the right).

Project 43: Machine Learning Accelerated Bayesian Inference of Early Universe Star Formation

Supervisor I: Dr Peter Sims (ps550@cam.ac.uk)

Supervisor II: Jiten Dhandha (jvd29@cam.ac.uk)

Supervisor III: Professor Anastasia Fialkov (afialkov@ast.cam.ac.uk)

UTO: Professor Anastasia Fialkov (afialkov@ast.cam.ac.uk)

Project summary

This project aims to precisely constrain astrophysical properties of the earliest galaxies and their impact on cosmic reionization by undertaking a comprehensive multi-wavelength Bayesian analysis. Building upon the work of Sims et al. (2025) and integrating the advanced flexible star formation efficiency (SFE) model and additional observational constraints from Dhandha et al. (2025), the project will employ the in-house semi-numerical code 21cmSPACE. A core focus will be to utilize information theoretic analyses and posterior predictive distributions to rigorously analyze the combined constraints from the full suite of early Universe observables. This will shed critical light on the processes governing the first billion years of cosmic history

Background

The Epoch of Reionization (EoR) and Cosmic Dawn (CD) represent a transformative period when the first stars and galaxies emerged, heating and ionizing the neutral intergalactic medium (IGM). Understanding the properties of these first sources is a key goal in modern cosmology. Observational probes of this era include the redshifted 21-cm signal of neutral hydrogen, which is sensitive to the IGM's thermal and ionization states, the UV luminosity functions (UVLFs) of high-redshift galaxies - a direct census of luminous sources, and Lyman line constraints on the neutral fraction of the IGM, which provides direct measurements of ionization history of the Universe. Additional constraints come from the Cosmic Microwave Background (CMB) power spectrum which gives us integral constraint on the EoR duration, and the diffused backgrounds such as the Cosmic X-ray Background (CXB) and Cosmic Radio Background (CRB).

Previous work by Sims et al. (2025) performed a joint Bayesian analysis of 21-cm power spectrum upper limits, CMB power spectrum measurements, and Lyman line constraints. This work utilized information theoretic tools such as Kullback-Leibler (KL) divergence and information triangle plots to quantify the constraining power of these individual and combined datasets. More recently, Dhandha et al. (2025) advanced the theoretical modeling of the early Universe by introducing a more flexible star formation model within the 21cmSPACE framework, and incorporated comprehensive UVLF, CXB, and CRB constraints. This project will build directly on the outlined advancements, bringing together the full

suite of observables and statistical methodologies to rigorously test and compare different theoretical models of early galaxy evolution.

Project details
After a thorough literature review, the student will complete the following key tasks:

1. Get familiar with the simulation code 21cmSPACE and Bayesian inference software.
2. Integrate the full suite of multi-wavelength constraints from Sims et al. (2025) (21-cm, Lyman line, CMB) and Dhandha et al. (2025) (UVLF, CXB, CRB) into a comprehensive joint analysis model.
3. Derive posterior distributions of key astrophysical parameters at Cosmic Dawn.
4. By generating and interpreting KL divergence-based information triangle plots you will be able to visualize and quantify the constraining power of different observable combinations on the early Universe parameters.
5. Time permitting, you will generate posterior predictive distributions for key observables (e.g., global 21-cm signal, 21-cm power spectrum, SFRD, UVLFs) and compare these predictions with existing and upcoming observational limits and discuss implications for future experiments (e.g., SKA, JWST).

Expected outcomes and potential impact
This project will yield state-of-the-art constraints on the parameters governing early galaxy star formation, offering a more nuanced understanding of how galaxies assembled and shaped the IGM during reionization. The quantitative model comparison using Bayesian evidence will provide crucial guidance for future theoretical modeling. Furthermore, the information-theoretic analysis will highlight the most informative future observations. The work will contribute directly to the ongoing effort to resolve current cosmological tensions and prepare for the deluge of data from next-generation observatories.

Prerequisite Skills required
Python or Matlab programming skills are essential.

Other skills used in the project
Experience with machine learning frameworks (e.g., TensorFlow/Keras), prior knowledge of cosmology and galaxy formation and evolution and an understanding of Bayesian inference, nested sampling, and statistical methods are valuable but can be learned during the project.

Acceptable Programming Languages
Python, Matlab

Useful References (List of important papers/review articles relevant to the project)

- Dhandha, J., et al. 2025, “Exploiting synergies between JWST and cosmic 21-cm observations to uncover star formation in the early Universe”, arXiv:2503.21687. doi:10.48550/arXiv.2503.21687 (Primary source for flexible SFE model, UVLF, CXB, CRB data).
- Handley, W. J., Hobson, M. P., Lasenby, A. N., 2015a, “POLYCHORD: next-generation nested sampling”, MNRAS, 453, 4384 (Bayesian inference and evidence computation).
- Sims, P. H., et al. 2025, “Rapid and Late Cosmic Reionization Driven by Massive Galaxies: a Joint Analysis of Constraints from 21-cm, Lyman Line & CMB Data Sets”, arXiv:2504.09725. doi:10.48550/arXiv.2504.09725 (Source for multi-wavelength data and information theoretic analysis).

General References (List of papers referred to in the project)

- https://www.cosmicdawnlab.com/21cmSPACE (A. Fialkov group website).
- Trotta, R. 2008, “Bayes in the sky: Bayesian inference and model selection in cosmology”, Contemporary Physics, 49, 71. doi:10.1080/00107510802066753 (Good general review of Bayesian methods in cosmology)

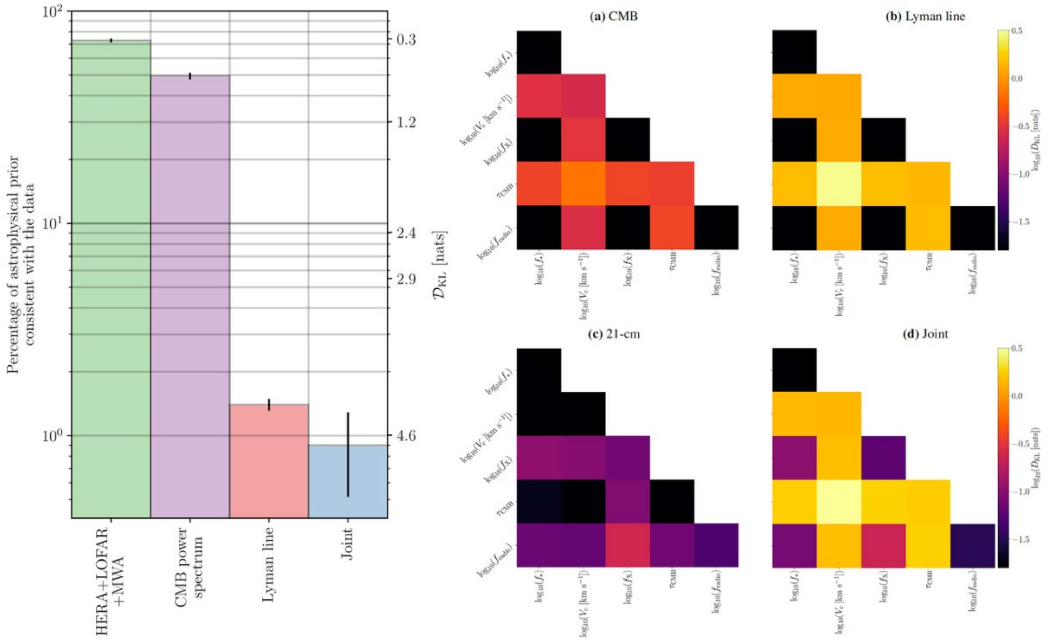


Figure 1: The astrophysical information content, in nats, illustrating the complementarity of data sets in three wavelength ranges – CMB, Lyman line, 21-cm – and of inferences drawn from them jointly. Figure credit: Sims et al. 2025.

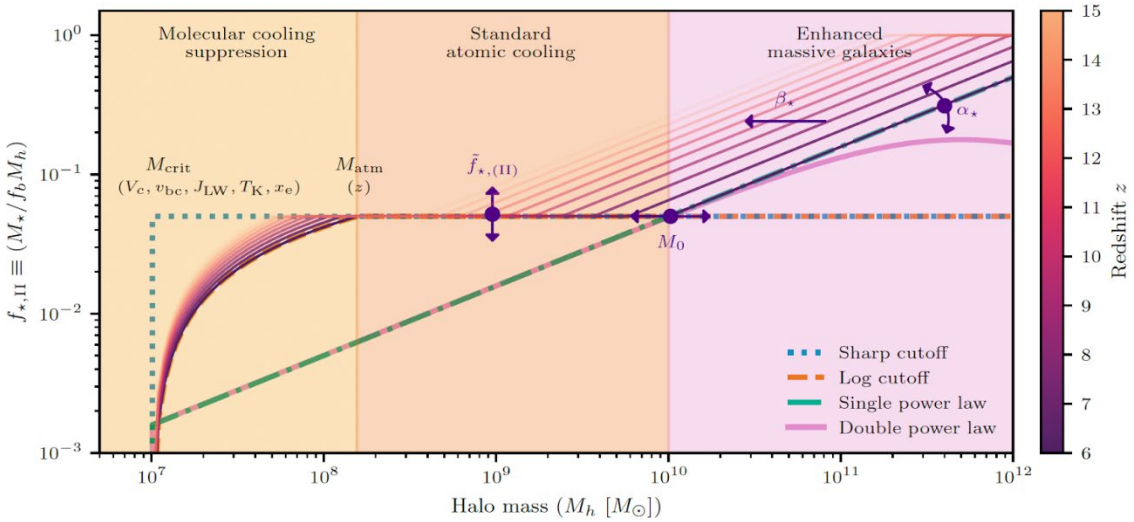


Figure 2: Schematic of the flexible redshift- and halo mass-dependent Population II star-formation efficiency model developed in Dhandha et al. 2025.

Project 44: Statistical and machine learning enhanced comparison of constraints from 21-cm Simulation codes

Supervisor I: Dr Harry Bevins (htjb2@cam.ac.uk)

UTO: Professor Anastasia Fialkov (af677@cam.ac.uk)

Project summary

The 21-cm signal from the Epoch of Reionization, Cosmic Dawn and Dark ages is a powerful cosmological probe which, if detected, promises to revolutionize our understanding of the formation of the first stars, properties of dark matter and the emergence of the first X-ray emitting objects. Numerous radio experiments currently target sky-averaged observations of this signal as well as its fluctuations. Existing upper limits from both the sky-averaged and power spectrum experiments (SARAS3, HERA, MWA and LOFAR) have been leveraged to constrain the physics of the first galaxies (Bevins et al. 2024, Pochinda et al 2024, Dhandha et al 2025). However, a significant challenge arises from the diversity of models of the predicted 21-cm signals. These models often employ different parameterizations, incorporate different astrophysical assumptions, or use distinct simulation approaches making a comparison of the constraints from each code difficult. This project aims to address this issue by developing a robust statistical framework to evaluate the similarity of constraints across these diverse 21-cm model space.

Background

A number of different methods for modelling the 21-cm signal from the early universe exist and are being used to perform inference on data sets from sky-averaged and power spectrum experiments. However, differences in modelling assumptions makes it hard for researchers to understand the relationships between the constraints derived with one model relative to those derived with another. The physics of the early universe is therefore often challenging to interpret and biases in the inferred parameters are likely. Researchers typically translate constraints on model parameters to constraints on signals and visually compare between the results of different codes. This project aims to address this issue by using information theory and machine learning tools. This will be the first time such an analysis has been performed ,and with new experiments like the SKA-Low coming online the project is very timely.

Project details

The project will involve generating a fiducial sky-averaged 21-cm signal with one simulation code (e.g. 21cmSPACE), adding noise of various levels to the signal and then performing Bayesian inference on it with two other simulation codes (e.g. Zeus21 and Echo21) with different parameterisations of the signal. Bayesian inference will give you a posterior on the parameters of the model $P(\theta_1|D, M_1)$

and using the simulation code this can be translated to a posterior in the functional space $P(T_{21}(\theta_1)|D, M_1)$

where M_1 represents model 1. The goal then is to make comparisons between $P(T_{21}(\theta_1)|D, M_1)$ and $P(T_{21}(\theta_2)|D, M_2)$which is often easier than directly comparing $P(\theta_1|D, M_1)$ and $P(\theta_2|D, M_2)$. Making comparisons like this is important because it can help us understand the biases between the simulation codes and any biases that are introduced in our inference by the particular choices of modelling we have made. To make the comparison in functional space we will use two different approaches; the Kullback-Leibler divergence and a machine learning tool called Classifier Two Sample Test. We will explore both of these metrics in the project.]

Prerequisite Skills required
Basic Python programming

Familiarity with machine learning concepts (e.g., neural networks)

Basic understanding of cosmology and statistical methods

Other skills used in the project
MATLAB programming for some simulation pipelines

High-performance computing (HPC) usage

Bayesian inference and likelihood-free methods

Radio astronomy and 21-cm cosmology

Acceptable Programming Languages
Python

Useful References (List of important papers/review articles relevant to the project)

- Zeus21 code https://zeus21.readthedocs.io/en/latest/
- Echo21 code https://echo21.readthedocs.io/en/latest/

General References (List of papers referred to in the project)

- Singh et al 2021 https://arxiv.org/abs/2112.06778
- Bevins et al. 2024 https://ui.adsabs.harvard.edu/abs/2024MNRAS.527..813B/abstract
- Pochinda et al 2024 https://ui.adsabs.harvard.edu/abs/2024MNRAS.531.1113P/abstract
- Dhandha et al 2025 https://ui.adsabs.harvard.edu/abs/2025arXiv250321687D/abstract
- Bevins et al 2022 https://ui.adsabs.harvard.edu/abs/2022NatAs...6.1473B/abstract

- Lopez-paz and Oquab 2016 https://arxiv.org/abs/1610.06545

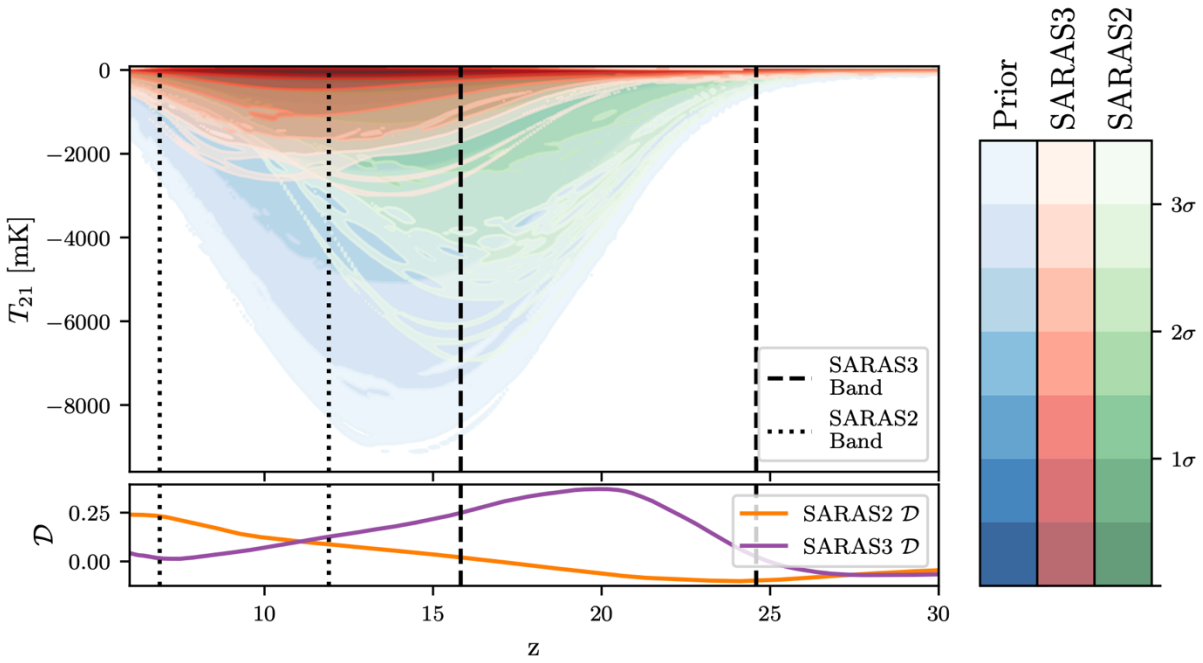


Figure 1: An example of the functional prior and posterior for constraints from the SARAS2 and SARAS3 instrument with the semi-numerical simulation code 21cmSPACE. The functional prior is also shown and the DKL in the bottom panel between the prior and the posteriors shows the difference between the distributions as a function of redshift. Figure from Bevins et al 2022.

Project 45: Probing the nature of dark matter with 21-cm cosmology

Supervisor I: Professor Anastasia Fialkov (afialkov@ast.cam.ac.uk)

Supervisor II: Jiten Dhandha (jvd29@cam.ac.uk)

Supervisor III: Professor Alexander Belyaev (a.belyaev@soton.ac.uk)

UTO: Professor Anastasia Fialkov (afialkov@ast.cam.ac.uk)

Project summary

This project aims to quantitatively link dark matter (DM) annihilation and decay scenarios to their signatures within high-redshift cosmological observables, specifically the 21 cm background and cosmic X-ray background. We will use a state-of-the-art code, micrOMEGAs, to calculate the energy injection from DM particles, which then will be incorporated into the energy budget of the Universe (calculated by the in-house 21-cm code 21cmSPACE), thereby influencing the predicted observable signals. The core objective is to explore these unique signatures to determine which fundamental DM properties can be constrained by future high-redshift observations.

Background

The early Universe, particularly the period spanning the Dark Ages, Cosmic Dawn (CD), and Epoch of Reionization (EoR), represents a crucial frontier in cosmology. During these epochs, the first stars and galaxies formed, profoundly altering the state of the intergalactic medium (IGM). The redshifted 21-cm line of neutral hydrogen serves as a powerful probe of this period, offering a unique window into the thermal and ionization history of the IGM [1]. Complementary to this, the cosmic X-ray background provides insights into early energetic processes.

Dark Matter (DM) constitutes approximately 85% of the Universe's matter content, yet its fundamental nature remains one of the most significant unsolved mysteries in physics. Beyond its gravitational effects, many theoretical models propose that DM particles could interact non-gravitationally, including through annihilation or decay. Such processes would inject energy into the IGM, leading to heating and ionization [2]. This energy injection would leave distinct, observable imprints on the 21-cm background and the cosmic X-ray background, deviating from standard cosmological predictions.

Current and upcoming high-redshift observations from instruments like HERA, LOFAR, and the future SKA-Low aim to detect these faint cosmological signals and have already provided upper limits that constrain early universe astrophysics. However, the potential contribution of DM annihilation or decay to the energy budget of the early Universe is a critical, yet often unconstrained, aspect of these analyses. By precisely modeling the energy deposition from various DM scenarios, we can leverage these cosmological observables to place constraints on DM particle properties, complementing

searches from particle accelerators and direct detection experiments. This project will build upon existing theoretical frameworks and computational tools to bridge the gap between particle physics models of DM and their observable cosmological consequences.

Project details

This project will employ a multi-step computational approach to investigate the cosmological signatures of Dark Matter. We will utilize the state-of-the-art particle physics code micrOMEGAs to scan a range of DM models (varying their parameters: mass, annihilation channels, and cross-sections) and calculate the energy spectra of injected particles. These outputs will then be post-processed to account for cosmological redshift evolution and incorporated into the 21cmSPACE code to calculate the resulting 21-cm signals. Finally, we will compare our derived 21-cm signals and associated constraints with existing literature to draw conclusions regarding the detectability of these DM signatures with future high-redshift observations, particularly those from the Square Kilometre Array (SKA) in a similar way it was done in [3] for the signatures of first stars.

Prerequisite Skills required

Programming skills e.g. with Python or Matlab. Background knowledge of cosmology.

Other skills used in the project

We will learn how to use micrOMEGAs, 21cmSPACE, analyse and visualize data and perform Bayesian analysis. An extension would include using machine learning tools, e.g. creating Neural Network emulators to emulate the DM signatures.

Acceptable Programming Languages

Python, Matlab.

Useful References (List of important papers/review articles relevant to the project)

- [1] Pritchard, J. R., & Loeb, A. (2012). *Reports on Progress in Physics*, 75(8), 086901.
- [2] Poulin, V., Serpico, P. D., & Lesgourgues, J. (2015). JCAP, 2015(03), 031.
- [3] Gessey-Jones et al. (2025) arxiv: 2502.18098

General References (List of papers referred to in the project)

- Furlanetto, Briggs, Oh (2006), PR 433, 181

Project 46: Exploring XRB stochasticity through Higher-Order Statistics and Image-Based Analyses of the 21-cm signal

Supervisor I: Professor Anastasia Fialkov (afialkov@ast.cam.ac.uk)

Supervisor II: Saswata Dasgupta (sd2023@cam.ac.uk)

UTO: Professor Anastasia Fialkov (afialkov@ast.cam.ac.uk)

Project summary

The redshifted 21-cm signal of neutral hydrogen is a powerful probe of the intergalactic medium during Cosmic Dawn (CD) and the Epoch of Reionization (EoR). Recent work has shown that stochasticity in astrophysical sources, such as X-ray binaries (XRBs), introduces significant fluctuations in the 21-cm brightness temperature field, manifested as distinct features in the small-scale ($k > 0.3$) power spectrum at $z = 20$-30. While traditional power spectrum analysis effectively captures two-point correlations, it is inherently blind to the non-Gaussian features and mode couplings that are likely induced by these rare, bright sources. This project aims to extend current analyses by exploring higher-order statistics, such as the Bispectrum, and image-domain techniques to extract additional information from the simulated 21-cm signal, particularly in scenarios involving stochastic X-ray heating. Furthermore, a crucial aspect of this project will be to assess how instrumental systematics, such as beam convolution, and foreground residuals, impact the robustness of these advanced statistical measures.

Background

The early Universe, spanning the Cosmic Dawn (CD) and Epoch of Reionization (EoR), is a period when the first stars and galaxies emerged, profoundly transforming the intergalactic medium (IGM). The redshifted 21-cm signal, originating from the spin-flip transition of neutral hydrogen atoms, offers a unique and powerful observational window into this era, tracing the thermal and ionization evolution of the IGM. This signal's brightness temperature, observed at radio frequencies, provides a direct measure of the IGM's state relative to the Cosmic Microwave Background (CMB). Among the earliest luminous sources, X-ray binaries (XRBs) play a crucial role. These are binary star systems consisting of a normal star orbiting a compact object—a white dwarf, neutron star, or black hole. As matter from the normal star accretes onto the compact object, it heats to extreme temperatures, emitting copious X-rays. These energetic X-ray photons from high-redshift XRBs efficiently penetrate and heat the surrounding IGM, driving the 21-cm signal from an initial absorption phase (cooler than the CMB) into emission (warmer than the CMB). The spatial distribution and episodic nature (stochasticity) of these XRBs are thus directly imprinted on the 21-cm brightness temperature field, leading to complex, non-Gaussian features.

Current and future low-frequency radio observatories, such as HERA and the Square Kilometre Array (SKA-Low), are specifically designed to detect these faint 21-cm signals. While the power spectrum (a two-point correlation function) is a standard tool for analyzing these signals, it does not fully capture the richer, non-Gaussian information encoded by stochastic sources like XRBs in the 21-cm maps. To unlock this additional information, higher-order statistics, such as the Bispectrum, become essential. The Bispectrum is a three-point correlation function in Fourier space that quantifies non-Gaussianity and mode coupling within a field. It provides unique insights into the non-linear processes and the statistical properties of the underlying sources. However, extracting these subtle non-Gaussian features from observational data is challenging due to inherent instrumental effects (like beam convolution) and bright astrophysical foregrounds, necessitating robust analysis techniques that can effectively disentangle the cosmological signal.

Project details

This project will extend existing work on XRB stochasticity in 21-cm semi-numeric simulations (Dasgupta et al. in prep.), which has primarily focused on power spectra. Higher order statistics (like Bispectrum) in 21-cm signal with stochastic X-ray binaries have not been tested and will be considered in this project. The student will, first, compute the 3-point correlation function (bispectrum) of the 21-cm signal across a range of redshifts. Next, the bispectra will be explored (e.g. the sensitivity of bispectrum shapes and amplitudes to the slope of the X-ray luminosity function, XRLF, will be related). We will then identify characteristic triangle configurations (e.g., equilateral, squeezed) that best capture stochastic signatures. We will simulate observational effects for instruments like SKA: beam convolution, uv-coverage, and noise; and propagate these effects through the bispectrum analysis. Finally, we will evaluate which statistical features survive observational filtering. If time permits we will repeat the analysis on the 21-cm brightness temperature maps in the image plane using morphological and topological measures (e.g., skewness, Minkowski functionals).

Prerequisite Skills required

1) Familiarity with Python and/or MATLAB for scientific computing.
2) Background in cosmology or astrophysics.

Other skills used in the project

21-cm cosmology; optional but beneficial: knowledge of Fourier methods, statistics, or interferometry.

Acceptable Programming Languages

Python, MATLAB

Useful References (List of important papers/review articles relevant to the project)

- Sartorio et. al, 2023

- Dijkstra et. al, 2012
- Gilfanov et. al, 2023
- Fragos et. al, 2013
- Fragos et. al, 2013b
- Liu et. al, 2023

General References (List of papers referred to in the project)

- Pritchard and Loeb, 2012]

Project 47: Extrasolar Trojans and Horseshoes

Supervisor I: Professor Mark Wyatt (wyatt@ast.cam.ac.uk)

Supervisor II: Tuhin Ghosh (tg536@cam.ac.uk)

UTO: Mark Wyatt (wyatt@ast.cam.ac.uk)

Project summary

The aim of this project is to create a model for the level and spatial distribution of dust created in the destruction of planetesimals trapped in a planet's Lagrangian equilibrium points L4 and L5 (i.e., Trojans), and those on horseshoe orbits that encompass both points. This model would be applied to the recent first detection of such populations orbiting the stars TWA7 and PDS70 (Balsalobre-Ruza et al. 2023; Lagrange et al. 2025).

Background

Debris disks are the disks of dust found around nearby main sequence stars created in the destruction of planetesimals (i.e., asteroids and comets) orbiting the star (Wyatt 2008). Their planetesimals are usually found to be confined to belts at 10s of au that are analogous to the Solar system's Kuiper belt. Structures within these belts, such as eccentricities, clumps, spirals or sharp inner edges have often been attributed to sculpting by gravitational interactions with unseen planets. Those planets typically remain unseen because observations are only sensitive to planets more massive than Jupiter, which are found to be rare at 10s of au, whereas it only requires a Saturn- or Neptune-mass planet to sculpt the disks. Disk structures thus provide a unique probe of that distant planetary population, and many studies have characterised the ways in which a planet's gravity perturbs a disk, such as through resonant or secular interactions (e.g., Murray & Dermott 1999; Wyatt et al. 1999), with a planet discovered using this technique in beta Pic (Lagrange et al. 2010).

One of the dynamical processes which has yet to be explored is the possibility that a planet retains a population of planetesimals in its 1:1 resonance. Yet these are a significant feature of the Solar system, with Jupiter's Trojan points (i.e., the Lagrangian equilibrium points L4 and L5 that lie 60 degrees in front of and behind the planet) particularly populated with asteroids that were captured there soon after the system formed, and have remained throughout the subsequent 4.5Gyr of evolution. Planetesimals can also follow horseshoe orbits that encompass both the L4 and L5 points, but these are relatively rare in the Solar system.

Direct imaging with JWST has opened up new parameter space to the detection of planets in the outer regions of planetary systems, and one of the first discoveries has been of a Saturn-mass planet orbiting within a gap in a debris disk in which there appears to be a concentration of dust in a horseshoe orbit (Lagrange et al. 2025; see Fig. 1). This follows a tentative detection of a Trojan dust population in the PDS70 system where a protoplanet is also seen (Balsalobre-Ruza et al. 2023; see Fig. 2). These systems present a perfect opportunity to test theories of planet-disk interaction, and to use the disk

structure to set further constraints on the planet's mass and orbit, as well as its formation mechanism. But to do so requires a thorough understanding of the dynamics of the 1:1 resonance, and the mechanism by which dust is created and subsequently evolves, to link that to the observables.

Project details

This project will develop a model for planetesimals trapped in 1:1 resonance with a planet, and the debris created from collisions within that population. The project will start with N-body simulations to follow the dynamical evolution of the Trojan planetesimals (e.g., Fig. 1 right). This will then be post-processed to determine the rate at which collisions occur at different points within the swarm. Simulations will also be run for the evolution of dust created from those planetesimals, by including radiation and gas drag forces, to determine the dust lifetime (e.g., Liu & Schmidt 2018a). This information will then be coupled to analytical considerations (Wyatt et al. 2011) to determine the predicted dust brightness, and used to make simulated images of the dust distribution (e.g., Fig. 2 right). This would be applied to both the TWA7 system (Fig. 1) and the PDS70 system (Fig. 2), to assess whether the dust detections can be explained by Trojan or horseshoe planetesimals in these systems, as well as to consider the observable properties of dust from such planetesimal populations more generally. The model can also be applied to the Solar system: the thermal emission of Jupiter Trojan dust is hard to see due to the ubiquitous dust in the inner Solar System (Kuchner et al. 2000), but has now been detected and analysed (Liu & Schmidt 2018a; see Fig. 2 right).

Prerequisite Skills required

Attendance at the Part III Planetary System Dynamics course would be beneficial to understand the processes being modelled.

Acceptable Programming Languages

The project would involve running N-body simulations using REBOUND (Rein & Liu 2012), and using code written in C to follow the collisional evolution. Programmes would be written to analyse the output, which can use any language.

Useful References (List of important papers/review articles relevant to the project)

- Balsalobre-Ruza O. et al. 2023, A&A, 675, A172
- Kuchner M. J. et al. 2000, Icarus, 145, 44
- Lagrange A.-M. et al. 2010, Science, 329, 57
- Lagrange A.-M. et al. 2025, Nature, in press, arXiv.org/abs/2502.15081
- Liu X., Schmidt J. 2018a, A&A, 609, A57
- Liu X., Schmidt J. 2018b, A&A, 614, A97

- Murray C. D., Dermott S. F. 1999, Solar System Dynamics (CUP)
- Rein H., Liu S. F. 2012, A&A, 537, A128
- Wyatt M. 2008, ARAA, 46, 339
- Wyatt M., Clarke C., Booth M. 2011, CeMDA, 111, 1
- Wyatt M. et al. 1999, ApJ, 527, 918

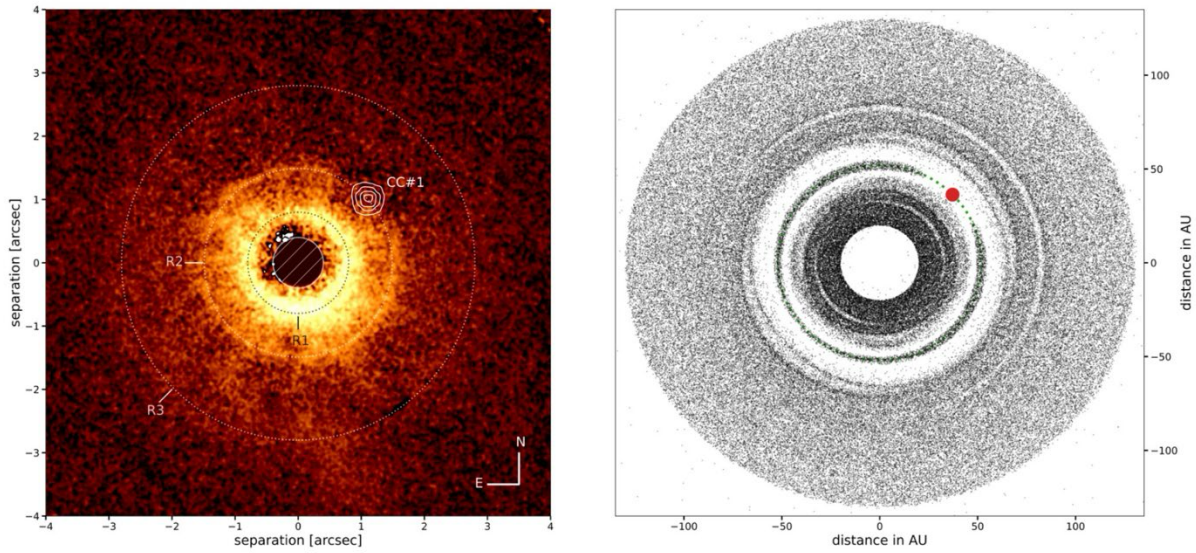


Figure 1: The TWA7 planetary system (Lagrange et al. 2025). (Left) The colour scale shows the SPHERE polarimetric image of micron-sized dust in the disk, in which an incomplete annular ring is seen. The contours show the JWST imaging detection of a Saturn-mass planet that sits in the gap in that ring. (Right) An N-body simulation of the distribution of planetesimals subject to perturbations from a Saturn-mass planet shows the same morphology as the image, with the ring being comprised of planetesimals in the planet's Trojan and horseshoe orbits.

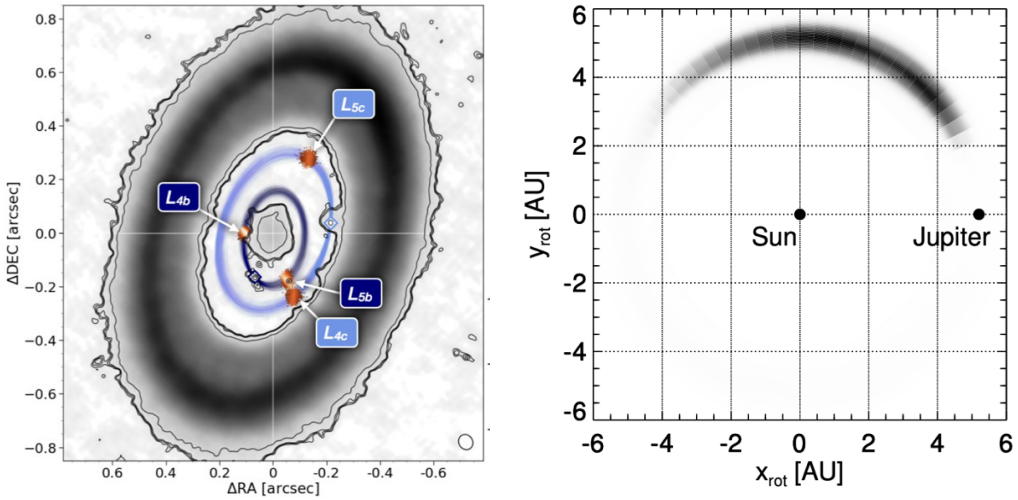


Figure 2: (Left) Observations of the PDS70 system showing the detection of dust in the L5 point of planet PDS70b (Balsalobre-Ruza et al. 2023). (Right) Simulations of the distribution of dust created in the L5 point of Jupiter in the Solar System (Liu & Schmidt 2018a).

Project 48: Accurate modelling of stellar heterogeneity for exoplanet transmission spectroscopy

Supervisor I: Dr Lalitha Sairam (ls2071@cam.ac.uk)

UTO: Professor Nikku Madhusudhan (nmadhu@ast.cam.ac.uk)

Project summary

Understanding the intrinsic spectral variability of M dwarfs due to magnetic activity is vital in the context of exoplanet transmission spectroscopy. The aim of this project is to create synthetic disk-integrated stellar spectra of M dwarfs (spectral types M3–M9), incorporating realistic surface heterogeneity due to unocculted starspots and faculae for application to exoplanet transmission spectroscopy. These stellar features due to magnetic activity, which remain outside the planetary transit chord, introduce wavelength-dependent distortions in observed transit depths and can bias atmospheric retrievals of exoplanets. Using existing libraries of stellar atmosphere models and a combination of empirical and machine learning approaches, we will simulate composite spectra across a range of temperatures, surface coverage fractions, and spot/faculae contrasts. The spectra will be generated over the 0.6–15 μm range and convolved with the response functions of JWST and ARIEL instruments. The end goal is a high-resolution, disk-integrated spectral grid for forward modelling and retrieval studies which accounts for heterogeneity-driven spectral contamination in M dwarf host stars.

Background

M dwarfs, particularly those of spectral type M3 to M9, are increasingly important targets for the atmospheric characterisation of small exoplanets. With the capabilities of high-precision space-based observatories such as JWST and the forthcoming ARIEL mission, transmission spectroscopy of exoplanet atmospheres around these cool stars is entering a new era of precision. However, M dwarfs are also magnetically active, exhibiting complex photospheric inhomogeneities in the form of dark starspots and bright faculae. These features introduce wavelength-dependent distortions in observed transit depths, which can bias inferences about planetary atmospheres if left uncorrected.

A particular concern arises when these active regions are unocculted during transit. In such cases, the stellar flux includes contributions from spots and faculae that are not occulted by the planet [Rackham et al. 2023]. This causes the measured transit depth to reflect the contrast between the transit chord and the spatially averaged stellar disk, rather than the true planet-to-star radius ratio. The resulting effect varies with wavelength and can be degenerate with key molecular absorption features in the atmosphere of an exoplanet, particularly in the near- and mid-infrared where water, carbon monoxide, and methane dominate (Zhang et al 2018) . Unocculted heterogeneities can therefore mimic or suppress atmospheric signatures, leading to potentially significant misinterpretation of the data. The

problem is especially pronounced in M dwarfs, where spot coverage can be extensive and temperature contrasts large.

This form of stellar contamination has been increasingly recognised as a major source of systematic error in exoplanet spectroscopy. While it is sometimes assumed that such effects average out over time, studies have shown that persistent magnetic structures can lead to non-negligible deviations in observed spectra (Oshagh et al. 2014, Rackham et al. 2018, Thompson et al. 2024). The extent of this contamination depends on both the temperature contrast between active regions and the photosphere, and their relative surface coverage.

Although theoretical work has highlighted the importance of addressing this issue, many previous studies have relied on blackbody approximations or a limited number of atmosphere models. A systematic and physically motivated spectral library covering the necessary range of parameters for M dwarf heterogeneity is currently lacking. The aim of this project is to construct such a library using existing models of stellar atmospheres, e.g. PHOENIX (Husser et al 2013), and incorporating a wide range of surface activity scenarios and instrument response effects. This resource will support more accurate modeling of stellar contamination in exoplanet transmission spectroscopy and improve the reliability of atmospheric retrievals from missions such as JWST and ARIEL.

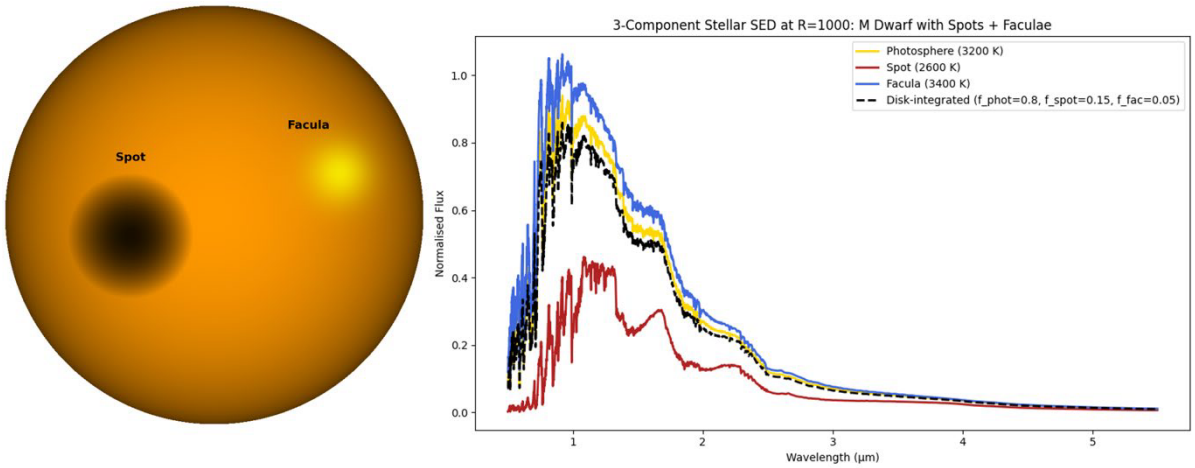


Figure 1: Left panel: Schematic of an M dwarf photosphere with a dark starspot and bright plage/facula region, including limb darkening. Right panel: Model SEDs for that are relevant to construction of the forward stellar model. The three solid-line SEDs correspond to the three temperature components that make up the surface of the active, heterogeneous stars considered in this work: the quiet photosphere, T_{phot} = 3200 K (yellow), the cooler starspot, T_{spot} = 2800 K (red), and the hotter facula, T_{fac} = 3400 K (blue). The black dashed-line SED represents the average, disk-integrated SED that would be observed, due to the weighted contributions of the quiet photosphere, spot, and facula, assuming filling factors of F_{phot} = 0.80, F_{spot} = 0.15, and F_{fac} = 0.05.

Project details

The project will begin by setting up access to to the PHOENIX model atmosphere library (Husser et al. 2013), followed by the development of a framework to generate disk integrated synthetic spectra for magnetic activity. The goal is to simulate stellar spectra composed of a quiescent photosphere, cooler starspots and hotter faculae, and examine how these surface inhomogeneties affect the observed stellar spectrum. The spectra will span the wavelength range of 0.6-15 μm, covering the infrared regions most relevant to exoplanet transmission spectroscopy with JWST and upcoming ARIEL mission.

The project steps are as follows:

Step 1: Build a synthetic spectral grid

- Generate base spectra for photosphere, spots, and faculae by interpolating PHOENIX models across Teff = 2300-3400 K. For each spectral type (M3-M9), produce. A quiescent photosphere spectrum, cooler spot spectra (ΔT = 200–1000 K in steps of 100), hotter facula spectra (ΔT = 50-500 K, in steps of 50). Techniques employed would involve standard numerical interpolation as well as machine learning techniques.
- Combine the spectra into disk-integrated models using an area-weighted sum. In this step also to vary spot and faculae coverage fractions (fs, ff) from 0% to 50% in steps of 5%. Construct a full synthetic spectral grid spanning the entire spectral types M3–M9, multiple combinations of fs, ff, and ΔT in the desired wavelength range.
- Convolve each model spectrum with the throughput profiles of JWST/NIRISS, JWST/NIRSpec, and JWST/MIRI to simulate realistic observational data.

Step 2: Use machine learning to emulate heterogeneities in 3D and predict impact on transmission spectra

- Use the synthetic spectral grid as training data for a machine learning model that maps stellar parameters (Teff, ΔT(spot), ΔT(faculae), fs, ff) directly to the full output spectrum.
- Explore machine learning algorithms, e.g. random forest or neural networks, to efficiently learn the relationship between input stellar parameters and spectral shape.
- Validate the emulator on a test set by comparing the predicted and true spectra and evaluating residuals.
- As an extension, introduce latitudinally or longitudinally dependent coverage functions to mimicking 3D variations.

Step 3: Fit observed spectra

- Fit the disk-integrated models to out-of-transit spectra of well-characterised M dwarfs such as TRAPPIST-1 or GJ 1214, using publicly available JWST data.
- As an extension, apply emulator on publicly available dataset
- Identify whether specific combination of spots and faculae coverage and temperature contrast can reproduce the observed stellar spectrum.

  Use the results to evaluate how well the synthetic spectral grid captures realistic photospheric conditions in M dwarf hosts.

Prerequisite Skills required

Basic programming skills, preferably in Python

Familiarity with stellar atmospheres, blackbody radiation, and radiative transfer concepts

Other skills used in the project

Data analysis, numerical methods and machine learning

Acceptable Programming Languages

Python

Useful References (List of important papers/review articles relevant to the project)

- Rackham, Apai & Giampapa (2018), AJ, 853, 122
  - foundational paper introducing the formalism for modeling stellar contamination in transmission spectroscopy.
- Rackham et al. (2023), RAS Techniques and Instruments, 2(1), 148–206
  - summarises the community-wide consensus on the importance of addressing stellar contamination
- Zhang et al. (2018), ApJ, 858, L8
  - explores the impact of photospheric spots/faculae on observed spectra and transmission observations of Trappist-1 system
- Oshagh et al. (2014), A&A, 568, A99
  - Foundational paper which quantifies how unocculted starspots bias transit light curves.
- Thompson et al. (2024), AJ, 167, 131
  - introduces retrieval approaches explicitly accounting for stellar contamination using real data for specific spot configuration.
- Husser et al. (2013), A&A, 553, A6
  - source of the atmospheric models used to simulate heterogeneous stellar spectra

Project 49: The Binary Fraction of Wolf-Rayet Stars

Supervisor I: Dr Simon Hodgkin (sth@ast.cam.ac.uk)

Supervisor II: Dr Avishai Gilkis (agilkis@ast.cam.ac.uk)

UTO: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

Project summary

Using Gaia DR3, you will:

1. Measure the binary fraction in known Wolf-Rayet (WR) stars to confirm that it is indeed significantly lower than that observed in other types of massive stars.
2. Search for new members of the class to increase the sample of known WR stars.

Background

Massive stars are key to many aspects in the history of the Universe. They are the probable progenitors of a huge diversity of spectacular transient events including supernovae and gamma-ray bursts. Their remnants are likely to be black holes and neutron stars, and if these arise in binary systems, then their eventual inspiral and merger could lead to the kinds of gravitational wave events seen with the LIGO-VIRGO experiments in recent years. Massive stars also show extreme outflows, and strong emission of ionising photons which have a decisive influence on their Galactic environment.

Many massive stars will pass through a Wolf-Rayet (WR) phase towards the end of their lives. WR stars were originally identified by their spectra, having strong emission lines (originating in their optically-thick winds), and extremely high temperatures between 20,000 and 200,000 K. Most massive stars are born in binaries, with an incidence of around 70%. But for Wolf Rayet stars, the reported binary incidence is much closer to 40%.

Project details

The goal of this project will be to investigate the properties of the known sample of WR stars (about 670 in total, from a catalogue provided by P. Crowther). You will explore their G, BP and RP photometry, as well as their optical XP spectra. You will measure their intrinsic luminosities, temperatures and emission line strengths and compare to state of the art models (see e.g. Hamann et al. 2019, who examined a small subset of 55 of the WN subclass of WR stars).

A key focus of the project will be to address the question of whether, and why, WR stars show such a significantly different binary fraction to their parent population of massive stars? You will use Gaia data to investigate their parameters, and examine their binary status. Perhaps previous measurements of the binary fraction have been underestimated? Or perhaps there are some physical insights to be gained.

One approach to studying binarity will be to use astrometric signatures such as the Renormalised Unit Weight Error (RUWE: Belokurov etc al. 2020). Another will be to explore the scatter in their radial velocity measurements from the Gaia Radial Velocity Spectrometer (e.g. Bashi et al. 2024).

Having identified the key properties of known WR stars, you will begin a search to look for more similar objects in Gaia Data Release 3. It is quite likely that previous surveys are incomplete, and that Gaia's astro-photo-spectro all-sky survey will enable you to discover new members of the class. This enlarged sample could then help to shed light on the mysterious class of objects.

Skills required

You will use the ESA Gaia database (and/or a local copy of those tables) which will require some basic knowledge of simple SQL (this can be taught by us). Some programming will be required to wrap the SQL queries, and generate plots. More coding will be required to access and analyse the Gaia XP spectra which are also held in relational databases. Python in a unix environment is recommended, but other approaches are possible (please discuss with us).

Useful References (List of important papers/review articles relevant to the project)

- Physical Properties of Wolf-Rayet Stars, Crowther 2007, https://ui.adsabs.harvard.edu/abs/2007ARA%26A..45..177C/abstract
- Why binary interaction does not necessarily dominate the formation of Wolf-Rayet stars at low metallicity, Shenar et al. 2020, https://ui.adsabs.harvard.edu/abs/2020A%26A...634A..79S/abstract
- Gaia Data Release 3. Summary of the content and survey properties, Gaia collaboration, Vallenari et al. 2023, https://ui.adsabs.harvard.edu/abs/2023A%26A...674A...1G/abstract
- Gaia Data Release 3. A golden sample of astrophysical parameters, Gaia collaboration, Creevey et al. 2023, https://ui.adsabs.harvard.edu/abs/2023A%26A...674A..39G/abstract
- Gaia Data Release 3. Stellar multiplicity, a teaser for the hidden treasure, Gaia collaboration, Arenou et al. 2023, https://ui.adsabs.harvard.edu/abs/2023A%26A...674A..34G/abstract
- Gaia Data Release 3. Properties and validation of the radial velocities, Katz et al. 2023, https://ui.adsabs.harvard.edu/abs/2023A%26A...674A...5K/abstract
- Gaia Data Release 3. Processing and validation of BP/RP low-resolution spectral data, Gaia collaboration, de Angeli et al. 2023, https://ui.adsabs.harvard.edu/abs/2023A%26A...674A...2D/abstract

General References (List of papers referred to in the project)

- Hamann et al. 2019, https://ui.adsabs.harvard.edu/abs/2019A%26A...625A..57H/abstract
- Belokurov etc al. 2020: https://ui.adsabs.harvard.edu/abs/2020MNRAS.496.1922B/abstract
- Bashi et al. 2024: https://ui.adsabs.harvard.edu/abs/2024arXiv240512335B/abstract

Figure 1:

Shells of cosmic dust created by the interaction of binary stars appear like tree rings around Wolf-Rayet 140. The remarkable regularity of the shells' spacing indicates that they form like clockwork during the stars' eight-year orbit cycle, when the two members of the binary make their closest approach to one another. In this image, blue, green, and red were assigned to Webb's Mid-Infrared Instrument (MIRI) data at 7.7, 15, and 21 microns (F770W, F1500W, and F2100W filters, respectively). Credits: NASA, ESA, CSA, STScI, NASA-JPL, Caltech

Project 50: A new exploration of multiple populations in globular cluster streams

Supervisor I: Sarah Kane (sgk27@cam.ac.uk)

Supervisor II: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

UTO: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

Project summary

Globular clusters (GCs), ancient groups of thousands to millions of stars, are distinguished by the presence of multiple populations among their member stars. Although the origin of these multiple populations is still debated, it is almost universal among GCs that there is a first generation (1G) of chemically typical stars and a second generation (2G) of anomalously nitrogen, sodium, and aluminium rich and oxygen, carbon, and magnesium depleted stars. The 2G stars are so anomalous that no site of origin besides GCs is yet known for stars with such a chemical abundance pattern, making the multiple population phenomenon unique to globular clusters. This uniqueness makes the presence of multiple populations a strong diagnostic with which to determine when a stellar stream, the artefact of a disrupted cluster or satellite dwarf galaxy, originated from a GC. Leveraging the newly released Dark Energy Spectroscopic Instrument (DESI) Data Release 1 alongside the Gaia low resolution XP spectra, this project will thus explore the presence of multiple populations in stellar streams suspected to originate from GCs. In particular, the project outlines the development of pseudo-colour “chromosome maps” that separate stars based upon their carbon and nitrogen abundances. The quantity of data provided by Gaia and DESI poses an exciting opportunity to explore the Milky Way’s tidally disrupted globular clusters, including fainter and more distant stars than was previously possible.

Background

Ancient, densely-packed, and numerous in the Galactic stellar halo, globular clusters (GCs) are relics of early star formation in the Milky Way and in other galaxies. One of the most unique features of GCs is the presence of multiple stellar populations that are distinguished by the elemental abundances of their constituent stars. The first generation (1G), thus called because it is thought to have formed first, contains stars with chemistry approximately consistent with non-cluster stars that formed at similar metallicities. In contrast, the second generation (2G) members of a cluster are enriched in nitrogen, sodium, aluminium, and sometimes helium or potassium and comparatively depleted in carbon, oxygen, and sometimes magnesium. The abundance patterns between these two populations form the so-called “light element anticorrelation” among a GC’s members (e.g., with sodium and oxygen abundances being anticorrelated between the 1G and 2G). In a sharp contrast to the light element anticorrelation, which is almost universal among GCs, within a typical cluster there is a very small spread in iron abundance among member stars, indicating no evidence of intra-cluster enrichment from supernova products. There are a variety of proposed mechanisms to produce the anomalous abundances of the 2G stars,

with theories ranging from enrichment from AGB stars to supermassive stars among the 1G members, but no consensus has yet been reached regarding which theory can adequately explain all features of the multiple populations phenomenon (see the review in Milone & Marino 2022 for more information on multiple populations in GCs).

The chemical differences between 1G and 2G stars are so stark that they can be distinguished via photometry alone. So-called “chromosome maps” (ChMs) are pseudo-colour-colour diagrams that are especially efficient at distinguishing between the populations of GC stars because their pseudo-colours utilize filters that cover carbon and nitrogen molecular features—elements crucial in distinguishing between 1G and 2G stars—as well as capturing colour variations that arise from the differences in helium content between 1G and 2G members. (The term “pseudo-colour” here merely indicates that the “colours” are the differences between magnitudes at different wavelengths, often outside of the visible spectrum, as opposed to “real” colours. Page 4 of Milone & Marino 2022 also introduces ChMs.) Among the most famous ChMs are those constructed with Hubble’s “magic filters,” UV and optical filters (Piotto, et al. 2015; Milone, et al. 2017) that include coverage of the carbon and nitrogen molecular features and thus can separate stars that differ in these abundances. As shown in Fig. 1 below, these narrow-band filters offer sufficient distinguishing power to separate the 1G and 2G stars into separate tracks on a colour-magnitude diagram, all the way from the main sequence to the red giant branch.

Recently, Mehta et al. (2025) used the low resolution Gaia XP spectra to develop filters centred on the CN and CH molecular features, thus allowing them to distinguish between stars with different carbon and nitrogen abundances from low resolution spectrophotometry. This is especially powerful for GCs, where second generation (2G) stars are markedly more nitrogen-enhanced and carbon-depleted than their first generation (1G) counterparts in the same cluster. Using these filters, Mehta et al. (2025) developed chromosome maps (see Fig. 2 below) for five clusters to identify their first and second generation members across a range of metallicities. This identification was then used to study the distribution of the generations within each cluster, with several of their studied clusters exhibiting a higher central concentration of 2G stars. This finding is consistent with the formation scenario wherein 2G stars formed in the innermost regions of the cluster from enriched gas from the 1G stars.

The globular clusters we observe today are only a fraction of those that once existed in the Milky Way and among its accreted structures. Many GCs have been significantly disrupted or dissolved, with Belokurov and Kravtsov (2023) estimating that 50% or more of the Milky Way’s pre-disk (halo) stars formed in GCs. Recent observations at high redshift with JWST suggesting that GCs may have played a significant role in the star formation histories of other galaxies have further increased community interest, making this an especially relevant time to study intact and disrupted GCs.

Stellar streams, tracks of stars that move along an orbit through the Galaxy, are the disrupted remnants of GCs or satellite dwarf galaxies (see the review in Bonaca & Price-Whelan 2025). One way to distinguish between these two origins—GC or galaxy—are via their chemistry, as only stars that originated in a GC would exhibit the distinctive light element anticorrelation. Nonetheless, getting detailed chemistry for streams can be difficult, meaning that the presence of multiple stellar populations has only been confirmed in a few streams. The goal of this project will thus be to construct narrow-band

photometric filters, as done in Mehta, et al. (2025), this time from both the Gaia XP spectra and the Dark Energy Spectroscopic Instrument (DESI)'s spectra. These narrow band filters will then enable us to construct chromosome maps and explore the presence of multiple populations in streams at a large scale.

Project details

This project will proceed in several stages. First, the student will develop the same colour filters used in Mehta et al. (2025), this time both with the low resolution Gaia XP spectra used in that work and for the first time with the somewhat higher resolution DESI spectra. With coverage from 3600 Å to 9800 Å (Koposov et al. 2025), DESI covers much of the same wavelength range as the Gaia XP spectra, making a reproduction of the methods of Mehta, et al. (2025) possible. From these photometric filters, ChMs can then be constructed. We will first apply them to GCs, which are known to host multiple populations, as a step to verify the ChMs. This will also be a useful stage at which to compare the ChMs to DESI's published stellar abundances (e.g., nitrogen, which has a unique distinguishing power between 1G and 2G stars). After the ChMs are verified on GCs, they will be applied to stream member stars to explore the presence of multiple populations. Also making a ChM for a confirmed dwarf galaxy stream may prove to be a useful comparison.

The benefit of this project is twofold. First, it will leverage the scope of Gaia and DESI to analyse many more stream stars than is possible with targeted high-resolution spectroscopy, which would require very long exposure times for such faint stars. Only a limited number of streams have confirmed presence of multiple populations consistent with origin in the GC (for instance, Usman et al. 2024). Successful identification of multiple populations would thus be an exciting contribution to our understanding of disrupted globular clusters and could motivate follow-up with higher resolution spectroscopy (i.e., a VLT proposal following the completion of the project). Moreover, with more stars available in DESI than is typically possible with high resolution spectroscopy, we can study the fraction and distribution of multiple populations in streams to an extent that has not yet been accomplished. Second, this work will provide an independent test of the newly released DESI stellar abundances, including both chemically "typical" (1G) and peculiar (2G) stars.

A possible extension to this project could include more formal abundance derivations from the DESI spectra for streams or clusters of particular interest, including multiple elements involved in the light-element anticorrelation (e.g., N, C, Mg, etc.). More detailed abundances could enable a comparison with the Galaxy's surviving GCs, marking differences between disrupted and in-tact clusters.

Acceptable Programming Languages

Python

Useful References (List of important papers/review articles relevant to the project)

- Bonaca, A. & Price-Whelan, A. M. (2025) "Stellar streams in the Gaia era." *New Astronomy Reviews*, 100, https://ui.adsabs.harvard.edu/abs/2025NewAR.10001713B/abstract

- Koposov, S. E., et al. (2025) “DESI Data Release 1: Stellar Catalogue.” https://ui.adsabs.harvard.edu/abs/2025arXiv250514787K/abstract
- Mehta, V. J., et al. (2025) “Spectrophotometry and radial distribution of multiple stellar populations in globular clusters from Gaia XP spectra.” MNRAS, 536, 2, https://ui.adsabs.harvard.edu/abs/2025MNRAS.536.1077M/abstract
- Milone, A. P. et al. (2017) “The Hubble Space Telescope UV Legacy Survey of Galactic globular clusters – IX. The Atlas of multiple stellar populations.” MNRAS, 464, 3, https://ui.adsabs.harvard.edu/abs/2017MNRAS.464.3636M/abstract

General References (List of papers referred to in the project)

- Belokurov, V. & Kravtsov, A. (2023) “Nitrogen enrichment and clustered star formation at the dawn of the Galaxy.” MNRAS, 525, 3, https://ui.adsabs.harvard.edu/abs/2023MNRAS.525.4456B/abstract
- Milone, A. P. & Marino, A. F. (2022) “Multiple Populations in Star Clusters.” *Universe*, 8, 7, https://ui.adsabs.harvard.edu/abs/2022Univ....8..359M/abstract
- Piotto, G., et al. (2015) “The Hubble Space Telescope UV Legacy Survey of Galactic globular clusters. I. Overview of the Project and Detection of Multiple Stellar Populations.” ApJ, 149, 3, https://ui.adsabs.harvard.edu/abs/2015AJ....149...91P/abstract
- Usman, S. A., et al. (2024) “Multiple populations and a CH star found in the 300S globular cluster stellar stream.” MNRAS, 529, 3, https://ui.adsabs.harvard.edu/abs/2024MNRAS.529.2413U/abstract

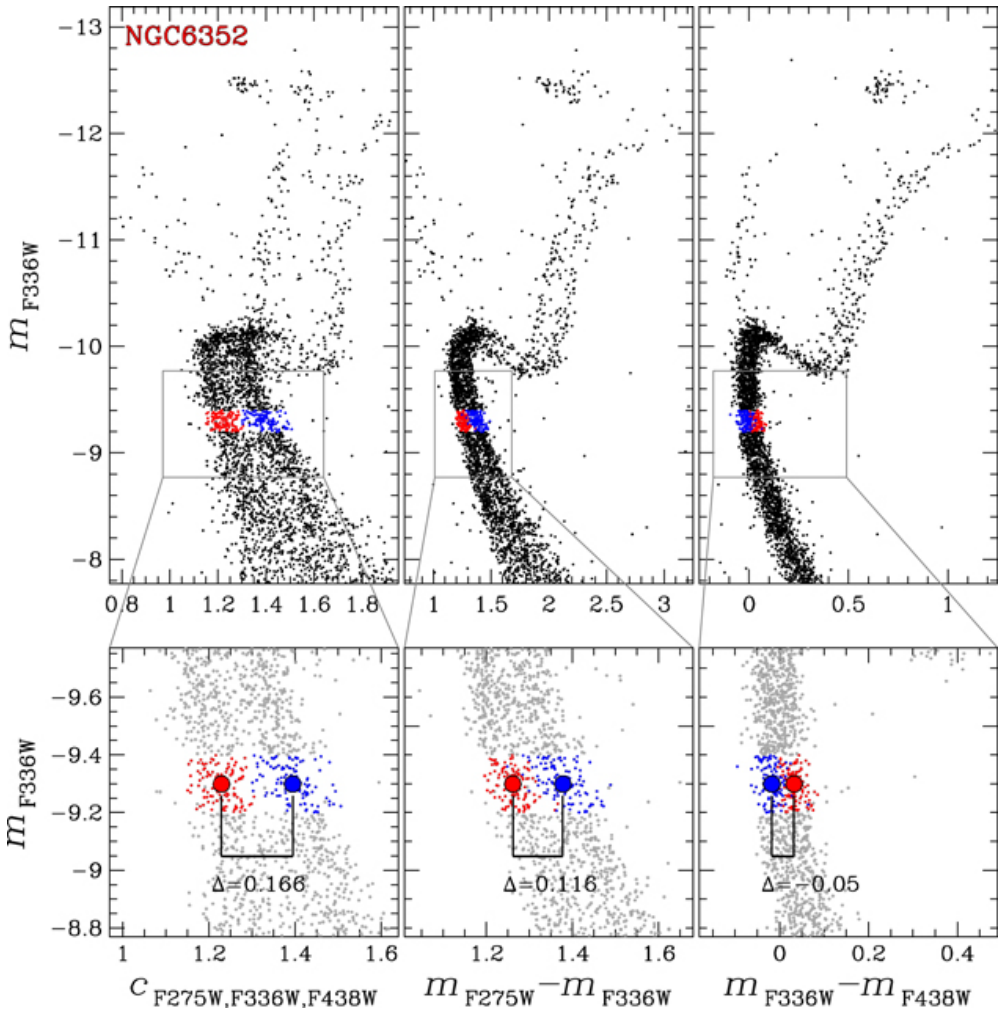


Figure 1: Reproduced from Fig. 2 of Piotto et al. (2015), these are the colour-magnitude diagrams of the globular cluster NGC 6352 constructed with the Hubble Telescope's "magic filters." The 1G stars are marked in blue, and the 2G stars are marked in red; note the clear separation of the two, which differs between which filter is used. The lower panels show a magnified portion the cluster's main sequence.

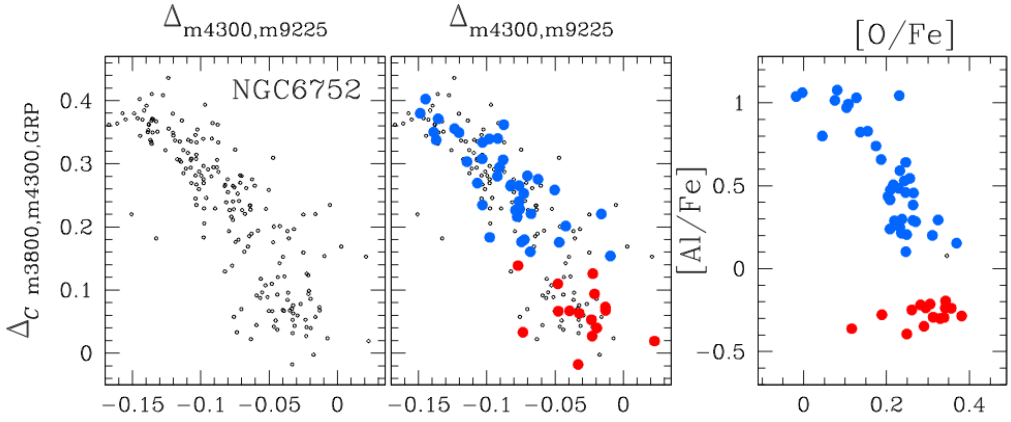


Figure 2: Reproduced from Fig. 7 of Mehta et al. (2025). In the left and middle panels is the chromosome map of RGB stars from the GC NGC 6752. The right panel shows the oxygen and aluminium abundances of a subset of the stars with higher resolution spectroscopy, with the aluminium-enhanced and oxygen-depleted 2G stars marked in blue and the 1G stars in red. The classifications for these stars are also marked in the chromosome map in the middle panel; note the relatively clear separation of the 1G and 2G stars in this colour-colour space.

Project 51: Accelerating Galactic Foreground Modeling for 21cm Cosmology with GPU-Enabled Bayesian Inference

Supervisor I: Dr Will Handley (wh260@cam.ac.uk)
Supervisor II: Dr Melis Irfan (mi385@cam.ac.uk)
UTO: Dr Will Handley (wh260@cam.ac.uk)

Project Summary
Detection of the cosmological 21cm signal from the Epoch of Reionization requires precise modeling of Galactic foregrounds, which are 10^3-10^5 times brighter than the target signal. The Bayesian Global Sky Model (B-GSM) framework of Carter and Handley (2025) performs simultaneous component separation and calibration of low-frequency radio data, but current CPU-based nested sampling limits model complexity. This project will reimplement B-GSM using BlackJAX's GPU-accelerated nested sampling to enable computationally intensive extensions including spatially varying spectral indices and improved noise modeling. The enhanced framework will be validated against existing results and applied to investigate astrophysical questions about Galactic emission processes.

Background
The 21cm hydrogen line provides a probe of cosmic structure formation during the cosmic dark ages and Epoch of Reionization (Furlanetto et al. 2006). Galactic synchrotron and free-free emission dominate over the cosmological signal by factors of 10^3-10^5 at frequencies below 200 MHz, requiring accurate foreground modeling for signal detection (Shaver et al. 1999).

Existing foreground models such as the Global Sky Model (GSM) employ Principal Component Analysis but lack uncertainty quantification and absolute calibration (de Oliveira-Costa et al. 2008; Zheng et al. 2017). The B-GSM framework addresses these limitations using Bayesian inference to simultaneously model foreground components and calibration parameters, conditioning on both diffuse sky maps and absolute temperature measurements from experiments like EDGES (Carter & Handley 2025a,b).

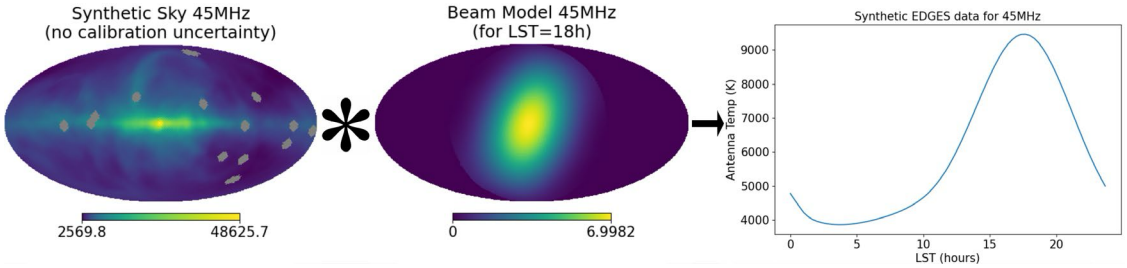
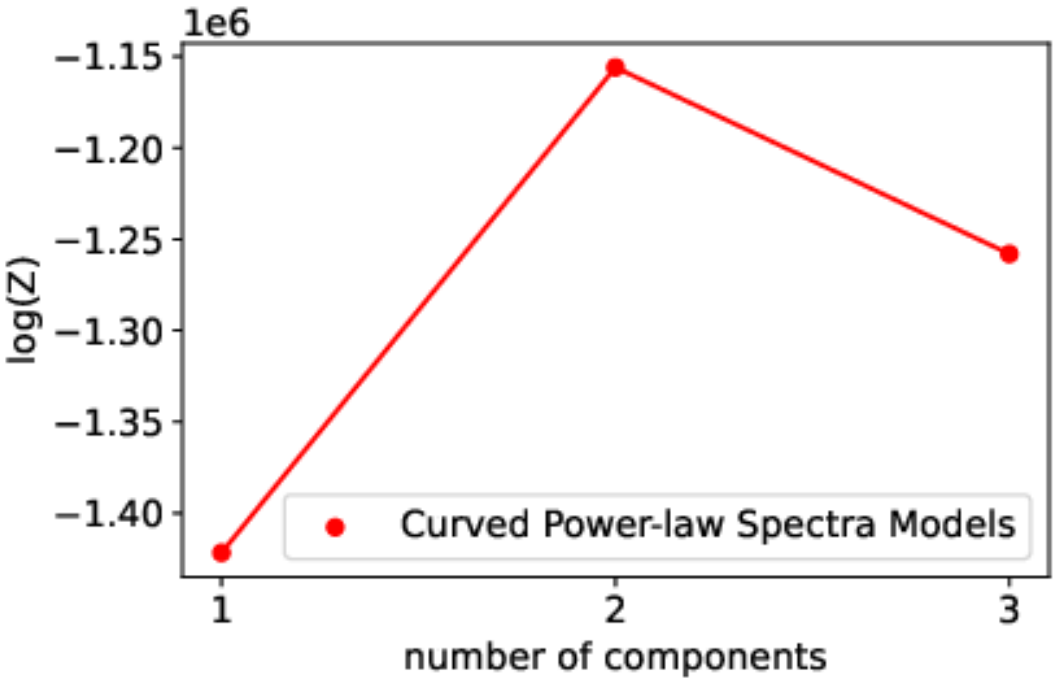


Figure
Figure 1: The B-GSM data generation pipeline showing synthetic sky map at 45 MHz, beam model, and resulting EDGES-like observations. This illustrates the relationship between sky model, instrument response, and observed data that the Bayesian framework aims to invert (Carter & Handley 2025).

The B-GSM uses nested sampling for model selection and parameter estimation. Analysis of data spanning 45-408 MHz identified a two-component model with curved power-law spectra: a steep-spectrum component (β_1 = -2.633 ± 0.002) consistent with synchrotron emission at high Galactic latitudes, and a flatter component (β_2 = -2.108 ± 0.008) dominant in the Galactic plane. Computational limitations of the current CPU implementation restrict the framework to simplified assumptions including uniform spectral indices and 5° resolution.



Figure

Figure 2: Bayesian evidence comparison for curved power-law spectral models with different numbers of components. The evidence peaks at 2 components, demonstrating the framework's ability to automatically select optimal model complexity while avoiding overfitting (Carter & Handley 2025).

BlackJAX provides GPU-accelerated nested sampling with demonstrated speedups of 10-1000× for cosmological inference problems. The JAX implementation enables automatic differentiation and efficient vectorization, with reported performance gains of 300× for CMB analysis and >1000× for high-dimensional cosmic shear problems. The slice sampling kernel operates without tuning parameters while maintaining robustness for complex likelihood surfaces.

Project Details

1. **Reimplement B-GSM Core Algorithm in JAX/BlackJAX**
   - Port the existing B-GSM likelihood computation from NumPy/SciPy to JAX for GPU acceleration
   - Implement the approximate marginal likelihood derived by Carter & Handley, ensuring numerical stability and gradient compatibility
   - Replace PolyChord nested sampling with BlackJAX nested sampling, validating identical posterior recovery on existing synthetic datasets
2. **Benchmark Computational Performance**

- Compare inference times between CPU (PolyChord) and GPU (BlackJAX) implementations across different model complexities
- Quantify scaling behavior with number of frequency maps, sky resolution, and spectral parameters
- Establish computational feasibility thresholds for advanced model extensions

3. **Implement Spatially Varying Spectral Index (SVSI) Model**
   - Extend the framework to model spectral parameters $\beta(\Omega)$ and $\gamma(\Omega)$ as functions of sky position using spherical harmonic expansion or Gaussian Process priors
   - Apply the enhanced model to existing datasets to address limitations such as negative temperatures in certain sky regions
4. **Evaluate the astrophysical interpretation of spatial variations in spectral behavior**
   - Enhanced Foreground Physics and Data Integration
   - Incorporate improved noise models including spatial correlations from telescope beams
   - Implement spatially varying calibration corrections for composite surveys like the Haslam 408 MHz map
   - If time permits: Include point source modeling as a third component with catalog-based priors from GLEAM or NVSS
5. **Assess Impact on 21cm Cosmology Applications**
   - Propagate B-GSM posterior uncertainty through 21cm power spectrum analysis pipelines
   - Quantify improvements in foreground subtraction accuracy relative to existing methods

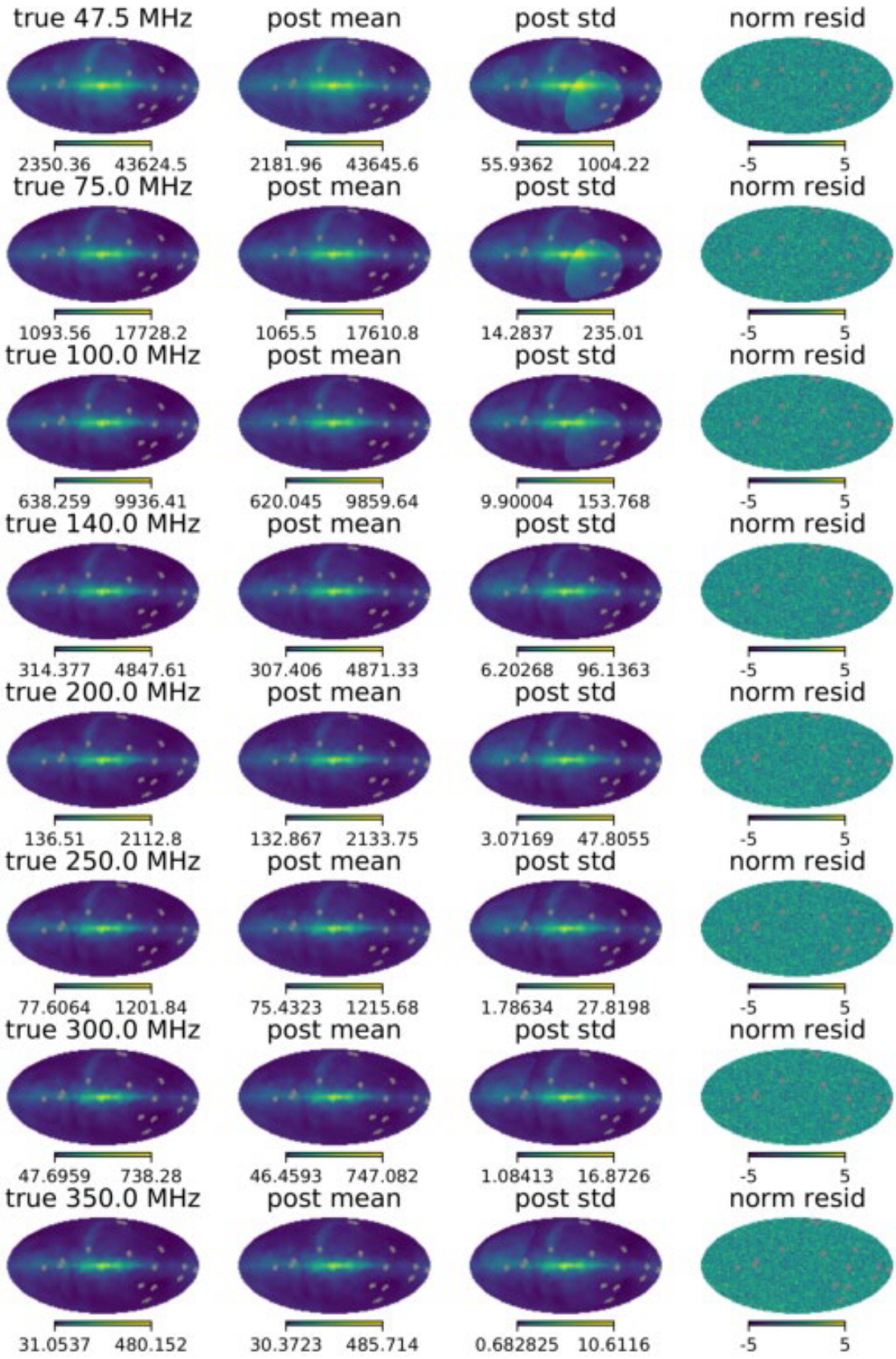


Figure 3: B-GSM posterior sky map reconstructions across multiple radio frequencies (47.5-350 MHz), showing true sky models, posterior means, posterior uncertainties, and normalized residuals. This demonstrates the framework's ability to reconstruct accurate sky models with proper uncertainty quantification across the frequency range relevant to 21cm cosmology (Carter & Handley 2025).

Skills Required

- **Programming:** Strong Python programming skills required, with experience in scientific computing libraries (NumPy, SciPy, matplotlib). Willingness to learn JAX for GPU acceleration during the project
- **Astrophysics Background:** Basic understanding of cosmology and radio astronomy; Part II course "Astrophysical Fluid Dynamics" or equivalent recommended
- **Statistical Methods:** Introduction to Bayesian inference helpful; advanced methods including MCMC and nested sampling will be taught as part of the project
- **Mathematics:** Part III-level mathematics including statistics, differential equations, and linear algebra

Group Resources and Support

Students will be supervised by Will Handley throughout the project, meeting one-on-one weekly during term time to discuss progress and plan next steps. Students are expected to attend weekly group meetings and interact with the rest of the Part III, PhD, and postdoc members of the group.

The project will utilize the group's GPU-accelerated computational infrastructure, including BlackJAX nested sampling with 10×-1000× speedups, JAX ecosystem tools, and access to large astronomical datasets. Students will benefit from the group's expertise in Bayesian inference, cosmological analysis, and machine learning applications in astrophysics.

If successful, the project could lead to a publication (with approximately 50% of previous Part III projects leading to publications).

Useful References

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Project 52: GPU-Accelerated Nested Sampling for Optimal Fiber Positioning in Astronomical Surveys

Supervisor I: Dr Will Handley (wh260@cam.ac.uk)
UTO: Dr Will Handley (wh260@cam.ac.uk)

Project Summary
Large-scale astronomical surveys like DESI and WEAVE depend on the efficient robotic positioning of thousands of optical fibers, a task that represents a complex, high-dimensional optimization problem. This problem is typically solved using heuristic methods like simulated annealing, which can be computationally intensive and may not find the globally optimal solution. This project will explore the use of nested sampling, a powerful Bayesian inference algorithm, as a superior alternative. Leveraging the cutting-edge **BlackJAX** library, which provides massive (10×-1000×) GPU acceleration for nested sampling, we will develop a new solver for the fiber assignment problem. The primary goal is to benchmark the performance and accuracy of nested sampling against established methods, demonstrating its potential to handle complex constraints and multi-modal optimization landscapes, thereby unlocking more efficient and scientifically productive astronomical observations.

Background
Modern observational cosmology is driven by Multi-Object Spectroscopy (MOS), which enables the simultaneous capture of spectra from thousands of celestial objects. The efficiency of instruments like DESI (5,000 fibers) and WEAVE (1,000 fibers) hinges on their robotic fiber positioning systems. The scale of these challenges is set to increase dramatically: the proposed Wide-field Spectroscopic Telescope (WST) will feature an unprecedented **20,000-22,000 robotic fiber positioners** - four times more than DESI. The core computational challenge is the **fiber assignment problem**: assigning fibers to scientifically valuable targets while respecting a complex set of constraints, including the limited “patrol area” of each robotic positioner and the critical need for collision avoidance.

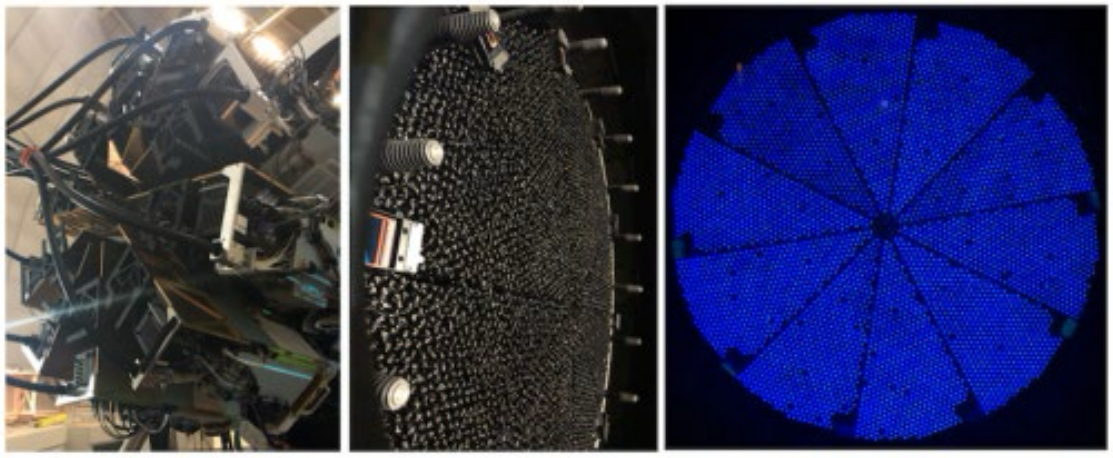

Figure Figure 1: The DESI fiber positioning system showing the mechanical structure, positioner tips, and fiber distribution across the focal plane. This demonstrates the scale and complexity of robotic fiber positioning that requires sophisticated optimization algorithms to maximize scientific throughput (DESI Collaboration 2022).

The assignment problem is a combinatorial optimization task of immense complexity. The standard approach, as detailed for the WEAVE survey, is to use a metaheuristic algorithm like simulated annealing (Hughes et al. 2022). While effective, simulated annealing is a heuristic method that explores the parameter space by randomly accepting worse solutions to escape local minima; it offers no guarantee of finding the global optimum and can be slow to converge for highly constrained problems.

This project proposes a fundamentally different approach by framing fiber assignment as a Bayesian optimization problem. **Nested sampling** (Skilling 2006) provides a powerful framework for exploring the full landscape of fiber configurations, returning not just a single optimal solution but a collection of high-quality assignments with different trade-offs. This enables multi-objective optimization where survey teams can select from diverse solutions based on real-time priorities. The algorithm's "dead points" (discarded configurations from earlier iterations) and posterior samples together provide a rich toolkit of near-optimal solutions. Nested sampling is renowned for its ability to navigate complex, multi-modal optimization landscapes and naturally handles the explicit inclusion of uncertainties, such as errors in target positions, which is not straightforward with traditional methods.

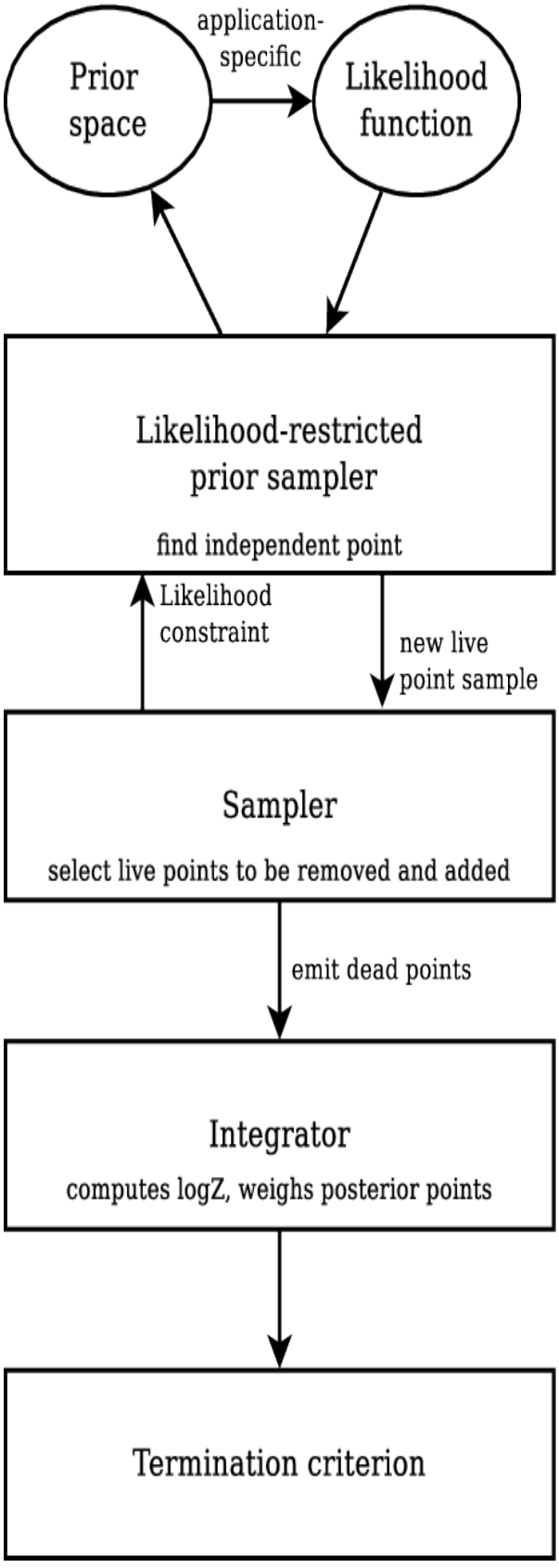

Figure 2: Overview of the nested sampling algorithm showing the flow from prior space through likelihood-restricted sampling to evidence computation. This general-purpose optimization framework can be adapted for fiber positioning by treating configurations as the parameter space and assignment quality as the likelihood function (Buchner 2021).

Until recently, the computational cost of nested sampling made it impractical for real-time problems like fiber positioning. This has been transformed by the development of **BlackJAX**, a JAX-based library for probabilistic modelling that includes a highly parallelized, GPU-accelerated implementation of nested sampling. In other astrophysical domains, such as the analysis of gravitational waves and the cosmic microwave background, BlackJAX has demonstrated performance improvements of 100×-1000× over traditional CPU-based methods. This project will be the first to apply this cutting-edge computational tool to the fiber positioning problem, creating an opportunity to significantly advance the state of the art.

Project Details
The student will undertake the following tasks:

1. **Problem Formulation:** The first step will be to formulate the fiber assignment problem as a Bayesian optimization. This involves defining a likelihood function that quantifies the scientific value of a given fiber-target assignment (e.g., based on target priorities) and priors that enforce the hard mechanical constraints of the system (patrol areas, collision avoidance).
2. **Implementation of a Nested Sampling Solver:** Using the BlackJAX library, the student will implement a nested sampling solver for the fiber assignment model. This will involve writing Python code within the JAX ecosystem to define the model and interface with the BlackJAX sampler.
3. **Benchmarking and Comparison:** The performance of the new nested sampling solver will be rigorously benchmarked against a standard simulated annealing algorithm. This will be tested on realistic, simulated target fields based on the configurations of surveys like WEAVE and DESI, with scalability tests extending to WST-scale problems (20,000+ fibers). Key metrics for comparison will include computational time, the diversity and quality of solution collections, and the ability to find multiple high-quality solutions in highly clustered or constrained fields.

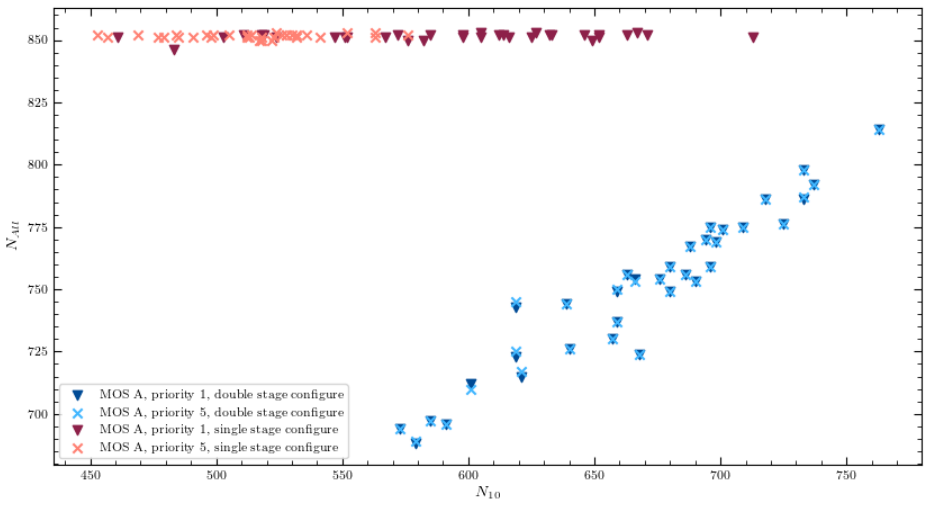

Figure 3: WEAVE fiber allocation efficiency showing the number of allocated fibers versus target density for different configuration strategies. The comparison between single and double stage configurations demonstrates optimization trade-offs that nested sampling could potentially improve by exploring multiple high-quality solutions simultaneously (Hughes et al. 2022).

4. **Analysis of Complex Scenarios:** The project will investigate the solver's performance in challenging scenarios where traditional methods struggle. This includes fields with high target density, complex overlapping patrol areas, and the presence of "forbidden zones" on the focal plane.
5. **If time permits:** The student can explore the full power of the Bayesian framework by incorporating uncertainties in target positions into the model. The collection of solutions produced by nested sampling can be used to assess the robustness of different assignments and quantify the probability of successful observations across diverse configurations. Additionally, the multi-objective nature of the approach can be exploited to optimize for competing survey objectives simultaneously.

Skills Required

- **Programming:** Strong Python programming skills required, with experience in scientific computing libraries (NumPy, SciPy, matplotlib). Enthusiasm for computational problem-solving and willingness to learn JAX during the project.
- **Mathematics and Statistics:** Basic understanding of optimization methods and statistics helpful. Introduction to Bayesian inference beneficial but advanced methods will be taught.
- **Astrophysics Knowledge:** Understanding of observational astronomy and survey science. Knowledge of multi-object spectroscopy and astronomical instrumentation will be developed as part of the project.
- **Courses:** Part II/III course in Computational Physics, Mathematical Methods, or Statistics recommended but not essential.
-

Group Resources and Support

Students will be supervised by Will Handley throughout the project, meeting one-on-one weekly during term time to discuss progress and plan next steps. Students are expected to attend weekly group meetings and interact with the rest of the Part III, PhD, and postdoc members of the group.

The project will utilize the group's GPU-accelerated computational infrastructure, including BlackJAX nested sampling with 10×-1000× speedups, JAX ecosystem tools, and access to large astronomical datasets. Students will benefit from the group's expertise in Bayesian inference, cosmological analysis, and machine learning applications in astrophysics.

If successful, the project could lead to a publication (with approximately 50% of previous Part III projects leading to publications).

Useful References

- Hughes, S. et al. (2022). “Optimisation of the WEAVE target assignment algorithm”. arXiv:2209.01145
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Project 53: GPU-Accelerated Bayesian Inference for Quasar Time Delay Cosmography

Supervisor I: Dr Will Handley (wh260@cam.ac.uk)
Supervisor II: Namu Kroupa (nk631@cam.ac.uk)
UTO: Dr Will Handley (wh260@cam.ac.uk)

Project Summary
This project will develop advanced computational methods to measure the expansion rate of the Universe using gravitationally lensed quasars. When light from a distant quasar is gravitationally lensed by an intervening galaxy, it creates multiple images with different light travel times, providing a direct probe of cosmological distances across billions of light-years. These time delay measurements offer an independent route to determining the Hubble constant and constraining dark energy—crucial for resolving current tensions in cosmology. However, extracting precise time delays from noisy astronomical data presents significant computational challenges. This project will develop next-generation GPU-accelerated methods to overcome these challenges, enabling the analysis of large samples of lensed quasars and realizing their potential as high-precision cosmological probes in the era of large-scale astronomical surveys.

Background
Quasars—the brilliant cores of distant galaxies powered by supermassive black holes—are among the most luminous objects in the Universe, visible across cosmic time. When a massive galaxy lies along our line of sight to a quasar, Einstein's general relativity predicts that the galaxy's gravity will bend the quasar's light, creating multiple images. Since the light paths have different lengths, brightness variations in the quasar appear at different times in each image, with delays ranging from days to years. These time delays encode fundamental information about cosmic geometry and the expansion rate of the Universe, offering a completely independent measurement of the Hubble constant—crucial for understanding the nature of dark energy that drives cosmic acceleration (Refsdal 1964; Suyu et al. 2017).

However, measuring these time delays from noisy, irregularly sampled light curves presents significant statistical challenges. Recent advances have employed Gaussian processes (GPs) to model the intrinsic quasar variability while simultaneously inferring time delays (Hojjati et al. 2013). This approach treats the multiple light curves as time-shifted, magnitude-shifted, and microlensed versions of a single underlying stochastic process, providing a principled framework for uncertainty quantification.

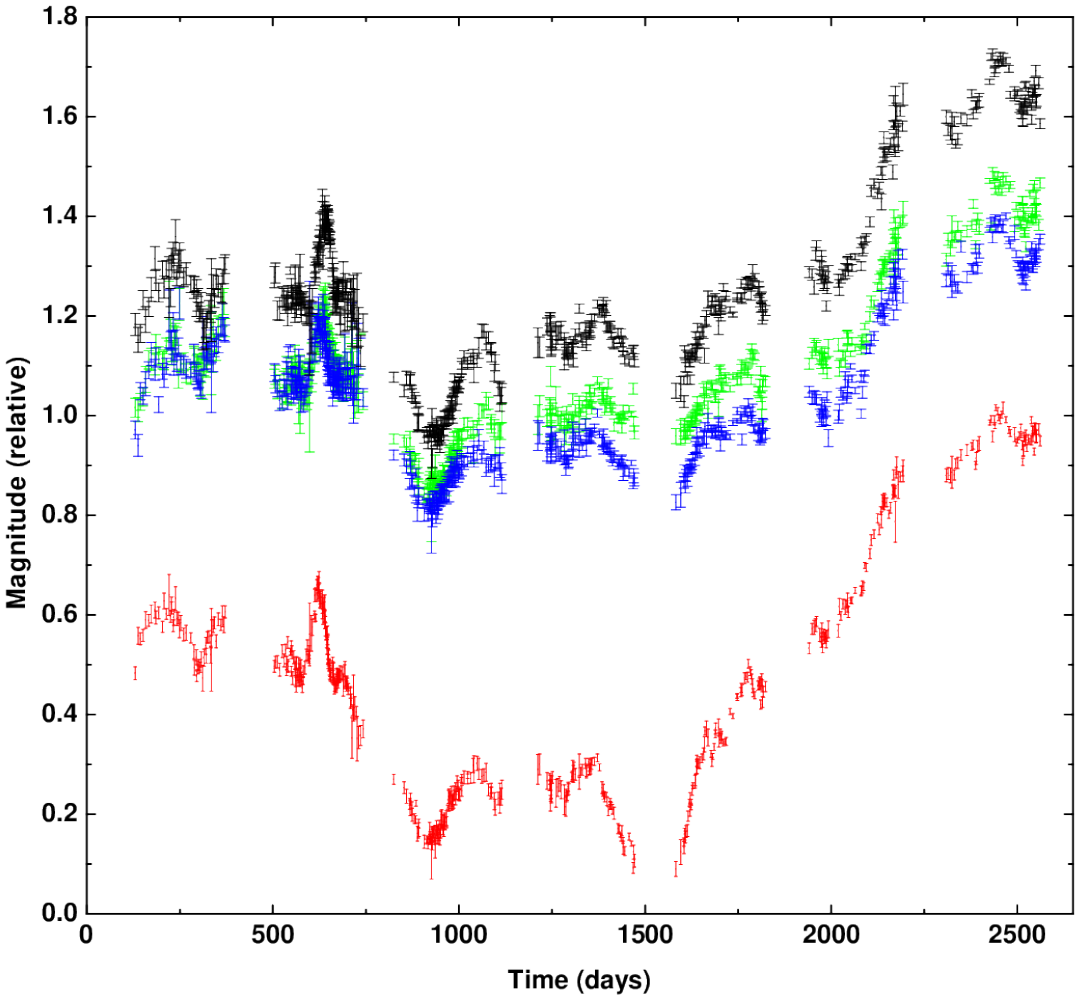

Figure 1: Multi-epoch light curves of a gravitationally lensed quasar showing four images (different colors) with characteristic time delays and microlensing variations. The similar patterns offset in time demonstrate the time delay phenomenon that encodes cosmological information, while the amplitude differences reveal microlensing effects that must be modeled (Hojjati et al. 2013).

Recent work has identified several fundamental challenges in GP-based time delay inference. Kroupa et al. discovered that the likelihood function exhibits a pathological “W-shaped” structure with a correct narrow peak at the true time delay but also rises gradually toward unphysically large delays, systematically causing nested sampling algorithms to fail. More fundamentally, the standard GP approach suffers from $O(N^3)$ computational scaling due to the Cholesky decomposition required for likelihood evaluation, severely limiting the size of datasets that can be analyzed. However, promising theoretical advances suggest this bottleneck can be overcome for one-dimensional GPs using state-space methods based on continuous autoregressive moving average (CARMA) kernels, which reduce computational complexity from $O(N^3)$ to $O(N)$ through Kalman filtering techniques (Kelly et al. 2014).

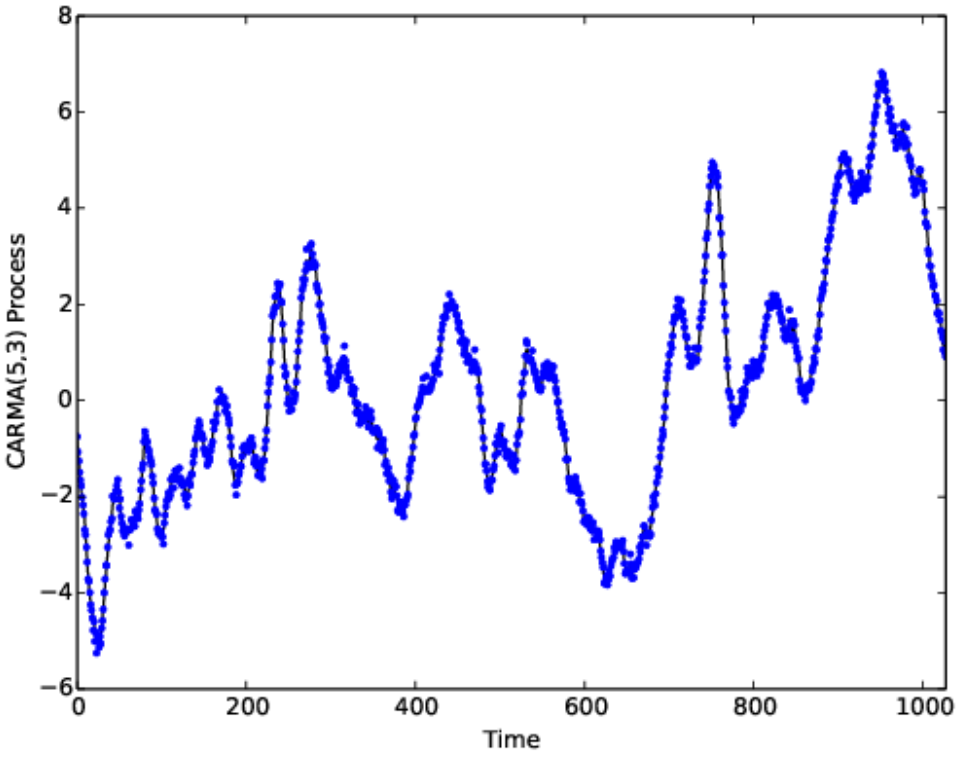

Figure 2: Example CARMA(5,3) time series demonstrating the complex, non-linear variability patterns characteristic of quasar light curves. CARMA processes provide a computationally efficient O(N) framework for modeling such data through state-space methods, enabling the analysis of large datasets that are prohibitive with traditional $O(N^3)$ Gaussian Process approaches (Kelly et al. 2014).

This computational bottleneck is particularly problematic because upcoming astronomical surveys like the Vera Rubin Observatory's Legacy Survey of Space and Time (LSST) will discover thousands of new lensed quasars, creating an unprecedented opportunity to map cosmic expansion with exquisite precision. However, current analysis methods are too slow and unreliable to handle this volume of data. Furthermore, quasar variability is extraordinarily complex—driven by accretion disk physics, relativistic jets, and microlensing by stars in the lensing galaxy—requiring sophisticated models that current computational approaches cannot explore efficiently.

Project Details

This project will develop and apply next-generation computational methods for time delay cosmography:

1. **Validate analysis on known systems**: Implement and validate the GP-based time delay inference framework on well-studied lensed quasars (WFI J2033–4723, B 1608 + 656, HE 0435–1223). Reproduce existing results and establish baseline performance for the GPU implementation while gaining experience with the interplay of quasar physics, gravitational lensing, and observational effects.

2. **Develop robust inference algorithms**: Design and implement novel sampling strategies to reliably handle the “pathological likelihood” identified by Kroupa et al. Test adaptive live point allocation, alternative proposal mechanisms, and tempering approaches to achieve reliable convergence without requiring prohibitive computational expense.
3. **Implement linear-scaling GP methods**: Investigate and implement state-space GP approaches that reduce computational complexity from $O(N^3)$ to $O(N)$. This includes developing CARMA kernel implementations (extending beyond the buggy tinygp library), augmented with flexible mean functions (polynomial, trigonometric, exponential), and potentially incorporating Bayesian change-point methods for piecewise linear mean functions. Address interpretable prior specification issues that have limited practical adoption.
4. **Scale to large survey data**: Benchmark the developed methods on larger samples of lensed quasars from current surveys (DES, LSST precursor data) to ensure efficient scaling to the data volume expected from the Vera Rubin Observatory, which will discover thousands of new lensed systems.
5. **Quantify cosmological impact**: Perform forecasting studies to determine how improved modeling and computational efficiency translate to tighter constraints on the Hubble constant, directly addressing the current tensions between different measurements of cosmic expansion.
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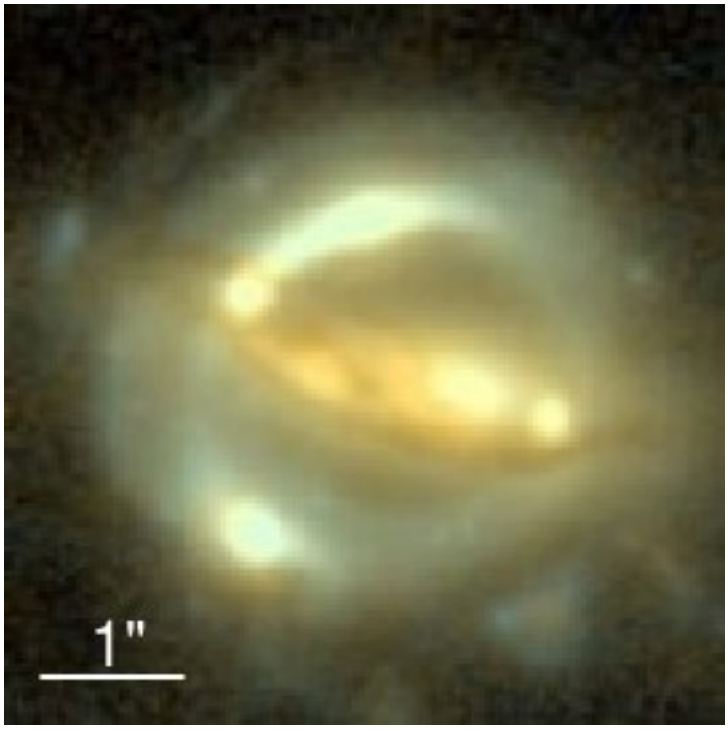

Figure 3: Example of cosmological constraints from gravitational lens time delays, showing the precision achievable with current methods. The enhanced computational approaches developed in this project aim to improve both the precision and reliability of such constraints by enabling robust analysis of much larger samples of lensed quasars (Suyu et al. 2017).

7. **If time permits**: Explore alternative approaches such as Gaussian mixture approximations of complex posterior surfaces (inspired by the LITMUS method for reverberation mapping), or investigate neural network-based approaches enabled by the linear-scaling GP implementation. Alternatively, develop real-time analysis capabilities for rapid parameter updates as new systems are discovered.

Skills Required

- **Programming**: Strong Python programming skills required, with experience in scientific computing libraries (NumPy, SciPy, matplotlib). Willingness to learn JAX for GPU computing during the project.
- **Statistical Knowledge**: Basic understanding of statistics and probability; introduction to Bayesian inference helpful. MCMC methods and advanced techniques will be developed as part of the project.
- **Coursework**: Part II "Principles of Statistics" or Part III "Statistical Theory" recommended; Part II cosmology course beneficial.
- **Physics Background**: Basic understanding of general relativity and cosmology helpful; knowledge of gravitational lensing will be developed during the project.

Group Resources and Support

Students will be supervised by Will Handley throughout the project, meeting one-on-one weekly during term time to discuss progress and plan next steps. Students are expected to attend weekly group meetings and interact with the rest of the Part III, PhD, and postdoc members of the group.

The project will utilize the group's GPU-accelerated computational infrastructure, including BlackJAX nested sampling with 10×-1000× speedups, JAX ecosystem tools, and access to large astronomical datasets. Students will benefit from the group's expertise in Bayesian inference, cosmological analysis, and machine learning applications in astrophysics.

If successful, the project could lead to a publication (with approximately 50% of previous Part III projects leading to publications).

Useful References

- Kroupa, N. et al. "Towards fully Bayesian time delay inference" - *In preparation*. Identifies the pathological likelihood problem and proposes marginalization solutions
- Kelly, B. C., et al. (2014). "Flexible and Scalable Methods for Quantifying Stochastic Variability in the Era of Massive Time-Domain Astronomical Data Sets". arXiv:1402.5978 - CARMA kernels and state-space methods for O(N) GP scaling

- Ashton, G., et al. (2022). “Nested sampling for physical scientists”. arXiv:2205.15570 - Comprehensive review of nested sampling methods
- Hojjati, A., Kim, A. G., & Linder, E. V. (2013). “Robust strong lensing time delay estimation”. arXiv:1304.0309 - Original GP-based time delay inference framework

General References

- Refsdal, S. (1964). “On the possibility of determining Hubble’s parameter and the masses of galaxies from the gravitational lens effect”. *Monthly Notices of the Royal Astronomical Society*, 128, 307. https://academic.oup.com/mnras/article/128/4/307/2601707
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Project 54: Improving the measurement of the masses of supermassive black holes in the first billion years of the Universe

Supervisor I: Professor Richard McMahon (rgm@ast.cam.ac.uk)

Supervisor II: Katherine Kauma (ktk25@cam.ac.uk)

Supervisor III: Member of Gaia or 4MOST teams, to be confirmed

UTO: Professor Richard McMahon (rgm@ast.cam.ac.uk)

Project summary

Quasi-stellar objects or quasars and Active Galactic Nuclei (AGN) are the directly observable manifestation of accreting supermassive ($>10^5$ to 10^{10} Solar mass) black holes that are located in the central regions of massive galaxies, including our own Milky Way galaxy. Due to their intrinsic luminosity, which is more luminous than the starlight in their host galaxies, quasars can be observed over enormous redshifts, distances, and look-back times and hence are a direct observational probe of the Universe at early times. This data analysis project will use multi-epoch optical (0.4-0.8micron) and mid infra-red (2-4micron) wavelength observations from the Zwicky Transit Facility, Gaia satellite (https://www.cosmos.esa.int/web/gaia/) and WISE/NEOWISE (https://science.nasa.gov/mission/neowise/) satellite to measure the flux variation timescales for up to 200,000 quasars with published black hole mass measurements, in order investigate whether these timescales can be used to infer the rest-frame light crossing time of the emission line regions surrounding the central supermassive black hole in order to improve measurement of the masses of the supermassive black holes at redshifts out to 5 and beyond, during the first one billion years of the Universe.

Background

The central engines of quasars or Active Galactic Nuclei (AGNs) are powered by accreting supermassive black holes, and while quasars and AGN are known to play an important role in galaxy evolution, some of the key physical processes occur on scales that are too small to be resolved spatially (aside from a few exceptional cases). Time variable ionizing radiation from close to the black hole drives correlated variability in surrounding gas/dust, but with a time delay due to the light travel time between the regions, allowing temporal mapping of flux variations to effectively replace spatial resolution of the UV/optical accretion disk, the more extensive emission line region and the dusty torus. See Figure 1 for a schematic that highlights the main observational components of the nuclear region of a galaxy hosting a supermassive black hole.

Quasars, powered by accretion onto supermassive black holes (SMBHs), exhibit significant variability, offering insights into the physics of accretion and the properties of the central engines. Recently building

on previous work such as Hook et al., 1994, Benati Goncalves et al, 2025 have analysed the photometric variability and its correlation with key quasar properties, including black hole mass, using a sample 915 quasars with $0 < z < 3.0$. Variability metrics were derived from approximately 6-year light curves provided by the Zwicky Transient Facility (ZTF), while SMBH masses and luminosities were obtained from the SDSS DR16 quasar catalogue of Wu and Shen (2022). Benati Goncalves et al. (2025) confirmed the results of previous work of a strong anti-correlation between variability amplitude and luminosity, which is expected to correlate with black hole mass.

Over the next few years, there will be a transformational explosion in the availability of time-domain observations of quasars, with the release of 5 years of monthly cadence observations from Gaia in late 2026 and the start of a 10-year observational program with the LSST Rubin and multi-epoch X-ray observations from the eRosita satellite. This project is an important and timely step in determining what will be possible with these new observations over the next decade.

Project details

You will analyse the time series photometric data from the Gaia and WISE satellites for a sample of around 700 quasars with redshifts(z) $0.0 < z < 3.0$ that have been recently studied using multi-epoch optical observations with the Zwicky Transient Facility (ZTF) by Benati Goncalves et al. (2025) in order to determine how accurately the Gaia and WISE time series data can be used to characterise the rest frame light crossing time and amplitude of the luminosity variations at widely separated wavelengths that sample different physical regimes. You will then investigate how these temporal variability measurements correlate with other observed and derived properties of these quasars, such as redshift, supermassive black hole (SMBH) mass, luminosity, accretion rate, Eddington luminosity and Eddington ratio. The goal is to determine if the Gaia and WISE data can be used to expand the sample size studied by Benati Gonclaves by a factor of over a 100 and to extend the work to higher redshifts. If time permits, you will then extend the work to a larger sample of up to 200,000 quasars in order to improve the statistical significance of your results or to higher redshifts at $4 < z < 6$

Prerequisite Skills required

Gaussian statistical methods ; Programming in Python

Acceptable Programming Languages

Python

Useful References (List of important papers/review articles relevant to the project)

- First Sagittarius A* Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole in the Center of the Milky Way, Event Horizon Telescope Collaboration, 2022; https://ui.adsabs.harvard.edu/abs/2022ApJ...930L..12E
- Exploring Quasar Variability with ZTF at $0 < z < 3$: A Universal Relation with Eddington Ratio, Benati Goncalves et al, 2025; https://ui.adsabs.harvard.edu/abs/2025arXiv250509779G

- The variability of optically selected quasars, Hook, McMahon et al, 1994 https://ui.adsabs.harvard.edu/abs/1994MNRAS.268..305H

General References (List of papers referred to in the project)

- Reverberation mapping of active galactic nuclei: from X-ray corona to dusty torus, Cackett, Bentz and Kara (2021), https://ui.adsabs.harvard.edu/abs/2021iSci...24j2557C
- Gaia Data Release 3. The first Gaia catalogue of variable AGN, Carnerero, et al, 2023; https://ui.adsabs.harvard.edu/abs/2023A&A...674A..24C
- Virial black hole mass estimates of quasars in the XQ-100 legacy survey, Lai et al, 2023, https://ui.adsabs.harvard.edu/abs/2023MNRAS.526.3230L
- A Catalog of Quasar Properties from Sloan Digital Sky Survey Data Release 16, Wu and Shen, 2022; https://ui.adsabs.harvard.edu/abs/2022ApJS..263...42W
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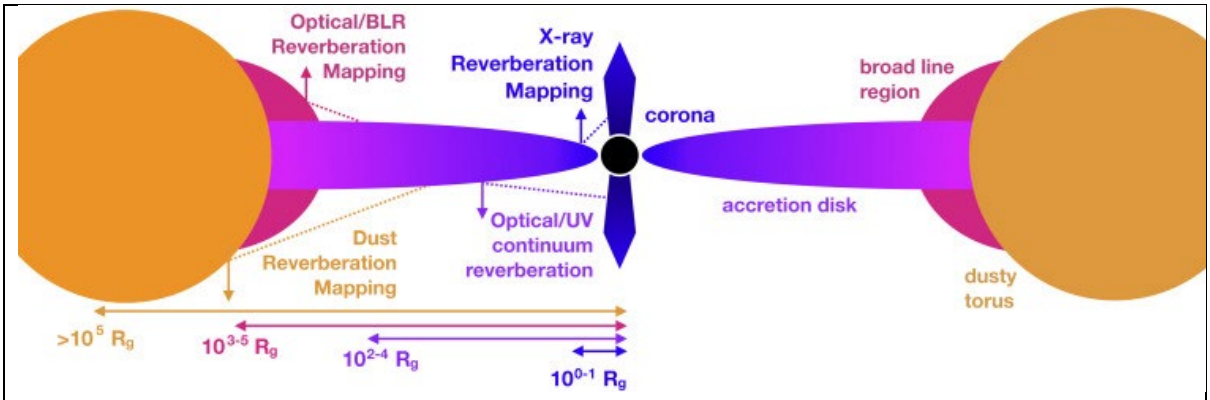

Figure 1: A cross-sectional schematic of an Active Galactic Nucleus (AGN). The schematic highlights the main observational components of the nuclear region of a galaxy hosting a supermassive(>10^5 Solar Masses) black hole (with general radial scales from the black hole indicated by labels. Figure from Cackett, Bentz and Kara (2021) (https://ui.adsabs.harvard.edu/abs/2021iSci...24j2557C)

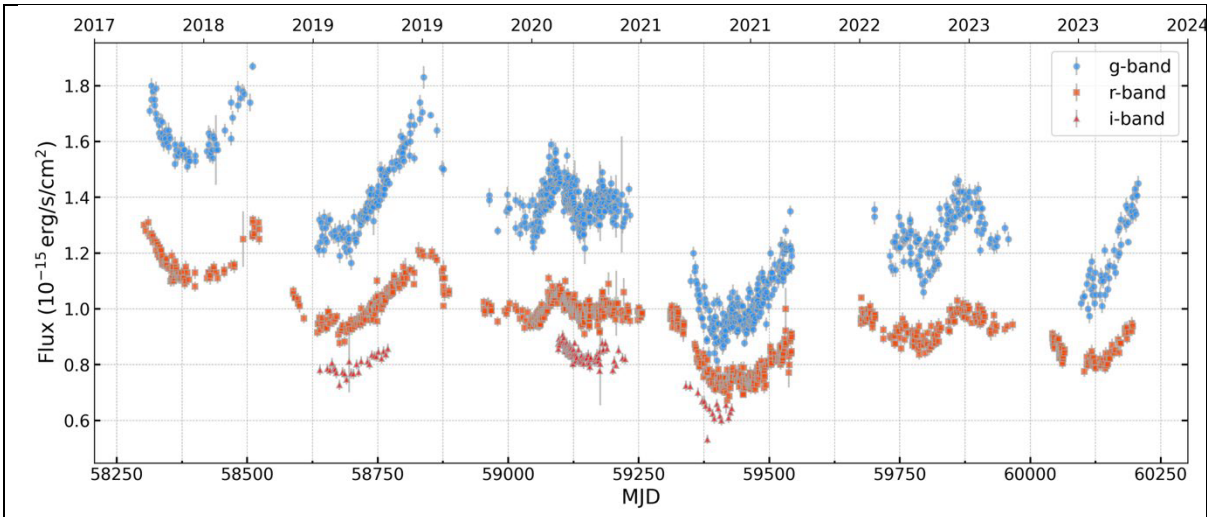

Figure 2 Example Zwicky Transit Facility light curves in the three wavebands (g-, r-, and i-bands) for one of the AGNs (SDSS J225307.4+194235) with a redshift z=0.28

Project 55: Spin–Orbit Alignment of Wide Binaries Across the Milky Way

Supervisor I: Dolev Bashi (db975@cam.ac.uk)

Supervisor II: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

Supervisor III: Professor Cathie J. Clarke (cclarke@ast.cam.ac.uk)

UTO: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

Project summary

Wide binaries (i.e. projected separation $\gtrsim 10^3$ au) preserve their original orbital planes unless perturbed. Comparing each binary's internal orbital-angular-momentum vector ($\boldsymbol{L}_{binary}$) with its Galactic orbital-angular-momentum vector ($\boldsymbol{L}_{gal}$) tell us how binary systems couple to large-scale Galactic rotation and how subsequent dynamical heating, like bar/halo torques, randomises or aligns systems over time.

The aim of this project would be to test whether there is a preferred alignment between **projected** binary poles of wide binaries and the z-component of the Galactic orbital-angular-momentum vector and whether that alignment differs between the thin disc, thick disc, halo, and tidal-stream populations. Alignment will be quantified via a Monte-Carlo forward model that propagates *Gaia* errors and RV incompleteness.

Background

Wide binaries occupy a unique dynamical regime as their binding energies are so low that they can form only in quiescent, low-density gas (Reipurth & Mikkola 2012) yet remain intact for ~Gyr if they avoid strong tidal encounters. Because each system's orbital pole is set at birth and evolves only through weak, cumulative torques, wide binaries have become powerful archaeological tracers of both star-formation physics and Galactic dynamical history.

Simulations show that cores forming inside globally rotating giant-molecular clouds tend to fragment into binaries whose orbital angular-momentum vectors are preferentially prograde with respect to the parent cloud's spin (Bate 2018). If molecular clouds themselves inherit angular momentum from Galactic shear, we expect an excess of binaries whose internal poles align with the Milky Way's external rotation axis. Detecting—or ruling out—such an excess therefore probes how faithfully angular momentum cascades from kiloparsec-scale gas down to $<10^2$ au protostellar discs.

After birth, binaries orbit the Galaxy and experience torques from giant molecular clouds, spiral arms and (in the inner disc) the Galactic bar. In an ideal, perfectly axisymmetric

potential only the z-component of angular momentum, $\mathbf{L}_{gal_z}$, is approximately conserved though, it is important to note that possible Bar/spiral torques can change $\mathbf{L}_{gal_z}$ to some extent, over a few Gyr. Measuring how any primordial spin–orbit alignment decays with stellar age, Galactocentric radius or metallicity constrains the efficiency of these perturbations and even the granularity of the dark-matter halo.

Gaia DR3 provides six-dimensional phase-space data for $\gtrsim 10^5$ high-purity wide binaries (El-Badry & Rix 2021) and radial-velocity measurements for >30% of their primaries (Katz et al. 2023). This unprecedented sample finally allows a statistically robust comparison between each binary's **projected** internal pole l_{binary} and the system's Galactocentric $\mathbf{L}_{gal_z}$, across the thin disc, thick disc, halo and known tidal streams. Any detected alignment—or its absence—will furnish direct evidence on whether wide binaries still `remember' the spin of their natal clouds or have been re-oriented by subsequent Galactic dynamical processes.

Project details

Starting with the Gaia eDR3 wide-binary catalogue of El-Badry & Rix (2021), we take for every pair the sky-projected separation vector:

$$\boldsymbol{r} = [\Delta\alpha, \Delta\delta]$$

and the relative proper-motion vector

$$\boldsymbol{v} = [\Delta\mu_{\alpha^*}, \Delta\mu_{\delta}]$$

where Gaia already supplies $\mu_{\alpha^*} \equiv \mu_{\alpha} \cos\delta$.

The associated **projected** orbital angular momentum is

$$l_{binary} \equiv \boldsymbol{r} \times \boldsymbol{v}$$

Cross-matching the binary's primary (and, when available, the secondary) with the Gaia DR3 RVS catalogue (or any other literature available RV catalogues) yields full 3-D space velocities. Combined with *Gaia* observables, we can then convert each star's six-vector into a Galactocentric Cartesian position $\boldsymbol{R} = [X, Y, Z]$ and velocity $\boldsymbol{V} = [U, V, W]$ from which we compute the system's **Galactic angular-momentum vector** L_{gal}.

$$\mathrm{L}_{gal} \equiv \boldsymbol{R} \times \boldsymbol{V}$$

To measure spin–orbit coupling across different stellar populations, we define an alignment metric

$$\cos\theta = \frac{\boldsymbol{l}_{binary} \cdot L_{gal}\boldsymbol{z}}{\left\| l_{binary} \right\| \left\| \mathbf{L}_{gal_z} \right\|}$$

where $\mathbf{L}_{gal_z}$ is the Galactic angular-momentum z-component.

Throughout we compare the projected binary pole $\boldsymbol{l}_{binary}$ to the Galactic $\mathbf{L}_{gal_z}$, because only $\mathbf{L}_{gal_z}$ is (approximately) conserved in the Milky-Way potential.

Any alignment we detect is therefore a lower limit on the primordial coupling.

so that $\theta \approx 0^{o}$ indicates prograde alignment and \ $\theta \approx 90^{o}$ implies orthogonality.

Examining how θ varies with binary separation, stellar age, metallicity, and Galactic location will reveal whether (and where) wide binaries retain memory of their birth angular momentum or are re-oriented by subsequent dynamical processes.

The observed sample provides a noisy estimate of $\cos\theta$ for each binary, because both $\boldsymbol{l}_{binary}$ and $\mathbf{L}_{gal_z}$inherit astrometric errors, including uncertainty in parallax and RV. To compare with theory, we will perform a few simulations, including:

1. Forward-model candidate parent distributions—e.g. isotropic, 10 % prograde excess, bar-shuffled—into predicted $\cos\theta$ histograms using Monte-Carlo realisations of the *Gaia* error ellipses and an RV-selection mask that mirrors the RVS completeness.

2. Fit the data with a hierarchical Bayesian model that marginalises over each binary's distance & RV posterior, yielding posterior constraints on the true spin–orbit distributions.

3. Assess robustness by repeating the forward-modelling with different Galactic-potential prescriptions (axisymmetric vs barred) to bracket the amount by which $\mathbf{L}_{gal_z}$ may have drifted since birth.

Prerequisite Skills required
Fundamental Keplerian motion, binary stars, Galactic-dynamics concepts. Statistical analysis. Ability to query and manipulate *Gaia* DR3 data

Other skills used in the project
Using TOPCAT can be efficient, but not mandatory

Acceptable Programming Languages
Python, C or similar

Useful References (List of important papers/review articles relevant to the project)

- The eccentricity distribution of wide binaries and their individual measurements

  https://ui.adsabs.harvard.edu/abs/2022MNRAS.512.3383H/abstract

- A million binaries from Gaia eDR3: sample selection and validation of Gaia parallax uncertainties El-Badry, Kareem; Rix, Hans-Walter ; Heintz, Tyler M. 2021

  https://ui.adsabs.harvard.edu/abs/2021MNRAS.506.2269E/abstract

- Gaia Data Release 3. Properties and validation of the radial velocities

  https://ui.adsabs.harvard.edu/abs/2023A%26A...674A...5K/abstract

General References (List of papers referred to in the project)

- galpy: A Python Library for Galactic Dynamics

  https://ui.adsabs.harvard.edu/abs/2015ApJS..216...29B/abstract

- Formation of the widest binary stars from dynamical unfolding of triple systems

  https://ui.adsabs.harvard.edu/abs/2012Natur.492..221R/abstract

- On the diversity and statistical properties of protostellar discs

  https://ui.adsabs.harvard.edu/abs/2018MNRAS.475.5618B/abstract

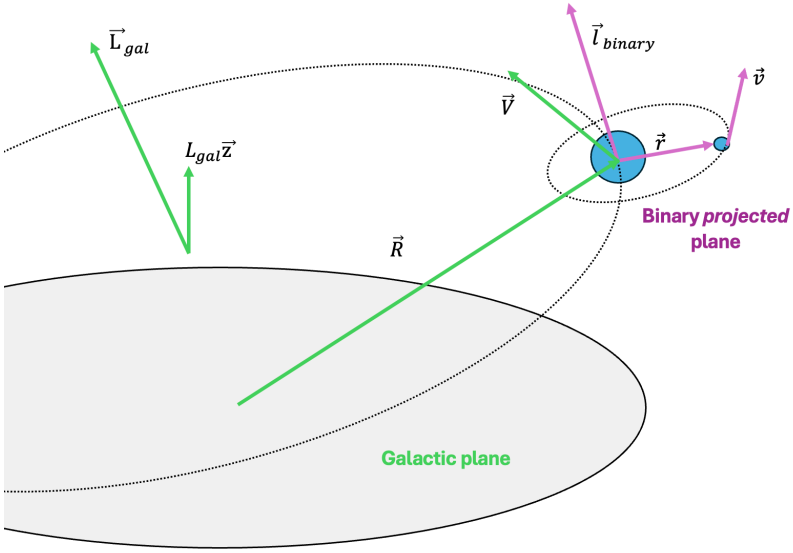

Figure 1: Geometry used to define the spin-orbit alignment angle θ for a wide binary. The binary's projected orbital plane is drawn as the small dotted ellipse; its relative-separation vector $\boldsymbol{r}$ and relative proper-motion vector $\boldsymbol{v}$ (both magenta) yield the projected internal angular-momentum vector $\boldsymbol{l}_{binary}$. The large dotted curve shows the system's trajectory around the Milky Way. From the binary's Galactocentric position $\boldsymbol{R}$ and centre-of-mass velocity $\boldsymbol{V}$ (green) we compute the Galactic angular momentum L_{gal}; only its nearly conserved z-component, $\mathbf{L}_{gal_z}$, is indicated. The grey disk marks the Galactic plane; Vector lengths are not to scale, and all quantities are shown in projection for clarity.

Project 56 Kinematic analysis of halo-WD origin: accreted versus in situ

Supervisor I: Dolev Bashi (db975@cam.ac.uk)

Supervisor II: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

Supervisor III: Na'ama Hallakoun (naama.hallakoun@weizmann.ac.il)

UTO: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

Project summary

Dating the Milky Way's oldest building blocks is notoriously hard; classical methods lose resolution beyond ~10 Gyr. We propose an alternative chronometer: the cooling ages of halo white dwarfs (WD). This project will deliver a population-level breakdown of in-situ versus accreted halo WDs by combining Gaia DR3 kinematics with WD catalogues. Estimating the cooling-age distributions of both components will provide a clean, homogeneous timeline for the Galaxy's major merger and star-formation events. In parallel, we will cross-validate kinematic membership fractions with totally independent WD tracers such as mass, spectral type and cooling age, offering a stringent internal consistency check on current halo-formation models.

Background

Dating the oldest Galactic stellar populations is inherently hard. Beyond $\simeq$10 Gyr the main-sequence (MS) turn-off becomes almost vertical in the Hertzsprung-Russell (HR) diagram, so small photometric errors translate into large age spreads (a few Gyrs), while grid-based asteroseismic fits show rapidly growing formal uncertainties and model degeneracies (Moser 2023). Even the most precise sub-giant or globular-cluster MS turn-off studies of local halo stars settle for 11–13 Gyr with ±1 Gyr error bars.

WDs offer a complementary clock. After nuclear burning ceases, a WD's luminosity decline is governed almost entirely by well-modelled cooling physics, so a pair of observable parameters—effective temperature $T_{\rm eff}$ and surface gravity $\log g$ —maps onto a cooling age almost one-to-one. This method of ` Cosmochronology' can, in principle, time star-formation bursts to ≤0.5 Gyr even at ages $\gtrsim$12 Gyr (Fontaine et al. 2001). Atmospheric classes (DA, DB), core composition, and the initial–final mass relation further tie WD ages to those of their progenitor populations.

The Milky Way's stellar halo contains both stars born in the Galaxy (in-situ) and stars deposited by merged dwarf galaxies (accreted). Chemical signatures and full 6-D kinematics have shown that major accretion events such as Gaia-Sausage-Enceladus (GSE) dominate the radially biased halo, whereas in-situ stars retain modest prograde rotation.

Gaia DR3 delivers precision astrometry for >1 billion sources, including tens of thousands of high-latitude WDs. The Gentile-Fusillo et al. (2021) EDR3 catalogue provides probabilistic WD identification

down to G ≃ 21, and the DESI Early Data Release and SDSS-V add ~15K spectroscopically confirmed WDs with radial velocities (RVs) and atmospheric parameters (e.g. Figure 1). Where RVs are still missing, a proposed population-level probabilistic algorithm (see next section) or others (e.g., Callingham & Helmi 2024) infer halo membership directly from 5-D phase space, recovering IoM distributions with high fidelity. Together, these resources make a population-level, kinematically tagged WD chronometer of the halo not just feasible but timely.

Project details

We propose a two-step project depending on the student's progress and the results we find:

1. A 6-D approach:

Compile a sample of WDs with 6-D (3-D positions; 3-D velocities) to affiliate WDs in the halo.

- Building the sample:
    - Query Gaia DR3 for $\frac{\sigma_\pi}{\pi} < 0.2$ and RUWE < 1.4; intersect with the Gentile-Fusillo WD list (PWD > 0.75) and attach RVs from DESI and SDSS, giving a complete-phase-space WDs. Compute energies (E) and angular momenta (L_z) in a MW-like potential. Estimate Accreted vs. in-situ fractions (e.g., Belokurov & Kravtsov 2023.
    - Compile broadband photometry (Gaia G, BP, RP + GALEX FUV/NUV for hot DAs + WISE W1 for cool DB/DCs). Fit $T_{\rm eff}$, $\log g$ and atmospheric type using some Python package like 'WDPhotTools' (or other available catalogue values).
    - Convert to cooling ages using `wdwarfdate'.

- Data exploration:
    - Assess halo components as a function of WD mass.
    - Perform differential cooling-age distributions—look for a narrow, older peak for GSE debris and a broader, younger tail for in-situ stars.
    - Constraints on the timing and duration of the last major halo merger.

* Optional: If time allows, the project may be extended to include a sample of WD-MS systems

2. A 5-D approach:

In this approach, we assume that no RV is available (i.e., 3-D positions and 2-D projected velocities). To affiliate the WDs in the halo, we propose a population-level approach using a control sample of MS halo stars based on the stellar tangential velocities.

- Building the sample:

- The Initial MS halo selection should be done by querying Gaia DR3 for sources with reliable astrometry (for $\frac{\sigma_\pi}{\pi} < 0.2$ and RUWE < 1.4) and tangential velocity ($V_t = 4.74\ \mu/\pi$ > 180 km/s to suppress disc contamination.
    - Use an HR diagram with a statistical/ ML approach to isolate two sequences that correspond to the 'blue metal-poor' and 'red metal-rich' halo tracks seen in MS stars. Validate the split with metallicity where available (LAMOST, APOGEE) and with orbital parameters using RV information (E, L_z) from Gaia RVS.
    - Fit the V_t distribution of each HR-defined group with a 2-component model (e.g. Gamma distribution) to derive a probabilistic membership function P(in-situ|V_t), P(accreted|V_t).
    - Cross-match the halo parent sample with a WD catalogue e.g. Gentile Fusillo et al. 2021. Retain WDs with secure classification probability and good photometry. Use Bayesian approach to solve for the global fractions of in-situ and accreted halos.

The remaining steps are identical to those in the 6D approach.

Prerequisite Skills required

- Galactic-dynamics
- Ability to query and manipulate *Gaia* DR3 data.
- Statistical analysis

Other skills used in the project

Using TOPCAT can be efficient, but not mandatory

Acceptable Programming Languages

Python, C or similar

Useful References (List of important papers/review articles relevant to the project)

- Uncertainties in asteroseismic grid-based estimates of the ages of halo stars https://ui.adsabs.harvard.edu/abs/2023A%26A...671A..78M/abstract
- The Potential of White Dwarf Cosmochronology https://ui.adsabs.harvard.edu/abs/2001PASP..113..409F/abstract
- A catalogue of white dwarfs in Gaia EDR3 https://ui.adsabs.harvard.edu/abs/2021MNRAS.508.3877G/abstract

- A magnitude-limited catalogue of unresolved white dwarf-main sequence binaries from Gaia DR3
  https://ui.adsabs.harvard.edu/abs/2025arXiv250515895R/abstract
- Co-formation of the disc and the stellar halo
  https://ui.adsabs.harvard.edu/abs/2018MNRAS.478..611B/abstract
- Nitrogen enrichment and clustered star formation at the dawn of the Galaxy
  https://ui.adsabs.harvard.edu/abs/2023MNRAS.525.4456B/abstract
- Filling in the blanks: A method to infer the substructure membership and dynamics of 5D stars
  https://ui.adsabs.harvard.edu/abs/2024A%26A...690A..98C/abstract
- Gaia Data Release 2. Observational Hertzsprung-Russell diagram
  https://ui.adsabs.harvard.edu/abs/2018A%26A...616A..10G/abstract

General References (List of papers referred to in the project)

- DESI Data Release 1: Stellar Catalogue
  https://ui.adsabs.harvard.edu/abs/2025arXiv250514787K/abstract
- galpy: A Python Library for Galactic Dynamics
  https://ui.adsabs.harvard.edu/abs/2015ApJS..216...29B/abstract
- The DESI Early Data Release white dwarf catalogue
  https://ui.adsabs.harvard.edu/abs/2024MNRAS.535..254M/abstract

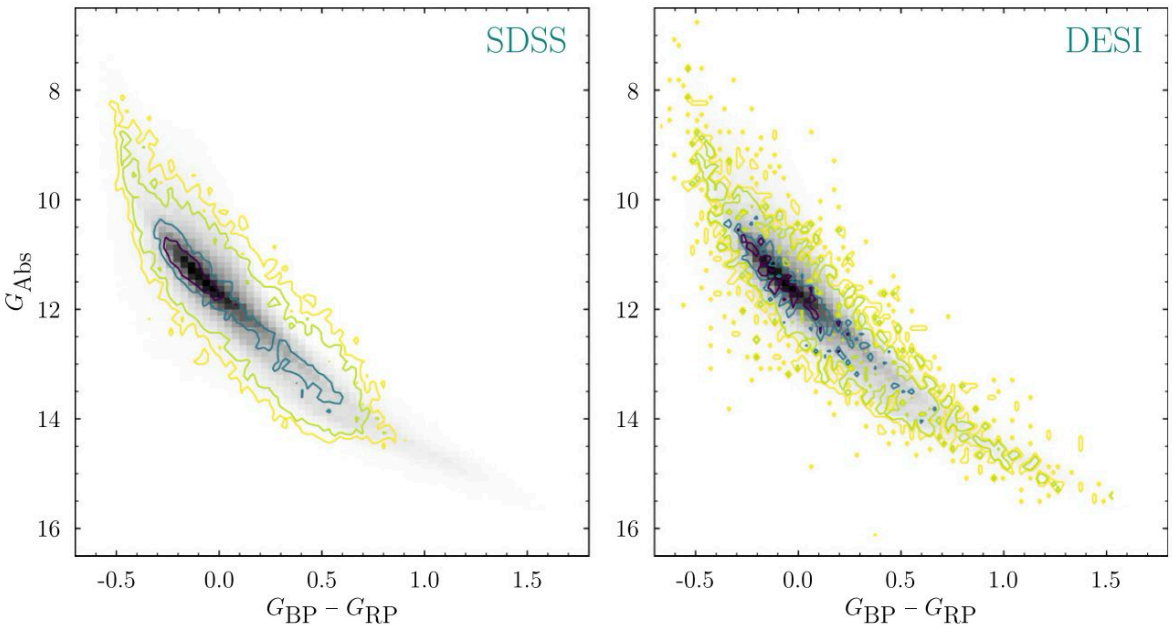

Figure 1: The distributions of the spectroscopically confirmed sample of Gaia–SDSS white dwarfs of GF21 (left) and the sample of DESI EDR white dwarfs (Figure from Manser et al. 2024).

Project 57: Super-Chandrasekhar White Dwarfs

Supervisor I: Professor Christopher Tout (cat@ast.cam.ac.uk)

UTO: Professor Christopher Tout (cat@ast.cam.ac.uk)

Project summary
The aim is to systematically model super-Chandrasekhar white dwarfs that are either rapidly rotating or have significant internal magnetic fields or both.

Background
Type Ia supernovae are thought to be exploding carbon-oxygen white dwarfs. A cold carbon-oxygen white dwarf accreting material degenerately ignites carbon in its core at about 1.38 solar masses. The conversion of about 1 solar mass of this material to mostly nickel-56 in nuclear statistical equilibrium that is expelled into the interstellar powers the supernova at a luminosity similar to an entire galaxy for a few moths. In practice SNeIa have a range of peak luminosities and a range of light curve shapes. A relation between the two allows them to be used as standard candles and their brightness makes them useful to map the expansion of the Universe. Much of the dispersion in luminosity can be explained by composition changes in the white dwarf but there are some examples of both brighter and fainter SNeIa that require ignition at a higher or lower mass. In this project we are concerned with the most luminous and hence most massive white dwarfs. A cold non-rotating white dwarf is limited by the Chandrasekhar mass, of about 1.44 solar masses, at which electrons reach become extremely relativistic. Two global structural changes can feasibly allow the white dwarf to exceed this mass and consequently ignite carbon at a higher mass. The first is rotation (Yoon & Langer 2004) which can reduce the gravitational force by adding a centrifugal component and the second is a strong magnetic field that can add a magnetic pressure (Gupta, Mukhopadhyay & Tout 2010) to the electron degeneracy pressure supporting the star against collapse.

Project details
You will use the Cambridge stellar evolution code STARS to model accreting white dwarfs. The first step will be to create a CO white dwarf of around a solar mass by evolving an intermediate mass star through red giant, core helium burning and asymptotic giant branch evolution with appropriate mass loss. You will then accrete material at various rates to confirm carbon ignition at 1.38 solar masses when the accretion rate is slow enough and the white dwarf remains effectively cold. You will then adapt the structure equations to incorporate a centrifugal component to the gravity with the white dwarf rotating as a solid body. It should be possible to delay the carbon ignition to reach higher mass before explosion. The code can be modified to add the centrifugal force to a differentially rotating white dwarf and optimize the mass that can be reached by varying the distribution of angular momentum. If time

allows you can also quantify the effect of adding a magnetic pressure for various magnetic field distributions to both non-rotating and rotating white dwarfs.

Prerequisite Skills required
Some experience with programming using a high-level language. The STARS code is written in FORTRAN 77 so you will need to become familiar with that language but it is straightforward to use by someone with an understanding of any language. The part III course on Structure and Evolution of Stars is useful but not essential.

Other skills used in the project
Ability to produce good quality instructive plots.

Acceptable Programming Languages
Any but FORTRAN 77 will be necessary to modify the code.

Useful References (List of important papers/review articles relevant to the project)
Martin R. G., Tout C. A., Lesaffre P., 2006, MNRAS, 373, 263

General References (List of papers referred to in the project)
Gupta A., Mukhopadhyay B., Tout C. A., 2020, MNRAS 496, 894

Yoon S.-C., Langer N., 2004, A&A 419, 623

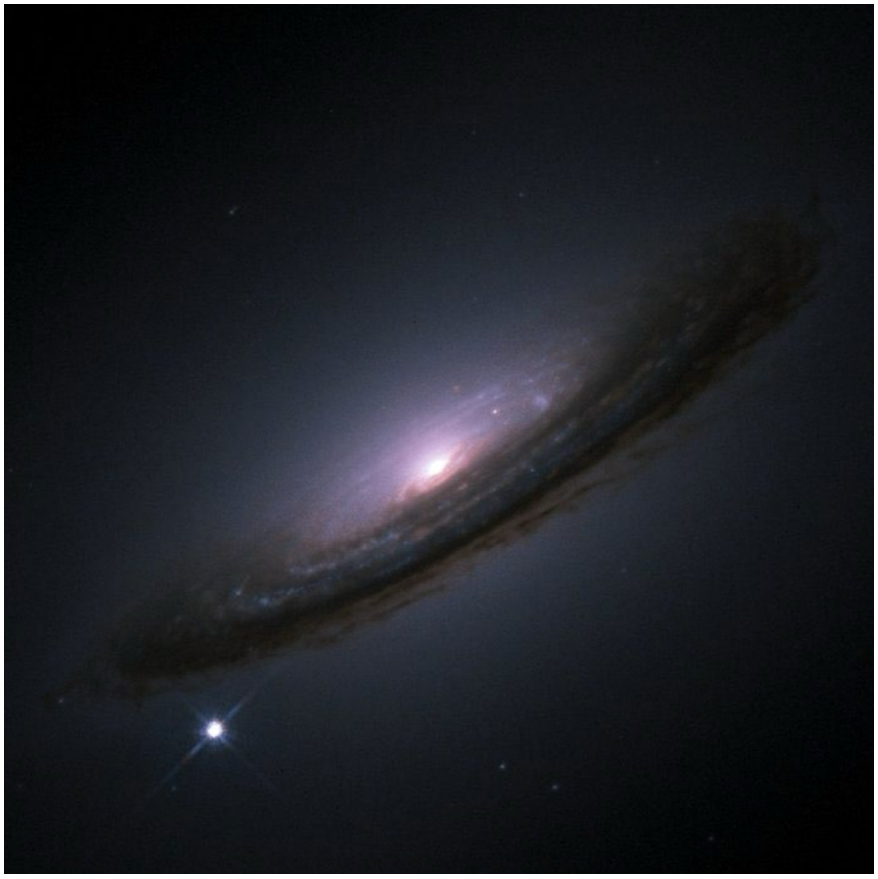

Figure 1: SN 1994D, a type Ia supernova in front of its host galaxy NGC 4526

Project 58: Illumination of the Debris around ZZ Piscium

Supervisor I: Professor Christopher Tout (cat@ast.cam.ac.uk)

Supervisor II: Professor James Pringle (jep@ast.cam.ac.uk)

UTO: Professor Christopher Tout (cat@ast.cam.ac.uk)

Project summary

In this project you will examine how the light from a white dwarf is absorbed and re-emitted by debris in orbit around the star. By modelling various possibilities, including simple circular rings and discs and elliptical and extended structures, you will investigate what can be learnt about the how the debris orbits the star from the combined light, its spectrum and variability, of the whole system.

Background

In his review article Farihi (2016) states, "Circumstellar disks of planetary debris are now known or suspected to closely orbit hundreds of white dwarf stars". Over 36 years have passed since the discovery of the infrared excess from circumstellar dust orbiting the white dwarf ZZ Piscium (G29-38) and, at distance of d = 17.5pc, it remains the nearest and the brightest of its class. ZZ Psc stands out as having the largest infra-red excess. Around 3 to 4 per cent of the stellar flux is re-radiated at IR wavelengths (see figure 1 of Reach et al. 2005). Reach et al. (2009) and Ballering et al. (2022) modelled the spectra of the dust's contribution but relied, in line with the common assumption exemplified by Farihi's statement above, on the simplifying hypothesis that the dust lies in a circular disc. Even then the size and the nature of the disc remains uncertain (see figure 6 of Reach et al. 2009).

The dust that gives rise to the IR excess is thought to originate from the tidal disruption of a planetesimal or asteroid which strayed too close to the white dwarf. Such disrupted material would first follow a highly eccentric orbit which then, it is supposed, circularises and shrinks closer to the white dwarf (e.g. Debes et al. 2012; Veras et al. 2014). Although the precise morphology of the orbiting dust might help answer fundamental questions of its dynamical origins and evolution, it remains only marginally constrained by existing data.

However ZZ Psc is also unusual in that it shows strong pulsations both at optical wavelengths, from the stellar surface, and in the IR, presumably caused by time-dependent heating of the dust by pulsations from the central star (see figure 5 of von Hippel et al. 2024). This gives us another means of estimating the morphology of the circumstellar dust.

Project details

The main aim of the project is to ask what can in principle be learned about the morphology of the circumstellar dust by examining the oscillations seen at optical and at iR wavelengths. Do the observations agree with the usual assumption of a simple circular disc? What observations are needed to test different possible dust morphologies? It is anticipated that this initial study can be the basis for future more detailed work and would lead to a future observing proposal to be submitted to JWST.

The oscillation modes are expected to manifest themselves as time-dependent variations in temperature on the white dwarf surface, with spatial distributions in the form of spherical harmonics $Y_m^l(\theta,\varphi)$ where θ and φ are spherical polar coordinates. The first aim of the project is to compute the apparent brightness of the star as a function of time as seen from an arbitrary point in space. Once this has been achieved, the main task will be to use this information to compute the integrated response of the dust emission at infrared wavelengths for various different possible dust morphologies. There will be two main lines of enquiry. First are there particular combinations of dust morphologies and stellar modes which are expected to give rise to particularly strong or particularly weak amplitudes in the IR responses? Secondly to what extent does the dust morphology give rise to modification of the underlying stellar oscillation either in terms of oscillation shape or in terms of time-delay?

Skills required

A desire to develop theoretical models with a view to future possible observations.

Ability to write numerical code (FORTRAN, Python, etc.).

The Planetary Systems Dynamics and Exoplanets Part III courses is useful but optional.

Useful References (List of important papers/review articles relevant to the project)

- General background to dust around white dwarfs is reviewed by Farihi (2016, New Astronomy Reviews, 71, 9). Investigation of optical and IR oscillations, and simple conclusions, is given by von Hippel et al. (2024, ApJ 963, 113).

General References (List of other papers referred to in the project)

- Ballering N. P. et al. 2022, ApJ, 939, 108
- Debes J. H., Walsh K. J., Stark C. 2012, ApJ, 747, 148
- Reach W. T. et al. 2005, ApJ, 635, L164
- Reach W. T. et al. 2009 ApJ, 693, 697
- Veras D., Leinhardt Z. M., Bonsor A., Gänsicke B. T. 2014, MNRAS, 445, 2244

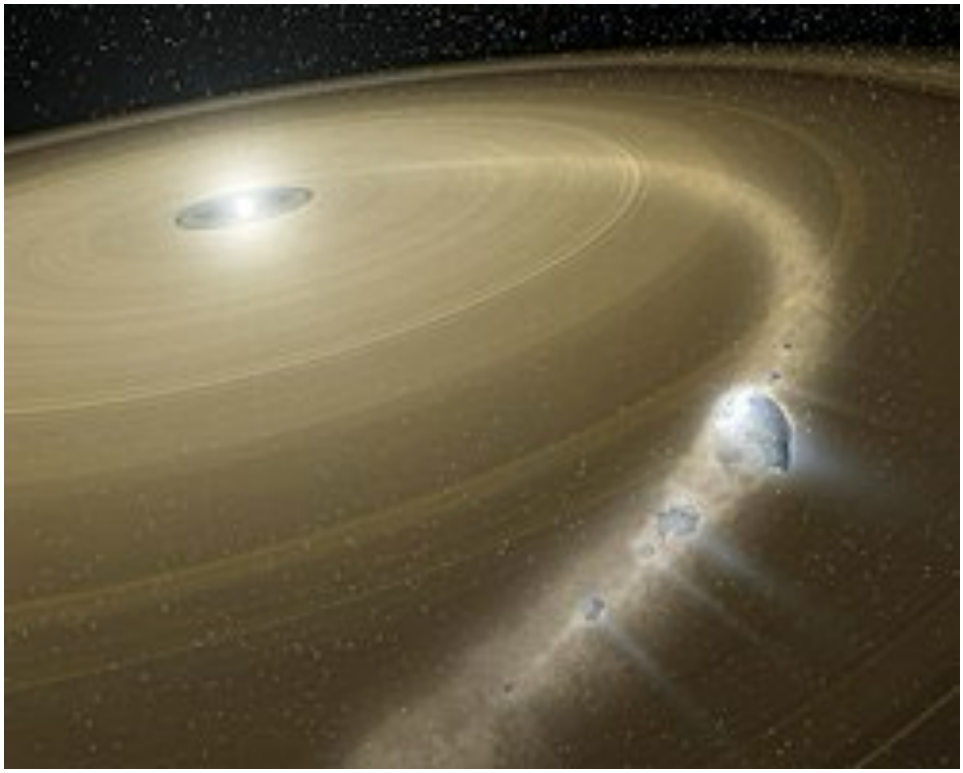

Figure 1: An artist's impression of the debris disc around the variable white dwarf ZZ Psc.

Project 59: Calling Time on Cosmic Chronometers

Supervisor I: Dr Will Handley (wh260@cam.ac.uk)
Supervisor II: Professor Hiranya Peiris (hiranya.peiris@ast.cam.ac.uk)
UTO: Dr Will Handley (wh260@cam.ac.uk)

Project Summary
Cosmic chronometers provide a model-independent approach to measuring the expansion history of the Universe through differential aging of passively evolving galaxies at different redshifts. This technique has achieved systematic uncertainties of 2-6% and produced 32 independent H(z) measurements (Moresco 2024), though questions remain about the astrophysical robustness of age determinations in real galaxy populations. This project will use Pop-Cosmos mock catalogues (Alsing et al. 2024, Thorp et al. 2025) to systematically test cosmic chronometer assumptions, examining how galaxy evolution, stellar population synthesis uncertainties, and observational selection effects impact age-redshift relations. The student will validate cosmic chronometer methodology against known simulation truth, investigating systematic biases that may affect cosmological parameter measurements.

Background
The cosmic chronometer method, pioneered by Jimenez & Loeb (2002), measures the Hubble parameter H(z) directly from the differential aging of passively evolving galaxies: $H(z) = -(1+z)^{-1} \cdot dz/dt$. Unlike distance-based cosmological probes, this approach is cosmology-independent, requiring no assumptions about the distance-redshift relation or specific cosmological models (Moresco et al. 2016).

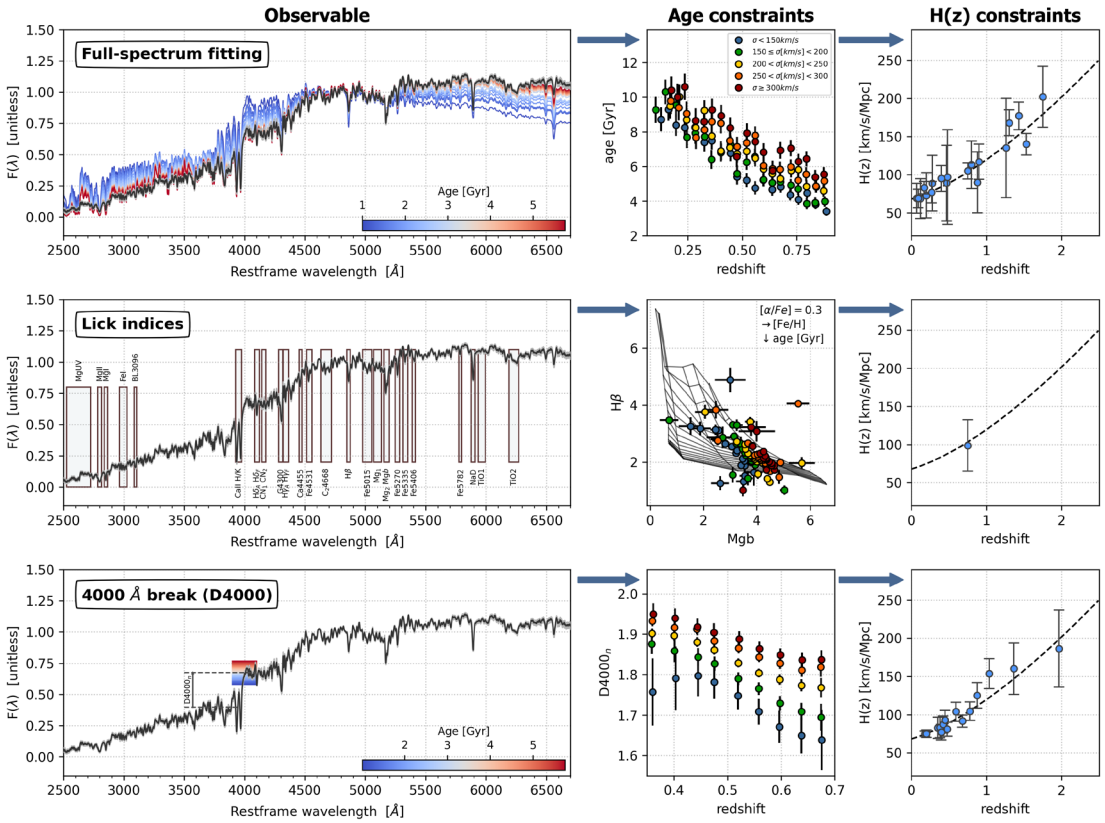

Figure 1: Overview of the three main cosmic chronometer methods from Moresco (2024). Left panels show typical galaxy spectra with key age-sensitive features. Middle panels display derived age constraints as a function of redshift or stellar mass. Right panels present the resulting H(z) measurements. From top to bottom: full-spectrum fitting, Lick indices, and 4000 Å break methods.

Cosmic chronometers have been developed as a cosmological tool with claimed systematic uncertainties of 2-6% through improved stellar population synthesis models (Moresco et al. 2020). Machine learning techniques have enabled photometric age estimation from samples of ~19,000 galaxies, representing a substantial increase over traditional spectroscopic approaches (Jimenez et al. 2023). These measurements span $0 < z < 2$ and yield constraints on dark energy properties, with reported values of $w = -0.989 \pm 0.049$ and $H_0 = 69.85 \pm 1.27$ km/s/Mpc (Jimenez et al. 2023).

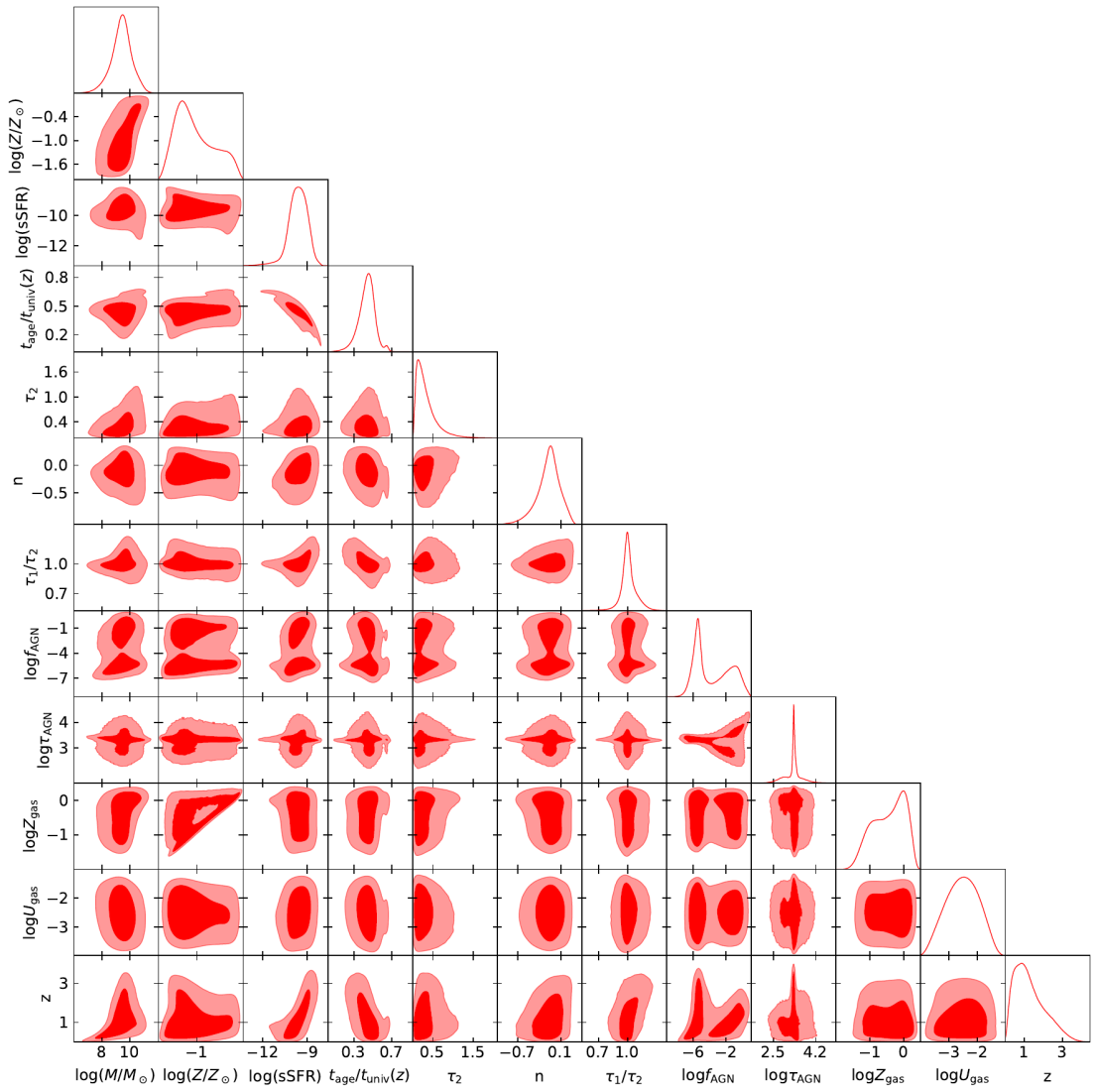

Figure 2: Parameter correlations in the Pop-Cosmos mock catalogue from Alsing et al. (2024). This triangle plot shows the distribution and correlations of key galaxy properties including the mass-weighted age parameter t_age/t_univ(z) (fourth row/column) that cosmic chronometers directly measure. The correlations with stellar mass, specific star formation rate, and redshift reveal potential systematic biases that could affect cosmic chronometer measurements, making Pop-Cosmos ideal for testing method robustness.

However, the claimed precision of cosmic chronometer measurements depends critically on understanding the astrophysical processes that govern galaxy aging. While current studies focus heavily on statistical methodology and systematic uncertainties from stellar population models, the underlying assumption that massive early-type galaxies evolve passively may not hold universally. Selection effects, contamination from younger stellar populations, and variations in star formation histories could introduce systematic biases that are difficult to quantify from observations alone. These challenges have been highlighted in studies of passive galaxy evolution and stellar population modeling (Conroy 2013).

The Pop-Cosmos framework (Alsing et al. 2024,Thorp et al. 2025) provides an opportunity to test cosmic chronometer assumptions against known truth. Pop-cosmos is a generative model of the galaxy population evolving over cosmic time, which has been calibrated to the 26-band photometry of ~420,000 galaxies in the COSMOS2020 catalog with deep selection of Spitzer IRAC Ch. 1 < 26. This generative model, encapsulated within an advanced machine learning model known as a score-based diffusion model, encodes realistic scaling relations governing galaxy formation physics within a sophisticated stellar population synthesis parameterization. Observational selection effects have already been inverted during the calibration of the model, enabling controlled experiments that are difficult to perform with real data. By comparing cosmic chronometer results from "observed" Pop-Cosmos galaxies with their known formation histories, we can assess the method's astrophysical robustness and identify potential systematic biases.

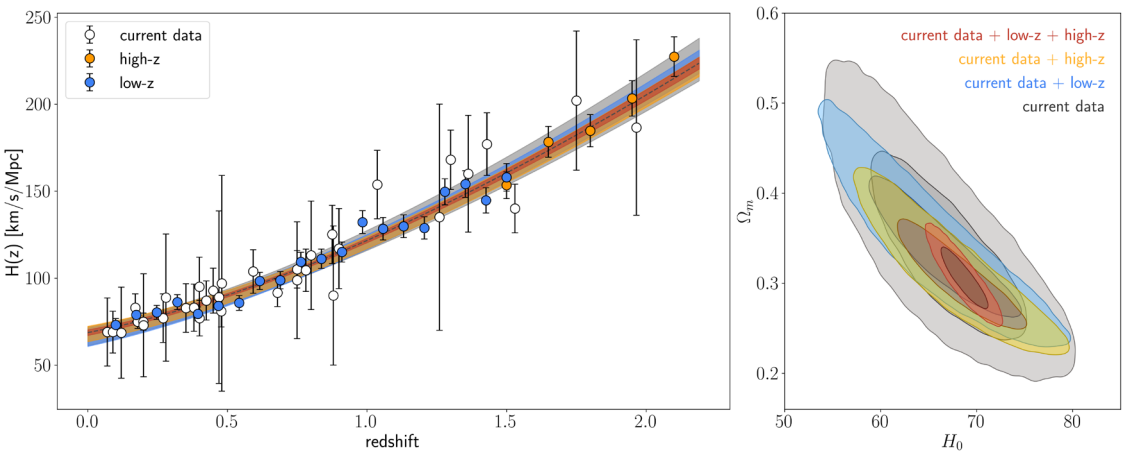

Figure 3: Cosmological constraints from cosmic chronometer measurements. Left: H(z) measurements across redshift showing current data (gray), low-z (blue), and high-z (orange) samples with ΛCDM model fits. Right: Parameter constraints in the Ω_m-H_0 plane demonstrating how different data combinations constrain fundamental cosmological parameters. From Moresco (2024).

Project Details

1. **Validate cosmic chronometer methodology using Pop-Cosmos mock catalogues**: Extract large samples of massive passive galaxies from Pop-Cosmos simulations. Isolate massive passively-evolving samples using cosmic chronometer selection criteria in the literature. Quantify and compare the actual scatter in the age-redshift distribution of the selected population, compared to assumptions or claims in the literature. Test whether methodological assumptions (such as simple Gaussian likelihoods for age uncertainties) are justified when compared to the known ground truth from Pop-Cosmos.

2. **Investigate astrophysical systematics in age-redshift relations**: Systematically explore how variations in star formation histories, metallicity evolution, and stellar population synthesis model assumptions affect age determinations. Identify which astrophysical properties most strongly correlate with age-dating errors.

3. **Quantify selection effect impacts on cosmic chronometer samples**: Analyze how observational selection criteria (e.g. colour selection vs specific star formation rate selection)

bias the resulting cosmic chronometer measurements. Develop improved selection strategies to minimize systematic errors and develop cleaner catalogues for cosmic chronometer studies.

4. **Construct a clean cosmic chronometer sample from COSMOS2020:** Inference has been performed under the pop-cosmos model for all COSMOS2020 galaxies. Using the selection criteria developed in previous steps, isolate a cosmic chronometer sample from this catalogue and derive H(z) measurements. Compare the inferences with the literature. Push to cosmological inferences if time allows.
5. If time permits: **Develop improved cosmic chronometer methodology**: Based on identified systematic effects, propose modifications to the cosmic chronometer approach that may reduce astrophysical uncertainties. Test the impact of these updated methods on the H(z) inferences from the COSMOS2020 catalogue.
6.

Skills Required

- **Programming**: Good Python skills are required for galaxy catalog analysis and stellar population modeling. Familiarity with or willingness to learn to use modern machine learning and Bayesian inference tools.
- **Background**: Part II Astrophysics or equivalent background in stellar evolution and galaxy formation is essential
- **Statistical analysis**: Experience with statistical data analysis and uncertainty quantification is beneficial
- **Courses**: Part II course ‘Stellar Dynamics and Structure of Galaxies’ is recommended

Group Resources and Support

Students will be supervised by Will Handley and Hiranya Peiris throughout the project, meeting one-on-one weekly with Will and as necessary with Hiranya during term time to discuss progress and plan next steps. Students are expected to attend weekly group meetings and interact with the rest of the Part III, PhD, and postdoc members of the group.

The project will be supported by the group’s computational infrastructure, allowing the student to analyze large, simulated datasets and focus on the key astrophysical questions. Students will benefit from the group’s expertise in galaxy evolution modeling, stellar population synthesis, and cosmological analysis. The project builds on the group’s existing Pop-Cosmos infrastructure and analysis pipelines.

If successful, the project could lead to a publication (with approximately 50% of previous Part III projects leading to publications).

Key References

- Alsing et al. (2024), “pop-cosmos: A comprehensive picture of the galaxy population from COSMOS data”, ApJS, 274, 12
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- Vagnozzi et al. (2020), “Eppur è piatto? The cosmic chronometer take on spatial curvature and cosmic concordance”, PhRvD, 102, 023518
- https://cosmo-pop.github.io

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- Jiao, K. et al. (2022), “New Observational H(z) Data from Full-Spectrum Fitting of Cosmic Chronometers in the LEGA-C Survey”, ApJS, 265, 48
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- Moresco, M. et al. (2020), “Setting the Stage for Cosmic Chronometers. II. Impact of Stellar Population Synthesis Models Systematics and Full Covariance Matrix”, ApJ, 898, 82
- Moresco, M. (2024), “Measuring the expansion history of the Universe with cosmic chronometers”, arXiv:2412.01994
- Di Valentino, E. et al. (2025), “The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics”, arXiv:2504.01669
- Stern, D. et al. (2010), “Cosmic Chronometers: Constraining the Equation of State of Dark Energy. I: H(z) Measurements”, ApJ, 700, 1
- Tomasetti, E. et al. (2023), “A new measurement of the expansion history of the Universe at z=1.26 with cosmic chronometers in VANDELS”, A&A, 679, A96
- Vagnozzi, S. et al. (2020), “Eppur è piatto? The cosmic chronometer take on spatial curvature and cosmic concordance”, PhRvD, 102, 023518

Project 60: Tracing galaxy mergers through observed and simulated shells

Supervisor I: Dr Elisabeth Sola (es2074@cam.ac.uk)

Supervisor II: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

UTO: Professor Vasily Belokurov (vasily@ast.cam.ac.uk)

Project summary

This project combines deep imaging and numerical simulations to investigate the properties of shells, that are remnant of radial galaxy mergers. The spatial properties of the simulated and observed shells will be compared to gather hints on the merger timelines and dark matter halo properties.

The observational part will rely on the analysis of 20000 nearby galaxies, visually classified for the presence of tidal features, to measure shell incidence, develop an automated shell-finding algorithm, and classify systems by morphology.

The simulation part will rely on the same detection method applied to snapshots of massive, early-type galaxies from the IllustrisTNG simulations. The evolution shell properties with merger age and viewing angle will be explored, and compared with the observed sample. Alongside scientific insights, you will gain hands-on experience in image processing and algorithm development.

Background

In hierarchical models of galaxy formation and evolution, galaxies grow by accreting gas and merging with other galaxies. The gravitational interactions during these events eject stars and gas, forming extended, faint tidal features around the host. The morphology and properties of these structures depend on the propeties of the collision that created them. Among these tidal features, shells are concentric, arc-shaped structures believed to arise primarily from nearly radial mergers. By analysing their positions and kinematics, shells provide a powerful probe of a galaxy's dark matter distribution (Figure 1).

Shell-galaxy systems can be classified by the spatial arrangement of their arcs (e.g., Quinn 1984, Prieur 1988, Figure 2). Type I shells form interleaved, concentric arcs on opposite sides of the galaxy and are well explained by radial mergers in spherical potentials; the number and radii of these shells can reveal the time since the encounter. Type II shells appear as randomly distributed arcs around the host, indicative of more eccentric or off-centre mergers. Type III systems exhibit only a few shells or display irregular, complex patterns, often reflecting a wider range of orbital parameters for the merger.

Idealised numerical studies (e.g., Ebrová et al. 2013; Bílek et al. 2020, Dong-Paez+2022) demonstrate that both the number and radius of shells increase with the time since the merger. Observational studies show that the environment in which galaxy reside impacts the fraction of shells, with reduced fractions in galaxy clusters (e.g. Malin & Carter 1983). Comparisons between statistical analyses of shells in cosmological simulations (e.g., Pop et al. 2018) and deep imaging surveys are essential to better understand the conditions under which shells form and to assess the constraints that shells can provide on the host galaxy's potential.

Project details

This project is composed of two complementary parts: it aims at characterising and comparing shells systems observations and simulations.

1) Observations

We will use a sample of 20000 galaxies in the nearby Universe (z<0.02) from the SGA-2020 catalogue classified by Sola+2025. They have been visually inspected for the presence of tidal features, including shells, using original + model + residual images from the DESI Legacy Imaging Survey. The goal is to study the incidence of galaxy with shells as a function of the host properties, and to develop an algorithm to find the position of the shells.

a- Incidence study

- Investigate how the fraction of galaxies with shells evolves as a function of the host galaxy stellar mass, morphological type, and environment.

b- Shell detection

- Develop an algorithm to find the shells in the images:
    - Compute the surface brightness radial profiles of the galaxies (averaged and in angular wedges)
    - Identify the prominent peaks corresponding to shells edges
    - Study the number and radii of these shells
    - Compare their orientation to the galaxy's, investigate the correlation with eccentricity and classify the shell system as Type I, II or III.
    - Segment the shell (e.g. skeletonization) using the radius information to help a detection algorithm

2) Simulations

We will analyse stellar surface density maps from the Illustris-TNG simulations. We will select massive, early type galaxies that exhibit shell features, and inspect snapshots before and after the merger event

that produced them. The same detection algorithm developed in Part 1 will be applied to these simulated systems.

a- Image retrieval

- Retrieve the snapshots of the massive, early-type galaxies with shells before and after mergers

b- Shell detection

- Apply the shell detection algorithm developed in Part 1 on the images to:
    - Identify the number and radii of shells
    - Determine their alignment with the galaxy's orientation and classify their type
    - Correlate shell properties with the time elapsed since the triggering merger Investigate the effect of the projection angle on the detectability of the shells

c- Projection Effects

- Study how viewing angle influences shell detectability and measured properties.

d- Observational Comparison

- Compare the results from the simulations with the observational findings to gather hints on the shell formation history.

In summary, this project will use deep imaging and simulations to study how shells reflect galaxy merger histories and dark matter distributions, while also building practical skills in image processing and algorithm development.

Acceptable Programming Languages
Python

Useful References (List of important papers/review articles relevant to the project)
Malin & Carter 1983, Quinn 1984, Prieur 1988, Ebrova 2013, Bílek+2020, Pop+2018, Dong-Paez+2022, Sola+2025]

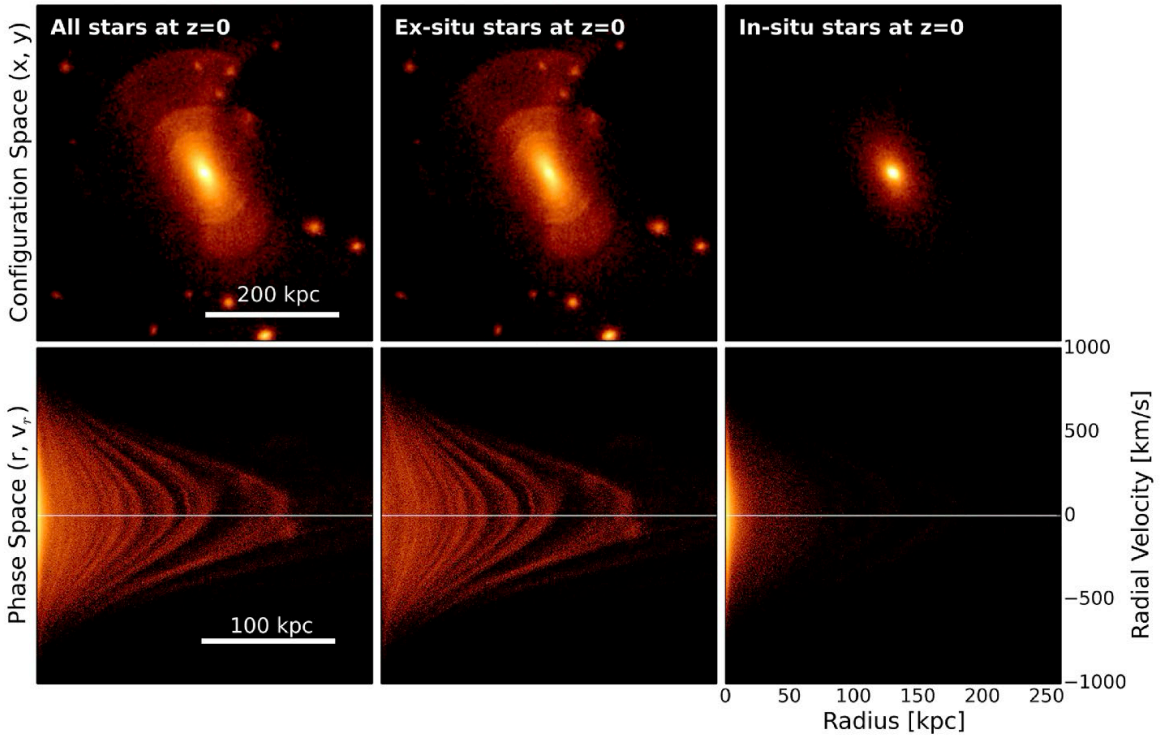

Figure 1: Example of shells in IllustrisTNG (Pop+2018). The top panel shows the spatial distribution of shells, the bottom panel shows the phase space of shells.

Figure 4: Galaxy-subtracted image of the type I shell galaxy NGC 7600 from Turnbull et al. (1999). North is up and east is to the left. The dark oval shape is an artifact of the subtraction process. The easternmost shell lies 215″ away from the galaxy center. The field of view is 9′. Acknowledgment: The Isaac Newton Group of Telescopes and the Royal Astronomical Society.

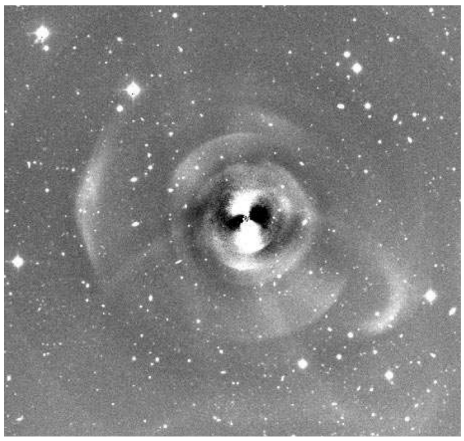

Figure 5: Galaxy-subtracted image of the type II shell galaxy NGC 474 from Turnbull et al. (1999). North is up and east is to the left. The easternmost shell is 202″ from the galaxy center. NGC 470 is located just off the frame, ∼300″ west. The field of view is 9′. Acknowledgment: The Isaac Newton Group of Telescopes and the Royal Astronomical Society.

Figure 2: Examples of a type I (left) and II (right) shell galaxies. Figures from Ebrova+2013 (PhD thesis)

Project 61: Disentangling Key Galaxy Evolution Markers Using High-dimensional Colour Information

Supervisor I: Dr. Sinan Deger (sinan.deger@ast.cam.ac.uk)

Supervisor II: Professor Hiranya Peiris (hiranya.peiris@ast.cam.ac.uk)

UTO: Professor Hiranya Peiris (hiranya.peiris@ast.cam.ac.uk)

Project summary

The era of large-scale surveys is here. With the Vera C. Rubin Observatory's Legacy Survey of Space and Time (Rubin-LSST) starting operations, we need to upgrade aspects of our analysis toolkit to meet the needs of the new Big Data era. The challenge is many-fold: Rubin-LSST will gather data unprecedented in volume but only in a handful of photometric bandpasses. This creates the need to understand a complex galaxy population evolving through cosmic time using only information captured in colours, or the flux difference between two bandpasses. This project will utilize a generative galaxy population model developed by our team to investigate the information content in colours about key galaxy evolution processes. This will be achieved by generating catalogues containing a comprehensive set of galaxy properties and accompanying colour information from our model, and investigating combinations of colours capable of disentangling key galaxy evolution markers.

Background

Galaxy spectral energy densities (SEDs) encode valuable information about the composition and physical processes within galaxies. The observed SED is an amalgamation of the information contained in light emitted and re-emitted by the stellar populations of galaxies through the assembly history of the galaxy, the central supermassive black hole and its status of activity, the gas component, and dust.

Spectroscopic observations of galaxies require large amounts of observational resources and time, and are often limited to brighter sources or limited in the wavelength range spanned by the instrument. Photometry, or the observation of the SED in bandpasses that integrate the light in relatively broad wavelength bins, is a resource-efficient alternative still capable of capturing critical information about galaxies. Much of our understanding of the evolution of galaxy properties over cosmic time comes from extremely successful multibandpass photometric surveys such as the The Cosmic Evolution Survey (COSMOS, Weaver et al. 2022). Notably, the Vera C. Rubin Observatory's Legacy Survey of Space and Time (Rubin-LSST) is a photometric survey that will create the most comprehensive galaxy catalogue to date.

A galaxy colour is the difference between the flux received by two different bandpasses. Specific combinations of galaxy colours can act as major indicators of particular galaxy properties, and help pinpoint the evolutionary stage of the galaxy. This is demonstrated in Figure 1, which shows the *NUV* -

r and *r* - *J* colour distributions of galaxies. The hexagonal bins are colour-coded by specific star formation rate (sSFR), which is the ratio of the star formation rate of the galaxy to its stellar mass (total amount of mass contained in stars). sSFR is an excellent marker of the significance of star formation activity in a galaxy, as it is measured relative to the stellar mass the galaxy already formed. Therefore even galaxies with higher than average star formation rates can be labeled as “low activity” if they are massive. Galaxies with low-to-no star formation activity or low sSFRs are referred to as “quiescent”, as opposed to “star-forming” galaxies with high sSFR values.

Figure 1 shows that the *NUV* - *r* and *r* - *J* space strongly correlates with sSFR. This is a noteworthy result: it shows that observations in only 3 bandpasses allow for distinguishing between quiescent and star-forming galaxies, which are key evolutionary stages for galaxies. This colour-colour space is used for this purpose in observational studies that often do not have the sSFR information, which requires observations in more bandpasses to accurately measure. We plot the quiescent/star-forming selection boundary commonly used in literature (e.g. Weaver et al. 2023) in Figure 1, where quiescent galaxies preferentially reside above the empirically-defined orange separator.

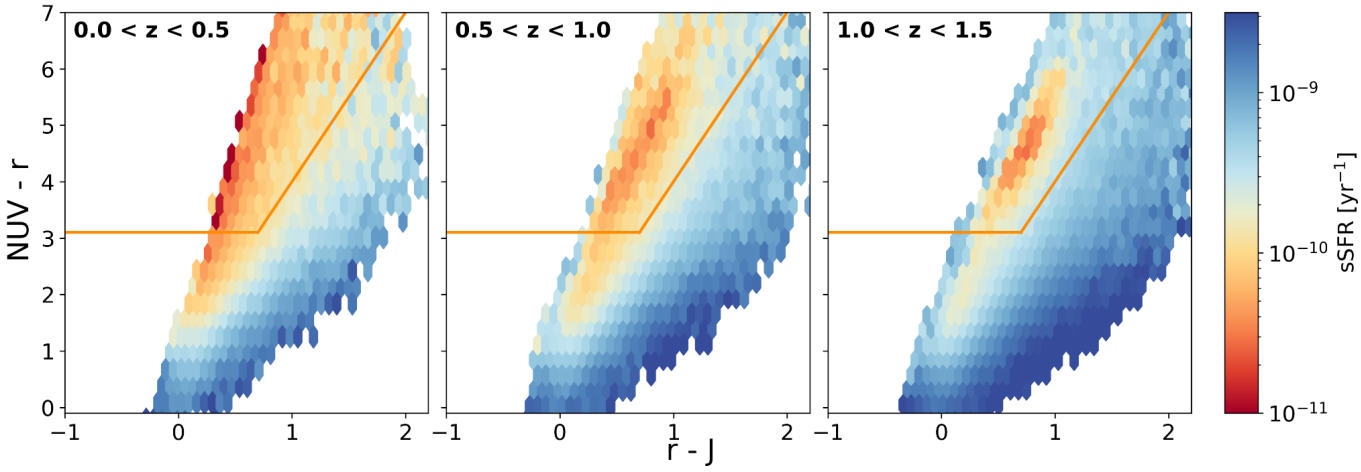

Figure 1: The NUV - r and r - J colour-colour diagram presented as a hexbin plot. The hexes are colour-coded by the specific star formation rate of the galaxies in the figure. Galaxies in different redshift (z) or distance ranges are plotted in different panels, indicated at the top left corner of each panel. Observational studies, in the absence of reliable sSFR information, identify galaxies remaining above the orange separator as “quiescent.”

The results shown in Figure 1 are generated using the *pop-cosmos* model developed by our team (Alsing et al. 2024, Thorp et al. 2025). pop-cosmos is a generative model for the evolving galaxy population which has been calibrated on 26-bandpass photometry from the COSMOS survey. Simulated data generated using pop-cosmos are representative of the galaxy population up to a redshift of 4, which spans 90% of the age of the Universe. pop-cosmos links a comprehensive set of galaxy properties such as the sSFR and stellar mass with the galaxy SED and hence the colours. We can therefore generate catalogues of the properties together with colours to study the building blocks of galaxy evolution imprinted in colour spaces in depth. Figure 1 reveals the benefit of this joint view: we find that the trend with sSFR in the left panel is different from the trends in the middle or right panels,

where each panel is showing galaxies at different redshift or distance ranges. The findings on this colour-colour space therefore also correlate with redshift. A key goal of the project will be to interpret the origin of such findings.

The aim of this project is to discover new combinations of colours (including selections involving more than two colours) that serve to distinguish the building blocks of galaxy populations, and potentially provide new observational proxies for galaxy evolution processes that are difficult to disentangle. We will quantify the correlations between colours and galaxy properties using measures from information theory (e.g. mutual information, see Piras et al. 2023) and statistics. This project will lead to a detailed census of the information captured in high-dimensional colours, which will be of critical importance in the era of massive-scale photometric surveys.

Project details

The project will begin by forming a working understanding of our galaxy population model, pop-cosmos (Alsing et al. 2024, Thorp et al. 2025). We expect the student to acquire a working knowledge of the galaxy evolution investigation presented in these papers. Next, the student will work with the galaxy properties and colours of galaxy populations generated using pop-cosmos. In the first instance the student will identify colours of interest based on the causal links between galaxy properties and their SEDs. Next, the student will carry out a comprehensive statistical analysis quantifying the correlations between one or multiple colours and the fundamental properties of the evolving galaxy population.

A starting workflow can be (but does not need to be limited to) the following:

- review of literature on the reference list and related papers,
- build familiarity with pop-cosmos (Alsing et al. 2024, Thorp et al. 2025) and the analysis presented in these papers,
- Identify a set of photometric bandpasses capturing key galaxy evolution markers
- Use the high-dimensional space created by the colours to carry out a statistical & information theoretical analysis, investigating colour combinations capable of disentangling interconnected galaxy evolution processes.

Prerequisite Skills required

A strong familiarity with scientific analysis in Python is required. A background and a keenness to develop skills in statistical data analysis is essential.

Other skills used in the project

The project will rely on an understanding of galaxies and their evolution. Industry towards learning more about this field would be important. Consequently, the Part III course “The Life and Death of Galaxies” is recommended, in case it is offered.

Acceptable Programming Languages

Our team develops our expanding codebase explicitly in Python and Python-based libraries. We have extensive experience in scientific analysis in Python as a result. We have a strong preference that the student works on the project using Python from the start, both to benefit from our expertise, and to be able to natively connect to our code repository and other Python-based libraries we frequently use in our research.

Useful References (List of important papers/review articles relevant to the project)

1. C. Conroy (2013) (https://arxiv.org/abs/1301.7095) – Review paper on galaxy SEDs
2. Alsing et al. (2024) (https://arxiv.org/abs/2402.00935) – Paper where we introduce pop-cosmos
3. Thorp et al. (2025) (https://arxiv.org/abs/2506.12122) - The next version of pop-cosmos fitted to a deeper galaxy sample
4. Weaver et al. (2022) (https://arxiv.org/abs/2110.13923) – Example extragalactic survey, COSMOS2020
5. Weaver et al. (2023) (https://arxiv.org/abs/2212.02512) Example galaxy evolution analysis on the COSMOS2020 data

General References (List of papers referred to in the project)

1. J. Alsing et al (2020) (https://arxiv.org/abs/1911.11778)
2. J. Alsing et al. (2023) (https://arxiv.org/abs/2207.05819)
3. B. Leistedt et al (2023) (https://arxiv.org/abs/2207.07673)
4. D. Piras et al. (2023) (https://arxiv.org/abs/2211.00024)

Project 62: Connecting Milky Way Stellar Content to LISA's Stochastic Gravitational Wave Foreground

Supervisor I: Dr Valeriya Korol (korol@mpa-garching.mpg.de)

Supervisor II: Dr Christopher J. Moore (cjm96@cam.ac.uk)

UTO: Dr Christopher J. Moore (cjm96@cam.ac.uk)

Project summary

The Laser Interferometer Space Antenna (LISA) is a next-generation space-based gravitational wave observatory scheduled for launch in the mid-2030s. Operating in the millihertz frequency range, LISA will open a new observational window to the gravitational-wave universe (Figure 1). It is particularly well suited to detecting short-period compact binary systems composed of stellar fossil remnants—such as white dwarfs, neutron stars, and black holes—within our Galaxy. While LISA is expected to individually resolve tens of thousands of such binaries, the vast majority will remain unresolved, collectively forming a confusion-limited stochastic gravitational wave foreground (Figure 2). The goal of this project is to investigate how the spectral shape of this unresolved foreground encodes the properties of the Milky Way, such as its stellar content and spatial structure.

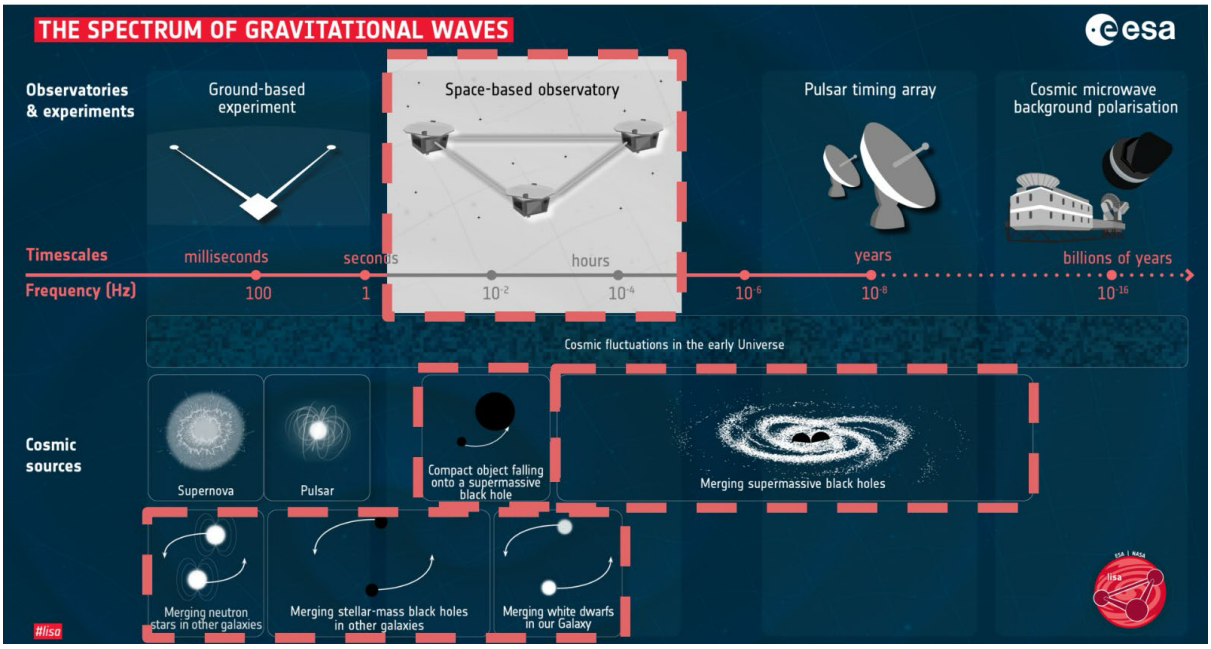

Figure 1: The LISA mission targets the millihertz band of gravitational waves, lying between the nanohertz regime probed by pulsar timing arrays and the decahertz regime accessible to ground-based detectors. Many types of astrophysical sources produce gravitational waves (GWs) in this band (see highlighted boxes), of particular importance for this project are the Galactic white dwarf binaries. Image credit: LISA Definition Study Report, Colpi et al. 2024

Background

LISA will be sensitive to gravitational waves in the frequency band of approximately 0.1–10 mHz. Within this range, especially around 1–3 mHz—the heart of the LISA band—the dominant source of noise is expected to be astrophysical rather than instrumental. The Milky Way hosts millions of short-period ($P \lesssim 1$ hour) compact object binaries, primarily double white dwarfs, along with some neutron stars and black holes (Colpi et al. 2024, arXiv:2402.07571). While LISA will individually resolve some of these systems, the majority will remain unresolved, creating a confusion-limited stochastic gravitational wave foreground (e.g., Korol et al. 2022, arXiv:2109.10972). Understanding this background is essential both for characterising the Galactic binary population and for isolating other astrophysical or cosmological gravitational wave signals.

Estimates of the Galactic gravitational wave foreground have so far relied largely on binary population synthesis methods – i.e. forward-modelling approaches that simulate the formation and evolution of binary stars based on simplified assumptions about binary evolution physics and the properties of the Milky Way. While these methods are powerful and computationally efficient, they involve many ad hoc parameters, often with strong correlations, which makes it difficult to isolate the features of the binary population that most significantly shape the gravitational wave foreground.

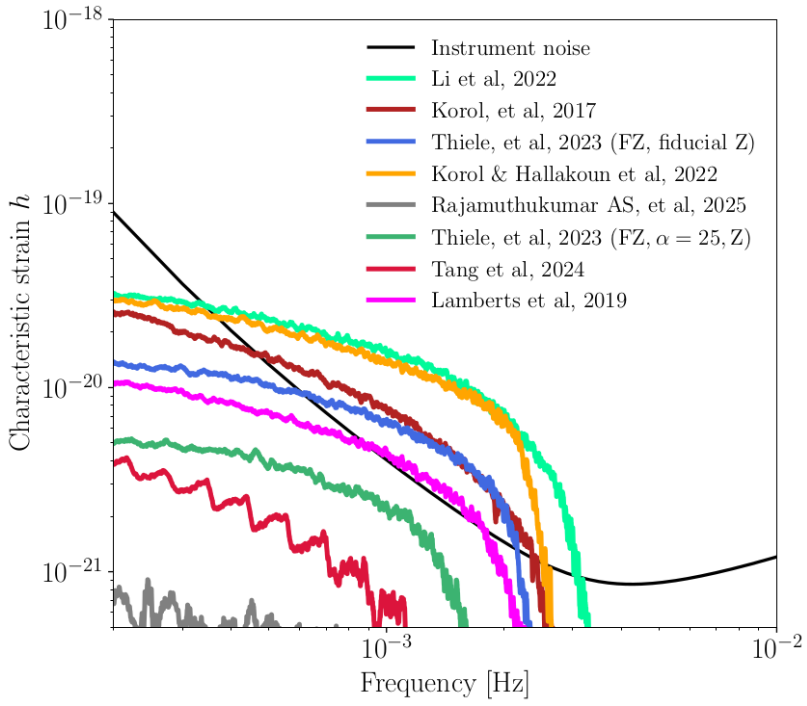

Figure 2: Compilation of Galactic gravitational wave foregrounds as predicted by theory- and observation-based models for LISA in recent years. These models are typically complex, involving many parameters related to both binary evolution and the structure of the Milky Way. As a result, it is difficult to directly link the shape of the foreground to the underlying stellar and binary population of the Galaxy. In this project, we will construct a simpler, more interpretable model of the Galaxy that will allow us to gain insight into the connection between the shape of the gravitational wave foreground and the properties of the Galactic compact binary population.

In this project, instead of relying on population synthesis, we adopt a more pedagogical approach. We will construct simplified catalogs of compact object binaries based on simple, controlled assumptions about the Milky Way's stellar content and spatial structure. This will enable us to disentangle how specific Galactic properties—such as the stellar mass distribution or binary formation efficiency—map onto features of the gravitational wave foreground observable by LISA.

Project Details

This project aims to investigate how the unresolved Galactic gravitational wave foreground in the LISA data is shaped by the underlying compact binary population of the Milky Way. Specifically, the student will:

- **Generate synthetic catalogs** of compact object binaries using simplified assumptions about the Milky Way's stellar content (Korol et al. 2022, arXiv:2109.10972).
- **Produce corresponding synthetic LISA data sets** based on these catalogs.
- **Apply iterative subtraction techniques** to estimate and characterise the astrophysical foreground (Karnesis et al. 2021, arXiv:2103.14598). A data analysis pipeline for this task has already been developed by a collaborator (see gitlab.in2p3.fr/Nikos/gwg). The student will primarily focus on applying the pipeline and interpreting the results, rather than developing the tools from scratch.
- **Compare the resulting stochastic backgrounds** across different population models to assess how variations in stellar content influence the Galactic gravitational wave foreground. These results will also be compared to foregrounds predicted by population synthesis models (see Figure 2).

This work will contribute to improving foreground modelling strategies for LISA and provide insights into the connection between the Milky Way's stellar content and gravitational wave observables (Colpi et al. 2024, arXiv:2402.07571). In the long term, the methods developed here could be extended into an inference framework, enabling the use of LISA data to constrain Galactic structure and binary population properties.

Prerequisite Skills required

Proficiency in Python programming is essential. Experience with GPU-accelerated methods in Python (e.g., using CuPy) is welcome and could be beneficial, but is not a requirement for this project.

Other skills used in the project

It is not essential to have all of the following skills already, but if not then these skills will be developed during this project.

- Gravitational wave data analysis

- Signal processing and noise modeling
- Working with astrophysical simulations and catalogs
- Scientific interpretation of model-(synthetic)data comparisons

Acceptable Programming Languages
Python.

Useful References (List of important papers/review articles relevant to the project)
Observationally-driven Galactic double white dwarf population for LISA, Korol et al. 2022, arXiv:2109.10972

Characterisation of the stochastic signal originating from compact binary populations as measured by LISA, Karnesis et al. 2021, arXiv:2103.14598

General References (List of papers referred to in the project)
LISA Definition Study Report, Colpi et al. 2024, arXiv:2402.07571

Project 63 Oscillating rings as a model for the dynamics of warped and broken discs

Supervisor I: Professor Gordon Ogilvie (DAMTP) (gio10@cam.ac.uk)

UTO: Professor Gordon Ogilvie (DAMTP) (gio10@cam.ac.uk)

Project summary
This theoretical project will investigate aspects of the gas dynamics of warped discs around stars or black holes using a convenient semi-analytical model. The oscillating ring model is motivated partly by the finding that tilted discs around a binary or a spinning black hole can break into independently precessing rings. As well as exploring the time-dependent behaviour of the model with different initial conditions and parameter values to reveal the physics of the system, there is scope for either (i) running direct numerical simulations using a hydrodynamic code to compare with the outcome of the model or (ii) developing theoretical descriptions of the behaviour that is found.

Background
Gaseous astrophysical discs consist of gas or plasma in orbital motion around single or multiple stars or black holes. They can often be warped, meaning that the orientation of the orbital plane varies with distance from the centre. Warps typically occur because of a misalignment in the system, for example between the plane in which the disc forms and the spin axis of a black hole, or the plane of a binary or planetary orbit. Such misalignments are commonplace in observations, numerical simulations and theoretical scenarios.

Understanding the gas dynamics of a warped disc has been a challenging theoretical problem for decades. A special property of Keplerian discs means that the vertical motions associated with the inclination of the orbits is accompanied by resonant horizontal motions within the disc. These lead to the radial transport of the warp, either as a bending wave or as an accelerated diffusive process. Numerical simulations of tilted discs around a binary or a spinning black hole often show that the disc breaks into two or more pieces that are only tenuously connected. In some cases multiple rings can break off and precess independently [e.g. 3]. This process has been partly understood as a consequence of the nonlinear dynamics of the warped disc.

It has also been found that warped discs can be very far from hydrostatic equilibrium. Indeed, a large-amplitude “bouncing” motion can be excited through a nonlinear resonance [2]. In large-scale numerical simulations this motion is accompanied by shocks that are highly dissipative and accelerate the accretion process [4]. Precessing tilted discs or rings and their associated oscillations offer the most

promising explanations of quasi-periodic oscillations seen in X-ray observations of accreting black holes and neutron stars.

We have recently developed a greatly simplified, yet precise, model for studying aspects of the dynamics of warped discs without the need for global, 3D numerical simulations. The model is local (following a small region of the disc in its orbital motion) and essentially 2D (capturing the azimuthal variation of the disc in its time-dependence). It consists of a cylinder with an elliptical cross-section, representing a narrow and thin disc or ring. A system of nonlinear ordinary differential equations (ODEs) describes how the shape and orientiation of the ellipse, and the accompanying velocity field, evolve in time as a result of pressure, Coriolis and tidal forces [1]. This model supports the principal types of fluid motion present in a warped disc [2].

Project details

In previous work [2] it was found that an ideal fluid ring (consisting of a perfect gas with no viscosity) in Keplerian rotation undergoes extreme vertical bouncing motion, as a result of a nonlinear resonance, when the initial warp exceeds a certain threshold. One of the aims of the project is test how this behaviour is modified (i) if the ring is slightly non-Keplerian (because it is close to a binary or a black hole) and (ii) if the fluid has some viscosity. To implement (i) it is only necessary to change one or two of the parameters of the model, while (ii) will require adding viscous terms to the ODEs as described in Appendix B of Callum Fairbairn's PhD thesis (https://www.repository.cam.ac.uk/handle/1810/353533). Viscosity could have a number of effects of interest, all of which should be investigated: moderation of the horizontal motions induced by the warp; moderation and dissipation of the bouncing motions; radial spreading of the ring, which may actually be reversed if the warp exceeds a threshold amplitude.

The project will certainly involve coding up the ODE system and carrying out a number of tests (e.g. of conserved quantities) before investigating the solutions of scientific interest. The student will need to decide on the best way to analyse and display the output to illustrate the physics at work and its dependence on the parameters.

If the student makes rapid progress with these investigations, two possible extensions are foreseen. One option is to set up and run local, 2D numerical simulations (as described in [1, 6]) to compare with the predictions of the oscillating ring model; when this was done using another (GRMHD) code [5], unexplained discrepancies were found when the model predicted extreme bouncing. The other option is to develop a theoretical description of the nonlinear behaviour that is found in the model, in the case of a thin ring with a small viscosity, where the transfer of energy between the different oscillations can be captured through orbit-averaging methods.

Prerequisite Skills required
The project will require some familiarity with orbital mechanics and the dynamics of compressible fluids. It is recommended to attend the Part III courses on *Astrophysical Fluid Dynamics* and *Dynamics of Astrophysical Discs*.

Other skills used in the project
The project will require high-quality numerical solutions of a system of nonlinear ODEs. This can be done using any appropriate computational method or software, provided that the method is sufficiently accurate to evolve nonlinear oscillations over a long timescale and sufficiently adaptive to capture extreme compressions where needed.

Acceptable Programming Languages
For the ODE solver, any appropriate scientific programming language or software can be used. If the student opts to carry out numerical simulations, a standard hydrodynamic code such as PLUTO, Idefix or Athena++ will be used, requiring a knowledge of C/C++ and Python.

Useful References (List of important papers/review articles relevant to the project)
[1] Fairbairn C. W., Ogilvie G. I., 2021a, MNRAS, 505, 4906

[2] Fairbairn C. W., Ogilvie G. I., 2021b, MNRAS, 508, 2426

General References (List of papers referred to in the project)
[3] Rabago I., Zhu Z., Lubow S., Martin R. G., 2024, MNRAS, 533, 360

[4] Kaaz N., et al., 2023, ApJ, 955, 72

[5] Kaaz N., Lithwick Y., Liska M., Tchekhovskoy A., 2025, ApJ, 979, 192

[6] Ogilvie G. I., 2022, MNRAS, 513, 1701

Figure 1: *Left*: Top-down view of a narrow ring in the global orbital plane. The numbered quadrants denote different locations around the anticlockwise orbit. *Right*: Local view of slices cutting through the ring in each quadrant location. The red arrows show the future evolution of the tilting ellipse as locally seen by an observer orbiting around the fixed global structure. Connecting these slices shows how a global warp is equivalent to alocal tilting mode. Reproduced from [1].

Figure 2: Snapshots of the density from a simulation of a tilted disc around a spinning black hole, showing the breaking of the disc and the formation of an independently precessing ring. Reproduced from [3].

Project 64 Defining requirements for a near infrared (NIR) low-resolution spectroscopy instrument to estimate stellar atmosphere parameters

Supervisor I: Dr Francesca De Angeli (fda@ast.cam.ac.uk)

UTO: Professor Sandro Tacchella (st578@cam.ac.uk)

Project summary

In order to derive instrument requirements from a set of scientific goals, we need to develop a tool simulating the observation and processing of the data until the final result. The tool can be very simple to start with, but should be built so that more sophisticated modules simulating the various stages of the process can be integrated as needed later on. The tool would use as input a library of synthetic spectra, for which true stellar atmosphere parameters are known, and a set of instrument parameters. The output should be the resulting stellar atmosphere parameters. We would then be able to run it multiple times with different instrument parameters to study the effect of these on the stellar atmosphere parameter accuracies.

Background

A science white paper for a new all-sky NIR astrometry mission (GaiaNIR, https://www.astro.lu.se/GaiaNIR) was selected as a potential future Large-class mission during ESA's Voyage 2050 call for new mission proposals. NIR astrometry and photometry are essential to probe the dynamically-important hidden regions of the Galaxy. Reaching magnitude ~23 in G could allow for a catalogue of 75 billion stars (~25% of the Galaxy).

Depending on the final design details, GaiaNIR could reach Gaia astrometric accuracies for a 5-years mission for B-type stars and significantly outperform Gaia for K and M-type sources, but for a much larger catalogue. When combined with the Gaia solution, GaiaNIR would allow to improve the astrometric accuracies by more than one order of magnitude for the sources in common.

A preliminary design of the GaiaNIR focal plane is shown in Figure 1. In this design, the third block of CCDs is dedicated to the acquisition of low-resolution spectroscopy in the range ~[800, 2300] nm. The main goal of this will be to derive stellar atmosphere parameters for all sources observed, focussing on gravity, effective temperature, reddening and metallicity. Figure 2 shows the synthetic spectra of a few stellar types covering the near infrared range of wavelengths.

Requirements for the instrument will be driven by the scientific goals. The design will then be defined to fulfil those requirements within the constraints imposed by technical feasibility and available funding.

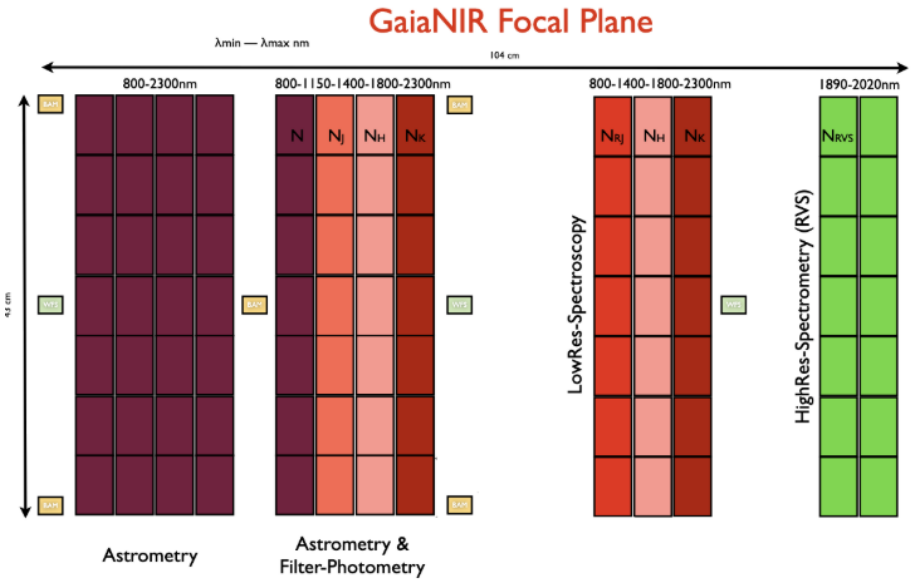

Figure 1: Preliminary design for the GaiaNIR focal plane. Each rectangle represents a CCD. Different blocks of CCDs have different purposes: after a first block fully dedicated to the astrometric measurements in a broad near infrared filter, comes a block of CCDs for photometric measurements in narrower bands required to correct the astrometry for chromaticity effects; the third and fourth blocks are dedicated to low and high resolution spectroscopy their main focus being stellar atmosphere parameters and radial velocities.

Credits: D. Hobbs, https://indico.ast.cam.ac.uk/event/8/contributions/43/attachments/22/30/CAM1_2025_GaiaNIR_Hobbs.pdf

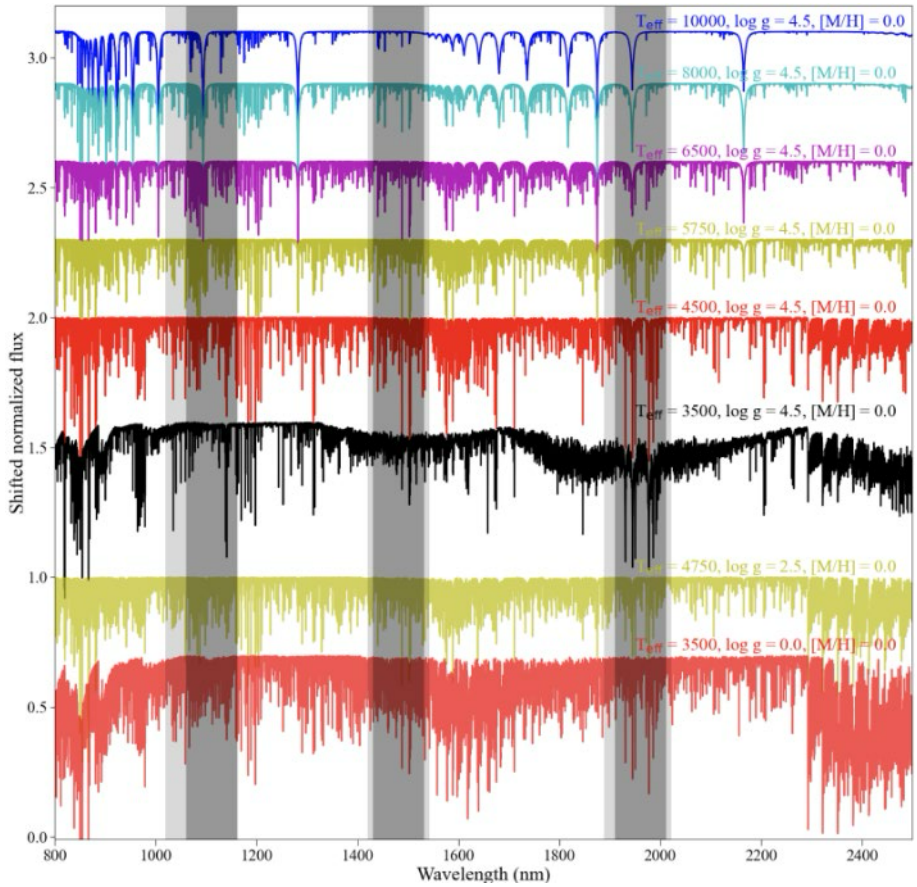

Figure 2: Spectral energy distribution for a range of spectral types drawn from the BOSZ synthetic spectra library, covering the near infrared wavelength range. The grey shaded region are different options being considered for the high-resolution spectrometer. Credits: S. Mészáros, https://indico.ast.cam.ac.uk/event/8/contributions/53/attachments/26/38/meszaros_gaianir.pdf

In order to define realistic requirements and to verify that a given design fulfills those requirements, we need to be able to simulate the resulting stellar atmosphere parameter accuracies given an instrument design. This will eventually be fully specified by a large number of instrument parameters, but a good starting point for this project could be to focus on a few parameters that are expected to be very significant for the stellar atmosphere parameter determination, such as the wavelength range covered by the low-resolution spectra and the spectral resolution. The goal of this project is to enable answering questions like: if we want to achieve a certain accuracy in effective temperature for a given type of source, what range of wavelength and what spectral dispersion would we need?

The simulation requires the adoption of an algorithm for the determination of the stellar atmosphere parameters. Many options are available in the literature. One possible approach for this project is known as “The Cannon” (Ness et al. ApJ 808 16, 2015). Alternatively, a neural network approach could be chosen, such as “The Payne” (Ting et al. ApJ 879 69, 2019). Different options could be discussed.

The simulation tool could be validated by a run where the instrument parameters are set to mimic the Euclid NISP instrument (also a low-resolution spectrograph in the NIR wavelength regime, https://www.esa.int/Science_Exploration/Space_Science/Euclid/Euclid_s_instruments). Results obtained from applying “The Cannon” to simulated data and real Euclid NISP spectra (from the Q1 release, https://www.cosmos.esa.int/web/euclid/euclid-q1-data-release) should agree within the accuracy expected, considering the rough approximations in the instrument set-up. This would be a nice addendum to the project if time allows and would provide the opportunity to test the approach on recently released data.

Project details

The project could include the following steps:

Develop software to simulate an observed spectrum given a spectral energy distribution (SED) and some instrument parameters, with a focus on wavelength range covered and spectral resolution.

Develop software to derive stellar atmosphere parameters applying the chosen approach to a set of observed spectra. This requires setting up a “reference dataset” consisting of observed spectra with known stellar atmosphere parameters or labels. This could be achieved using a synthetic stellar spectral library, e.g. BOSZ (Mészáros et al., A&A, 688, A197, 2024). “The Cannon” is a data-driven approach that learns from the known labels of reference stars how the continuum-normalized spectra depend on these labels by fitting a flexible model at each wavelength. ”The Cannon” makes no use of stellar models nor any list of spectral lines, but requires that the reference dataset covers the parameter space. “The

Payne" instead relies on stellar models for training the neural network. The model then gets applied to spectra of other sources to derive stellar atmosphere parameters. Implementations of both these algorithms are available in Python, so this step should be quite straightforward. Ideally, both approaches could be implemented and results compared.

Run the software multiple times for a range of plausible values of the instrumental parameters to analyse the effects of these changes on the derived stellar atmosphere parameter accuracies.

If time allows, a possible validation of the approach could be obtained by running the full process with instrument parameters similar to the Euclid NISP instrument and comparing the resulting accuracies with those obtained running "The Cannon" on some Euclid Q1 spectra of sources with similar stellar atmosphere parameters.

Prerequisite Skills required

Basic coding skills are required. Python would be an obvious choice given the existence of many data analysis and scientific libraries as well as tutorials, e.g. for accessing Euclid Q1 data.

Useful References (List of important papers/review articles relevant to the project)

The updated BOSZ synthetic stellar spectral library, Mészáros et al., A&A, 688, A197, 2024

A New Stellar Atmosphere Grid and Comparisons with HST/STIS CALSPEC Flux Distributions, Bohlin et al., AJ, 153, 234, 2017

The Cannon: A data-driven approach to stellar label determination, Ness et al., ApJ, 808, 16, 2015

CDF Study Report GaiaNIR, https://sci.esa.int/documents/34923/36148/1567260328914-GaiaNIR_CDF_study_report_v2.pdf

Euclid Q1 resources available at https://www.euclid-ec.org/science/q1/

Project 65: Improving a Distributed Strategy for Scalable Bayesian Inference in Supernova Cosmology

Supervisor I: Professor Kaisey Mandel (km723@cam.ac.uk)

Supervisor II: Dr Harry Bevins (htjb2@cam.ac.uk)

Supervisor III: Dr Matthew Grayling (mg2102@cam.ac.uk)

UTO: Professor Kaisey Mandel (km723@cam.ac.uk)

Project summary
To improve & apply distributed strategies for Bayesian inference in supernova cosmology via computing and combining “nuisance-free likelihoods” of sub-datasets.

Background
Bayesian models are used in Type Ia supernova (SN Ia) cosmology for principled statistical inference. Hierarchical Bayesian models (HBMs) enable simultaneous probabilistic inference of both the latent variables (e.g. magnitudes, colours, distances, dust extinction) describing individual supernovae, and the hyperparameters describing their population distributions (e.g. Mandel+2009, 2011; March+2011; Rubin+2015; Mandel+2022). Modern statistical analysis of SN Ia datasets with these models is usually accomplished using Markov Chain Monte Carlo methods to sample the joint posterior distribution over a high-dimensional parameter space, comprising the latent variables of each individual SN Ia in the sample along with the population hyperparameters and cosmological parameters. Such methods can analyse current datasets of over ~10^3 SNe Ia using state-of-the-art Hamiltonian Monte Carlo algorithms leveraging GPUs and auto-differentiation codes (e.g. Grayling+2024).

However, with SN Ia datasets soon to increase by orders of magnitude with the imminent Vera C. Rubin Observatory’s Legacy Survey of Space and Time (LSST), we are motivated to develop computational methods that are even more scalable and efficient with increasing sample size. While most of the parameters of the statistical models are “nuisance parameters” pertaining to individual supernovae or surveys, the primary parameters of interest are a much smaller set of population parameters and/or cosmological parameters. Thus, it would be computationally advantageous to analyse smaller subsets of supernovae (SNe Ia) in a parallel, distributed strategy, marginalise out the nuisance parameters in each, and then combine these separate inferences together to compute final inferences of the primary parameters of interest from the full dataset. It would also be scientifically interesting to perform inference on separate surveys, or subsets of SNe Ia differentiated by physical properties, e.g. host galaxy characteristics or spectral sub-type, to analyse potential systematic tensions between them, before combining their information for final inferences.

Project details

In this project, the student will implement, improve, and apply a distributed strategy, as outlined by Bevins+2022 (with other astrophysics applications described in Bevins+2023), for Bayesian inference with SNe Ia. The approach involves analysing SN Ia subsets (as separate "experiments") by sampling their separate full posteriors (including both nuisance parameters and the primary parameters of interest) and computing their Bayesian evidences using nested sampling. The resulting posterior samples are used to train a normalising flow to approximate the posterior distribution of the primary parameters marginalised over the nuisance parameters. These ingredients are then used to compute for each subset a "nuisance-free likelihood" (essentially a prior-weighted average of each full likelihood, with the nuisance parameters marginalised over). The marginalised likelihoods for each subset can be computed separately in parallel and then compared and subsequently multiplied to compute the final combined inferences on the primary parameters.

The student will build on previous work that implemented this approach for a toy model for simulated SNe Ia as simple standard candles with a Gaussian absolute magnitude distribution with some population mean and intrinsic variance. In this simplified setting, this computational approach can be compared to a (mostly) analytic solution. The computational trade-offs of the method will be investigated. As time allows, the method can be further developed and applied. One avenue is to apply the method to use this simple model to analyse conventional standardized SN Ia magnitudes from real current datasets, e.g. Pantheon+ (Scolnic+22), Dark Energy Survey (DES Collaboration+24), or realistic simulated future datasets (e.g. LSST). Another avenue is to develop the method for more realistic, complex hierarchical Bayesian models with latent variables for individual SNe Ia (e.g. Mandel+2017, see Fig 1). An additional challenge is accounting for systematic error covariance between the supernovae subsets.

Prerequisite Skills required

An aptitude/interest in statistical computation, data analysis, and numerical programming.

Other skills used in the project

An aptitude/interest in Bayesian statistical modelling and related mathematical analysis. The Part III Astrostatistics course will be useful.

Acceptable Programming Languages

Python and Python-based numerical computing frameworks.

Useful References (List of important papers/review articles relevant to the project)

- Bevins, H. et al. 2022. "Marginal Bayesian Statistics Using Masked Autoregressive Flows and Kernel Density Estimators with Examples in Cosmology". https://arxiv.org/abs/2207.11457

- Bevins, H. et al. 2023. “Marginal Post-Processing of Bayesian inference products with normalizing flows and kernel density estimators”. MNRAS, 526, 4613.
- Mandel, K. et al. 2017. “The Type Ia Supernova Color-Magnitude Relation and Host Galaxy Dust: A Simple Hierarchical Bayesian Model”. ApJ, 842, 93.
- Scolnic et al. 2022. “The Pantheon+ Analysis: The Full Data Set and Light-curve Release”. ApJ, 938, 113.
- DES Collaboration et al. 2024. “The Dark Energy Survey: Cosmology Results with ∼1500 New High-redshift Type Ia Supernovae Using the Full 5 yr Data Set”. ApJL, 917, L14.

General References (List of papers referred to in the project)

- Mandel, K. et al. 2009. “Type Ia Supernova Light-Curve Inference: Hierarchical Bayesian Analysis in the Near-Infrared”. ApJ, 704, 629.
- Mandel, K. et al. 2011. “Type Ia Supernova Light Curve Inference: Hierarchical Models in the Optical and Near-infrared”, ApJ, 731, 120.
- Mandel, K. et al. 2022. “A hierarchical Bayesian SED model for Type Ia supernovae in the optical to near-infrared”. MNRAS, 510, 3939.
- March, M. et al. 2011. “Improved Constraints on cosmological parameters from Type Ia supernova data”. MNRAS, 418, 2308.
- Rubin, D. et al. 2015. “UNITY: Confronting Supernova Cosmology’s Statistical and Systematic Uncertainties in a Unified Bayesian Framework”. ApJ, 813, 137.
- Grayling, M. et al. 2024. “Scalable hierarchical BayeSN inference: investigating dependence of SN Ia host galaxy dust properties on stellar mass and redshift”. MNRAS, 531, 952.

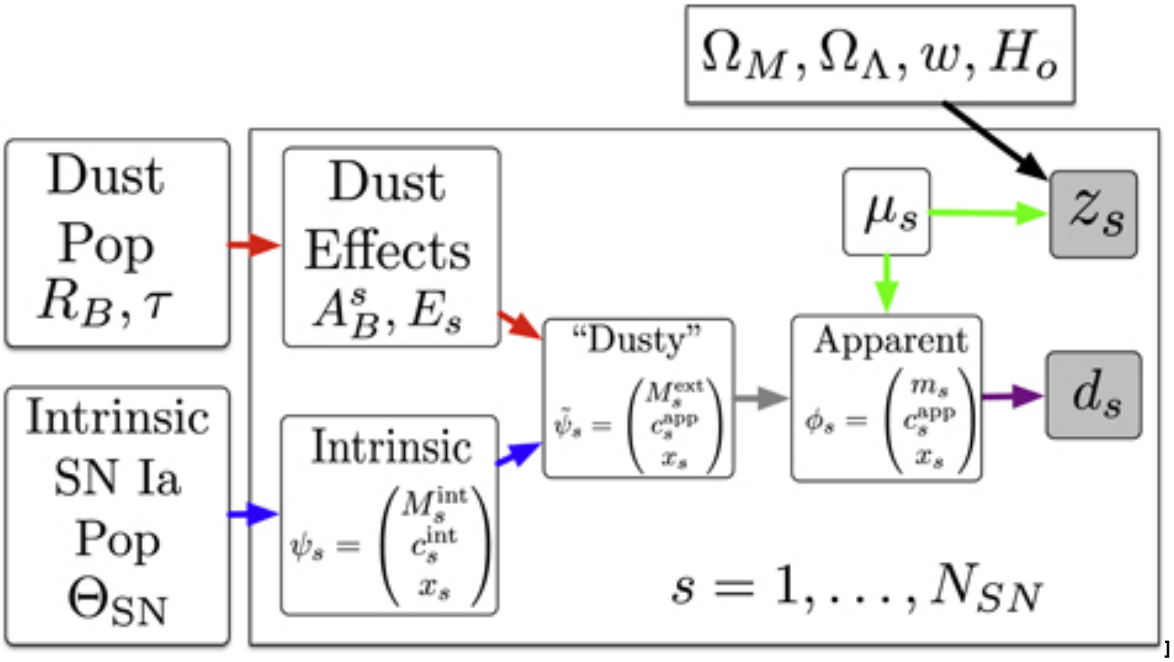

Figure 1: A probabilistic graphical model describing a hierarchical Bayesian model for Type Ia supernovae (Mandel+2017). A goal of this project is to explore distributed inference with models of this type through "nuisance-free likelihood" analysis of subsets of the full supernova dataset.

Project 66: X-ray Variability of Narrow-Line Seyfert 1 Galaxies in the 4XMM-DR14 Catalogue

Supervisor I: Dr David Homan (dsh54@cam.ac.uk)

UTO: Professor Richard McMahon (rm@ast.cam.ac.uk)

Project summary

Narrow-Line Seyfert 1 galaxies (NLSy1s) represent an extreme class of active galactic nuclei (AGN) powered by relatively low-mass supermassive black holes (~10^6–10^7 solar masses) accreting close to or above the Eddington limit. These systems exhibit unusually rapid and large-amplitude X-ray variability, making them key laboratories for probing the physics of accretion on human observable time-scales. It also makes NLSy1s very suitable to study the behaviour and geometry of the X-ray-emitting region in AGN, known as the corona. The 2024 release of the stacked source catalogue for XMM-Newton, and ESA X-ray satellite, has made available long-baseline, pre-computed X-ray fluxes and spectral diagnostics for over 300,000 sources, including many NLSy1s. This has opened up an opportunity for large-scale, multi-epoch studies of NLSy1 variability in the soft X-ray regime.

This project will exploit the catalogue's (4XMM-DR14) data set to construct and analyse a sample of NLSy1 galaxies cross-matched with a newly published, homogeneous optical catalogue based on SDSS DR17. The aim is to characterise how their X-ray variability amplitude and spectral hardness evolve over time, and to test whether these highly accreting AGN follow or diverge from the trends observed in more typical broad-line Seyfert galaxies. By establishing statistical relationships between variability, black hole mass, and accretion rate, the project will deliver new insight into the mechanisms governing high-Eddington accretion and X-ray emission in AGN.

Background

NLSy1s are distinguished by their narrow Balmer emission lines, strong optical Fe II emission, and weak [O III] lines, a combination which is typically interpreted as due to both low black hole masses and very high accretion rates. Their compact central engines imply shorter timescales than in standard Seyfert 1 galaxies, making them ideal for studying accretion disc instabilities and coronal heating/cooling on observationally tractable time baselines. Their rapid variability also makes them an important link between AGN and Galactic black hole X-ray binaries, potentially unifying our understanding of accretion across mass scales.

Despite their importance, systematic X-ray studies of NLSy1s have been limited by sample size and energy coverage. Most recent population-level work has relied on Swift data, which are typically soft-band limited and relatively shallow. The new 4XMM-DR14s catalogue—comprising stacked and multi-epoch fluxes, hardness ratios, and variability diagnostics derived from 20+ years of XMM-Newton

observations—offers a transformative opportunity to study NLSy1 X-ray behaviour at both higher energies and higher signal-to-noise than previously possible.

This effort is further enabled by the recent publication of a clean, homogeneous SDSS DR17-based catalogue of over 22,000 NLSy1s, providing accurate optical classifications, redshifts, and black hole mass proxies. Together, these data sets make it possible to construct a statistically significant sample of NLSy1s with well-characterised X-ray and optical parameters, as well as to compare them directly with a control sample of broad-line Seyferts selected from the same parent survey.

Project details

The student will begin by cross-matching the SDSS DR17 NLSy1 catalogue with the 4XMM-DR14s source list, using positional matching and X-ray characteristics to ensure reliable identification. This will yield a sample of NLSy1s with high-quality flux measurements and hardness ratios. The student will next define a subset of objects with multiple epochs of XMM data, resulting in a dataset with X-ray light curves across multiple epochs, in many cases spanning a baseline of ten to twenty years. A corresponding control sample of broad-line Seyfert 1s (also drawn from the SDSS-based catalogue) will be constructed to match the redshift and luminosity distributions of the NLSy1s, enabling differential comparisons free from major observational biases.

The core of the project will focus on quantifying and interpreting the ensemble variability properties of the NLSy1 population. The student will use the catalogue-provided measures of fractional variability, χ^2 variability probabilities, and hardness ratios to explore how the amplitude and spectral character of X-ray variability depend on black hole mass, Eddington ratio, and other optical parameters. In particular, they will investigate whether NLSy1s deviate from the well-established variability–mass relation observed in broader AGN samples, and whether changes in spectral hardness correlate with flux variations, suggesting shifts in coronal temperature or obscuration.

A possible extension of the project will focus on variability on short timescales. For NLSy1s in XMM-DR14s with sufficiently long individual exposures, the student will access the XMM-Newton science archive and search for the short-term light curves extracted for these sources. Based on these light curves it is possible to define similar statistics of variability as calculated for the long-term variability in XMM-DR14s (FVAR, FRATIO). Similar to the initial part of the project, these measures can be used to establish potential correlations among physical parameters derived from the optical spectra and X-ray variability. Another possible line of additional investigation is to further crossmatch the catalogues with Gaia data, with the aim of comparing the X-ray data with measures of optical photometric variability.

If time permits, the brightest subset of the sample may be also used for limited spectral modelling. This will mean first accessing the underlying X-ray data and extracting the spectra. These spectra can then be modelled in Xspec, possibly using the Bayesian fitting algorithms of BXA. A variety of models may be tests, starting with a simple absorbed power-law and testing potential additional components. These models probe the underlying emission mechanisms and their evolution over time in more detail.

The result will be a robust, data-driven study of one of the most variable classes of AGN in the X-ray sky, contributing both to our understanding of accretion physics and to future target selection for high-resolution X-ray missions.

Prerequisite Skills required
A familiarity with common statistical methods and basic coding skills.

Other skills used in the project
Much of the project will be working with large datasets, in table format, working on crossmatching catalogues; visualising the analysis will also play an important role in the project. Time permitting and based on student interest, it will be possible to develop skills in the analysis of X-ray spectra. This is not a requirement for starting the project, however.

Acceptable Programming Languages
Python (familiarity with R is also acceptable)

Useful References (List of important papers/review articles relevant to the project)

- X-ray perspective of Narrow-line Seyfert 1 galaxies [https://ui.adsabs.harvard.edu/abs/2018rnls.confE..34G/abstract]
- Supermassive Black Holes in X-rays: From Standard Accretion to Extreme Transients
- [https://ui.adsabs.harvard.edu/abs/2025arXiv250322791K/abstract]
- **General References (List of papers referred to in the project)**
- The XMM-Newton serendipitous survey. X. The second source catalogue from overlapping XMM-Newton observations and its long-term variable content [https://ui.adsabs.harvard.edu/abs/2020A%26A...641A.137T/abstract]
- Narrow-line Seyfert 1 galaxies in Sloan Digital Sky Survey: a new optical spectroscopic catalogue [https://ui.adsabs.harvard.edu/abs/2024MNRAS.527.7055P/abstract]
- The variability and soft X-ray excess properties of narrow-line Seyfert 1 galaxies: The Swift view [https://ui.adsabs.harvard.edu/abs/2022A%26A...659A.172D/abstract

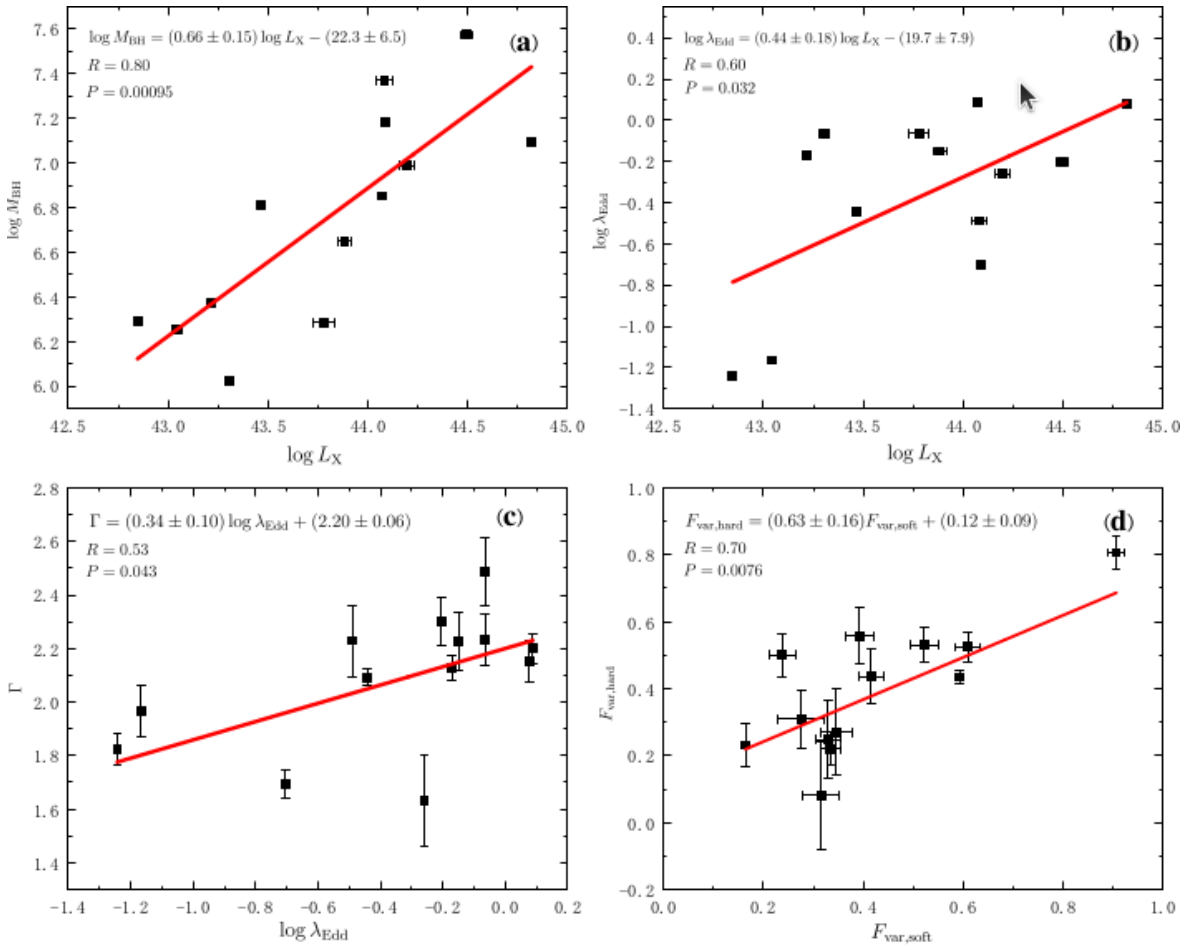

Figure 2: Correlations between important parameters for NLSy1s. These values derived from an older SDSS sample and Swift-XRT data. a) Mass of the central black hole vs. X-ray luminosity. b) Eddington ratio (a measure of accretion rate) vs. X-ray luminosity. c) X-ray power-law index (a spectral measurement) vs. Eddington ratio. d) X-ray variability vs. Eddington ratio. We will aim to combine optical and X-ray information in a similar way, but using a larger dataset. Figure from Ding et al., 2021,A&A 659, A172.

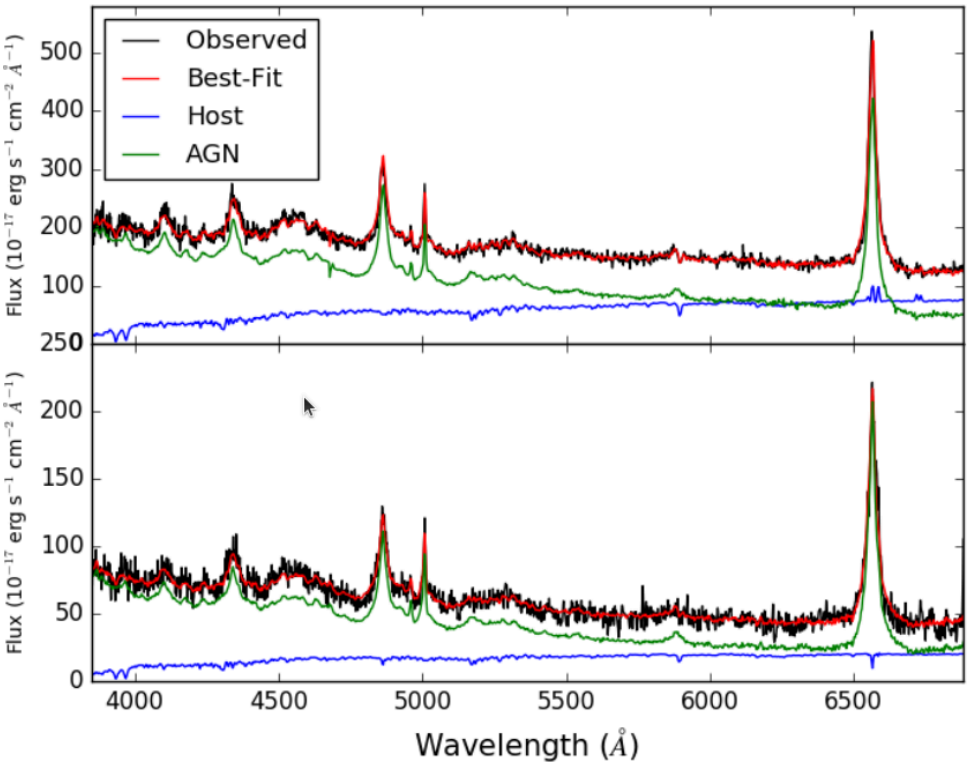

Figure 3: Typical optical spectra for NLSy1s. Note the significant difference in the relative strength of Hbeta and [OIII], around 5000 Angstrom. Figure from Chen et al., 2019, A&A, 615, A167.